CompTIA Network+ — Detailed Summary Notes

Network+ Exam Overview and Study Resources

Exam domains mentioned

Domain 4 — Network Security: 14% exam weight.

Domain 5 — Network Troubleshooting: 24% exam weight.

Recommended study resources

- CompTIA Network+ Certification Objectives: A free document from CompTIA containing the exam objectives and the topics candidates need to know.
- Professor Messer's website: professormesser.com/objectives
- Network+ books: Additional study material to support the video course.
- Professor Messer Discord: Community for discussing A+, Network+, Security+, and other certifications, including ad hoc study sessions.
- Additional Network+ resources: professormesser.com/netplus
- YouTube: Course replays, monthly live study groups, interactive Q&A, and aftershows. Live sessions can be accessed through the website calendar.

OSI Model (Open Systems Interconnection)

Introduction to the OSI model

The OSI model is a reference model used to describe how data travels through a network.

- It provides a broad overview of network communication rather than a detailed description of the data itself.
- It is not the OSI protocol suite.
- Most modern network protocols are based on TCP/IP, but the OSI model can be applied to TCP/IP and many other protocols.
- Multiple protocols can operate at a single OSI layer. Each layer may contain tens or hundreds of protocols.

Purpose of the OSI model

The OSI model provides a common language for IT professionals to discuss network operations and troubleshooting.

For example:

- Layer 1 problem: A physical connection or signal issue.
- Layer 4 problem: A transport-related issue.
- Layer 7 problem: An application-related issue.

This terminology allows IT professionals in different organizations to understand the type of network problem being discussed.

The seven OSI layers

The OSI model contains seven layers, numbered from Layer 1 at the bottom to Layer 7 at the top.

OSI model — Top to bottom

Mnemonic, from Layer 7 down to Layer 1:

All People Seem To Need Data Processing

Application → Presentation → Session → Transport → Network → Data Link → Physical

Layer 1 — Physical layer

The physical layer describes the physical signals used to transmit data across a network.

Main components

- Copper cables.
- Fiber-optic connections.
- Wireless signals.
- Network adapter cards and physical interfaces.

Function

It is responsible for transmitting a signal from one point of the network to another.

At this layer, the focus is on whether the physical signal can successfully travel through the medium.

Common Layer 1 problems

- Damaged or faulty cables.
- Faulty fiber connections.
- Wireless interference.

- Defective network adapter cards.
- Physical device or connection failures.

Troubleshooting methods

- Loopback testing.
- Testing cables and fiber connections.
- Checking network adapter cards.
- Checking the physical operation of network devices.

Key identification: Cables, electrical signals, fiber, wireless signal problems and physical connectivity belong to Layer 1.

Layer 2 — Data Link layer

The Data Link layer provides communication between devices on a network and handles hardware addressing and frame forwarding.

MAC address

MAC stands for Media Access Control.

A MAC address is the hardware address associated with a network adapter, such as an Ethernet or wireless adapter.

It is not referring to an operating system.

The Data Link Control (DLC) layer is associated with network cards and hardware addressing.

Address types mentioned:

- MAC addresses.
- EUI-48 addresses.
- EUI-64 addresses.

EUI stands for Extended Unique Identifier.

Switching

Network switches use destination MAC addresses to determine where to forward traffic.

For this reason, Layer 2 is also known as the switching layer.

Common Layer 2 subjects

- Ethernet frames.

- Source and destination MAC addresses.
- Network switching.
- Ethernet and wireless network adapters.

Key identification: MAC addresses, Ethernet frames and switching belong to Layer 2.

Layer 3 — Network layer

The Network layer is responsible for IP addressing and routing traffic between different networks.

Routing

Routers operate at Layer 3 and use the destination IP address to determine the next hop for network traffic.

They forward data between different IP subnets.

IP addressing

Layer 3 includes:

- IP addresses.
- Subnet masks.
- Routing decisions.
- Packet fragmentation.

Fragmentation

When data must cross a network that supports smaller frames than the local network, the data can be divided into smaller pieces.

These pieces travel across the network and are reassembled at the destination.

Key identification: IP addresses, subnet masks, routing and fragmentation belong to Layer 3.

Layer 4 — Transport layer

The Transport layer is responsible for transporting information between devices.

It can be thought of as the post office layer, responsible for delivering information from one endpoint to another.

Main protocols

TCP: Transmission Control Protocol.

UDP: User Datagram Protocol.

TCP and UDP: Both are commonly used to transport information contained within IP packets.

Functions

- Transporting data between devices.
- Dividing large amounts of data into smaller pieces.
- Reassembling the pieces at the destination.
- Using port numbers to identify communication endpoints.

Ports

TCP and UDP port numbers are associated with Layer 4.

For example, TCP port 443 is commonly associated with HTTPS communication.

Key identification: TCP, UDP and port numbers belong to Layer 4.

Layer 5 — Session layer

The Session layer manages communication sessions between two endpoints.

Main functions

- Initiating sessions.
- Managing communication sessions.
- Ending sessions.
- Restarting sessions.
- Supporting certain control protocols and tunneling functions.

A session establishes and manages the communication between point A and point B.

Key identification: Session initiation, termination, restarting and communication management belong to Layer 5.

Layer 6 — Presentation layer

The Presentation layer prepares data into a format that can be understood and presented to the user or application.

Main functions

- Character encoding.
- Data formatting.

- Application-level encryption.
- Decryption.

The presentation layer is commonly associated with SSL/TLS encryption and decryption in the OSI model described in the video.

It is also closely associated with the Application layer.

Key identification: Data representation, encoding, encryption and decryption belong to Layer 6.

Layer 7 — Application layer

The Application layer is the layer closest to the user and the applications they interact with.

Main functions

- Providing application-level network services.
- Supporting communication between applications.
- Handling application protocols and messages.

Protocols mentioned

HTTP / HTTPS: Used for web communication.

FTP: Used for file transfer between systems.

DNS: Used for resolving domain names into IP addresses.

POP3: An application-layer protocol used for receiving emails.

The application itself, its messages, and the information displayed to the user are associated with Layer 7.

Key identification: Application protocols, web services and user-facing application communication belong to Layer 7.

OSI Model in Real-World Networking

Mapping real-world components to OSI layers

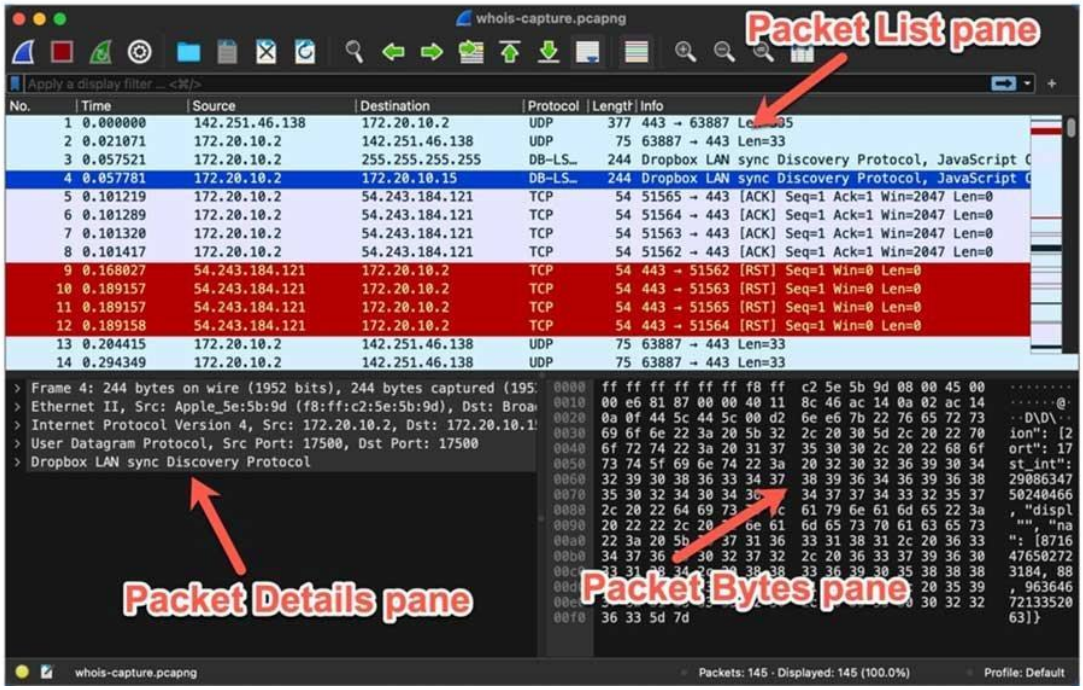
- Layer 1 — Physical: Examples include copper cables, fiber, and wireless signals. Main identifiers are physical signals and connectivity.
- Layer 2 — Data Link: Examples include Ethernet, wireless adapters, and switches. Main identifiers are MAC addresses and Ethernet frames.
- Layer 3 — Network: Examples include routers and IP networks. Main identifiers are IP addresses and subnet masks.

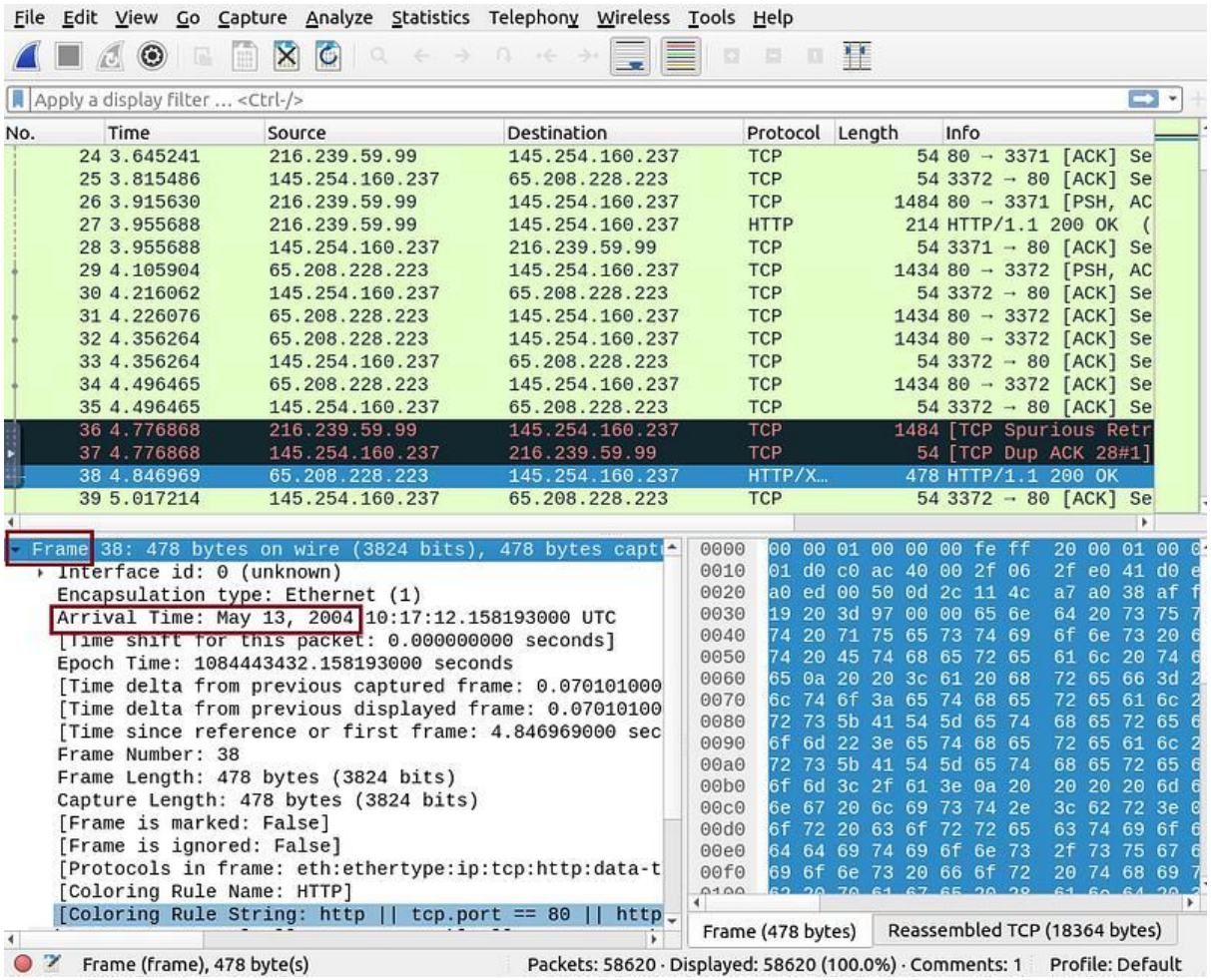
- Layer 4 — Transport: Examples include TCP and UDP communication. Main identifiers are TCP/UDP port numbers.
- Layer 5 — Session: Examples include session management, control protocols, and tunneling. Main identifiers are session initiation and termination.
- Layer 6 — Presentation: Examples include SSL/TLS encryption, decryption, and data encoding. Main identifiers are data format and encryption.
- Layer 7 — Application: Examples include HTTP, HTTPS, FTP, DNS, and applications. Main identifiers are application data and messages.

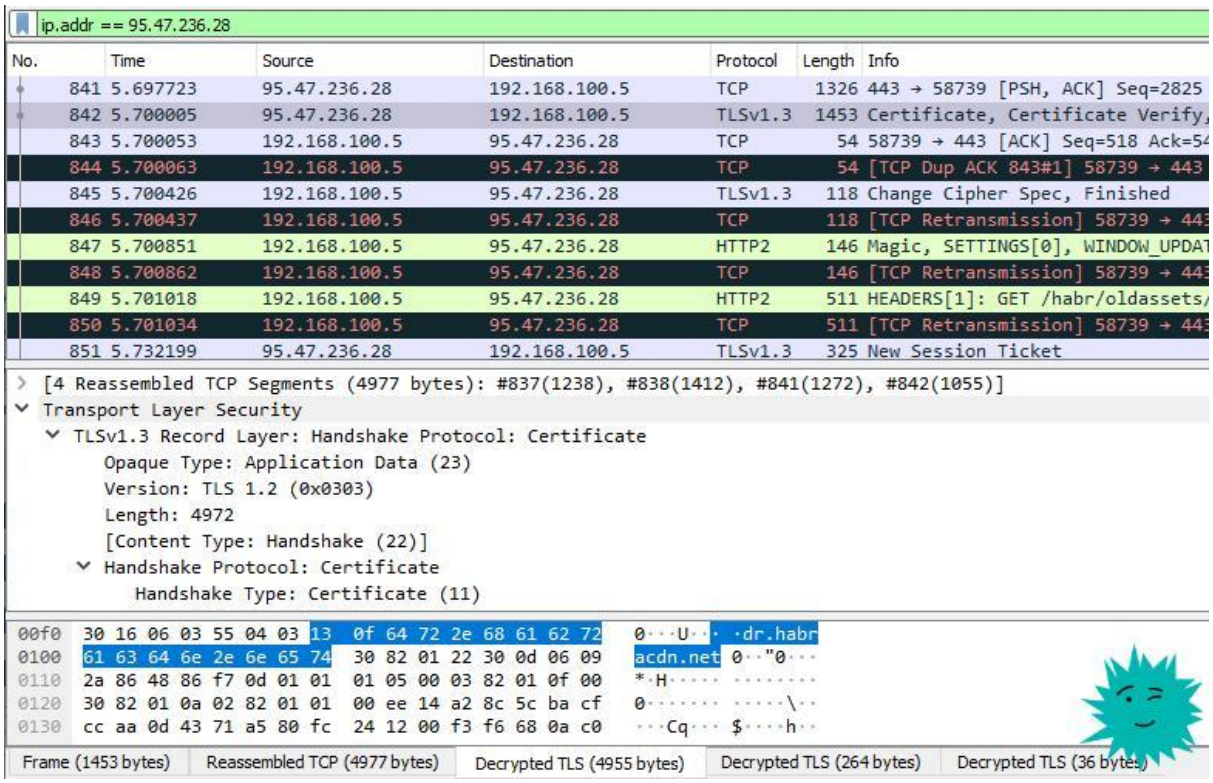
Understanding OSI encapsulation using Wireshark

Wireshark can capture network traffic and display the protocols and information contained within a frame.

The video describes three main Wireshark windows:




File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
24 3.645241 216.239.59.99 145.254.160.237 TCP 54 80 → 3371 [ACK] Se
25 3.815486 145.254.160.237 65.208.228.223 TCP 54 3372 → 80 [ACK] Se
26 3.915630 216.239.59.99 145.254.160.237 TCP 1484 80 → 3371 [PSH, AC
27 3.955688 216.239.59.99 145.254.160.237 HTTP 214 HTTP/1.1 200 OK (
28 3.955688 145.254.160.237 216.239.59.99 TCP 54 3371 → 80 [ACK] Se
29 4.105904 65.208.228.223 145.254.160.237 TCP 1434 80 → 3372 [PSH, AC
30 4.216062 145.254.160.237 65.208.228.223 TCP 54 3372 → 80 [ACK] Se
31 4.226076 65.208.228.223 145.254.160.237 TCP 1434 80 → 3372 [ACK] Se
32 4.356264 65.208.228.223 145.254.160.237 TCP 1434 80 → 3372 [ACK] Se
33 4.356264 145.254.160.237 65.208.228.223 TCP 54 3372 → 80 [ACK] Se
34 4.496465 65.208.228.223 145.254.160.237 TCP 1434 80 → 3372 [ACK] Se
35 4.496465 145.254.160.237 65.208.228.223 TCP 54 3372 → 80 [ACK] Se
36 4.776868 216.239.59.99 145.254.160.237 TCP 1484 [TCP Spurious Retr
37 4.776868 145.254.160.237 216.239.59.99 TCP 54 [TCP Dup ACK 28#1]
38 4.846969 65.208.228.223 145.254.160.237 HTTP/X... 478 HTTP/1.1 200 OK
39 5.017214 145.254.160.237 65.208.228.223 TCP 54 3372 → 80 [ACK] Se
Frame 38: 478 bytes on wire (3824 bits), 478 bytes capt
Interface id: 0 (unknown)
Encapsulation type: Ethernet (1)
Arrival Time: May 13, 2004 10:17:12.158193000 UTC
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1084443432.158193000 seconds
[Time delta from previous captured frame: 0.070101000
[Time delta from previous displayed frame: 0.07010100
[Time since reference or first frame: 4.846969000 sec
Frame Number: 38
Frame Length: 478 bytes (3824 bits)
Capture Length: 478 bytes (3824 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:http:data-t
[Coloring Rule Name: HTTP]
[Coloring Rule String: http || tcp.port == 80 || http
0000 00 00 01 00 00 00 fe ff 20 00 01 00
0010 01 d0 c0 ac 40 00 2f 06 2f e0 41 d0
0020 a0 ed 00 50 0d 2c 11 4c a7 a0 38 af
0030 19 20 3d 97 00 00 65 6e 64 20 73 75
0040 74 20 71 75 65 73 74 69 6f 6e 73 20
0050 74 20 45 74 68 65 72 65 61 6c 20 74
0060 65 0a 20 20 3c 61 20 68 72 65 66 3d
0070 6c 74 6f 3a 65 74 68 65 72 65 61 6c
0080 72 73 5b 41 54 5d 65 74 68 65 72 65
0090 6f 6d 22 3e 65 74 68 65 72 65 61 6c
00a0 72 73 5b 41 54 5d 65 74 68 65 72 65
00b0 6f 6d 3c 2f 61 3e 0a 20 20 20 20 6d
00c0 6e 67 20 6c 69 73 74 2e 3c 62 72 3e
00d0 6f 72 20 63 6f 72 72 65 63 74 69 6f
00e0 64 64 69 74 69 6f 6e 73 2f 73 75 67
00f0 69 6f 6e 73 20 66 6f 72 20 74 68 69
Frame (478 bytes)
Reassembled TCP (18364 bytes)
Frame (frame), 478 byte(s)
Packets: 58620 · Displayed: 58620 (100.0%) · Comments: 1
Profile: Default


ip.addr == 95.47.236.28
No. Time Source Destination Protocol Length Info
841 5.697723 95.47.236.28 192.168.100.5 TCP 1326 443 → 58739 [PSH, ACK] Seq=2825
842 5.700005 95.47.236.28 192.168.100.5 TLSv1.3 1453 Certificate, Certificate Verify,
843 5.700053 192.168.100.5 95.47.236.28 TCP 54 58739 → 443 [ACK] Seq=518 Ack=54
844 5.700063 192.168.100.5 95.47.236.28 TCP 54 [TCP Dup ACK 843#1] 58739 → 443
845 5.700426 192.168.100.5 95.47.236.28 TLSv1.3 118 Change Cipher Spec, Finished
846 5.700437 192.168.100.5 95.47.236.28 TCP 118 [TCP Retransmission] 58739 → 443
847 5.700851 192.168.100.5 95.47.236.28 HTTP2 146 Magic, SETTINGS[0], WINDOW_UPDAT
848 5.700862 192.168.100.5 95.47.236.28 TCP 146 [TCP Retransmission] 58739 → 443
849 5.701018 192.168.100.5 95.47.236.28 HTTP2 511 HEADERS[1]: GET /habr/oldassets/
850 5.701034 192.168.100.5 95.47.236.28 TCP 511 [TCP Retransmission] 58739 → 443
851 5.732199 95.47.236.28 192.168.100.5 TLSv1.3 325 New Session Ticket
[4 Reassembled TCP Segments (4977 bytes): #837(1238), #838(1412), #841(1272), #842(1055)]
Transport Layer Security
TLSv1.3 Record Layer: Handshake Protocol: Certificate
Opaque Type: Application Data (23)
Version: TLS 1.2 (0x0303)
Length: 4972
[Content Type: Handshake (22)]
Handshake Protocol: Certificate
Handshake Type: Certificate (11)
00f0 30 16 06 03 55 04 03 13 0f 64 72 2e 68 61 62 72 0···U··· ·dr.habr
0100 61 63 64 6e 2e 6e 65 74 30 82 01 22 30 0d 06 09 acdn.net 0··"0···
0110 2a 86 48 86 f7 0d 01 01 01 05 00 03 82 01 0f 00 *·H····· ·····
0120 30 82 01 0a 02 82 01 01 00 ee 14 a2 8c 5c ba cf 0····· ·····\··
0130 cc aa 0d 43 71 a5 80 fc 24 12 00 f3 f6 68 0a c0 ···Cq··· $····h··
Frame (1453 bytes)
Reassembled TCP (4977 bytes)
Decrypted TLS (4955 bytes)
Decrypted TLS (264 bytes)
Decrypted TLS (36 bytes)

1. Top window — Frame-by-frame breakdown of captured network traffic.
2. Middle window — Detailed protocol information for the selected frame.
3. Bottom window — Hexadecimal and ASCII representation of the data.

Example: An HTTPS communication to Google Mail

Wireshark displays multiple protocol layers within the same captured frame.

Frame 88: Shows 2,005 bytes on the wire and captured — **Layer 1 (Physical Layer)**.

Ethernet II: Shows source and destination MAC addresses — **Layer 2 (Data Link Layer)**.

Internet Protocol (IP): Shows source and destination IP addresses — **Layer 3 (Network Layer)**.

Transmission Control Protocol (TCP): Shows source and destination TCP ports — **Layer 4 (Transport Layer)**.

SSL/TLS: Shows encrypted communication and session-related information — **Layers 5–6 (Session and Presentation Layers)**.

HTTPS / Google Mail: Shows application-level communication — **Layer 7 (Application Layer)**.

The example also shows the IP address 72.14.247.19 associated with googlemail.l.google.com.

The destination TCP port is 443, which Wireshark identifies as HTTPS communication.

How to interpret the protocol decode

- The frame information describes the captured frame and its size.
- Ethernet II identifies the Data Link layer information.
- IP identifies the Network layer information.
- TCP identifies the Transport layer information.
- SSL/TLS is associated with session and presentation functions in the video's OSI explanation.
- Google Mail and HTTPS represent the application-level communication.

The protocol decode shows that a single communication contains information associated with multiple OSI layers.

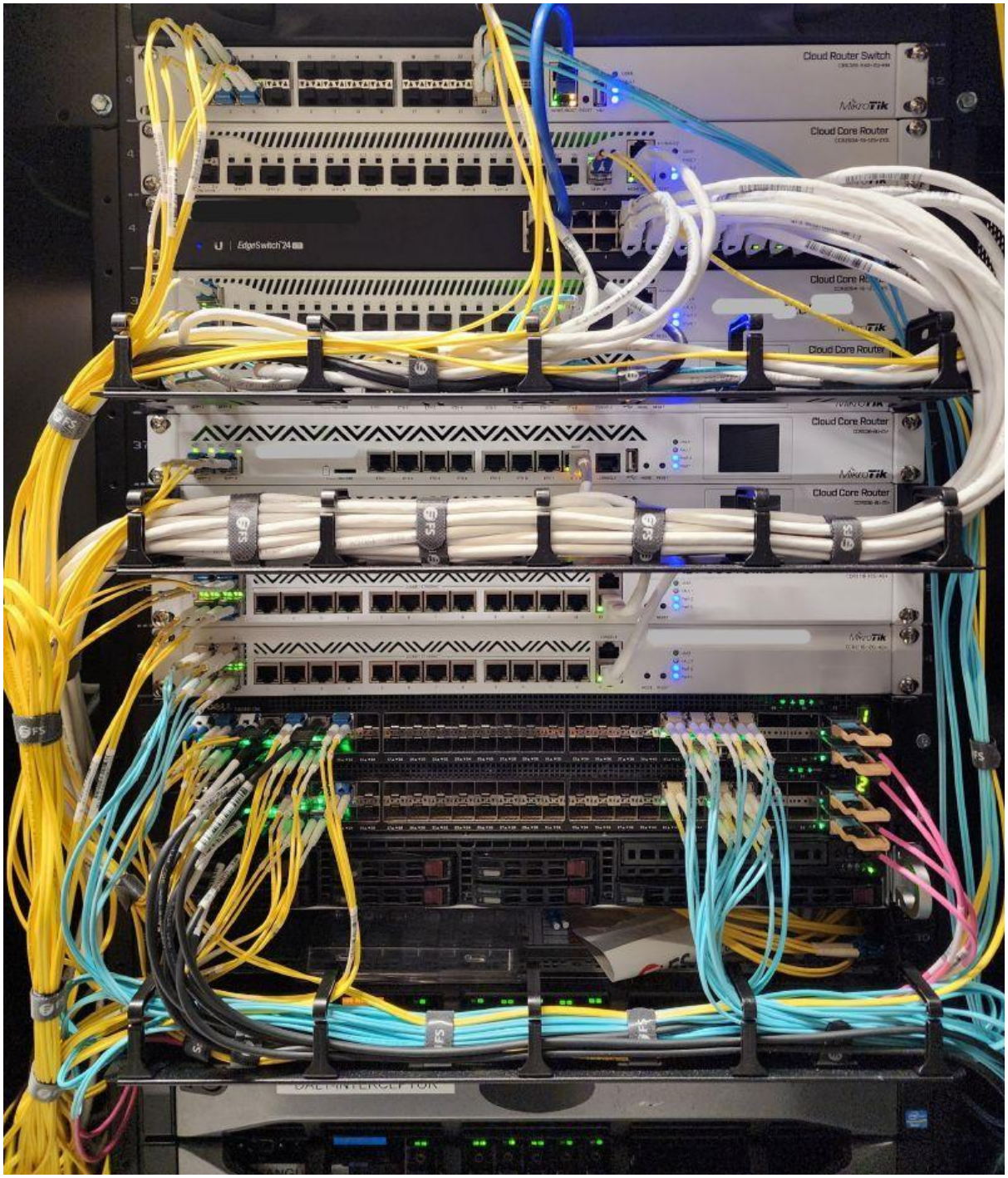
Important concept: As you move through the encapsulated protocol information, you can identify the functions associated with the different OSI layers.

The OSI model is a conceptual reference for describing network communication and troubleshooting, rather than a requirement that every protocol operates exclusively within one perfectly isolated layer.

Network Devices and Their Functions

Network devices work together to transfer data between different parts of a network. Each device is installed to perform a particular function.

Router



A router forwards traffic between different IP subnets.

Functions

- Connects different IP networks.
- Uses destination IP addresses to determine the next hop.
- Routes traffic between local networks and wide area networks.
- Connects copper-based and fiber-based networks.

- Can connect networks in different geographical locations.

OSI layer

Layer 3 — Network layer.

Layer 3 switch

A Layer 3 switch combines two functions in one physical device:

- Layer 2 switching based on MAC addresses.
- Layer 3 routing based on IP addresses.

The device includes both switching and routing functionality. The term Layer 3 switch refers to this combination.

Network switch



A network switch forwards traffic using MAC addresses.

Functions

- Forwards Ethernet traffic to the appropriate destination.
- Operates primarily in hardware.
- Uses an Application-Specific Integrated Circuit (ASIC) for hardware-based functionality.

- Connects devices within a local network.

Power over Ethernet (PoE)

PoE allows power to be delivered over the same Ethernet cables used for network communication.

This supports devices that can receive both power and network connectivity through the Ethernet connection.

OSI layer

Layer 2 — Data Link layer.

A switch that also performs routing can be referred to as a Layer 3 switch.

Firewall



A firewall manages and filters network traffic between different parts of a network.

Traditional firewall

A traditional firewall filters traffic based on TCP or UDP port numbers.

Next-Generation Firewall (NGFW)

An NGFW can identify applications traversing the network and control whether those applications are allowed.

It provides application-aware traffic management.

Additional firewall functionality

VPN: Creates encrypted tunnels between remote users or sites and networks.

NAT: Translates network addresses between internal and external networks.

Routing: Forwards traffic between different networks.

Dynamic Routing: Uses dynamic routing protocols to automatically update routes.

Traffic Filtering: Controls communication between internal and external networks.

VPN example

A firewall at one remote site can establish an encrypted VPN tunnel with a firewall at another remote site.

This allows communication between the two networks over the tunnel.

Firewall placement

Firewalls are commonly positioned between the internal network and the internet connection.

They manage inbound and outbound traffic between the internal and external networks.

Key identification: Traffic filtering, VPN, NAT and application-aware filtering are associated with firewall functionality.

Intrusion Detection System (IDS) and Intrusion Prevention System (IPS)



IDS and IPS inspect network traffic for attacks and known vulnerabilities.

They can identify attack patterns involving operating systems and applications.

Attacks mentioned

- Buffer overflow attacks.
- Cross-site scripting (XSS).
- Exploits targeting known vulnerabilities.

IDS vs IPS

IDS — Intrusion Detection System

Detects suspicious or malicious traffic and generates alerts.

Detection and alerting

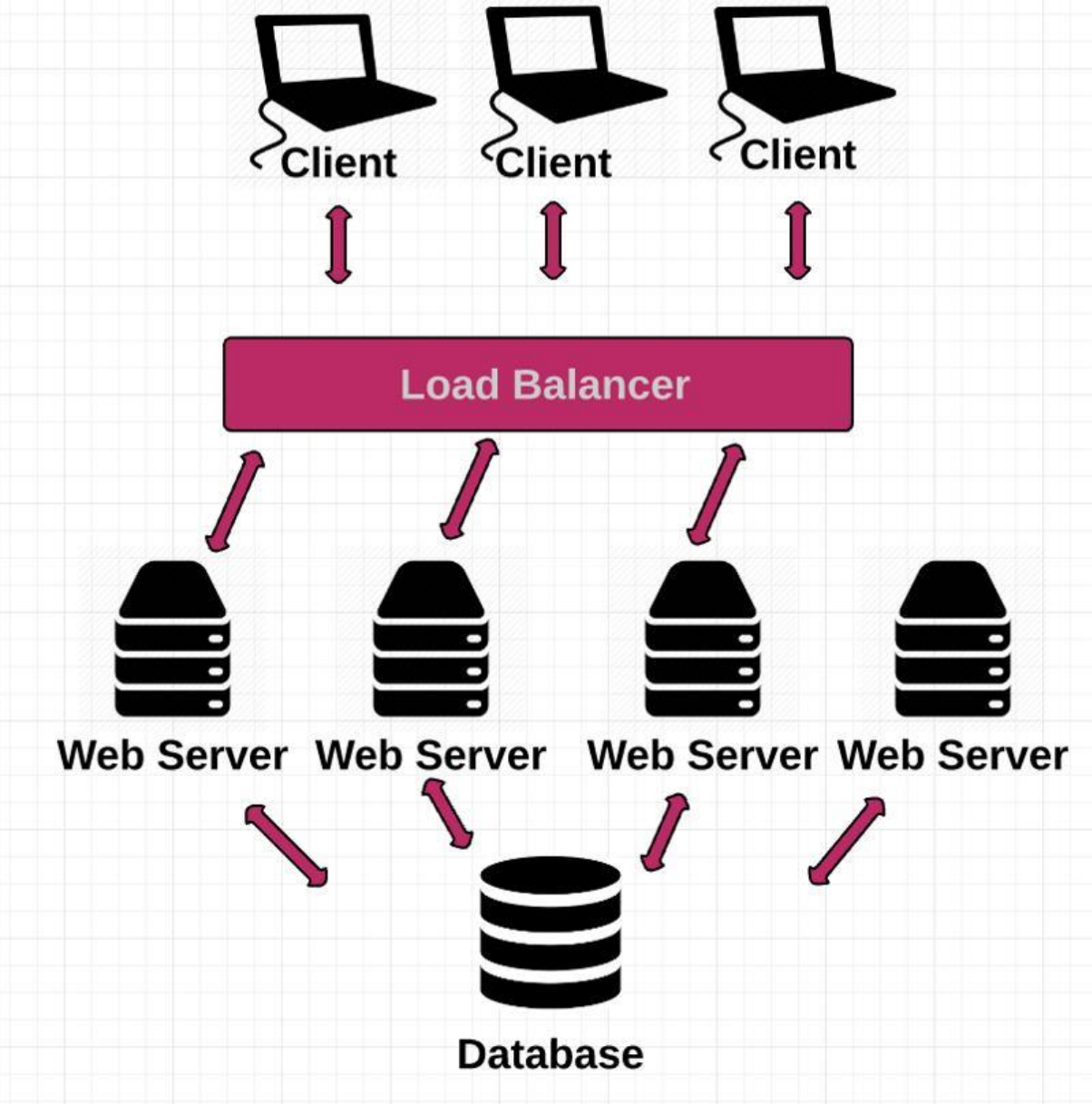
IPS — Intrusion Prevention System

Detects attacks and can block the identified malicious traffic before it reaches the internal network.

Detection and blocking

The material notes that IDS and IPS functionality can be integrated into modern next-generation firewalls or provided through standalone devices.

Load balancer



A load balancer distributes incoming traffic across multiple servers.

Main purpose

To support service availability and distribute the workload among multiple physical servers.

An end user may believe they are accessing a single server, while the load balancer distributes their requests among several servers in the data center.

Server health monitoring

The load balancer can identify server outages.

When a server fails because of hardware or software problems, the load balancer can:

1. Recognize the failed server.
2. Remove it from the active server rotation.
3. Continue directing requests to the remaining available servers.

Additional load balancer functions

TCP Offloading: Handles TCP communication to optimize communication with backend servers.

SSL Offloading: Performs SSL/TLS encryption and decryption instead of requiring each backend server to handle it.

Caching: Stores data to respond to repeated requests without contacting the backend server.

Quality of Service (QoS): Prioritizes certain types of network traffic.

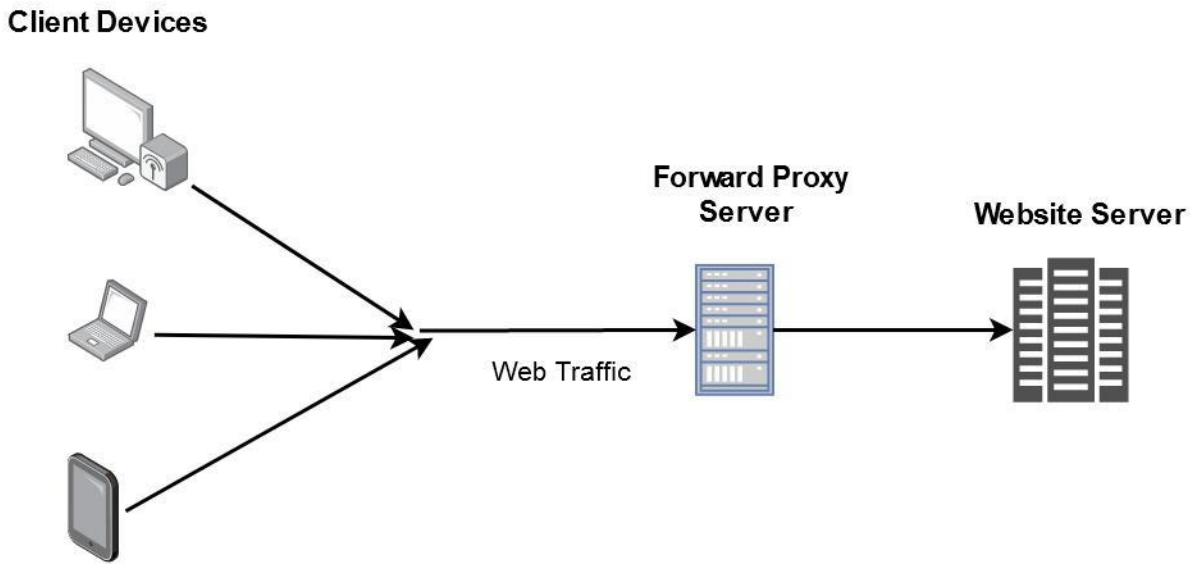
Application-Centric Load Balancing: Directs requests for specific pages or applications to designated servers.

Example

Internet users access a service through a load balancer, which distributes requests across a group of web or database servers.

The load balancer helps maintain availability when an individual server becomes unavailable.

Proxy server



A proxy sits between the user and the external server and makes requests on the user's behalf.

How a proxy works

1. The user sends a request to the proxy.
2. The proxy makes the request to the destination server.
3. The proxy receives the response.
4. The proxy can inspect the response for malicious software or code.
5. The proxy provides the response to the user.

Proxy functions

- Caching frequently requested information.
- User authentication using usernames and passwords.
- URL filtering.
- Content scanning.
- Checking for malicious software.
- Managing user access to internet resources.

Types of proxies

- Explicit proxy: Requires the operating system or application to be configured to use the proxy.
- Transparent proxy: Operates without requiring changes to the operating system or applications.
- Proxy: Acts as an intermediate point for managing user requests and responses.

Network-Attached Storage (NAS)



NAS provides centralized file-level storage over a network.

File-level access

NAS provides access to files as complete files.

When a system needs to modify information within a file, the file is transferred into the system's memory. After modification, the file is written back to the NAS.

Characteristics

- Centralized file storage.
- File-level access.
- Transfers the entire file for the described read and modification process.
- Can be placed on an isolated, high-bandwidth network.

Storage Area Network (SAN)



A SAN provides block-level access to storage.

Block-level access

Instead of transferring an entire file to modify a small part, a SAN allows the system to modify only the relevant storage blocks.

This can be more efficient when working with large files.

Wireless access point (AP)



An access point provides wireless connectivity between wireless devices and the wired network.

Functions

- Provides wireless communication.
- Bridges an 802.11 wireless network to an 802.3 Ethernet network.
- Commonly connects to the wired network through Ethernet.

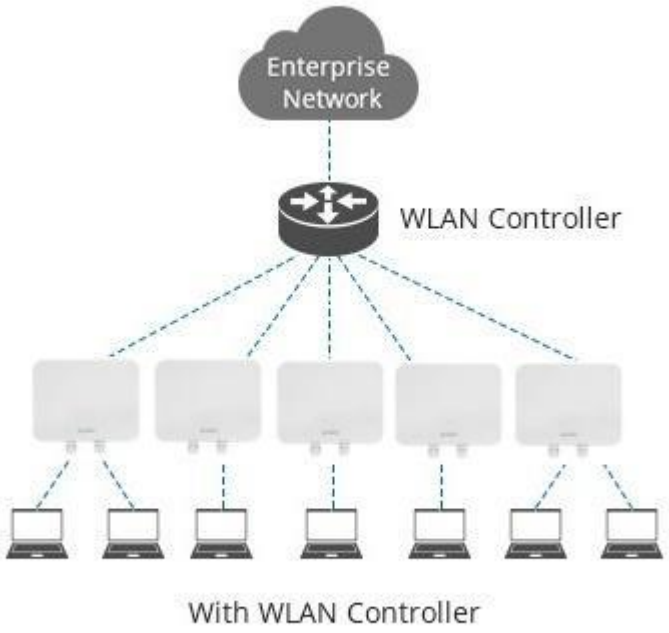
OSI layer

Layer 2 — Data Link layer.

An enterprise access point is typically a dedicated wireless device, unlike a home wireless router that may combine routing, switching and wireless access point functions.

Wireless LAN controller (WLC)

A Wireless LAN Controller centrally manages multiple wireless access points.



Main functions

- Centrally configures access points.
- Deploys full configurations to new access points.
- Manages wireless security settings and access policies.
- Supports seamless roaming between access points.
- Monitors wireless performance and security.
- Generates usage reports.
- Allows configuration changes to be deployed across multiple access points.
- Helps identify whether access point locations need to be updated.

A WLC provides centralized management through a single interface, often described as a single pane of glass.

Proprietary systems

Wireless LAN controllers are often proprietary. Access points from a particular manufacturer are commonly managed by a controller from the same manufacturer.

Cloud Computing and Network Virtualization

Cloud computing

Cloud computing changes how computing resources and applications are deployed and managed.

Key characteristics

Cloud computing provides on-demand deployment of applications and services, scalability to increase or decrease resources, elasticity to adjust resources based on demand, global access from anywhere in the world, and multitenancy, where multiple customers share the same infrastructure, improving resource utilization and cost efficiency. Cloud infrastructure supports virtual servers and virtual networks in addition to physical servers.

Network Function Virtualization (NFV)

Network Function Virtualization replaces physical networking devices with virtualized versions running in a virtualized environment.

Example: Migrating a data center to the cloud

A data center may contain 100 physical servers, redundant networking infrastructure, multiple firewalls and load balancers.

When these servers are migrated to cloud infrastructure:

1. The physical servers are replaced by virtual servers running on cloud hardware.
2. Networking functions are also virtualized.
3. Virtual routers, switches and firewalls provide the same basic functions as their physical counterparts.
4. Network interfaces and configurations can be managed through the hypervisor.
5. New virtual networking devices can be deployed and configured on demand.

Virtual network appliances

Examples include:

- Virtual routers.
- Virtual switches.

- Virtual firewalls.
- Virtual load balancers.

NFV provides flexibility when deploying applications, configuring network connectivity and connecting cloud infrastructures in different locations.

Virtual Private Cloud (VPC)

A Virtual Private Cloud is an isolated virtual network environment within a cloud infrastructure.

A cloud application instance may contain:

- Web servers.
- Database servers.
- Load balancers.
- Virtual switches.
- Virtual routers.
- Virtual firewalls.

Organizations may create separate VPCs for different applications or business units.

This allows the systems to remain separated while being managed as individual virtualized environments.

Cloud networking components

Transit gateway

A transit gateway connects multiple VPCs and enables communication between them.

It can be thought of as a cloud router.

Its purpose is to provide connectivity between separate virtual private clouds.

VPN connection

A VPN can connect a remote site or user to a transit gateway.

This allows remote users and sites to access private VPC networks through an encrypted tunnel.

Internet gateway

An internet gateway allows applications in a VPC to be accessible from the internet.

It is used when an application must be available to users anywhere on the internet.

NAT gateway

A VPC NAT gateway allows private network resources to communicate outbound over the internet using network address translation.

It allows private IP addresses to be translated to public IP addresses for outbound communication.

The configuration can permit outbound internet access without necessarily allowing inbound connections initiated by external devices.

VPC endpoint

A VPC endpoint provides a direct connection to resources in a VPC without requiring those resources to be accessible through the public internet.

It can be used to connect services or resources to private cloud networks.

For example, cloud storage may need to communicate with a virtual machine in a private subnet. A VPC endpoint can provide the connection without exposing the private subnet to the internet.

Cloud network security

Cloud providers provide security groups and network security lists to control traffic to and from cloud-based resources.

Network security lists

Network security lists define inbound and outbound traffic rules for virtual cloud networks.

Rules can specify:

- TCP or UDP protocols.
- Port numbers.
- Individual IP addresses.
- IP address ranges.
- CIDR blocks.
- IPv4 or IPv6 ranges.

The material explains that network security lists can be automatically assigned to multiple virtual cloud networks or subnets.

Limitation of network security lists

Network security lists may lack granularity because a rule can apply to multiple virtual cloud networks even when individual networks require different rules.

This can increase the difficulty of managing network security.

Network security groups (NSGs)

Network security groups provide more granular control.

They allow rules to be assigned to individual Virtual Network Interface Cards (VNICs).

For example:

- Network Security Group A: Inbound TCP port 443.
- Network Security Group B: Inbound TCP port 22.

These groups can be assigned to different network interfaces within the same subnet.

Cloud Deployment Models

Organizations can use public clouds, private clouds or a combination of both, depending on application access requirements.

Public cloud

A public cloud is suitable for applications that need to be accessed by users across the internet.

Private cloud

A private cloud is suitable for applications intended for internal use.

The organization may deploy these applications in its own virtualized local data center.

Hybrid cloud

A hybrid configuration uses both public and private cloud environments.

Organizations may use a combination of the two for different applications and requirements.

Cloud Service Models

There are three main cloud service models discussed in the material: SaaS, IaaS and PaaS.

Software as a Service (SaaS)

SaaS provides a complete, ready-to-use application through the cloud.

Characteristics

- Users log in through a browser or application.
- The provider manages the application.
- The provider manages the underlying infrastructure.
- Users do not need to install or upgrade the underlying software themselves.
- Application data is stored within the cloud service.

Examples mentioned

- Google Mail.
- Office 365.

SaaS allows users to access software without performing application development or infrastructure management.

Infrastructure as a Service (IaaS)

IaaS provides computing infrastructure through the cloud.

It is suitable for organizations that want to deploy and manage their own applications.

Characteristics

The cloud provider provides the underlying infrastructure, including:

- Physical data center.
- Physical network.
- Physical host hardware.

The customer is responsible for:

- Installing and managing software.
- Managing applications.
- Managing application data.
- Upgrading the software.
- Configuring and controlling the applications running on the infrastructure.

IaaS provides more control over the application environment and data than SaaS.

The material also refers to IaaS as Hardware as a Service (HaaS) in some contexts.

Platform as a Service (PaaS)

PaaS provides a cloud platform and development tools for building customized applications.

Characteristics

- The cloud provider manages the underlying platform and application engine.
- The customer develops and customizes applications.
- The customer maintains the applications built on the platform.
- The provider supplies the tools and building blocks needed for development.

Example mentioned

Salesforce provides tools for building customized applications on its platform.

PaaS is positioned between SaaS and IaaS because it provides a managed platform while allowing the customer to develop and manage their own applications.

Cloud Shared Responsibility Model

The shared responsibility model identifies which tasks are managed by the cloud provider and which are managed by the customer.

The division of responsibility depends on the service model.

On-premises infrastructure

In an on-premises environment, the customer manages everything, including:

- Data.
- Applications.
- Network infrastructure.
- Physical hosts.
- Physical data center.

Infrastructure as a Service

The cloud provider manages the physical data center, physical network and physical host infrastructure.

The customer manages the other components of the application environment, including software, applications and data.

Platform as a Service

The provider manages the underlying platform.

Some responsibilities, such as network controls, applications and identity or directory infrastructure, may be shared between the provider and customer.

The customer remains responsible for accounts, devices, information and data, as described in the material.

Software as a Service

The provider manages the application and the underlying infrastructure.

The customer remains responsible for the responsibilities assigned to them, including their accounts, devices and information or data.

Responsibility comparison

Who manages what?

The following summarizes the responsibility division described in the video.

The more managed the cloud service is, the more of the underlying infrastructure and application management is handled by the provider.

The customer still has responsibilities for the resources, accounts and data they use.

Final Revision Sheet

OSI Layer 1: Physical signals, cables, fiber, and physical troubleshooting.

OSI Layer 2: MAC addresses, Ethernet frames, and switching.

OSI Layer 3: IP addresses, subnet masks, and routing.

OSI Layer 4: TCP, UDP, and port numbers.

OSI Layer 5: Session management.

OSI Layer 6: Encoding, encryption, and decryption.

OSI Layer 7: Application protocols and user-facing applications.

Router: Routes traffic between IP subnets using destination IP addresses.

Switch: Forwards traffic using MAC addresses.

Firewall: Filters traffic and may provide NAT, VPN, and routing.

IDS: Detects attacks and generates alerts.

IPS: Detects and blocks attacks.

Load Balancer: Distributes requests among servers and removes failed servers from rotation.

Proxy: Makes requests on behalf of users and can filter, cache, and inspect content.

NAS: Provides file-level storage access.

SAN: Provides block-level storage access.

Access Point: Bridges wireless 802.11 and wired 802.3 networks.

WLC: Centrally manages multiple wireless access points.

NFV: Replaces physical network functions with virtualized network appliances.

Transit Gateway: Connects multiple VPCs.

Internet Gateway: Provides internet connectivity to VPC applications.

NAT Gateway: Enables outbound internet access through address translation.

VPC Endpoint: Provides private connectivity to cloud resources.

SaaS: Uses a complete application managed by the provider.

PaaS: Allows application development on a provider-managed platform.

IaaS: Provides provider-supplied infrastructure while the user manages software and applications.

Internet Protocol (IP) and Network Communication

The primary goal of networking is to transfer data from one device to another across a network. Internet Protocol (IP) is responsible for moving data packets from the source device to the destination device.

A moving-truck analogy helps explain this process:

- Network: Represents the roads used to transport data, such as wired Ethernet, wireless networks, and wide-area networks.
- Internet Protocol (IP): Represents the moving truck that carries data across the network.
- TCP/UDP: Represent the boxes placed inside the truck, containing application data.
- Application data: Represents the actual items inside the boxes that need to be delivered.

Data is packaged inside protocols, which are then encapsulated within larger protocols. This process is called encapsulation and is fundamental to network communication.

Encapsulation in an Ethernet frame

When a client communicates with a server, the data is organized into multiple layers:

1. Ethernet header: Contains Ethernet-level information.
2. IP header: Contains IP-level information within the Ethernet payload.
3. TCP or UDP header: Contains transport-layer information within the IP payload.
4. Application data: Contains information such as HTTP or website data.
5. Ethernet trailer: Appears at the end of the Ethernet frame.

The Ethernet frame contains the IP packet, which contains the TCP or UDP segment or datagram, which carries the application data.

A protocol decode of a network frame reveals the Ethernet header, IP header, TCP or UDP header, application data, and Ethernet trailer. This structure allows data to be transferred between devices using different protocols.

TCP and UDP — Transport Layer

TCP and UDP operate at OSI Layer 4, the Transport Layer. They allow multiple applications to communicate simultaneously between the same devices. This process is called multiplexing, which enables efficient communication between different applications over a network.

Transmission Control Protocol (TCP)

TCP is a connection-oriented protocol that establishes communication between two devices before transferring data and formally terminates the session when communication is complete.

TCP provides reliable delivery through acknowledgments. When a sender transmits data, the receiving device acknowledges successful receipt. If an acknowledgment is not received, the sender can assume that the data may not have arrived and retransmit it.

TCP uses sequence numbers to identify packets and help the receiver determine which data has been received and which data may be missing. This enables recovery of lost information without unnecessarily retransmitting all previously transmitted data.

TCP also supports flow control. The receiving device can communicate with the sender to adjust the transmission rate, allowing data to be sent faster or slower depending on the receiver's ability to process it.

Key TCP characteristics:

- Connection-oriented communication.
- Connection establishment and termination.
- Acknowledgments for reliable delivery.
- Sequence numbers to track data.
- Error recovery and retransmission.
- Flow control between sender and receiver.

User Datagram Protocol (UDP)

UDP is a connectionless protocol that does not require a formal session setup or teardown. A device can send UDP packets to another device without establishing a connection first.

UDP does not provide acknowledgments for delivered data. Therefore, the sender cannot guarantee that the destination received the information. UDP does not provide built-in retransmission, error recovery, or receiver-controlled flow control.

The term unreliable does not mean UDP inherently has a higher probability of communication failure than TCP. Both protocols can transfer data across the network with the same underlying probability of successful transmission. The difference is that UDP does not provide the acknowledgment and recovery mechanisms that TCP provides.

Key UDP characteristics:

- Connectionless communication.
- No formal session establishment or termination.
- No delivery acknowledgments.
- No built-in retransmission or error recovery.
- No receiver-controlled flow control.
- Minimal communication overhead compared with TCP.

IP Addresses, Ports, and Sockets

IP delivers packets to the destination device using its destination IP address. However, a single device can run many applications simultaneously, so the data must also be delivered to the correct application.

This is achieved using TCP or UDP port numbers.

The moving-truck analogy explains the distinction:

- IP address: Identifies the destination house.
- TCP/UDP port number: Identifies the room inside the house where the data must be delivered.
- Application: Receives the data delivered to its designated port.

For example, port 80 is commonly associated with HTTP, port 443 with HTTPS, port 123 with NTP, and port 25 with SMTP.

IP sockets

An IP socket is a combination of an IP address, a transport protocol, and an application port number.

A communication between a client and a server uses the following information:

- Client IP address.
- Client TCP or UDP protocol and port.
- Server IP address.
- Server TCP or UDP protocol and port.

Once both sides know these details, they can exchange information between the appropriate applications.

Well-known and ephemeral ports

Port numbers range from 0 to 65,535 for both TCP and UDP.

Non-ephemeral ports — 0–1,023: Commonly used as permanent or well-known service ports.

Ephemeral ports — 1,024–65,535: Commonly used as temporary client-side ports.

Servers commonly use non-ephemeral ports, while clients usually use ephemeral ports for temporary communication sessions. However, this is not a strict rule, and either a client or server can use any available port number.

TCP and UDP have separate port spaces. TCP port 80 and UDP port 80 are different endpoints, allowing separate applications to use the same numerical port with different transport protocols.

Port numbers and security

Port numbers identify applications and direct traffic to the appropriate service. They are not a security mechanism.

Changing an application's port number does not automatically increase or decrease network security. Firewalls are responsible for deciding whether traffic is allowed or denied.

For example, a web browser normally connects to web servers using TCP port 80 for HTTP or TCP port 443 for HTTPS. If a web server is configured to use a different port, clients must know and specify the new port to connect successfully.

Example of simultaneous application communication

Consider a network with a client at 10.0.0.1 and a server at 10.0.0.2.

The server provides three services:

Web Server — TCP 80: Used for web services.

Voice over IP — UDP 5004: Used for Voice over IP communication.

Email Service — TCP 143: Used for email services.

The client can use all three services simultaneously because each communication is distinguished by its transport protocol and port numbers.

For example:

- Web: Client source port 3000 → Server destination TCP port 80.
- Voice over IP: Client source port 7100 → Server destination UDP port 5004.
- Email: Client source port 4407 → Server destination TCP port 143.

When the server sends data back to the client, it reverses the source and destination IP addresses and the source and destination port numbers. This ensures that the response returns to the correct client application.

This combination of IP addresses, protocols, and ports allows network communication to scale across many devices and applications worldwide.

Common Network Protocols and Port Numbers

The following protocols are used for file transfer, remote access, email, network configuration, monitoring, and other networking functions.

File transfer and remote access

- FTP — TCP 20 and 21: File Transfer Protocol.
    - Port 20: Commonly associated with file transfer data.
    - Port 21: Used for control information.
- SSH — TCP 22: Secure Shell provides encrypted remote command-line access to devices.
- SFTP — TCP 22: Secure File Transfer Protocol transfers and manages files through SSH encryption.
- Telnet — TCP 23: Provides remote terminal communication but sends information in clear text without encryption.
- TFTP — UDP 69: Trivial File Transfer Protocol transfers simple files without authentication or login requirements by default.

FTP supports authentication with usernames and passwords, directory listings, file transfers, renaming, deletion, and other file-management operations.

SFTP uses SSH as its underlying protocol, allowing secure file transfers and file management through TCP port 22.

Telnet provides functionality similar to SSH for remote terminal access, but its lack of encryption means that captured packets can expose login credentials and other transmitted information. SSH is used instead when encrypted communication is required.

TFTP is designed for simple, efficient transfers with minimal overhead. One example is a voice-over-IP device obtaining an IP address through DHCP and then downloading its configuration file from a server using TFTP.

Email protocols

SMTP — TCP 25: Transfers email between mail servers and can also be used to send email from clients.

SMTP with TLS — TCP 587: Commonly used for encrypted email submission.

IMAP — TCP 143: Used to receive and manage email messages and inboxes.

POP3: Port not specified in the material; used to receive email messages.

SMTP stands for Simple Mail Transfer Protocol and is commonly used for server-to-server email transfers. SMTP on TCP port 25 is described as plain-text communication by default, while TCP port 587 is commonly used for SMTP with TLS encryption.

IMAP and POP3 are used for receiving email messages and managing access to an email inbox.

DNS and automatic IP configuration

DNS — UDP 53: Commonly handles small domain-name queries.

DNS — TCP 53: Commonly handles larger transfers between DNS servers.

DHCP — UDP 67 and 68: Automatically configures IP address settings for network devices.

NTP — UDP 123: Synchronizes clocks across network devices.

DNS (Domain Name System) translates fully qualified domain names (FQDNs) into IP addresses. Instead of memorizing IP addresses, users access resources through domain names. A DNS server responds to queries with the IP address required to communicate with the destination server.

DNS is a critical network service because devices rely on it to resolve domain names into usable IP addresses.

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP configuration to devices connecting to a network. A DHCP server maintains a pool of available IP addresses and assigns addresses to clients.

DHCP also uses lease times, which determine how long a device can use an assigned address. Clients can renew their leases when they are still using the address.

DHCP reservations allow specific devices to receive the same IP address each time they connect, commonly by associating the address with the device's MAC address.

NTP (Network Time Protocol) synchronizes the time and date of network devices, including operating systems, routers, and switches. It commonly uses UDP port 123.

Accurate time synchronization is important for correlating log files and reconstructing communications across different devices during investigations. Devices can synchronize their clocks multiple times per day, and the material describes approximately one millisecond of difference between devices on the same network.

Network management and monitoring

SNMP — UDP 161: Queries network devices and retrieves management information.

SNMP Traps — UDP 162: Sends notifications from devices to management stations.

LDAP — TCP 389: Queries and retrieves information from directory databases.

LDAPS — TCP 636: Secure version of LDAP.

Syslog — UDP 514: Transfers log data to a centralized logging system.

SNMP (Simple Network Management Protocol) allows network administrators to monitor and manage routers, switches, and other network devices.

A management station can query a device to retrieve information such as the amount of traffic that has passed through it. The collected information can be stored and used to create graphs, reports, and summaries.

SNMP versions have different capabilities:

- SNMPv1: Supports individual queries and responses, without encryption.
- SNMPv2: Supports bulk transfers of multiple variables, improving efficiency, but does not provide encryption.
- SNMPv3: Provides message integrity, authentication, and encryption.

SNMP traps allow devices to proactively send notifications to management stations, rather than waiting for a management station to initiate a query.

LDAP (Lightweight Directory Access Protocol) provides access to directory databases containing information about users, devices, and organizational resources. It uses a hierarchical structure with organizational units and entries such as users, databases, and storage devices.

LDAPS provides a secure version of LDAP.

Syslog is a standardized protocol for transferring log information across the network. It is commonly used to consolidate logs from routers, switches, firewalls, servers, and other devices into a central location.

Syslog is often used with a Security Information and Event Management (SIEM) system, which collects and stores logs in a centralized database for analysis and long-term retention.

File sharing, databases, and remote desktops

SMB — TCP 445: Provides file sharing, printer sharing, authentication, and permissions.

Microsoft SQL Server — TCP 1433: Commonly used for Microsoft SQL Server database communication.

RDP — TCP 3389: Provides remote desktop access to Windows devices.

SMB (Server Message Block) is integrated into Microsoft Windows and provides file sharing, printer sharing, file locking, authentication, and permissions management. It allows users to access network file shares and remote printers without requiring separate FTP software.

Earlier Windows versions used NetBIOS for this communication, while modern versions commonly communicate directly over IP using TCP port 445. SMB is also referred to as CIFS (Common Internet File System).

Microsoft SQL Server uses structured query language (SQL) to query and retrieve information from databases. Microsoft SQL Server commonly uses TCP port 1433, although other database products may use different port numbers.

RDP (Remote Desktop Protocol) allows users to view and interact with a remote desktop. It is commonly associated with Windows devices and uses TCP port 3389 by default. RDP clients are available for Windows, macOS, Linux, mobile devices, and other operating systems.

Voice over IP and session initiation

SIP stands for Session Initiation Protocol and commonly uses TCP ports 5060 and 5061.

SIP is used to establish, manage, and terminate communication sessions. For example, when a user picks up a phone and dials a number, SIP can initiate the call and disconnect the session when the call ends.

SIP can also support video conferencing, instant messaging, and file transfers.

ICMP, GRE, and VPN Technologies

Internet Control Message Protocol (ICMP)

ICMP is an IP-carried protocol that does not use TCP or UDP. It is commonly used for network diagnostics, connectivity testing, and reporting network errors.

The ping command uses ICMP to determine whether a destination device is reachable and responding.

ICMP can also report network conditions, such as:

- Destination network unreachable.

- Time to live (TTL) exceeded.
- Other administrative or network error conditions.

An ICMP Time Exceeded message indicates that the TTL value of a packet has expired.

Generic Routing Encapsulation (GRE)

GRE is a tunneling protocol used to encapsulate information within an IP packet so that it can travel through a tunnel between two endpoints.

GRE is commonly used to create tunnels between network locations, including in VPN configurations.

However, GRE does not provide encryption by itself. Additional VPN protocols are needed to encrypt or protect the data transported through the tunnel.

VPN concentrators and site-to-site VPNs

A VPN (Virtual Private Network) provides encrypted communication between remote users or network sites over a public network, such as the internet.

A VPN concentrator performs the encryption and decryption required for VPN connections. It may be a dedicated hardware appliance or a software implementation running on an operating system.

VPN concentrators are often integrated into firewalls or routers. Dedicated hardware devices may include specialized encryption hardware to improve efficiency and throughput.

A site-to-site VPN connects two corporate networks across a public network. VPN concentrators at each site establish the connection and encrypt traffic passing between them.

IPsec

IPsec (Internet Protocol Security) is a widely used set of protocols for securing IP communications over VPNs. It can provide confidentiality through encryption, integrity through authentication mechanisms, and anti-replay protection.

IPsec can connect devices from different manufacturers because it is a standardized protocol.

The two primary IPsec protocols described are:

- Authentication Header (AH): Provides integrity and authentication-related protection, but does not encrypt the original data. The information remains visible in clear text.

- Encapsulating Security Payload (ESP): Provides encryption and can also provide authentication and integrity protection.

Internet Key Exchange (IKE)

Before IPsec can transfer protected data, the two endpoints must establish a tunnel and agree on the cryptographic keys and security settings.

This process is called Internet Key Exchange (IKE).

IKE establishes a Security Association (SA), which defines the security parameters used for communication.

The material describes two phases:

1. Phase 1: Uses Diffie-Hellman to establish a shared secret and commonly uses UDP port 500. This is associated with ISAKMP, the Internet Security Association and Key Management Protocol.
2. Phase 2: Negotiates encryption algorithms, key sizes, and inbound and outbound security associations for the IPsec tunnel.

Once the negotiation is complete, the endpoints can transfer protected data through the IPsec tunnel.

IPsec transport mode and tunnel mode

IPsec can operate in transport mode or tunnel mode.

- Transport Mode:

- Original IP header remains visible.
- Original data is protected by IPsec.
- Does not replace the original IP header.
- Commonly used to protect communication between endpoints.

- Tunnel Mode:

- Original IP header is encrypted as part of the original packet.
- Protects both the original data and original IP header.
- Adds a new outer IP header.
- Commonly used for site-to-site VPN tunnels.

In transport mode, the IPsec header is inserted between the original IP header and the data. The original IP header remains visible, so someone capturing the traffic can see the original source and destination IP addresses.

In tunnel mode, the entire original IP packet, including its header and data, is protected inside a new packet. A new outer IP header identifies the IPsec tunnel endpoint or concentrator. This prevents an observer from directly seeing the original packet's destination from the encrypted inner packet.

Tunnel mode is commonly used to protect the original traffic in VPN implementations.

Network Communication Types

Network communication can be classified according to how many devices send and receive information.

Unicast

Unicast is a one-to-one communication method in which one device sends data directly to another device.

Only the intended destination participates in that communication. Other devices on the network are not intended recipients.

Examples include accessing websites, transferring files, and checking email.

Unicast is commonly used with both IPv4 and IPv6. However, sending the same information to many devices requires separate communication streams and separate copies of the data.

Multicast

Multicast is a one-to-many communication method in which one device sends information to multiple subscribed recipients simultaneously.

Instead of sending a separate copy of the same data to every recipient, multicast allows the data to be distributed to a group of interested devices.

Examples include:

- Multimedia distribution.
- Stock exchange information feeds.
- Routing updates.

Multicast requires network equipment to support multicast communication and is used in suitable network environments where distributing data to multiple recipients is efficient.

Anycast

Anycast is a one-to-nearest or one-to-one-of-many communication method.

Multiple devices are configured to respond to the same destination IP address. When a device sends data to that address, the network directs it to one of the available devices, commonly the closest according to routing.

A common example is anycast DNS, where a DNS query can be handled by a nearby data center.

Anycast is supported by both IPv4 and IPv6.

Broadcast

Broadcast is a one-to-all communication method in which a device sends a packet or frame to all devices within its local broadcast domain.

Broadcast traffic is limited to the local broadcast domain and is not forwarded across the entire internet.

Common uses include ARP requests and certain network discovery or configuration processes.

IPv4 supports broadcast communication, whereas IPv6 removes broadcast and uses multicast communication instead.

Unicast: One-to-one communication; sends data to one specific device.

Multicast: One-to-many communication; sends data to a subscribed group.

Anycast: One-to-nearest-of-many communication; sends data to one suitable destination among several.

Broadcast: One-to-all communication; sends data to all devices in the local broadcast domain.

Wireless Networking Standards

Wireless networking allows devices to connect to networks without wired connections. International standards ensure that wireless devices from different manufacturers can communicate with each other.

IEEE 802.11 and Wi-Fi

The Institute of Electrical and Electronics Engineers (IEEE) maintains the 802 series of networking standards. IEEE 802.11 specifically defines wireless networking standards.

The Wi-Fi Alliance tests devices for interoperability and provides the Wi-Fi trademark to devices that meet its requirements.

Wireless standards evolve over time, with newer generations providing improved performance and higher theoretical link rates.

- 802.11a: No formal Wi-Fi generation name.
- 802.11b: No formal Wi-Fi generation name.
- 802.11g: No formal Wi-Fi generation name.
- 802.11n: Wi-Fi 4.
- 802.11ac: Wi-Fi 5.
- 802.11ax: Wi-Fi 6.
- 802.11be: Wi-Fi 7.

The material identifies 2.4 GHz, 5 GHz, and 6 GHz as commonly used frequency ranges for wireless networking.

The number of antennas, radios, and other implementation details influences the theoretical link rate of a wireless network. Newer standards generally support higher link rates than older standards.

Cellular networking: 4G LTE and 5G

Mobile phones and tablets commonly use cellular networks provided by mobile network operators.

4G LTE stands for Long-Term Evolution. LTE unified different GSM and CDMA-based technologies into a common mobile networking standard.

The material describes the following approximate download rates:

- LTE: Approximately 150 Mbps.
- LTE Advanced: Approximately 300 Mbps.

5G is a newer mobile networking technology introduced in 2020, designed to improve mobile network performance and bandwidth.

The material identifies a theoretical or target throughput of up to 10 Gbps in suitable configurations, while other implementations may provide speeds in the range of approximately 100–900 Mbps.

Higher bandwidth improves mobile connectivity and enables more data to be transferred in less time. This also supports the growth of Internet of Things (IoT) devices and cloud-based processing.

Satellite networking

Satellite networking provides connectivity to remote locations that lack traditional terrestrial internet access.

A satellite dish can be installed at a remote site to send and receive data through a satellite.

The material identifies the following characteristics:

- Satellite connectivity can provide approximately 100 Mbps download and 5 Mbps upload speeds in common implementations.
- Traditional satellite networking may have around 250 milliseconds of latency in each direction, resulting in approximately 500 milliseconds for a round trip.
- Newer technologies such as Starlink advertise latency around 40 milliseconds, with efforts to reduce it further.
- Satellite networking requires a direct line of sight to the satellite.

Satellite networking can experience reduced performance or connectivity loss when obstacles interfere with the signal.

Rain fade occurs when rain or severe weather affects satellite communication. Connectivity may decrease or be interrupted during a thunderstorm and may recover once the storm passes.

Satellite networking also introduces additional cost and complexity compared with traditional terrestrial connectivity.

Ethernet Networking

Ethernet is one of the most widely used networking standards and is supported by a broad range of devices.

Ethernet networks use standards developed by the IEEE 802.3 committee. These standards define how devices communicate over Ethernet, including supported speeds, cabling, connectors, and equipment requirements.

Ethernet supports both twisted-pair copper cabling and fiber-optic connections.

Common Ethernet standards

1000BASE-T: Uses twisted-pair copper cabling with a speed of 1 Gbps.

10GBASE-T: Uses twisted-pair copper cabling with a speed of 10 Gbps.

1000BASE-SX: Uses fiber-optic cable with a speed of 1 Gbps.

These examples demonstrate how Ethernet supports different transmission speeds and physical media.

Understanding Ethernet naming conventions

Ethernet standard names contain information about the technology, although the complete meaning depends on the specific standard.

For example, 1000BASE-T can be broken down into three parts:

- 1000: Indicates a speed of 1,000 Mbps, or 1 Gbps.
- BASE: Refers to baseband signaling, in which the communication uses a single channel rather than multiple broadband frequencies.
- T: Indicates twisted-pair copper cabling.

Other naming conventions include:

- F: Commonly associated with fiber-optic media.
- SX: Commonly associated with short-wavelength light used over fiber-optic connections.

The name of an Ethernet standard may provide a general indication of its speed and media type, but it does not necessarily describe every technical detail.

To determine the exact capabilities and requirements of a particular Ethernet standard, the complete IEEE specification must be consulted rather than relying solely on its name.

Final Revision: Key Concepts

- IP: Moves packets between source and destination devices using IP addresses.
- Encapsulation: Packages application data inside transport, network, and data-link protocols.
- TCP: Connection-oriented, reliable delivery with acknowledgments, retransmissions, and flow control.
- UDP: Connectionless communication without built-in acknowledgments or retransmission.
- Ports: Identify applications and services running on a device.

- Sockets: Combine IP addresses, transport protocols, and port numbers to identify communication endpoints.
- DNS: Resolves domain names into IP addresses.
- DHCP: Automatically assigns IP configuration to network devices.
- NTP: Synchronizes device clocks.
- SNMP: Monitors and manages network devices.
- ICMP: Supports diagnostics and network error reporting.
- GRE: Encapsulates traffic inside tunnels but does not provide encryption by itself.
- IPsec: Protects IP communication using encryption and integrity mechanisms.
- Unicast, multicast, anycast, broadcast: Describe different communication delivery patterns.
- IEEE 802.11: Defines wireless networking standards.
- IEEE 802.3: Defines Ethernet networking standards.
- Satellite networking: Provides connectivity to remote locations but can experience latency and weather-related interference.

Detailed Summary Notes: Network Cabling, Connectors, Transceivers, and Topologies

These notes cover the material you provided, organized by topic while retaining the technical details, examples, standards, and explanations from the text.

Fiber-Optic Cabling

Purpose and advantages of fiber optics

- Fiber-optic connectivity transmits network traffic using light.
- It is often used in highly secure networks because tapping into a fiber-optic connection is difficult without some type of notification.
- Copper-based networks use radio frequency (RF) signals and are relatively easy to tap and obtain information from.
- Fiber optics are used over very long distances because their signals do not degrade as quickly as radio-frequency signals.
    - A signal that can travel approximately 100 meters over copper might travel kilometers over fiber.

- Light is not susceptible to radio-frequency interference (RFI).
- Fiber optics are useful in environments with interference because they avoid problems caused by radio-frequency interference.
- Fiber-optic networks have different standards and connector types, just like copper networks.
- Always check the device configuration and documentation to ensure that the connected fiber uses the correct:
    - Fiber type.
    - Connector type.

Structure of a fiber-optic cable

A fiber-optic communication system contains a light source at one end and a receiver at the other.

- Light source: An LED (Light Emitting Diode) or laser sends light through the fiber.
- Core: The highly reflective central part through which light travels, bouncing along the core toward the receiver.
- Cladding: A low-reflective layer surrounding the core.
- Buffer coating: The outer protective coating around the fiber-optic run.
- Ferrule: A protective component around the fiber at the connector's end.

  The core is extremely small and can be difficult to see at the center of a fiber-optic connector.

Types of fiber-optic cable

There are two main types of fiber-optic cables:

Multimode Fiber (MMF)

- Commonly used for relatively short-range communication.
- Supports distances up to approximately 2 kilometers, according to the material.
- Often uses an inexpensive light source, such as an LED.
- Has a relatively large core, larger than the wavelength of the light.
- Allows multiple modes of light to propagate through the fiber.
- Light travels through different paths, bouncing off the edges of the fiber.

- It is called multimode because multiple light modes travel through the same fiber.

  Single-Mode Fiber (SMF)

- Used for communication over much longer distances.
- Some fiber Ethernet standards can communicate up to 100 kilometers without regenerating the signal.
- May use a very bright, intense LED or a laser to extend the signal over greater distances.
- Has a smaller core than multimode fiber.
- Allows only a single mode of light to propagate through the fiber.

Multimode Fiber

- Typical distance: Up to approximately 2 km
- Core size: Relatively large
- Light propagation: Multiple modes
- Light source: Often an inexpensive LED

Single-Mode Fiber

- Typical distance: Some standards up to 100 km
- Core size: Smaller
- Light propagation: Single mode
- Light source: Intense LED or laser

Importance of proper cabling

- Cabling is the foundation of networking.
- Routers, switches, wireless access points, and other infrastructure devices depend on cabling for communication.
- Once installed, cabling is difficult to remove and replace.
- The correct cabling must be selected for the infrastructure.
- Even wireless networks eventually connect to a cable, so the right cables must be installed in the right locations to provide the required throughput.

Copper Cabling

Twisted-pair copper cable

- One of the most common cable types in networking.
- Used for practically all wired Ethernet connections.
- Consists of multiple wires twisted together within a single outer sheath.
- The wires are arranged in pairs.

Differential signaling

- One wire in a pair carries a signal, while the other carries the reverse form of that signal.
- Signals may be identified as:
    - Transmit positive (Transmit+).
    - Transmit negative (Transmit−).
    - Receive positive (Receive+).
    - Receive negative (Receive−).
- Sending the same signal in different forms allows the receiver to recognize and correct interference.
- Twisting the two wires means that at least one wire is always moving away from the interference.
- At the receiving end, the signals are compared to determine what the actual signal should look like.

Different twist rates

- Different cable pairs have different twist rates.
- Some pairs are twisted more tightly than others.
- The different twist rates can be observed by examining the individual cable pairs.

Cable speed and Ethernet standards

- A cable itself does not have a speed.
- Copper is simply the medium that supports a signal.
- The Ethernet standard being transmitted over the cable determines how much data can be transferred.
- Ethernet standards specify the signal and the expected throughput over the cable.
- The IEEE 802.3 standards specify:

- The minimum cable type permitted for an Ethernet standard.
- The type of signal used.
- The total expected throughput over that cable.

Cable categories

Cable categories identify the type of twisted-pair cable.

Examples include:

- Category 5 (Cat 5).
- Category 6 (Cat 6).
- Category 7 (Cat 7).

The Ethernet standard specifies the minimum cable category required.

Example: 1000BASE-T

- The 1000BASE-T standard specifies Category 5 as the minimum cable category.
- Category 5 or any category above Category 5 can be used with 1000BASE-T, according to the provided material.

Coaxial and Twinaxial Cables

Coaxial cable

- Another type of Ethernet cable used in networking.
- Contains two or more forms that share a common axis.
- Its components share the same central axis.

A coaxial cable consists of:

1. A single wire conductor in the center.
2. An insulator.
3. Shielding.
4. An outer jacket.

RG-6 coaxial cable:

- A common type of coaxial cable used with Ethernet and Internet connections.
- Often used for cable modem connections.

- A cable modem connection brought into a data center may use coaxial cable.

Twinaxial cable (Twinax)

- A twinaxial cable contains two conductors inside the cable.
- Commonly referred to as Twinax.
- Commonly associated with 10-gigabit-per-second Ethernet.
- Often installed as part of an SFP+ interface.

Characteristics:

- Supports full-duplex communication over a single cable.
- Limits the signal distance to approximately 5 meters.
- Has a relatively low cost compared with fiber optics.
- Often has much lower latency than a twisted-pair connection.

Plenum Cabling and Cable Ratings

What is a plenum?

- In some buildings, the visible ceiling is a drop ceiling with acoustic tiles.
- Above the drop ceiling is a space between the drop ceiling and the actual ceiling.
- This space is called the plenum.
- Network cables, heat sensors, power cables, and other infrastructure may run through this space.

Plenum and air circulation

- Some buildings have ductwork above the drop ceiling for heating and cooling.
- Supply air flows through ductwork.
- Return air may use additional ductwork to return air to the heating or cooling system.
- If air is transferred only through ductwork, there is no shared plenum air space in that configuration.
- If the same air space is shared with networking cables and other infrastructure without ductwork, it must be treated as a shared plenum.

Why plenum-rated cables are required

- A fire in a building can cause cables to produce smoke and potentially hazardous fumes.
- In a shared plenum, smoke and fumes can spread throughout the building.
- Therefore, the correct type of cable must be installed in a shared plenum.

Plenum-rated cable materials

Ethernet cables have an outer jacket, and many are manufactured using Polyvinyl Chloride (PVC).

For installation in a plenum, plenum-rated cables are required. These are usually manufactured using:

- Fluorinated Ethylene Polymer (FEP).
- Low-smoke Polyvinyl Chloride (PVC).

  Plenum-rated cables are designed to be safer during a fire than cables not designed for plenum use.

Installation considerations

- Plenum-rated cables may be less flexible than non-plenum-rated cables.
- They can be more difficult to install when the cable route includes many twists and turns.
- When installing new cables or replacing existing ones, check where the cable will run and use the correct cable type for that location.

Network Transceivers

What is a transceiver?

- The word transceiver combines the terms transmitter and receiver.
- A transceiver combines transmitting and receiving functions within the same piece of equipment or component.
- Transceivers provide modularity by allowing network interfaces to support different media and network types.

How transceivers work

- A network switch may have several open interfaces.
- An appropriate transceiver can be inserted into an interface to configure it for the desired media or network type.
- For example:

    - A copper Gigabit Ethernet transceiver can provide a copper Gigabit Ethernet connection.
    - A fiber 10-gigabit transceiver can provide a 10-gigabit fiber connection.
- Each interface on a switch can use a different media type, depending on the installed transceiver.

Transceiver compatibility

- Different transceivers are available for different media, form factors, and network types.
- An Ethernet switch requires an Ethernet transceiver.
- A Fibre Channel switch requires a Fibre Channel transceiver.
- Fibre Channel transceivers cannot be used in an Ethernet switch, and Ethernet transceivers cannot be used in a Fibre Channel switch.

Modularity and cost

- Copper and fiber transceivers allow the network to use the media appropriate for the configuration.
- A copper transceiver can be removed and replaced with a fiber transceiver if the connection requirements change during installation.
- This modularity provides flexibility but often comes at an additional cost.

SFP and SFP+ transceivers

SFP — Small Form-factor Pluggable

- A common transceiver type.
- Can be used with fiber connections by inserting a fiber SFP.
- Can also support copper connections by inserting a copper SFP and connecting an RJ45 connector.
- Commonly associated with Gigabit Ethernet.
- Supports 1 gigabit per second in the material.

SFP+ — Enhanced Small Form-factor Pluggable

- An enhanced version of SFP.
- Has the same physical size as SFP.

- Supports higher speeds, up to 16 gigabits per second, according to the material.
- Commonly used for 10-gigabit connections.

QSFP and QSFP+

QSFP stands for Quad Small Form-factor Pluggable.

- Designed to provide more connectivity and throughput within limited data-center rack space.
- Data-center equipment is generally 19 inches wide.
- Rack space is limited, so more connectivity must be provided within a small physical area.
- QSFP effectively provides four times the throughput of a single SFP in a similar space.

QSFP

- Consists of four SFP channels.
- If one SFP provides 1 Gbps, a four-channel QSFP can provide 4 Gbps total throughput.

QSFP+

- A four-channel version of SFP+.
- If a single SFP+ provides a 10 Gbps connection, QSFP+ can support four such channels.
- Total throughput: 40 Gbps.

Advantages of QSFP and QSFP+

- Efficient use of physical space.
- Can extend four separate links over a single fiber connection, as described in the material.
- Provides cost benefits for equipment and cabling media.

Form-factor comparison

- SFP and SFP+ share the same form factor, although their transceivers differ.
- QSFP and QSFP+ also share the same form factor.
- QSFP/QSFP+ are slightly larger than SFP/SFP+.

- They are not four times the size of SFP, providing improved space efficiency.

  - SFP

- Channels: 1
- Throughput: 1 Gbps

  - SFP+

- Channels: 1
- Throughput: 10 Gbps

  - QSFP

- Channels: 4
- Throughput: 4 Gbps

  - QSFP+

- Channels: 4
- Throughput: 40 Gbps

Fiber-Optic Connectors

Fiber-optic networks use many different connector types. The correct connector must be selected for the connection.

SC connector

- SC stands for Subscriber Connector.
- Also referred to as:
  - Square connector, because of its square shape.
  - Standard connector, although other fiber connector types also exist.
- Commonly used in data centers.

Connection mechanism

- Pushes into the interface and snaps into place with a locking mechanism.
- Does not easily slip out accidentally.
- Must be pulled to unlock before removal.

Fiber arrangement

- Each connector supports an individual fiber.
- Connectors are often combined into a pair.
- One fiber is commonly used for transmitting and the other for receiving.

LC connector

- LC stands for Local Connector.
- Also called Lucent connector or little connector.
- Slightly smaller than an SC connector.

Connection mechanism

- Has a clip on the top that locks the connector in place.
- Prevents accidental disconnection.
- Pressing down on the clip releases the lock and allows removal.

Fiber arrangement

- LC connectors can be combined into a pair.
- Both connectors can be plugged in simultaneously.
- One fiber is used for transmit and the other for receive.

ST connector

- ST stands for Straight Tip.
- Uses a bayonet-style connector.

Connection mechanism

1. Push the connector into the interface.
2. Give it a slight twist to lock it into place.
3. Reverse the twist to unlock and remove it.

- The locking mechanism helps prevent accidental disconnection.
- This is useful in racks with many cables and fiber connections, where cables can easily be dislodged.

Construction

- Has a protective ferrule around the fiber.
- Uses a visible bayonet connector that is pushed and twisted into place.

MPO connector

- MPO stands for Multi-fiber Push On.
- Designed to reduce the amount of space required by fiber connections.
- Contains 12 individual fibers within a single cable connection.

Connection mechanism

- Uses a locking mechanism similar to the SC connector.
- The connector is pushed into place.
- It must be pulled slightly to unlock it from the interface.

MTP

- MPO may also be referred to as MTP.
- MTP is a name provided by Corning.
- Corning refers to it as an MTP MPO connector.

Construction

- Has a locking mechanism on the top.
- Individual fibers appear as small dots within the connector.
- When light is sent through the fibers and the connector is viewed directly, all 12 fibers can be seen more clearly.

Fiber connector comparison

- SC (Subscriber Connector)

- Locking: Push and snap
- Key feature: Square connector

- LC (Local Connector)

- Locking: Top clip
- Key feature: Smaller than SC

- ST (Straight Tip)

- Locking: Push and twist
- Key feature: Bayonet connector

- MPO (Multi-fiber Push On)

- Locking: Push and pull to unlock
- Key feature: 12 fibers in one connection

Copper Connectors

RJ11

- RJ11 stands for Registered Jack type 11.
- A six-position connector.
- Uses only two conductors.
- Also described as 6P2C (6 positions, 2 conductors).

Common uses

- Analog telephone connections.
- DSL connections, which use the same wires as telephone connections.

  The connector has six positions, but only the two conductors in the middle are used.

RJ45

- RJ45 stands for Registered Jack type 45.
- Used for Ethernet connections.
- Has eight positions and eight conductors.
- Uses all eight conductors available in the connector.
- Slightly larger and wider than RJ11.

RJ11 versus RJ45

- RJ11
- Positions: 6
- Conductors: 2
- Common use: Analog telephone and DSL

- RJ45
- Positions: 8
- Conductors: 8
- Common use: Ethernet

- A device may contain both RJ11 and RJ45 connectors.

- DSL connections use RJ11.
- LAN or Ethernet connections use RJ45.

F-connector

- Commonly used with coaxial cable and cable modem connections.
- A standard connector type.
- Usually has threads that allow it to be securely fastened to the device.

Common use and terminology

- Used to connect cable television infrastructure to a cable modem.
- Also referred to as a DOCSIS connector in the material.
- DOCSIS stands for Data Over Cable Service Interface Specification.
- The signal travels through the single copper conductor inside the coaxial cable.
- The F-connector is screwed into the cable modem.

BNC connector

- BNC is a bayonet-style connector commonly used with coaxial cable.
- The connector is pushed into the interface and twisted slightly to lock it in place.
- It does not easily disconnect unless it is untwisted first.

Full form

- B: Bayonet.
- NC: Neill-Concelman.

  The name refers to Paul Neill of Bell Labs and Carl Concelman of Amphenol, who were involved in creating the connector standard.

Common uses

- WAN connections.
- Other types of coaxial connections.

  The locking mechanism allows the connector to be securely attached to the device.

Network Topologies

Network topologies describe how networks connect to each other and how devices and locations communicate.

They are useful for:

- Network planning and design.
- Understanding how networks are connected.
- Troubleshooting network connectivity.
- Visualizing how data travels from one network to another.

The material covers star, mesh, hybrid, spine-and-leaf, and point-to-point topologies.

Star topology (Hub and Spoke)

- One of the most popular network topologies.
- Sometimes called a hub-and-spoke network.
- A central networking device connects all other devices.
- Every device connects to the same central device.
- Devices communicating with each other must send their traffic through the central hub.

  Example: Switched Ethernet
- An Ethernet switch sits in the middle of the network.
- All other devices connect to that switch.
- Communication between devices passes through the central switch.

Mesh topology

- A device or location connects to another device or location through more than one network connection.
- Data can follow one path or an alternate path to reach the destination.

Redundancy and fault tolerance

- If one link in the mesh fails, another link can be used to complete communication.
- This maintains connectivity even when a network connection becomes unavailable.

Load balancing

- Traffic can be distributed across multiple links.
- For example, half the data can travel over one connection and the other half over a different connection.

Common deployment

- Mesh networks can be used in local area networks.
- They are often deployed in wide area networks (WANs).
- They help maintain connectivity to remote sites regardless of which network connection is available.

Hybrid topology

- A hybrid network combines multiple network architectures.
- Large enterprise networks often use different topologies in different parts of the network.
- These separate architectures are combined into a single hybrid network.

  Example from the material:

  A large enterprise network may contain:
- One section using a star topology.
- Another section using point-to-point connections.
- Another section using a mesh topology.

  Combining these architectures creates a hybrid network.

Spine-and-leaf architecture

- Commonly used in data centers.
- Contains spine switches at the top and leaf switches below them.
- Devices connect to the leaf switches.

Architecture and connections

- Every leaf switch connects to all spine switches.
- Leaf switches do not connect directly to each other.
- Spine switches do not connect directly to each other.
- Devices connect through the leaf switches.

Top-of-rack switching

- Many data centers use spine-and-leaf architecture with top-of-rack switching.
- Each physical rack has a leaf switch at the top.
- All physical devices within that rack connect to its leaf switch.
- Cabling is kept simple because the connections are contained within the same rack.

Advantages

- Redundancy: Each rack connects to multiple spine switches.
- Performance: Devices are no more than one switch away from another device in the data center, according to the material.
- Simplified cabling: Devices within a rack connect to the leaf switch in that rack.

Cost considerations

- For a data center with only a few racks, the architecture may be relatively inexpensive.
- With tens, hundreds, or thousands of racks, a separate leaf switch is required for every rack.
- The cost increases as the number of racks increases.

Point-to-point topology

- A point-to-point connection connects one single point directly to another single point.
- Commonly used in wide area networks.

Historical WAN examples

- Older WANs commonly used T1 or T3 connections.
- These were described as point-to-point T1 or point-to-point T3 connections.

Campus network example

- Point-to-point connections can also be used in local area networks.
- On a campus, two buildings may be connected through two connections.
- The connection between those two buildings is a point-to-point connection.

Final Revision Table

- Fiber Optics: Uses light, supports long distances, and is immune to RF interference.
- Multimode Fiber: Larger core, multiple light modes, typically up to 2 km.
- Single-mode Fiber: Smaller core, one light mode, some standards reach up to 100 km.
- Twisted-pair Copper: Uses twisted wire pairs, differential signaling, and supports Ethernet standards.
- Cable Categories: Cat 5, Cat 6, Cat 7; Ethernet standards specify the minimum category.
- Coaxial Cable: Contains a central conductor, insulator, shielding, and jacket; RG-6 is a common example.
- Twinax: Associated with 10 Gbps Ethernet, SFP+, and approximately 5 m distance.
- Plenum Cable: Uses FEP or low-smoke PVC for safer installation in shared air spaces.
- SFP: Small Form-factor Pluggable; commonly associated with 1 Gbps Ethernet.
- SFP+: Enhanced SFP; commonly used for 10 Gbps connections.
- QSFP: Four SFP channels; example throughput of 4 Gbps.
- QSFP+: Four SFP+ channels; example throughput of 40 Gbps.
- SC / LC / ST / MPO: Fiber connector types with different locking mechanisms and arrangements.
- RJ11: 6P2C; used for analog telephone and DSL connections.
- RJ45: 8P8C; used for Ethernet.
- F-connector: Used for coaxial cable and cable modem connections.
- BNC: Bayonet-style coaxial connector.
- Star Topology: All devices connect through a central device.
- Mesh Topology: Multiple paths provide redundancy and load balancing.
- Hybrid Topology: Combines multiple network topologies.
- Spine-and-Leaf: Data-center architecture with leaf switches connected to all spine switches.

- Point-to-Point: Direct connection between two points; common in WANs and campus networks.

Three-Tier Network Architecture

A three-tier architecture is a common enterprise network design consisting of three layers: Core, Distribution, and Access.

Core Layer

- The central point for network resources.
- Contains critical resources such as:
    - Servers
    - Applications
    - Databases
    - Other critical network resources
- Core routers connect to other resources within the core.

Distribution Layer

- Acts as the midpoint between users and core resources.
- Usually consists of a series of switches.
- Provides connectivity between access switches and the core.
- Provides redundancy through multiple connections.

Access Layer

- The layer where end users connect to the network.
- Uses switches located close to users, often on the same floor of a building.
- Connects users to the distribution layer, which connects to the core.

Traffic Path

Core Layer = Core routers, servers, applications, databases

Distribution Layer = Distribution switches and redundant connections

Access Layer = Access switches connected to users

Traffic from users passes through the access layer, then distribution, and finally reaches the core resources.

City Analogy

The three-tier architecture can be compared to a city:

- Core: Downtown, where major resources are located.
- Distribution: Highways connecting homes to downtown.
- Access: Local roads connecting homes to highways.

Redundancy and Multiple Buildings

- Multiple links may exist between access and distribution switches.
- Multiple links may also exist between distribution and core layers.
- If one component fails, another network path can complete the communication.
- In a campus with multiple buildings:
    - Users on each floor connect to access switches.
    - Access switches connect to one or more distribution switches.
    - Distribution switches connect to the core, potentially located in a central data center.

Collapsed Core Architecture

A collapsed core is a two-tier architecture that combines the core and distribution layers into a single layer.

- The access layer remains unchanged.
- Suitable for smaller organizations that do not need a large three-tier architecture.

  Advantages:

- Simplified network design.
- Easier troubleshooting.
- Lower implementation cost because fewer devices need to be purchased and installed.

  Limitation:

- Fewer redundancy options.
- May be less resilient if individual components fail.

Data Center Traffic Flows

Traffic flow is described based on where the traffic originates and where it is going.

East-West Traffic

- Traffic that originates and terminates within the same data center.
- Remains within the local network.
- Usually has good response times because devices are located within the same local network.

  Example: A file server sending information to an image server within the same data center.

North-South Traffic

- Traffic entering or leaving the data center from an external source.
- Includes traffic coming from or going to the internet.
- Has a different security posture from east-west traffic because:
  - Outbound traffic leaves the data center.
  - Inbound traffic may originate from potentially anywhere.

East-West Traffic

  - Traffic direction: **Within the data center**
  - Example: **File server → Image server**
  - Network scope: **Local data center**
- **North-South Traffic**
  - Traffic direction: **Into or out of the data center**
  - Example: **Data center → Internet**
  - Network scope: **External and internal boundary**

Binary Numbering System

Binary is a numbering system that uses only two digits: 0 and 1.

Bits and Bytes

- Bit: A single binary digit, either 0 or 1.
- Byte: A group of 8 bits.
- Octet: Another term for an 8-bit byte, used to make clear that the group contains exactly eight bits.

An IPv4 address contains 32 bits, divided into four octets.

Binary-to-Decimal Conversion Chart

Each position in an 8-bit binary number represents a power of 2. Starting from the right, each position is double the previous one.

1st bit → 1

2nd bit → 2

3rd bit → 4

4th bit → 8

5th bit → 16

6th bit → 32

7th bit → 64

8th bit → 128

The chart can be extended beyond eight bits by continuing to double the values: 256, 512, and so on.

The corresponding powers of 2 are:

- 20=12^0=120=1
- 21=22^1=221=2
- 22=42^2=422=4
- 23=82^3=823=8
- 24=162^4=1624=16
- 25=322^5=3225=32
- 26=642^6=6426=64
- 27=1282^7=12827=128

Converting Binary to Decimal

Method:

1. Write the binary number.
2. Place the decimal conversion values above the corresponding bits.
3. For every bit containing 1, select the corresponding decimal value.
4. For every bit containing 0, add nothing.

5. Add the selected values.

Example 1: Binary 00000010

Only the bit corresponding to decimal 2 is set to 1.

$2=22=22=2$

Answer: 2 decimal

Example 2: Binary 10000010

The bits with value 1 correspond to 128 and 2.

$128+2=130128+2=130128+2=130$

Answer: 130 decimal

Example 3: Binary 11111111

All eight bits are 1.

$128+64+32+16+8+4+2+1=255128+64+32+16+8+4+2+1=255128+64+32+16+8+4+2+1=255$

Answer: 255 decimal

Therefore, the decimal range of an 8-bit value is 0–255.

Converting Decimal to Binary

Method:

1. Start with the decimal value to convert.
2. Check the largest available binary place value.
3. If adding that value to the selected values does not exceed the target, write 1.
4. Otherwise, write 0.
5. Continue through every position from left to right.

Example: Convert 154 to Binary

The selected values add up to:

Place value 128 $\rightarrow 128 \leq 154 \rightarrow$ **1**

Place value 64 $\rightarrow 128 + 64 = 192$, exceeds $154 \rightarrow$ **0**

Place value 32 $\rightarrow 128 + 32 = 160$, exceeds $154 \rightarrow$ **0**

Place value 16 $\rightarrow 128 + 16 = 144 \rightarrow$ **1**

Place value 8 → 144 + 8 = 152 → **1**

Place value 4 → 152 + 4 = 156, exceeds 154 → **0**

Place value 2 → 152 + 2 = 154 → **1**

Place value 1 → 154 + 1 exceeds 154 → **0**

128+16+8+2=154128+16+8+2=154128+16+8+2=154

Answer: 10011010

Number of Possible Binary Combinations

The total number of possible combinations depends on the number of bits.

2 bits → 4 combinations → Decimal range: **0–3**

3 bits → 8 combinations → Decimal range: **0–7**

4 bits → 16 combinations → Decimal range: **0–15**

5 bits → 32 combinations → Decimal range: **0–31**

6 bits → 64 combinations → Decimal range: **0–63**

7 bits → 128 combinations → Decimal range: **0–127**

8 bits → 256 combinations → Decimal range: **0–255**

Formula:

Possible combinations=2n\text{Possible combinations}=2^nPossible combinations=2n

Here, nnn is the number of bits.

IPv4 Address Configuration

IPv4 Address

- An IPv4 address is a unique identifier assigned to a device that communicates using IP.
- Written as four decimal numbers separated by periods.
- Example: 192.168.1.165
- Each number is an octet, containing 8 bits.
- Total length: 32 bits, or 4 bytes.

Each octet has a maximum decimal value of 255.

Subnet Mask

- A subnet mask is a four-octet value used with an IP address.
- Example: 255.255.255.0
- Determines which IP addresses belong to the local subnet and which are outside it.
- Used by the local device to determine whether a destination is local or outside its subnet.
- The subnet mask is not usually transmitted across the network.

Default Gateway

- The default gateway is the IP address of a device on the local subnet that allows communication outside that subnet.
- It is commonly the address assigned to the local router.
- Must belong to the same local network as the device.

  Example configuration:

  IP address: 192.168.1.165

  Subnet mask: 255.255.255.0

  Default gateway: 192.168.1.1

  The default gateway allows the device to communicate with networks outside its local subnet.

Loopback Address

- Used to reference the local device without requiring its specific network IP address.
- Loopback range given in the text: 127.0.0.1 through 127.255.255.254.
- Common loopback address: 127.0.0.1.
- Used to test whether the local IP stack is functioning correctly.

  Example:

  ping 127.0.0.1

Reserved IP Addresses

- Reserved addresses are set aside for future use or testing.
- They should not be assigned to ordinary devices.
- The text gives the reserved range as 240.0.0.1 through 254.255.255.254.

- Class E addresses are also described as reserved.

Virtual IP Address (VIP)

- A VIP is not associated with a physical Ethernet adapter.
- It can be assigned internally to reference a device.
- Allows a device to be referenced using a virtual IP address.
- Examples:
    - A virtual machine.
    - An internal address associated with a router.

IPv4 and the OSI Model

- IP and IPv4 operate at OSI Layer 3.
- Layer 3 is the Network layer.
- Working with IP addresses means working at the Network layer.

Automatic IP Address Configuration

DHCP – Dynamic Host Configuration Protocol

- Automatically assigns IP configuration settings to devices.
- Used when connecting to wired Ethernet or wireless networks.
- Provides the configuration required for network communication.
- Commonly supplies:
    - IP address
    - Subnet mask
    - Default gateway
    - DNS servers

  Without DHCP, IP configuration may need to be performed manually.

APIPA – Automatic Private IP Addressing

APIPA automatically assigns a link-local IPv4 address when a device cannot obtain an address from a DHCP server.

- Link-local addresses allow communication with devices on the same local subnet.
- They cannot be routed outside the local subnet.

- A device with an APIPA address cannot use that address to communicate with the internet.

APIPA Address Range

- Overall range given: 169.254.0.1 through 169.254.255.254.
- The first and last 256 addresses are reserved and not assigned to end stations.
- Functional range given: 169.254.1.0 through 169.254.254.255.

APIPA Assignment Process

1. The device does not receive an IP address from DHCP.
2. It uses ARP to check whether another device on the local network is using the proposed link-local address.
3. If no device responds to the ARP query, the address is assigned.
4. The address is selected randomly from the functional address range.

Private IPv4 Addresses and NAT

IPv4 Address Exhaustion

- There are more devices in the world than available IPv4 addresses.
- All available IPv4 address blocks have been used up.
- Private IP address ranges help extend the use of IPv4.

Private IP Address Ranges

- Private IP addresses are intended for use inside networks.
- They cannot be routed across the public internet.
- They are commonly used within corporate networks.
- A device accessing the internet may have a private IP address.

Network Address Translation (NAT)

NAT allows devices using private IP addresses to communicate over the public internet.

- Converts a private IP address into a public IP address.
- The public IP address is used for communication across the public network.

RFC 1918 Private Address Ranges

RFC 1918 defines three ranges for private IP addressing.

10.0.0.0 – 10.255.255.255

- CIDR notation: **/8**
- Subnet mask: **255.0.0.0**
- Host bits: **24**

172.16.0.0 – 172.31.255.255

- CIDR notation: **/12**
- Subnet mask: **255.240.0.0**
- Host bits: **20**

192.168.0.0 – 192.168.255.255

- CIDR notation: **/16**
- Subnet mask: **255.255.0.0**
- Host bits: **16**

Range 1: 10.0.0.0/8

- Described as one Class A block.
- Contains over 16 million addresses.
- Subnet mask: 255.0.0.0.
- Host ID length: 24 bits.

Range 2: 172.16.0.0/12

- Covers 16 contiguous Class B address ranges.
- Contains just over 1 million addresses.
- Subnet mask: 255.240.0.0.
- Host ID length: 20 bits.

Range 3: 192.168.0.0/16

- Covers 256 contiguous Class C address ranges.
- Contains over 65,000 addresses.
- Subnet mask: 255.255.0.0.

- Host ID length: 16 bits.

IPv4 Address Classes

Classful Addressing

- Classful addressing was a method of describing IP addresses based on their subnet masks.
- It has not been used since 1993.
- Class A, B, and C addresses are still referred to in casual networking conversations.
- Class-based addressing provides a starting point for understanding subnetting.

Class A

- First octet range: **0–127**
- Default subnet mask: **255.0.0.0**
- Network bits: **8**
- Host bits: **24**

Class B

- First octet range: **128–191**
- Default subnet mask: **255.255.0.0**
- Network bits: **16**
- Host bits: **16**

Class C

- First octet range: **192–223**
- Default subnet mask: **255.255.255.0**
- Network bits: **24**
- Host bits: **8**

Class D

- First octet range: **224–239**
- Default subnet mask: **Not applicable**
- Network bits: **—**

- Host bits: —

Class E

- First octet range: **240–255**
- Default subnet mask: **Reserved**
- Network bits: —
- Host bits: —

Identifying Classes Using Binary

- Class A: First bit is 0.
- Class B: First two bits are 10.
- Class C: First three bits are 110.
- Class D: First four bits are 1110.
- Class E: First four bits are 1111.

Purpose of Each Class

- Class A: Default network portion is the first 8 bits.
- Class B: Default network portion is the first 16 bits.
- Class C: Default network portion is the first 24 bits.
- Class D: Used for multicast communication; not assigned to individual devices.
- Class E: Reserved address range.

Identifying Classes Using Examples

17.22.90.7 → First octet: 17 → Class A

220.10.77.40 → First octet: 220 → Class C

165.245.0.1 → First octet: 165 → Class B

128.90.10.2 → First octet: 128 → Class B

191.77.24.250 → First octet: 191 → Class B

192.1.12.5 → First octet: 192 → Class C

Key point: To identify an IP address class, look at its first octet.

Four Important IPv4 Subnet Values

When calculating a subnet, four values are important.

Network Address

- Identifies the subnet.
- Calculated by setting all host bits to 0.

First Usable Host Address

- The first usable IP address on the subnet.
- One number higher than the network address.

Broadcast Address

- The last IP address of the subnet.
- Calculated by setting all host bits to 1.

Last Usable Host Address

- The last usable IP address on the subnet.
- One number lower than the broadcast address.

Classful Subnet Calculation Examples

Example 1: 10.74.222.11

- First octet 10 → Class A.
- Default subnet mask: 255.0.0.0.
- Network portion: First octet.
- Host portion: Remaining three octets.
- **Network address:** 10.0.0.0
- **First usable host:** 10.0.0.1
- **Broadcast address:** 10.255.255.255
- **Last usable host:** 10.255.255.254

Example 2: 172.16.88.200

- First octet 172 → Class B.
- Default subnet mask: 255.255.0.0.
- Network portion: First two octets.
- Host portion: Last two octets.
- **Network address:** 172.16.0.0

- **First usable host:** 172.16.0.1
- **Broadcast address:** 172.16.255.255
- **Last usable host:** 172.16.255.254

Example 3: 192.168.4.77

- First octet 192 → Class C.
- Default subnet mask: 255.255.255.0.
- Network portion: First three octets.
- Host portion: Last octet.
- Network address: 192.168.4.0
- First usable host: 192.168.4.1
- Broadcast address: 192.168.4.255
- Last usable host: 192.168.4.254

The same process can be used with class-based and classless subnetting.

Classless Interdomain Routing (CIDR)

Classless Subnetting

- Since 1993, classless subnetting has replaced classful subnetting.
- Also known as Classless Interdomain Routing (CIDR).
- Allows subnet masks to use a variable number of network bits instead of fixed Class A, B, or C arrangements.
- Allows networks to be divided according to their requirements.

CIDR Block Notation

CIDR notation represents the number of 1 bits in a subnet mask.

Examples:

/8 — Subnet mask: 255.0.0.0; Network bits: 8; Host bits: 24.

/12 — Subnet mask: 255.240.0.0; Network bits: 12; Host bits: 20.

/16 — Subnet mask: 255.255.0.0; Network bits: 16; Host bits: 16.

/24 — Subnet mask: 255.255.255.0; Network bits: 24; Host bits: 8.

/26 — Subnet mask: 255.255.255.192; Network bits: 26; Host bits: 6.

Example: 192.168.1.44/24

- IP address: 192.168.1.44
- /24 indicates 24 network bits.
- Subnet mask: 255.255.255.0.

Subnet Mask Configuration

When configuring a network device, common settings include:

- IP address
- Subnet mask
- Default gateway
- DNS servers

Different devices may accept different subnet mask formats:

- Windows may use decimal subnet masks.
- Routers and switches may use CIDR notation.

Always check the device documentation for the required notation.

Binary Subnet Mask Rules

A subnet mask consists of:

- A contiguous sequence of 1s on the left.
- A sequence of 0s on the right.

The 1s represent the network portion, while the 0s represent the host portion.

To calculate CIDR notation, count the total number of 1s.

Example 1: /16

Binary mask:

11111111.11111111.00000000.00000000

- First octet: 8 ones.
- Second octet: 8 ones.
- Total: 16 ones.
- CIDR: /16.
- Network bits: 16.

- Host bits: 16.

Example 2: /26

Binary mask:

11111111.11111111.11111111.11000000

- First three octets: 24 ones.
- Last octet: 2 ones.
- Total: 26 ones.
- CIDR: /26.
- Network bits: 26.
- Host bits: 6.

Example 3: /12

Binary mask:

11111111.11110000.00000000.00000000

- First octet: 8 ones.
- Second octet: 4 ones.
- Total: 12 ones.
- CIDR: /12.
- Network bits: 12.
- Host bits: 20.

Binary-to-Decimal Subnet Mask Conversion

00000000 → 0

10000000 → 128

11000000 → 192

11100000 → 224

11110000 → 240

11111000 → 248

11111100 → 252

11111110 → 254

11111111 → 255

Example 1: Convert /12 to Decimal

Binary:

11111111.11110000.00000000.00000000

- First octet: 255.
- Second octet: 240.
- Third and fourth octets: 0.

  Answer: 255.240.0.0

  CIDR: /12

  Network bits: 12; host bits: 20.

Example 2: Convert /19 to Decimal

Binary:

11111111.11111111.11100000.00000000

- First octet: 255.
- Second octet: 255.
- Third octet: 224.
- Fourth octet: 0.

  Answer: 255.255.224.0

  CIDR: /19

  Network bits: 19; host bits: 13.

Example 3: Convert /26 to Decimal

Binary:

11111111.11111111.11111111.11000000

Answer: 255.255.255.192

Network bits: 26; host bits: 6.

Example 4: Convert /20 to Decimal

Binary:

11111111.11111111.11110000.00000000

Answer: 255.255.240.0

Network bits: 20; host bits: 12.

Variable Length Subnet Masking (VLSM)

What Is VLSM?

- VLSM stands for Variable Length Subnet Mask.
- Allows different networks to use different subnet mask lengths.
- Unlike classful subnetting, the subnet mask does not have to use a fixed number of network bits.
- Allows network administrators to create networks of appropriate sizes.
- Provides control over the number of subnets and hosts required for each network.

For example, a Class A network such as 10.0.0.0 normally has a default /8 mask, but it can be subnetted using /24, /26, or other mask lengths.

10.2 Borrowing Subnet Bits

When subnetting a network:

- The default network bits form the original network portion.
- Additional bits can be borrowed from the host portion to create subnet bits.
- The remaining bits are used for host addresses.

Example:

Original Class A network:

- Network: 10.0.0.0
- Default mask: /8
- Network bits: 8
- Host bits: 24

After borrowing 16 additional bits:

- Subnet mask: /24
- Network bits: 8 original bits.
- Subnet bits: 16 borrowed bits.

- Host bits: 8 remaining bits.
- Decimal subnet mask: 255.255.255.0.

Calculating Number of Subnets and Hosts

Two important formulas are used.

Number of subnets

2number of subnet bits2^{{number of subnet bits}}2number of subnet bits

Usable hosts per subnet

2number of host bits−22^{{number of host bits}}-22number of host bits−2

The subtraction of 2 accounts for the network address and broadcast address.

VLSM Calculation Example 1: 10.1.1.0/24

Class A network with a /24 mask.

- Default Class A network bits: 8.
- New subnet mask: /24.
- Borrowed subnet bits: 16.
- Remaining host bits: 8.

  Number of subnets:

  216=65,5362^{16}=65,536216=65,536

  Usable hosts per subnet:

  28−2=256−2=2542^8-2=256-2=25428−2=256−2=254

  Result:
- Number of subnets: 65,536.
- Usable hosts per subnet: 254.

VLSM Calculation Example 2: 192.168.1.0/26

- Class C default network bits: 24.
- Subnet mask: /26.
- Borrowed subnet bits: 2.
- Host bits: 6.

  Number of subnets:

22=42^2=422=4

Usable hosts per subnet:

26−2=64−2=622^6-2=64-2=6226−2=64−2=62

Result:

- Number of subnets: 4.
- Usable hosts per subnet: 62.

VLSM Calculation Example 3: 172.65.5.21/21

- Class B default network bits: 16.
- Subnet mask: /21.
- Borrowed subnet bits: 5.
- Host bits: 11.

Number of subnets:

25=322^5=3225=32

Usable hosts per subnet:

211−2=2048−2=20462^{11}-2=2048-2=2046211−2=2048−2=2046

Result:

- Number of subnets: 32.
- Usable hosts per subnet: 2,046.

VLSM Revision Table

- 10.1.1.0/24

- Default class: A
- Subnet bits borrowed: 16
- Host bits: 8
- Number of subnets: 65,536
- Usable hosts per subnet: 254

- 192.168.1.0/26

- Default class: C
- Subnet bits borrowed: 2

- Host bits: 6
- Number of subnets: 4
- Usable hosts per subnet: 62

  - 172.65.5.21/21
- Default class: B
- Subnet bits borrowed: 5
- Host bits: 11
- Number of subnets: 32
- Usable hosts per subnet: 2,046

Magic Number Method

Purpose

The magic number method is a shortcut for calculating subnet information without repeatedly converting IP addresses and subnet masks between binary and decimal.

It helps determine four important values:

1. Network address (subnet ID) — first address in the subnet.
2. Broadcast address — last address in the subnet.
3. First usable host address.
4. Last usable host address.

The method involves identifying the interesting octet, calculating the magic number, and finding the address range.

Important Subnetting Charts

A. CIDR notation and decimal subnet masks

The following chart helps convert CIDR notation into a decimal subnet mask and identify the interesting octet.

- **Octet 2:** CIDR range /9 to /16
- **Octet 3:** CIDR range /17 to /24
- **Octet 4:** CIDR range /25 to /30

  Examples:
- /9 has 128 in the second octet.

- /17 has 128 in the third octet.
- /25 has 128 in the fourth octet.

  The subnet mask values in the interesting octet are:

  128, 192, 224, 240, 248, 252, 254, 255

  /31 and /32 are not used in its default host-range chart because /31 has no traditional usable host range and /32 leaves zero host bits.

B. Address block boundaries

An address block divides an octet into smaller ranges.

- Block size 128: 0–127, 128–255.
- Block size 64: 0–63, 64–127, 128–191, 192–255.
- Block size 32: 0–31, 32–63, 64–95, and so on.
- Block size 16: 0–15, 16–31, 32–47, and so on.
- Block size 8: 0–7, 8–15, 16–23, and so on.

  These boundaries help identify the subnet containing a particular IP address.

Steps in the Magic Number Method

Follow these steps for each subnetting problem.

1. Convert the subnet mask to decimal if it is given in CIDR notation.
2. Identify the interesting octet — the octet whose mask is neither 255 nor 0.
3. Calculate the magic number:

   256−interesting octet mask256-\ {interesting octet mask}256−interesting octet mask
4. Use the magic number to determine the address block.
5. Identify the network address as the first address in the block.
6. Identify the broadcast address as the last address in the block.
7. Calculate the first and last usable host addresses.

   The magic number represents the size of the address block in the interesting octet, including the network and broadcast addresses.

Example 1: Subnetting 165.245.77.14/20

Given:

- IP address: 165.245.77.14
- Subnet mask: 255.255.240.0
- CIDR: /20

Step 1: Identify the interesting octet

The third octet has a subnet mask of 240.

Therefore, the third octet is the interesting octet.

Step 2: Calculate the magic number

256−240=16256-240=16256−240=16

The block size is 16 addresses in the third octet.

Step 3: Determine the subnet range

The third octet of the IP address is 77.

The ranges are:

- 0–15
- 16–31
- 32–47
- 48–63
- 64–79
- 80–95

Since 77 falls between 64 and 79, the subnet starts at 64.

Step 4: Calculate the network address

For the network address:

- If the subnet mask is 255, copy the IP address octet.
- If the subnet mask is 0, use 0.
- For the interesting octet, use the starting value of its block.

  Network address:

  165.245.64.0

Step 5: Calculate the broadcast address

For the broadcast address:

- If the mask is 255, copy the IP address octet.
- If the mask is 0, use 255.
- For the interesting octet, add the magic number to the network's starting value and subtract 1.

  64+16−1=7964+16-1=7964+16−1=79

  Broadcast address:

  165.245.79.255

Step 6: Calculate the usable host range

First usable host:

165.245.64.0+1=165.245.64.1165.245.64.0+1=165.245.64.1165.245.64.0+1=165.245.64.1

Last usable host:

165.245.79.255−1=165.245.79.254165.245.79.255-1=165.245.79.254165.245.79.255−1=165.245.79.254

Example 1 — Final Results

- Network address: 165.245.64.0
- First usable host: 165.245.64.1
- Broadcast address: 165.245.79.255
- Last usable host: 165.245.79.254

Example 2: Subnetting 10.180.122.244/21

Given:

- IP address: 10.180.122.244
- Subnet mask: 255.248.0.0
- CIDR: /13

The mask 255.248.0.0 corresponds to /13, not /21. The example in the transcript uses a second-octet mask of 248.

Step 1: Identify the interesting octet

The second octet has the mask 248.

Step 2: Calculate the magic number

256−248=8256-248=8256−248=8

The block size is 8 in the second octet.

Step 3: Determine the network range

The second octet is 180.

The relevant block boundaries include:

- 160–167
- 168–175
- 176–183
- 184–191

The value 180 falls in the range 176–183.

Step 4: Calculate the network address

- First octet: 10 (mask 255, copy the IP address).
- Second octet: 176 (start of the block).
- Third and fourth octets: 0 (mask 0).

  Network address:

  10.176.0.0

Step 5: Calculate the broadcast address

The last value in the second-octet block is:

176+8−1=183176+8-1=183176+8−1=183

The third and fourth octets use 255.

Broadcast address:

10.183.255.255

Step 6: Calculate usable hosts

- First usable host: 10.176.0.1
- Last usable host: 10.183.255.254

Example 2 — Final Results

- Network address: 10.176.0.0
- First usable host: 10.176.0.1
- Broadcast address: 10.183.255.255

- Last usable host: 10.183.255.254

Example 3: Subnetting 172.16.242.133/27

Given:

- IP address: 172.16.242.133
- CIDR: /27
- Subnet mask: 255.255.255.224

Step 1: Identify the interesting octet

The fourth octet has a mask of 224.

Step 2: Calculate the magic number

256−224=32256-224=32256−224=32

Each subnet contains 32 total addresses.

Step 3: Determine the network range

The fourth octet is 133.

The relevant ranges are:

- 96–127
- 128–159
- 160–191

133 falls between 128 and 159.

Step 4: Calculate the network address

The first three octets remain unchanged, and the fourth octet becomes 128.

Network address:

172.16.242.128

Step 5: Calculate the broadcast address

128+32−1=159128+32-1=159128+32−1=159

Broadcast address:

172.16.242.159

Step 6: Calculate usable hosts

- First usable host: 172.16.242.129

- Last usable host: 172.16.242.158

Example 3 — Final Results

- Network address: 172.16.242.128
- First usable host: 172.16.242.129
- Broadcast address: 172.16.242.159
- Last usable host: 172.16.242.158

General Rules for the Magic Number Method

- Subnet Mask Octet = 255:
    - Network Address: Copy the IP octet.
    - Broadcast Address: Copy the IP octet.
- Subnet Mask Octet = 0:
    - Network Address: Use 0.
    - Broadcast Address: Use 255.
- Interesting Octet:
    - Network Address: Use the starting value of the block.
    - Broadcast Address: Use the last value of the block.

  Once the network address and broadcast address are known:
- First usable host = Network address + 1.
- Last usable host = Broadcast address − 1.

7-Second Subnetting Shortcut

Overview

The 7-second subnetting method is another shortcut for quickly calculating subnet information during an exam.

It is similar to the magic number method, but the main difference is that it relies on a predefined chart rather than calculating the magic number each time.

The method requires:

- No repeated binary-to-decimal conversions.
- No magic number calculation during each problem.

- Only adding 1 to the network address and subtracting 1 from the broadcast address to determine usable host addresses.

The method is intended to work with either an online exam whiteboard or a physical whiteboard at a testing center.

Creating the 7-Second Subnetting Chart

The chart contains three important columns:

1. CIDR notation and subnet mask.
2. Number of networks.
3. Number of addresses per subnet.

A. Number of networks

Start with 2 and double the value repeatedly:

2, 4, 8, 16, 32, 64, 128, 256

B. Addresses per subnet

Start with 128 and divide by 2 repeatedly:

128, 64, 32, 16, 8, 4, 2, 1

C. CIDR and subnet mask chart

- /9: Mask 128 (2nd octet) → 8,388,608 addresses
- /10: Mask 192 (2nd octet) → 4,194,304 addresses
- /11: Mask 224 (2nd octet) → 2,097,152 addresses
- /12: Mask 240 (2nd octet) → 1,048,576 addresses
- /13: Mask 248 (2nd octet) → 524,288 addresses
- /14: Mask 252 (2nd octet) → 262,144 addresses
- /15: Mask 254 (2nd octet) → 131,072 addresses
- /16: Mask 255 (2nd octet) → 65,536 addresses
- /17: Mask 128 (3rd octet) → 32,768 addresses
- /18: Mask 192 (3rd octet) → 16,384 addresses
- /19: Mask 224 (3rd octet) → 8,192 addresses
- /20: Mask 240 (3rd octet) → 4,096 addresses
- /21: Mask 248 (3rd octet) → 2,048 addresses

- /22: Mask 252 (3rd octet) → 1,024 addresses
- /23: Mask 254 (3rd octet) → 512 addresses
- /24: Mask 255 (3rd octet) → 256 addresses
- /25: Mask 128 (4th octet) → 128 addresses
- /26: Mask 192 (4th octet) → 64 addresses
- /27: Mask 224 (4th octet) → 32 addresses
- /28: Mask 240 (4th octet) → 16 addresses
- /29: Mask 248 (4th octet) → 8 addresses
- /30: Mask 252 (4th octet) → 4 addresses

  The chart allows the CIDR notation to be converted into a decimal subnet mask and shows the number of addresses in each block.

  For example:

- /26 corresponds to 255.255.255.192.
- The fourth octet has 64 addresses per block.
- /27 corresponds to 255.255.255.224.
- The fourth octet has 32 addresses per block.

3.3 Subnet Boundary Chart

A separate boundary chart can be useful for finding the start and end of each subnet.

- 128 addresses per block: boundaries at 0 and 128.
- 64 addresses per block: boundaries at 0, 64, 128, and 192.
- 32 addresses per block: boundaries at 0, 32, 64, 96, 128, 160, 192, and 224.
- 16 addresses per block: boundaries at multiples of 16.
- 8 addresses per block: boundaries at multiples of 8.
- 4 addresses per block: boundaries at multiples of 4.

  The subnet address is the starting boundary that contains the given IP address. The broadcast address is the last address before the next boundary.

Four Steps of the 7-Second Subnetting Method

1. Convert the IP address and subnet mask to decimal notation.

2. Determine the subnet or network address.
3. Determine the broadcast address.
4. Calculate the first and last usable IP addresses.

Rules for calculating the network address

- If the subnet mask is 255, copy the corresponding IP address octet.
- If the subnet mask is 0, use 0.
- If the subnet mask is an intermediate value, use the subnet boundary containing the IP address.

Rules for calculating the broadcast address

- If the subnet mask is 255, copy the corresponding IP address octet.
- If the subnet mask is 0, use 255.
- For an intermediate mask, use the last address before the next subnet boundary.

3.5 Example 1: 165.245.12.188/24

Given:

- IP address: 165.245.12.188
- CIDR: /24
- Subnet mask: 255.255.255.0

Network address

The first three octets have a mask of 255, so copy them. The last octet has a mask of 0, so use 0.

Network address: 165.245.12.0

Broadcast address

Copy the first three octets and use 255 for the last octet.

Broadcast address: 165.245.12.255

Usable host range

- First usable host: 165.245.12.1
- Last usable host: 165.245.12.254

Final results — /24

- Network: 165.245.12.0
- First host: 165.245.12.1
- Broadcast: 165.245.12.255
- Last host: 165.245.12.254

Example 2: 165.245.12.188/26

Given:

- IP address: 165.245.12.188
- CIDR: /26
- Subnet mask: 255.255.255.192

Step 1: Determine the address block

The fourth octet has 64 addresses per block.

The ranges are:

- 0–63
- 64–127
- 128–191
- 192–255

  The IP address ends in 188, which falls within the range 128–191.

Step 2: Network address

The first three octets remain unchanged. The fourth octet uses the starting boundary, 128.

Network address: 165.245.12.128

Step 3: Broadcast address

The next subnet starts at 192. The preceding address is 191.

Broadcast address: 165.245.12.191

Step 4: Usable host range

- First usable host: 165.245.12.129
- Last usable host: 165.245.12.190

  Final results — /26
- Network: 165.245.12.128

- First host: 165.245.12.129
- Broadcast: 165.245.12.191
- Last host: 165.245.12.190

Example 3: 165.245.12.188/20

Given:

- IP address: 165.245.12.188
- CIDR: /20
- Subnet mask: 255.255.240.0

Step 1: Determine the interesting octet

The third octet has a mask of 240.

The address block size is 16.

Step 2: Find the subnet boundary

The third octet is 12.

The value 12 falls within the range 0–15.

Therefore, the starting boundary is 0.

Step 3: Network address

The first two octets remain unchanged. The third octet becomes 0, and the fourth octet becomes 0.

Network address: 165.245.0.0

Step 4: Broadcast address

The next third-octet boundary is 16. The last address before that boundary has a third octet of 15 and a fourth octet of 255.

Broadcast address: 165.245.15.255

Step 5: Usable host range

- First usable host: 165.245.0.1
- Last usable host: 165.245.15.254

  Final results — /20
- Network: 165.245.0.0
- First host: 165.245.0.1

- Broadcast: 165.245.15.255
- Last host: 165.245.15.254

Example 4: 18.172.2.77/11

Given:

- IP address: 18.172.2.77
- CIDR: /11
- Subnet mask: 255.224.0.0

Step 1: Identify the interesting octet

The second octet has a mask of 224.

The block size is 32 addresses in the second octet.

Step 2: Find the address range

The second octet is 172.

The relevant boundaries are:

- 128–159
- 160–191
- 192–223

  172 falls within the range 160–191.

Step 3: Network address

The first octet is copied as 18. The second octet uses the starting boundary 160. The remaining octets become 0.

Network address: 18.160.0.0

Step 4: Broadcast address

The next boundary is 192. The preceding second-octet value is 191.

The remaining octets are 255.

Broadcast address: 18.191.255.255

Step 5: Usable host range

- First usable host: 18.160.0.1
- Last usable host: 18.191.255.254

  Final results — /11

- Network: 18.160.0.0
- First host: 18.160.0.1
- Broadcast: 18.191.255.255
- Last host: 18.191.255.254

Example 5: 18.172.2.77/17

Given:

- IP address: 18.172.2.77
- CIDR: /17
- Subnet mask: 255.255.128.0

Step 1: Identify the interesting octet

The third octet has a mask of 128.

Each block contains 128 addresses in the third octet.

Step 2: Find the subnet range

The third octet is 2.

The possible ranges are:

0–127

- 128–255

  The value 2 falls in the first range, 0–127.

Step 3: Network address

Network address: 18.172.0.0

Step 4: Broadcast address

The last address in the block has a third octet of 127 and a fourth octet of 255.

Broadcast address: 18.172.127.255

Step 5: Usable host range

- First usable host: 18.172.0.1
- Last usable host: 18.172.127.254

  Final results — /17
- Network: 18.172.0.0

- First host: 18.172.0.1
- Broadcast: 18.172.127.255
- Last host: 18.172.127.254

Important Exam Tips

- Practice writing the subnetting chart before starting an exam.
- For an online exam, practice entering the chart into a notepad or using the online whiteboard.
- For an in-person exam, check the testing center's rules regarding dry-erase markers.
- The magic number method, 7-second subnetting, and other shortcuts follow similar principles.
- Choose a method that works for you and practice it until the calculations become familiar.

  The main concept is dividing a network into smaller address blocks and identifying the network address, broadcast address, and usable host range.

Software-Defined Networking (SDN)

Overview

Virtualization is not limited to servers and operating systems. It can also be applied to routers, switches, firewalls, and other networking devices.

Networking functionality can be divided into three planes:

1. Data plane (infrastructure layer).
2. Control plane.
3. Management plane (application layer).

Separating these functions allows networking capabilities to be implemented in software and deployed in virtual environments.

Data Plane

The data plane is responsible for forwarding traffic and moving data from one interface to another.

It performs the heavy lifting of the networking device.

Functions include:

- Forwarding traffic.

- Transferring information between interfaces.
- Trunking.
- Encryption.
- Network Address Translation (NAT).

The data plane is also referred to as the infrastructure layer.

Control Plane

The control plane determines how traffic should be forwarded.

It contains information and tables used to make forwarding decisions.

Examples include:

- Routing tables.
- Switching tables.
- Network Address Translation tables.
- Other information that determines how the device operates.

The control plane provides the logic that guides the data plane.

Management Plane

The management plane is used by network administrators to manage and configure devices.

It is also referred to as the application layer.

Examples include:

- Connecting to a device using SSH.
- Managing a device through its console.
- Using a web-based management interface.

The management plane allows administrators to control and manage the networking device.

Example: Physical Firewall

A physical firewall can be divided into the three planes.

Firewall Components

Data plane — Interfaces

Connects networks and forwards data from one interface to another.

Control plane — Internal tables

Contains routing, switching, and filtering information that determines which traffic is forwarded or blocked.

Management plane — Console and web interface

Allows administrators to configure and manage the firewall.

When these functions are implemented through software, they can be deployed in virtualized environments.

Software-Defined Wide Area Network (SD-WAN)

What Is SD-WAN?

SD-WAN stands for Software-Defined Wide Area Network.

It is a wide area networking technology designed to manage the complexity of cloud-based environments.

Traditionally, organizations kept their applications and data in a centralized data center. Remote offices connected to that data center through WAN links.

With cloud computing, services such as email, databases, and applications may be distributed across multiple cloud providers and locations.

SD-WAN helps connect users and remote sites to these distributed resources.

Traditional WAN vs. Cloud-Based WAN

Traditional WAN

- Services were located in a centralized data center.
- Remote locations connected to the central data center using WAN links.
- Users accessed email, databases, and other applications through that data center.
- The network architecture was relatively straightforward because resources were centralized.

Cloud-Based Environment

- Email services may be hosted in the cloud.
- Applications and databases may be distributed across multiple cloud providers.
- Resources may be located anywhere in the world.
- Application locations can change over time.

This creates a need for a WAN that can identify applications and direct traffic to the appropriate resources.

Application Awareness

An important characteristic of SD-WAN is its ability to identify the application being transmitted over the network.

For example:

- If a user accesses email, SD-WAN can direct traffic toward the appropriate email service.
- If a user accesses a database, SD-WAN can direct traffic toward the relevant database service.

Application awareness helps the network determine where application data should be forwarded.

Zero-Touch Provisioning

Zero-touch provisioning allows remote routers and switches to update themselves automatically when network information or application locations change.

Characteristics:

- Remote devices learn how to reach services.
- Network changes can be distributed automatically.
- Application location changes can be reflected across connected systems.
- Manual user intervention is not required for each individual device update.

This allows SD-WAN devices to adapt when services move or network configurations change.

Transport Independence

SD-WAN is designed to be agnostic to the underlying transport technology.

Different cloud providers or remote locations may use different connectivity types, including:

- High-speed fiber.
- 5G.
- DSL.

SD-WAN can support connectivity across these different transport types.

Centralized Policy Management

Instead of configuring policies individually on every SD-WAN router, administrators can use a central management console.

The process is:

1. Configure policies through the central management console.
2. Make the required network changes centrally.
3. Push those changes to SD-WAN routers automatically.

This centralizes network policy management.

SD-WAN Traffic Flow

SD-WAN allows remote locations to access resources based on where those resources reside.

- Resources in the data center can still be accessed directly through the data center.
- Email and database services hosted in the cloud can be accessed directly through SD-WAN.
- Traffic can be directed toward the appropriate cloud-based application.

Key takeaway: SD-WAN provides application-aware connectivity, centralized policy management, zero-touch provisioning, and support for different transport technologies in distributed cloud environments.

Data Center Interconnection (DCI) and VXLAN

What Is Data Center Interconnection?

Data Center Interconnection (DCI) is a way to connect multiple data centers seamlessly.

Organizations and service providers may have applications, servers, and other resources distributed across several data centers located in different geographical areas.

DCI allows these environments to operate as connected resources rather than isolated data centers.

Why DCI Is Needed

Different data centers may have:

- Different IP addressing schemes.
- Different network infrastructures.

- Different connectivity types.
- Different bandwidth capabilities.

For example:

- A metropolitan data center may have high-speed fiber connectivity and large amounts of bandwidth.
- Another data center may use Metro Ethernet or copper-based connectivity.

Applications should be able to operate across these environments without depending on the specific addressing scheme or physical connectivity of each data center.

What Is VXLAN?

VXLAN stands for Virtual Extensible LAN.

It is designed to connect virtual networks across data centers, including large cloud-based environments.

VXLAN allows applications and virtual machines to communicate across different data centers without being concerned about the underlying network infrastructure.

VLAN vs. VXLAN

- VLAN:
    - Network Layer: Layer 2.
    - Number of Virtual Networks: Approximately 4,000.
    - Routing: VLANs themselves are not Layer 3 routed.
    - Scope: Limited Layer 2 environment.
- VXLAN:
    - Network Layer: Layer 2 overlay over a Layer 3 network.
    - Number of Virtual Networks: Up to approximately 16 million.
    - Routing: VXLAN tunnels can traverse Layer 3 networks.
    - Scope: Can connect virtual networks across data centers.

VXLAN allows virtual networks to communicate across Layer 3 infrastructure, including networks using the public internet as transport.

VXLAN Data Center Example

Consider two data centers.

Data Center A

Contains:

- A virtualization server.
- A virtual switch.
- Three virtual machines: A1, B1, and C1.
- A top-of-rack switch with a VXLAN Tunnel Endpoint (VTEP).

VTEP IP address: 1.1.1.1

Data Center B

Contains:

- A virtualization server.
- A virtual switch.
- Three virtual machines: A2, B2, and C2.
- A top-of-rack switch with its own VTEP.

VTEP IP address: 2.2.2.2

Virtual Network Associations

The virtual machines are associated as follows:

- A1 communicates with A2.
- B1 communicates with B2.
- C1 communicates with C2.

Each data center supports the same VXLAN Network Identifiers (VNIs):

- VNI 2000.
- VNI 3000.
- VNI 4000.

The objective is to connect matching VNIs across the two data centers.

- **Data Center A:**
    - VNI 2000
    - VNI 3000

    - VNI 4000
- **Data Center B:**
    - VNI 2000
    - VNI 3000
    - VNI 4000

VXLAN Tunnel and Encapsulation

VXLAN uses an existing IP network to transport encapsulated Ethernet frames between data centers.

The process works as follows:

1. A virtual machine generates an Ethernet frame in the first data center.
2. The original Ethernet frame contains an Ethernet header, an IP header, and the payload.
3. The frame is encapsulated inside a VXLAN header.
4. The VXLAN header is carried within a UDP header, an outer IP header, and an Ethernet frame.
5. The encapsulated data travels through the VXLAN tunnel over the existing IP network.
6. The receiving VTEP decapsulates the information.
7. The original Ethernet frame is delivered into the second data center.

VXLAN Encapsulation

Original Ethernet Frame

Ethernet header + IP header + payload

VXLAN Encapsulation

Outer Ethernet + IP + UDP + VXLAN + original frame

VXLAN Tunnel

Transport across the existing IP network

Decapsulation

The original Ethernet frame is recovered and delivered to the destination virtual machine.

VXLAN allows virtual machines in separate data centers to behave as if they were directly connected within the same virtual network.

Zero Trust Security

What Is Zero Trust?

Traditionally, organizations focused on protecting the network perimeter. This controlled who could enter or leave the network, while internal network resources were often broadly accessible.

Zero Trust is a holistic security approach that treats every user, device, and application as inherently untrusted.

The objective is to verify traffic and ensure that only authorized users can access the appropriate resources.

Zero Trust applies to every user and application across the network.

Zero Trust Security Controls

Zero Trust may involve:

- Authentication.
- Encryption.
- Additional firewalls.
- Network monitoring.
- Other security controls.

These technologies help verify users and protect access to network resources.

Authentication

Authentication verifies the identity of a user.

A typical authentication process may involve:

- Username.
- Password.
- Additional authentication factors.

Organizations can also use policy-based authentication, including adaptive identity.

Adaptive Identity

Adaptive identity evaluates several factors during authentication to determine the risk associated with a login attempt.

These factors include:

User identity

- Is the user a long-term employee?
- Is the user a recently onboarded vendor?

Geographic location

- Is the user authenticating from a familiar geographic area?
- Is the user authenticating from another country?

IP address and connection

- What IP address is being used?
- What type of connection is being used?
- Is the connection coming through a VPN?

The organization can combine these factors to determine whether additional authentication or restrictions are necessary.

Adaptive Authentication Examples

Example 1: Familiar connection

> A user connects from inside the corporate building during normal working hours.
>
> The organization may allow authentication using a username and password.

Example 2: Unusual connection

> A user connects from an unfamiliar IP address in another country using a VPN.
>
> The organization may require additional authentication factors.

Example 3: Unusual time or location

> A user provides the correct username and password but connects at an unusual time or from an unusual location.
>
> The organization may deny the authentication.

Adaptive identity allows authentication policies to respond to contextual information rather than relying only on credentials.

Authorization and Access Control

Authentication verifies who a user is. Authorization determines what resources that user is allowed to access.

Once authentication succeeds, the organization determines the user's access rights and permissions.

Example: Help Desk Access

- Help Desk Employee: May view the hardware database.
- Help Desk Manager: May view and modify the hardware database.
- User Outside the Help Desk: May have no access to the hardware database.

Authorization may also depend on:

- User role.
- Geographic location.
- Device being used.
- Whether the device is company-provided.
- Whether the device's certificate can be verified.

A company-managed laptop with a verified certificate may receive additional permissions.

Principle of Least Privilege

A key security best practice is to provide users with only the permissions required for their jobs.

Examples:

- If a user requires read access to a database, they should not receive permission to modify it.
- Administrative privileges should not be provided simply to access a particular resource.
- Users should operate with limited rights and permissions.

Why Administrative Privileges Are Risky

Administrators may have access to data and systems throughout the organization.

If a device with administrative access becomes infected with malware, the malware may inherit those administrative privileges.

This can potentially give malware access to a much larger set of systems and data.

The principle of least privilege reduces this risk by limiting the permissions available to users.

Secure Access Service Edge (SASE)

What Is SASE?

SASE stands for Secure Access Service Edge.

It provides network and security services through the cloud, close to where application data resides.

It can be thought of as a next-generation virtual private network that combines networking and security functionality.

SASE supports users regardless of where they are located and where the applications reside.

Why SASE Is Needed

Users may work from:

- Corporate offices.
- Home.
- Remote locations around the world.

Applications may reside in:

- Public cloud environments.
- Private data centers.
- Internet-accessible services.

A secure communication mechanism is needed regardless of the user's location or the application's location.

SASE helps provide this secure access.

SASE Components and Capabilities

A SASE environment may include:

- **Network as a Service (NaaS):** Provides network connectivity and services.
- **Quality of Service (QoS):** Supports traffic handling and prioritization.
- **Routing:** Directs traffic toward the appropriate resources.

- **Zero Trust Network Access (ZTNA):** Controls access based on identity and authorization.
- **Firewall as a Service (FWaaS):** Provides firewall security through the cloud.
- **DNS Security:** Provides security controls related to DNS

These services are delivered through the cloud-based SASE environment.

SASE Client and Operation

A SASE client can be installed on each user's device.

The client allows the user to access network and security services regardless of location.

The process is designed to be automatic:

1. The user uses their usual device.
2. The user connects to the application they need.
3. The SASE client provides access to the secure service.
4. Network and security functionality operate as part of the access process.

Users do not need to manually turn SASE functionality on or off for each connection.

Key takeaway: SASE brings networking and security services closer to cloud applications and provides secure access for users working from different locations.

Infrastructure as Code (IaC)

What Is Infrastructure as Code?

Infrastructure as Code (IaC) is a method of describing infrastructure configurations using code.

Instead of manually building, installing, and configuring every system, administrators create configuration files that describe the infrastructure.

These files can define:

- Servers.
- Network devices.
- Firewalls.
- Switches.
- Routers.

- Applications.

The cloud or infrastructure platform can then use the code to build systems according to the specified configuration.

What Can IaC Configuration Files Contain?

A configuration file can describe the details required to create and configure an environment.

Examples include:

- Hostnames.
- CPU specifications.
- Network configuration.
- IP addressing.
- Applications to install.
- Server and database configurations.
- Other infrastructure requirements.

Example

An IaC configuration might define a host called mail.example.com.

It could also define child systems such as:

- Web servers.
- Database servers.

  The configuration file describes the systems and their required settings.

Automated Infrastructure Deployment

IaC allows infrastructure to be deployed repeatedly using the same configuration.

The process is:

1. Create an infrastructure configuration file.
2. Deploy the configuration.
3. Make changes to the code when necessary.
4. Redeploy the configuration.
5. Apply the changes to the infrastructure.

IaC can also be used to create an identical environment in another data center by applying the same configuration.

This provides flexibility in where infrastructure can be deployed.

Infrastructure Duplication

One of the key benefits of IaC is the ability to reproduce infrastructure.

For example:

- Create an application environment in one data center.
- Reuse the same configuration file in another data center.
- Deploy an identical application instance.

This allows organizations to create consistent infrastructure across different locations.

Configuration Drift

Configuration drift occurs when different application instances or systems develop differences in their configurations.

For example, administrators might manually make minor changes to different systems, resulting in inconsistent configurations.

IaC helps reduce configuration drift by ensuring that systems are created and configured using the same definitions.

This supports consistent infrastructure deployment and configuration management.

Compliance and Standardization

IaC can help organizations maintain compliance with defined standards.

Because systems are created using the same configuration definitions, organizations can:

- Apply consistent settings.
- Maintain standardized infrastructure.
- Document configurations.
- Deploy systems according to required standards.

IaC can also ensure that a testing environment matches the production environment.

A configuration used for testing can be applied to production to create an identical environment.

Infrastructure Upgrades and Changes

IaC can be used to make changes to existing infrastructure.

For example:

1. Modify the configuration file.
2. Deploy the updated configuration.
3. The system identifies the differences.
4. The required configuration changes are applied.

This can be used for software upgrades, configuration changes, and other infrastructure updates.

Documenting Existing Infrastructure

IaC can also be used to document the configuration of an existing system.

The process involves:

1. Inspecting the existing system.
2. Recording its configuration in an IaC definition file.
3. Using that file as documentation.
4. Reusing the file to reproduce the system when required.

This provides both documentation and a reusable infrastructure definition.

Source Control and Version Control

Why Source Control Is Needed

In large environments, infrastructure definition files become extremely important.

Administrators need to ensure that the latest, approved configuration is used when deploying systems.

Source control helps manage changes to these files and ensures that configuration changes are tracked and controlled.

It prevents multiple administrators from making untracked changes to different copies of a configuration.

Git and Version Control

Git is a popular version control system used to maintain source code and configuration files across multiple users.

A centralized repository can store infrastructure configuration files and track changes made by different users.

The repository allows teams to:

- Track configuration changes.
- Maintain versions of configuration files.
- Test changes before deployment.
- Combine changes from multiple contributors.
- Manage the final production configuration.

Working with Multiple Contributors

Multiple administrators may work on different parts of the same infrastructure configuration.

Each person can make changes to the section relevant to their work.

Those changes can then be merged into a centralized configuration.

This allows several people to contribute to infrastructure development while maintaining a common version of the configuration.

Merge Conflicts

A merge conflict occurs when multiple people make different changes to the same line of code or configuration.

For example:

- Administrator A changes a particular configuration line.
- Administrator B makes a different change to that same line.
- The changes cannot be merged automatically without resolving the conflict.

Version control software identifies these conflicts and provides ways to manage them.

The conflict may be resolved automatically or may require an administrator to decide which change should be included in the final configuration.

Branching

Branching allows administrators to create separate versions of configuration files for testing and development.

A typical workflow is:

1. Start with the production configuration.
2. Create a separate branch for testing.
3. Make changes and perform testing.
4. Validate the changes.
5. Merge the tested changes back into the original configuration.
6. Deploy the updated production configuration.

This allows testing and optimization without directly changing the production configuration.

Infrastructure as Code Workflow

1. Create IaC configuration

   Define servers, networks, applications, and settings.

2. Store in version control

   Track changes and maintain configuration versions using Git.

3. Create a testing branch

   Make changes and test the updated configuration.

4. Resolve conflicts and merge

   Combine approved changes into the final configuration.

5. Deploy to production

   Apply the approved configuration to production infrastructure.

Playbooks and Security Orchestration

What Is a Playbook?

A playbook is a predefined series of steps used to investigate, resolve, or respond to a particular issue.

Playbooks allow organizations to automate processes that would otherwise be performed manually.

Examples include:

- Investigating a data breach.

- Recovering from ransomware.
- Responding to malware infections.
- Resolving recurring security issues.

Example: Ransomware Response Playbook

A system may identify malware and reference a playbook to determine the response actions.

The playbook may contain steps to:

1. Remove the infected device from the network.
2. Delete the contents of the storage drive.
3. Reimage the system.
4. Redeploy the system into the infrastructure.

The same playbook can be reused whenever the organization encounters the same type of issue.

SOAR

SOAR stands for Security Orchestration, Automation, and Response.

A SOAR platform provides a centralized management console for security operations.

It can be used to:

- Centralize security operations.
- Implement playbooks.
- Automate response processes.
- Monitor playbook execution.

Playbooks can be deployed and managed through SOAR platforms.

Relationship Between IaC, Playbooks, and SOAR

- Infrastructure as Code (IaC): Defines and deploys infrastructure configurations using code.
- Playbooks: Defines repeatable steps for resolving or responding to issues.
- SOAR: Centralizes and automates security operations and playbook execution.
- Git / Version Control: Tracks and manages changes to configuration files.

Together, these technologies support consistent infrastructure deployment and repeatable security response processes.

Quick Revision: Key Concepts

- Magic Number: 256 minus the subnet mask value in the interesting octet.
- Network Address: First address in the subnet.
- Broadcast Address: Last address in the subnet.
- First Usable Host: Network address + 1.
- Last Usable Host: Broadcast address − 1.
- SDN (Software-Defined Networking): Separates data, control, and management functions.
- SD-WAN (Software-Defined WAN): Application-aware WAN with centralized policy management.
- DCI (Data Center Interconnect): Connects multiple data centers.
- VXLAN: Extends virtual networks across Layer 3 infrastructure.
- Zero Trust: Every user, device, and application is inherently untrusted.
- Adaptive Identity: Uses contextual information to evaluate authentication risk.
- Least Privilege: Provides only the permissions required for a user's job.
- SASE (Secure Access Service Edge): Cloud-delivered networking and security services.
- IaC (Infrastructure as Code): Describes and deploys infrastructure through code.
- Configuration Drift: Differences between configurations of systems.
- Git: Tracks and manages changes to code and configuration.
- Playbook: Repeatable steps for responding to an issue.
- SOAR (Security Orchestration, Automation, and Response): Centralized security orchestration, automation, and response.

IPv4 Address Exhaustion and IPv6

IPv4 Address Exhaustion

- The number of devices connected to the internet is estimated at approximately 20–30 billion.

- IPv4 supports approximately 4.29 billion unique addresses because it uses a 32-bit address.
- IPv4 address allocation has been exhausted, meaning no more IPv4 addresses are available for general allocation from the exhausted pools.
- Network Address Translation (NAT) helps extend the use of IPv4 by allowing many devices to communicate using a limited number of public IPv4 addresses.
- Home and office networks commonly use NAT to connect hundreds or thousands of devices to the internet.
- Although NAT supports most applications, some applications work better without NAT because it introduces additional communication complexity.

Introduction to IPv6

IPv6 was introduced to overcome the limited address space of IPv4.

IPv4 vs IPv6

- **Address Length**
  - IPv4: 32 bits
  - IPv6: 128 bits
- **Address Space**
  - IPv4: Approximately 4.29 billion addresses
  - IPv6: Extremely large address space
- **Representation**
  - IPv4: Decimal numbers separated by periods
  - IPv6: Hexadecimal groups separated by colons

- IPv6 provides a much larger address space for assigning addresses to internet-connected devices.
- The transcript illustrates the scale by stating that every grain of sand on Earth could have approximately 45 quintillion unique IPv6 addresses.
- IPv6 resolves the address-space limitation associated with IPv4.

IPv6 Address Structure

Example IPv6 address:

fe80::5d18:652:6ffd:8f52

- IPv6 addresses use hexadecimal values rather than the decimal notation used in IPv4.
- Colons separate the groups of an IPv6 address.
- A complete IPv6 address contains 128 bits.
- An IPv6 address is divided into eight groups, each containing 16 bits.
- Each group is equivalent to:
    - 2 bytes.
    - 2 octets.
- Total address length:
    - 128 bits.
    - 16 bytes.

IPv6 Address Representation

- Full form: All 128 bits are represented using eight hexadecimal groups.
- Binary form: Each group contains 16 binary bits, represented using 0s and 1s.
- Compressed form: Consecutive zeros and leading zeros can be shortened to make the address easier to read.

IPv6 Address Compression Rules

There are two main rules for compressing IPv6 addresses.

Rule 1: Remove Leading Zeros

- Leading zeros can be removed from each individual hexadecimal group.
- Only zeros at the beginning of a group can be removed.
- Zeros occurring between nonzero digits must remain.

Example:

04C3 becomes 4C3

0066 becomes 66

Rule 2: Replace Consecutive Zero Groups with ::

- Two or more consecutive groups containing only zeros can be replaced by a double colon (::).

- The double colon can be used only once in an IPv6 address.
- This rule simplifies the representation of an address containing multiple consecutive zero groups.

Example 1: IPv6 Compression

Original address:

2600:DDDD:0000:0000:0000:1111:0000:0001

Step 1: Remove leading zeros from each group.

2600:DDDD:0:0:0:1111:0:1

Step 2: Replace the longest consecutive sequence of zero groups with ::.

Compressed address:

2600:DDDD::1111:0:1

Example 2: IPv6 Compression

Original address:

2601:04C3:4002:00BE:0000:0000:0000:0066

Step 1: Remove leading zeros.

2601:4C3:4002:BE:0:0:0:66

Step 2: Replace the consecutive zero groups with ::.

Compressed address:

2601:4C3:4002:BE::66

IPv4 and IPv6 Interoperability

- IPv4 and IPv6 use different address formats and cannot directly communicate with each other without a mechanism that bridges the protocols.
- Legacy systems may support only IPv4, while newer networks and devices may support IPv6.
- Three methods are used to support communication between IPv4 and IPv6 networks:
    1. Tunneling.
    2. Dual stack.
    3. Translation.

A. Tunneling

- Tunneling encapsulates one IP protocol inside another.
- IPv6 traffic can be carried over an IPv4 network.
- IPv4 traffic can also be carried over an IPv6 network.
- It allows communication across networks that do not natively support the protocol being transported.
- Tunneling was used extensively during the early stages of IPv6 deployment.
- It was intended primarily as a short-term migration solution.

6to4 Tunneling

- Allows IPv6 traffic to travel over an existing IPv4 network.
- Creates an IPv6 address based on the existing IPv4 address.
- Requires a specialized relay router to transfer the data.
- Does not support Network Address Translation (NAT).
- Because of its limitations and increased native IPv6 support, 6to4 is no longer commonly used.
- The transcript states that this functionality has been removed from all Windows versions.

4in6 Tunneling

- Carries IPv4 traffic over an existing IPv6 network.
- It is the reverse of carrying IPv6 traffic over IPv4.
- Like 6to4, it was used as a relatively short-term migration solution.
- Both 6to4 and 4in6 are uncommon on modern enterprise networks.

B. Dual Stack

- Dual stack allows IPv4 and IPv6 to operate simultaneously on the same system.
- A network adapter can have both an IPv4 address and an IPv6 address.
- Applications can use IPv4 or IPv6, depending on their requirements and network support.

How dual stack works:

1. The system has both IPv4 and IPv6 addresses assigned to its network adapter.
2. IPv4 applications use the IPv4 address and IPv4 routing table.
3. IPv6 applications use the IPv6 address and IPv6 routing table.
4. IPv4 and IPv6 routing tables operate independently.
5. The configuration allows applications to communicate using either protocol.

- Dual stack is described in the transcript as a practical middle ground for IPv4-to-IPv6 migration.

C. Translation

- Translation converts communication between IPv4 and IPv6.
- A specialized router performs the translation between the two protocols.
- NAT64 is an example of this translation method.

NAT64 and DNS64

NAT64

- NAT64 enables IPv6-only devices to communicate with IPv4-only devices.
- It uses a NAT64-capable router to translate IPv6 traffic into IPv4 traffic and vice versa.
- The router translates the address information so that each side can communicate using its supported IP version.

DNS64

- DNS64 is a specialized DNS mechanism used with NAT64.
- It translates DNS responses to provide an IPv6-compatible address for an IPv4-only destination.
- It helps an IPv6 client reach a server that supports only IPv4.

NAT64 Communication Process

Example: An IPv6 client wants to access professormesser.com, which supports only IPv4.

1. The IPv6 client sends a DNS request to the DNS64 server.
2. The DNS64 server makes a DNS request on behalf of the client to the DNS server associated with the IPv4 network.

3. The IPv4 DNS server returns the IPv4 address of the destination server.
4. DNS64 creates an IPv6 representation of that address that directs traffic to the NAT64 router.
5. The IPv6 client receives what appears to be the IPv6 address of the destination.
6. The browser sends its initial TCP SYN packet to that IPv6 address, which leads to the NAT64 router.
7. The NAT64 router translates the IPv6 packet into IPv4 so that the IPv4 web server can understand it.
8. The web server sends its response back.
9. The NAT64 router translates the response back into IPv6 and forwards it to the client.

Result: An IPv6-only client and an IPv4-only server can communicate through NAT64 and DNS64.

Routing Fundamentals

Purpose of a Router

- A router forwards traffic between different IP subnets.
- It examines the destination IP address in an incoming packet.
- It consults its routing table to determine the appropriate route.
- It forwards the packet through the appropriate interface toward its destination.

Basic Routing Process

1. The router receives a packet.
2. It identifies the destination IP address.
3. It examines its routing table to find a matching route.
4. If the destination network is directly connected, the packet is forwarded onto that local subnet.
5. If the destination is on a remote network, the router identifies the next hop.
6. The packet is forwarded to the next router through the appropriate interface.

7. Each subsequent router repeats the process until the packet reaches its destination.

- The next hop is the IP address of the next router along the route.
- If the routing table does not contain a valid route or matching default route, the router discards the packet.

Routing Table

- A routing table contains information used to determine where packets should be forwarded.
- Routers, workstations, and servers maintain routing tables.
- Routing tables contain information about directly connected networks and routes learned or configured for remote networks.
- When troubleshooting routing problems, the routing table is a key source of information.

A routing table helps identify:

- The destination network.
- The prefix length.
- The next-hop IP address.
- The outgoing interface.
- The source of the route.

Directly Connected Networks

Consider Router 1 connected to three networks:

- 10.10.10.0/24
- 10.10.40.0/24 — network between Router 1 and Router 2.
- 10.10.50.0/24 — network between Router 1 and Router 3.

Router 1 automatically knows about these directly connected networks.

However, Router 1 does not automatically know about remote networks behind Router 2 or Router 3 unless routes are configured or learned.

Remote networks:

- 10.10.20.0/24 — behind Router 2.
- 10.10.30.0/24 — behind Router 3.

If Sam sends a packet to Jack at 10.10.20.2:

1. Sam sends the packet to Router 1.
2. Router 1 checks the destination IP address.
3. It looks for a matching route to 10.10.20.0/24.
4. If the route is missing, Router 1 discards the packet.
5. The packet never reaches Router 2 or Jack.

This demonstrates why routers need complete and valid routing tables.

Static Routing

Definition

- Static routing is the administrative process of manually configuring routes in routers.
- The network administrator specifies the destination network and the next hop.
- Static routes remain unchanged until an administrator modifies or removes them.

Advantages of Static Routing

- Quick and straightforward to configure on a small network.
- Does not require dynamic routing protocol updates.
- Uses no additional CPU or memory for dynamic route-discovery processing.
- Provides predictable routing when the network topology is stable.
- Suitable for stub networks, such as remote locations with a single internet connection.
- Avoids the overhead associated with dynamic routing protocols.

Disadvantages of Static Routing

- Routes must be manually configured on every relevant router.
- Configuration becomes difficult on networks with hundreds or thousands of routers.
- Administrators must ensure that routes are correctly configured.
- Incorrect routes can create routing loops.

- Static routes do not automatically change when the network topology changes.
- If a router or link fails, routes must be manually updated to use another path.
- There is no automatic rerouting when a configured path becomes unavailable.

Configuring Static Routes: Example

Router 1 knows its three directly connected networks but needs routes to:

- 10.10.20.0/24 behind Router 2.
- 10.10.30.0/24 behind Router 3.

The administrator configures the following routes on Router 1.

Route 1: Network Behind Router 2

- Destination network: 10.10.20.0/24
- Next hop: 10.10.40.2

When Router 1 receives traffic destined for 10.10.20.0/24, it forwards the packet to 10.10.40.2.

Router 2 then recognizes the destination network as directly connected and forwards the packet.

Route 2: Network Behind Router 3

- Destination network: 10.10.30.0/24
- Next hop: 10.10.50.2

When Router 1 receives traffic destined for 10.10.30.0/24, it forwards the packet to 10.10.50.2.

Router 3 then forwards the packet to the destination network.

Result: Router 1 can forward traffic to both remote networks because the administrator has added the necessary static routes.

Dynamic Routing

Definition

- Dynamic routing allows routers to automatically discover routes and exchange routing information.

- Routers use dynamic routing protocols to learn about remote networks and update their routing tables.
- Network administrators do not need to manually configure every route on every router.
- When a router is added or removed, routing information can be updated automatically.

How Dynamic Routing Works

1. Routers listen for routing updates from other routers on their local networks.
2. Routing information is exchanged directly between routers, often using multicast.
3. Each router builds or updates its own routing table using the received information.
4. Routers advertise the routes they know to neighboring routers.
5. Receiving routers determine whether a new route is better than an existing route or should be retained as an alternative.
6. When network infrastructure changes, routers exchange updates so that routing tables reflect the new topology.

Dynamic Routing Example Using EIGRP

Consider Router 1, Router 2, and Router 3.

Router 1 is directly connected to three subnets but does not initially know about the remote networks behind Routers 2 and 3.

EIGRP Update from Router 2

- Router 2 sends an EIGRP routing update to Router 1.
- Router 1 learns a route to 10.10.20.0/24.
- The next hop is 10.10.40.2.
- Router 1 updates its routing table automatically.

EIGRP Update from Router 3

- Router 3 sends an EIGRP routing update to Router 1.
- Router 1 learns a route to 10.10.30.0/24.
- The next hop is 10.10.50.2.

- Router 1 updates its routing table.

When routers or links are added or removed, further EIGRP updates can add or remove routes automatically.

Dynamic Routing Overhead

Dynamic routing requires additional processing within routers.

- Routers must listen for routing updates.
- They must process and interpret information received from neighboring routers.
- They must determine whether a route is better than an existing route.
- They must update their routing tables.
- They must inform other routers when network changes occur.

This requires CPU and memory resources. Routers may need additional monitoring to ensure they can handle the processing load.

Choosing a Dynamic Routing Protocol

Different routing protocols use different criteria to select routes.

Possible criteria include:

- Number of hops to a destination.
- Link availability and whether a connection is up or down.
- Connection speed.
- Link cost.
- How quickly the protocol can respond to topology changes.

Other considerations include:

- Router hardware and manufacturer.
- Network size and topology.
- Convergence time.
- Compatibility with other network devices.

EIGRP — Enhanced Interior Gateway Routing Protocol

- EIGRP is commonly associated with Cisco networking environments.
- It is relatively easy to configure on Cisco routers.

- It requires a few configuration options to enable.
- It tends to converge relatively quickly when network changes occur.
- It can identify and prevent routing loops.
- It identifies other EIGRP-enabled routers and exchanges routing information efficiently.
- Its efficient updates help minimize routing traffic and preserve bandwidth.

Although EIGRP is available on some other manufacturers' devices, certain aspects are proprietary to Cisco, so it is commonly encountered in Cisco-centric environments.

OSPF — Open Shortest Path First

- OSPF is a standardized dynamic routing protocol.
- It is available on devices from many manufacturers.
- It is commonly used within an organization's autonomous system.
- An autonomous system (AS) is a network under a common administrative control.
- For example, an organization's WAN containing 50 routers may operate as a single autonomous system.

OSPF Characteristics

- OSPF is a link-state routing protocol.
- It evaluates link state and availability between OSPF routers.
- It assigns costs to individual links.
- Different links may have different costs.
- Link cost can reflect throughput, availability, and the time required to traverse a connection.
- OSPF selects the route with the lowest total cost.
- If multiple paths have identical costs, many implementations support load balancing across those paths.

BGP — Border Gateway Protocol

- BGP is an external gateway routing protocol.
- It is used to route traffic between autonomous systems.
- It is commonly used on WANs and internet connections.

- BGP supports dynamic routing between organizations and across the internet.
- It was designed to address the complexity of maintaining and updating routes across the internet.

If an organization has one or more internet connections and needs to dynamically route traffic to external networks, BGP is a protocol used for that purpose.

Routing Table Entries and Route Selection

Understanding Routing Table Codes

A routing table may contain routes that are directly connected or learned through dynamic routing protocols.

Example route codes:

- C — Directly connected route.
- R — Route learned through RIP.
- D — Route learned through EIGRP.

A routing table may also include routes learned through OSPF, BGP, and other routing methods.

Components of a Routing Table Entry

Example:

R 10.10.30.0/24 [120/1] via 10.10.50.2, ...

The entry contains several important fields:

- Route Code (R): Indicates that the route was learned through RIP.
- Destination Network (10.10.30.0/24): The remote network and its prefix length.
- Administrative Distance (120): Indicates the preference of the route source.
- Metric (1): The routing protocol's value for evaluating the route.
- Next Hop (10.10.50.2): The IP address of the next router.
- Timer: Indicates how long the route has been active in the routing table.
- Outgoing Interface: The interface used to forward traffic toward the next hop.

The exact fields displayed depend on the router and routing protocol.

How Routers Select Routes

When a router receives a packet, it compares the destination IP address with the network addresses and prefix lengths in its routing table.

Route selection involves:

1. Finding matching destination networks.
2. Selecting the most specific matching route.
3. If competing routes have the same prefix length, considering administrative distance.
4. If routes learned from the same routing protocol have equivalent destinations and prefix lengths, using the protocol's routing metric to select a route.

Longest Prefix Match

- A destination IP address may match multiple routes in a routing table.
- The router selects the route with the longest matching prefix.
- A longer prefix identifies a smaller and more specific network range.

Example

Destination IP address:

192.168.1.6

The routing table contains:

- 192.168.0.0/16
- 192.168.1.0/24
- 192.168.1.6/32

All three routes match the destination.

However:

- /16 identifies a larger network.
- /24 identifies a more specific network.
- /32 identifies the individual IP address 192.168.1.6.

Therefore, the router selects:

192.168.1.6/32

If the /32 route is unavailable, the router selects 192.168.1.0/24 rather than 192.168.0.0/16.

Key point: The longest matching prefix is selected before comparing the preference of competing route sources.

Administrative Distance

Administrative distance is used to compare routes to the same destination learned from different sources.

- A lower administrative distance indicates a more preferred route.
- Administrative distance is used by the router to select between competing route sources.
- It is not the same as a routing metric.

Cisco Administrative Distance Values from the Transcript

1. Directly Connected: 0
2. Static Route: 1
3. EIGRP: 90
4. OSPF: 110
5. RIP: 120

- Directly connected routes have an administrative distance of 0 because the network is physically connected to the router.
- Static routes have an administrative distance of 1.
- EIGRP has an administrative distance of 90.
- OSPF has an administrative distance of 110.
- RIP has an administrative distance of 120.

These are the Cisco administrative distance values discussed in the supplied transcript.

Routing Metrics

- A routing metric is a value used by a routing protocol to evaluate routes.
- Different routing protocols use different metric calculations.
- Metrics cannot be directly compared across different routing protocols.

Examples:

RIP:

- Uses hop count as its routing metric.
- A route with one hop has a metric of 1.

EIGRP:

- Uses its own routing metric calculation.
- Its metric differs from RIP's hop-count metric.

OSPF:

- Uses link costs to determine the lowest-cost path.

BGP:

- Uses its own routing attributes and route-selection process.

Important Difference

- Administrative distance compares the preference of routes learned from different sources.
- A routing metric helps a routing protocol select among routes within its own routing process.

For example, RIP may advertise a route with administrative distance 120 and metric 1, while EIGRP may advertise the same destination with administrative distance 90 and a different metric.

The administrative distance and metric serve different purposes.

First Hop Redundancy Protocols (FHRP)

Default Gateway and Redundancy

- A default gateway is the local router that allows a device to communicate outside its local IP subnet.
- A device's IP configuration typically specifies one default gateway.
- If the default gateway fails, the device may lose connectivity to other subnets and the internet.
- To maintain connectivity, networks can use redundant routers.

Virtual IP Address (VIP)

- A Virtual IP (VIP) is an IP address shared by a group of redundant routers.

- Instead of configuring each workstation with different router addresses, devices use the virtual IP as their default gateway.
- If the primary router fails, another router can take over the virtual IP.
- This allows workstations to continue using the same default gateway without requiring changes to their configurations.

FHRP Operation

First Hop Redundancy Protocols (FHRPs) provide gateway redundancy.

Example configuration:

- Router 1: Active router.
- Router 2: Standby or backup router.
- Virtual IP: Default gateway used by the network devices.

Normal Operation

1. Router 1 is the active router.
2. Router 2 remains on standby.
3. Network devices send their traffic to the virtual IP.
4. Router 1 forwards the traffic to other networks or the internet.

Failover Operation

1. Router 1 fails or becomes unreachable.
2. Router 2 detects that the active router is no longer available.
3. Router 2 becomes the active router.
4. Router 2 takes control of the virtual IP address.
5. Network devices continue using the same default gateway.
6. Traffic is forwarded through Router 2.

Key point: FHRP allows a network to maintain gateway availability without manually changing the default gateway on every workstation.

Router Subinterfaces and Inter-VLAN Routing

Subinterfaces

- A subinterface is a logical or virtual interface created within a physical interface.

- A single physical Ethernet interface can be divided into multiple subinterfaces.
- Each subinterface can have its own configuration, including an IP address and subnet mask.
- Subinterfaces allow multiple IP subnets or VLANs to use one physical connection.

Example:

A physical interface may be named:

Ethernet 1/1

Subinterfaces may include:

- Ethernet 1/1.10
- Ethernet 1/1.20
- Ethernet 1/1.100

The subinterface identifiers correspond to separate logical interfaces configured on the same physical connection.

Router-on-a-Stick Configuration

- A switch connects to a router through a single physical Ethernet cable.
- The switch port connecting to the router is configured as a trunk.
- The trunk carries traffic from multiple VLANs.
- The router uses separate subinterfaces for the individual VLANs.
- Each subinterface can have its own IP address, subnet mask, and routing configuration.

Example

A network contains three VLANs:

- Red VLAN.
- Green VLAN.
- Blue VLAN.

The switch carries all three VLANs over a single trunk connection to the router.

On the router:

1. Create a separate subinterface for each VLAN.

2. Configure the appropriate VLAN association.
3. Assign an IP address and subnet mask to each subinterface.
4. Use the subinterfaces to route traffic between the VLANs.

Result: One physical router connection can provide routing functionality for multiple VLANs.

Network Address Translation (NAT)

Definition and Purpose

- Network Address Translation (NAT) translates IP addresses as traffic passes through a router.
- A common use case is translating private IPv4 addresses into public IPv4 addresses.
- Private IP addresses are not routable across the public internet.
- NAT allows devices using private addresses to communicate with public internet servers.

NAT is used for several purposes, but the supplied transcript focuses on translating private IPv4 addresses into public IPv4 addresses.

Private IPv4 Address Ranges

RFC 1918 defines the following private IPv4 address ranges.

1. 10.0.0.0 – 10.255.255.255
   - CIDR: /8
   - Subnet Mask: 255.0.0.0
2. 172.16.0.0 – 172.31.255.255
   - CIDR: /12
   - Subnet Mask: 255.240.0.0
3. 192.168.0.0 – 192.168.255.255
   - CIDR: /16
   - Subnet Mask: 255.255.0.0

- The 10.0.0.0/8 range is commonly used in large enterprise networks.
- The 192.168.0.0/16 range is commonly used in home networks.

- Private addresses cannot be routed directly across the public internet.
- NAT translates these addresses when devices need to communicate with public internet destinations.

Basic NAT Operation

Example:

- Internal device IP: 10.10.20.15
- Destination web server IP: 104.20.19.63
- Public IP address assigned by the NAT router: 94.111.x.x (illustrative notation from the transcript).

Outbound Communication

1. The internal device sends a packet to the public web server.
2. The packet contains the private source IP address 10.10.20.15.
3. The NAT router identifies the source address as private.
4. The router replaces the private source IP with a public IP address.
5. The packet is forwarded to the internet.
6. The web server receives the packet with the public source address.

Return Communication

1. The web server sends a response to the public IP address.
2. The NAT router receives the response.
3. The router identifies the corresponding translation.
4. It replaces the destination public IP address with the original private IP address.
5. The packet is forwarded to the internal device.

Result: A private-addressed device can communicate with a public internet server through NAT.

NAT Overload / Port Address Translation (PAT)

Purpose

- Basic NAT may require multiple public IP addresses when many internal devices communicate simultaneously.

- NAT overload, also called Port Address Translation (PAT), allows multiple internal devices to share a single public IP address.
- PAT translates both IP addresses and port numbers.
- The NAT router maintains a translation table to track each connection.

PAT Example 1

Internal device:

- Private IP: 10.10.20.15
- Source port: 3233

Destination server:

- Public IP: 104.20.19.63
- Destination port: 80

The NAT router translates:

10.10.20.15:3233 → 94.111.x.x:1055

- The source IP address becomes the public IP.
- The source port is changed to a unique port.
- The router records the mapping in its NAT translation table.

PAT Example 2

Another internal device:

- Private IP: 10.10.20.70
- Source port: 5782

This device connects to the same web server on port 80.

The NAT router can use the same public IP address but assign a different public-side port.

For example:

10.10.20.70:5782 → 94.111.x.x:1056

PAT Benefits

- Multiple internal devices can share a single public IPv4 address.
- Port numbers distinguish different connections.

- The NAT translation table tracks private and public address-and-port mappings.
- This makes more efficient use of the limited IPv4 public address space.

VLANs and Broadcast Domains

Local Area Network (LAN)

- A Local Area Network (LAN) is a group of devices within the same broadcast domain.
- Devices connected to the same switch and assigned to the same VLAN can communicate within that broadcast domain.
- Separate broadcast domains provide separation between groups of devices.

Example

- Red network: Devices connected to the red switch share one broadcast domain.
- Blue network: Devices connected to the blue switch share another broadcast domain.
- If the two switches are not connected, devices on the red network cannot directly communicate with devices on the blue network.

Virtual Local Area Network (VLAN)

- A VLAN divides a physical switch into multiple logical networks.
- Devices connected to the same physical switch can belong to different broadcast domains.
- VLANs provide logical separation without requiring separate physical switches.
- Each switch interface can be assigned to a particular VLAN.

Example

A single switch can contain:

- VLAN 1
- VLAN 2
- VLAN 3

Devices connected to interfaces assigned to VLAN 1 communicate within VLAN 1's broadcast domain.

Devices assigned to VLAN 2 or VLAN 3 belong to their respective broadcast domains.

Key point: VLANs allow multiple logically separated networks to share the same physical switch.

VLAN Trunking

Problem

Suppose two switches each have:

- VLAN 100.
- VLAN 200.

The objective is to allow VLAN 100 on one switch to communicate with VLAN 100 on the other switch, and do the same for VLAN 200.

One option is to use a separate physical connection for each VLAN.

However, this becomes inefficient when connecting many VLANs across multiple switches.

Solution: VLAN Trunk

- A VLAN trunk allows multiple VLANs to share a single physical connection between switches.
- It is also called an 802.1Q trunk or an 802.1Q link.
- The trunk carries traffic from multiple VLANs.
- VLAN tags identify which VLAN each frame belongs to.

IEEE 802.1Q VLAN Tagging

- A standard Ethernet frame contains:

- Preamble.
- Start Frame Delimiter.
- Destination MAC address.
- Source MAC address.
- Type.
- Payload.
- Frame Check Sequence.
- IEEE 802.1Q adds a VLAN tag to the Ethernet frame.

- The tag is inserted after the source MAC address.
- The VLAN ID is 12 bits long.
- After excluding reserved VLAN IDs, 4,094 VLAN IDs are available for use.

How VLAN Tagging Works

1. A device sends an Ethernet frame within VLAN 200.
2. The switch forwards the frame toward the trunk interface.
3. The switch adds a VLAN tag identifying VLAN 200.
4. The tagged frame travels across the trunk.
5. The receiving switch reads the VLAN tag.
6. The receiving switch removes the tag and forwards the frame to the appropriate VLAN 200 device.

Result: Multiple VLANs can communicate across a single trunk connection without requiring a separate cable for each VLAN.

ISL and 802.1Q

- ISL stands for Inter-Switch Link.
- It was a proprietary trunking technology.
- ISL is relatively outdated.
- IEEE 802.1Q is the standard trunking method described in the transcript and is the one commonly encountered.

Default VLAN and Native VLAN

Default VLAN

- A switch has a default VLAN.
- VLAN 1 is commonly the default VLAN.
- When a switch is initially configured, interfaces may belong to VLAN 1 unless assigned to another VLAN.

Native VLAN

- The native VLAN is a VLAN that can traverse a trunk without having its traffic tagged.
- Frames belonging to the native VLAN are transmitted untagged over the trunk.

- It can be used by devices or administrative functions that require untagged communication.
- Some management traffic and switch notification messages may use the native VLAN.

Native VLAN Configuration

- The native VLAN should be configured consistently on both switches connected through the trunk.
- If the native VLAN settings do not match, switch logs may display error messages.

Layer 3 Switches and Inter-VLAN Routing

Layer 2 Switch

- A Layer 2 switch operates at the data link layer of the OSI model.
- It forwards Ethernet frames based on MAC addresses.
- VLANs can separate devices into different broadcast domains on the same physical switch.

Layer 3 Switch

- A Layer 3 switch combines Layer 2 switching and Layer 3 routing functionality in one device.
- Switching decisions are still based on MAC addresses.
- Routing decisions are based on destination IP addresses.
- It can route traffic between VLANs.
- Combining switching and routing functions can save space, power, and administrative effort.

Layer 3 Switch Operation

1. Devices are assigned to separate VLANs.
2. The switch provides Layer 2 switching within each VLAN.
3. Routing interfaces are configured to allow traffic between VLANs.
4. The Layer 3 functionality forwards packets between the appropriate networks.

Switched Virtual Interfaces (SVIs)

- A Switched Virtual Interface (SVI) is a logical Layer 3 interface associated with a VLAN.
- SVIs provide routing functionality for VLANs on a Layer 3 switch.
- They allow the switch to route traffic between different VLANs.

Configuration Considerations

- Layer 3 routing functionality may not be enabled by default on some switches.
- It may need to be enabled through the switch configuration.
- Some switches require a restart after the routing functionality is enabled.

Layer 3 Switch vs. Standalone Router

- A Layer 3 switch combines switching and routing capabilities.
- A standalone router may provide more extensive routing functionality.
- The transcript describes the industry principle as: a switch needs to switch, and a router needs to route.
- Layer 3 switches are commonly used in smaller remote sites and some home or internet gateway environments.

Voice VLANs and Data VLANs

Voice over IP (VoIP)

- Before Voice over IP, networks commonly used separate cabling for data and telephone communication.
- Ethernet cables connected computers to switches.
- Telephone cables connected telephones to a PBX (Private Branch Exchange).
- VoIP allows voice and data traffic to use the same Ethernet connection.

Voice and Data VLAN Separation

Voice and data have different network requirements.

- Data traffic often uses bursty bandwidth.
- Voice communication requires consistent network access.
- Network congestion can affect voice communication and reduce call quality.

To address this, separate VLANs can be used:

- Voice VLAN: Carries telephone traffic.
- Data VLAN: Carries computer traffic.

Voice and Data on a Single Cable

- A VoIP telephone connects to the switch.
- A computer connects to the Ethernet port on the back of the telephone.
- Both voice and data traffic use the same physical Ethernet connection to the switch.
- The switch and telephone support the appropriate VLAN configuration.

Example

- Voice traffic is assigned to VLAN 200.
- Computer traffic is assigned to VLAN 100.

The switch separates the traffic into the appropriate VLANs.

Result: A user can download data on a computer while simultaneously making a telephone call, with voice and data traffic logically separated over the same physical connection.

Quick Revision Summary

1. IPv4 and IPv6

- IPv4: 32-bit addressing; approximately 4.29 billion addresses.
- IPv6: 128-bit addressing; eight 16-bit hexadecimal groups.
- IPv6 Compression: Remove leading zeros and replace consecutive zero groups with :: once.
- 6to4: Enables IPv6 over IPv4; requires relay support and does not support NAT.
- Dual Stack: IPv4 and IPv6 operate simultaneously with separate routing tables.
- NAT64: Translates communication between IPv6 and IPv4.
- DNS64: Provides IPv6-compatible DNS responses for IPv4 destinations.

2. Routing

- Routing Table: Contains routes used to forward packets toward their destinations.
- Static Routing: Routes are manually configured by the administrator.

- Dynamic Routing: Routers automatically learn and update routes.
- EIGRP: Cisco-associated protocol with rapid convergence and efficient updates.
- OSPF: Link-state protocol that selects paths based on cost.
- BGP: Routes traffic between autonomous systems.
- Longest Prefix Match: Selects the most specific matching route.
- Administrative Distance: Compares route-source preference; lower values are preferred.
- Routing Metric: Helps a routing protocol evaluate routes.

3. Gateway Redundancy and Interfaces

- FHRP: Provides gateway redundancy through a virtual IP.
- Subinterfaces: Multiple logical interfaces on one physical interface.

4. NAT and PAT

- NAT: Translates private addresses into public addresses.
- PAT: Allows multiple internal devices to share one public IP using port translations.

5. VLANs and Switching

- VLAN: Creates separate logical broadcast domains on a physical switch.
- 802.1Q: Adds VLAN tags to Ethernet frames on trunks.
- Native VLAN: Traffic is carried untagged across a trunk.
- Layer 3 Switch: Combines switching and routing functionality.
- SVI: Provides a logical Layer 3 interface for a VLAN.
- Voice VLAN: Separates voice traffic from data traffic.

Interface Configuration

When configuring a router interface, switch interface, or local computer interface, several settings must be considered.

Ethernet Speed

- Ethernet speed determines how fast the connection operates.
- Common speeds mentioned:
    - 10 Mbps

    - 100 Mbps
    - 1000 Mbps / 1 Gbps
    - 10 Gbps
    - Faster connections
- The speed configuration should match on both sides of the connection.
- Example:
    - If a computer connects to a switch, the speed configured on both devices should match.
- Many devices can be configured for automatic speed negotiation.
- With automatic configuration, the switch can automatically adjust to the configured speed of the connected device.

Duplex

- Ethernet interfaces can use:
    - Half duplex
    - Full duplex
- Duplex is commonly configured automatically.
- If configured manually, both sides should use the same duplex setting.
- A speed mismatch can prevent the connection from working and may result in no link light.
- A duplex mismatch can allow the connection to operate but with poor efficiency and performance, especially when significant traffic is transmitted.
- When troubleshooting Ethernet, speed and duplex are among the first configurations to check.

IP Configuration

After verifying Ethernet connectivity, attention can shift to the Internet Protocol (IP) configuration.

IP configuration may be used for:

- Ethernet connections
- VLAN configurations

- Management interfaces

Common IP configuration settings include:

- IP address
- Subnet mask
- Default gateway
- DNS settings
- Other IP configurations

Configuration Validation

- These values are normally provided by a network administrator.
- The configured values should match the values provided by the administrator.
- An incorrect default gateway or subnet mask can prevent communication with other IP devices on the network.
- Configuration can be validated by comparing:
    - The assigned configuration
    - The configuration currently entered on the system

Link Aggregation

Link aggregation allows multiple interfaces to be combined between two devices.

Other Names

- Port bonding
- LAG — Link Aggregation

Purpose

- Connecting multiple individual interfaces normally could create a network loop.
- When interfaces are configured as link aggregation, the switch treats them as one larger connection.
- Example:
    - Four 1-Gigabit Ethernet connections can provide 4 Gbps of throughput between two switches.

LACP

- Some switches support automatic link aggregation using **LACP**.
- LACP stands for **Link Aggregation Control Protocol**.
- LACP allows the interfaces to automatically configure the underlying link aggregation between the devices.
- LACP traffic can be observed between the switches when capturing traffic.

MTU and Packet Fragmentation

MTU

- MTU stands for **Maximum Transmission Unit**.
- The goal is to send packets without fragmentation along the communication path.
- Fragmentation can make communication less efficient.
- If one fragment is lost, the entire frame may need to be resent.
- The optimal MTU is normally determined automatically when connecting to a device.
- Filtering and firewalls can sometimes prevent the automatic process from working correctly.
- In such situations, the MTU may need to be manually configured.

Jumbo Frames

Jumbo frames are used when the goal is to increase the size of transmitted Ethernet frames.

- Standard Ethernet frames are described in the text as 1500 bytes.
- Jumbo frames allow larger frames to be transmitted.
- Jumbo frames can reach up to **9,216 bytes of MTU** on many devices.
- Some devices use **9,000 bytes** as the accepted upper limit.
- The example configuration in the text supports frame sizes from:
    - 1,280 bytes
    - Up to 9,000 bytes

Advantages

- A larger frame can contain the same amount of information as several normal frames.
- This can reduce the number of packets that switches and routers need to process.
- More information can be transferred in a shorter amount of time.

Requirement

- Every device along the communication path must understand jumbo frames.
- This includes:
    - Switches
    - Routers
    - Other communicating devices
- If a device in the path does not understand jumbo frames, it can drop the frames and communication may fail.

Network Loops

Connecting two switches with multiple Ethernet cables without proper configuration can create a network loop.

Problems Caused by Loops

- Ethernet frames can continuously circulate around the loop.
- There is no mechanism at the MAC address level to count how many times a frame has been seen.
- More and more frames can be added to the network.
- The switches can eventually become overwhelmed.
- Network problems can appear within a short period of time.

Manual Resolution

- Disconnecting one of the cables removes the loop.
- Once the loop is removed, network operation can return to normal.
- However, manually removing cables is not always practical.

Spanning Tree Protocol

Spanning Tree Protocol (STP) provides an automated method to recognize and prevent network loops.

- Standard: **IEEE 802.1D**
- STP can configure switch interfaces into different states.

Blocking

- A blocking interface does not forward traffic.
- Blocking is commonly used to prevent loops.
- If a second cable creates a loop, Spanning Tree can recognize the loop and place one interface into a blocking state.

Listening

- The listening phase is used to clear existing tables and restart the process.
- STP listens to understand:
    - Where other switches are located
    - Whether loops already exist

Learning

- STP goes through a learning process while determining the network topology.

Forwarding

- A forwarding interface can transmit traffic.

Administratively Disabled

- An administratively disabled interface does not participate in the Spanning Tree process.

Spanning Tree Port Types

The text identifies three main types of ports in the example network.

Root Port (RP)

- The root port is the interface on a switch that connects ultimately to the single root bridge on the network.

Designated Port

- A designated port can forward traffic.
- It is not the port directly connected to the root bridge.

Blocked Port

- A blocked port is disabled by Spanning Tree from sending or receiving traffic.
- Blocked ports help prevent loops.

Topology Changes

- If a connection fails, Spanning Tree can recognize the change.
- It clears the existing configuration and relearns the network topology.
- A previously blocked port can be opened.
- Root ports can change.
- This allows communication to continue through another path without creating a network loop.

Rapid Spanning Tree Protocol

Modern networks commonly use **Rapid Spanning Tree Protocol (RSTP)**.

- Standard: **IEEE 802.1W**
- RSTP operates similarly to the original Spanning Tree Protocol.
- Its main advantage is faster convergence.

Convergence

- Original Spanning Tree may take approximately **30–50 seconds** to converge.
- Rapid Spanning Tree can reduce convergence to approximately **6 seconds**.
- This allows connectivity to return much faster after a network disruption.

Compatibility

- Rapid Spanning Tree is backwards compatible with the original Spanning Tree Protocol.
- Devices supporting the different standards can operate together.
- RSTP includes shortcuts and additional configuration settings but maintains familiarity with the original STP operation.

IEEE 802.11 Wireless Networking

Wireless networking standards are primarily developed through the **IEEE 802.11 committee**.

Wi-Fi Naming

The text describes the following relationships:

- **802.11ac → Wi-Fi 5**
- **802.11ax → Wi-Fi 6**
- **802.11ax extended version → Wi-Fi 6**
- **802.11be → Wi-Fi 7**

Future major releases are expected to continue increasing the Wi-Fi number.

Wireless Frequency Ranges

802.11 wireless networking commonly uses:

- 2.4 GHz
- 5 GHz
- 6 GHz

An access point may support:

- 2.4 GHz and 5 GHz
- 2.4 GHz, 5 GHz, and 6 GHz

Wireless Channels

Channels are used to identify portions of the available frequency ranges.

Example:

- 2.4 GHz range:
    - Channel 6
    - Centered on 2.437 GHz
- 5 GHz range:
    - Channel 44
    - Centered on 5.220 GHz

Using a channel number is easier than repeatedly specifying the center frequency.

Wireless Bandwidth

Wireless bandwidth refers to the amount of frequency being used.

Common bandwidths mentioned:

- 20 MHz
- 40 MHz
- 80 MHz
- 160 MHz

2.4 GHz

- The text identifies three non-overlapping channels in the 2.4 GHz range.
- The example uses 20 MHz bandwidth.

5 GHz

- Provides many more available frequencies.
- Different bandwidths can be used, ranging from 20 MHz to 160 MHz.

6 GHz

- Provides even more available frequencies.
- This gives wireless networks additional options and can reduce conflicts with nearby wireless networks.

Choosing Wireless Frequencies

Not every wireless frequency is necessarily the best choice.

The choice can depend on:

- The surrounding environment
- Other access points already using certain frequencies
- The capabilities of the wireless device
- Whether older devices only support 2.4 GHz
- Whether modern devices support 5 GHz or 6 GHz

A device may support multiple frequency ranges and choose between them.

Band Steering

Band steering allows an access point administrator to control which frequency a wireless device uses when multiple options are available.

Example:

- A phone supports both 2.4 GHz and 5 GHz.
- The access point supports both frequencies.
- Without band steering, the device may choose the frequency with the strongest signal.
- The strongest signal does not always provide the best throughput.
- With band steering, the administrator can steer the device toward a particular frequency, such as 5 GHz.

Wireless Frequency Management

Different countries make decisions about how wireless bandwidth is used.

FCC

- In the United States, the **Federal Communications Commission (FCC)** manages frequency allocations.
- The FCC works with other countries to help manage important wireless frequencies.

IEEE 802.11h

- The text describes 802.11h as providing interoperability features for wireless networks.
- It helps multiple wireless networks operate in the same area.

ITU

- The **International Telecommunications Union (ITU)** publishes guidelines for wireless communications.
- It provides a worldwide approach to managing different wireless technologies.

Dynamic Frequency Selection

DFS stands for **Dynamic Frequency Selection**.

- DFS allows wireless access points to use frequencies that do not conflict with other access points already operating.
- The access point can automatically examine the wireless environment.
- It can select frequencies that avoid conflicts.
- If the wireless environment changes and the access point needs to change channels:
    - The access point informs devices on the network.
    - The devices move along with the access point.

Transmit Power Control

TPC stands for **Transmit Power Control**.

- TPC places transmit power control in the hands of the access point.
- The access point can tell a client to use less transmission power.
- This allows devices to operate at appropriate power levels.
- It can help avoid conflicts with other services.

IBSS / Ad Hoc Wireless Networking

IBSS stands for **Independent Basic Service Set**.

It is also referred to as an **ad hoc connection**.

- Devices can communicate directly with each other without an access point.
- One device can be configured for an ad hoc connection.
- Another device can also be configured for an ad hoc connection.
- The two devices can communicate directly over wireless frequencies.

IoT Example

- An IoT device such as a door lock or light can initially use an ad hoc connection.
- Wireless network information can be transferred to the IoT device.
- Security configuration information can also be provided.

- The IoT device can then connect to the larger wireless network through an access point.

SSID and BSSID

SSID

SSID stands for **Service Set Identifier**.

- It is the name of the wireless network.
- Example:
  - SGC1
- The SSID appears in the list of available wireless networks.

The same SSID can be configured on multiple access points.

BSSID

BSSID stands for **Basic Service Set Identifier**.

- It identifies a particular access point using a hardware address.
- Multiple access points can use the same SSID while having different BSSIDs.
- Example BSSID from the text:
  - 60:3D:26:11:22:33

ESSID and Wireless Roaming

ESSID stands for **Extended Service Set Identifier**.

- Multiple access points can share the same wireless network name.
- This allows users to move between different areas while maintaining wireless connectivity.
- Each access point can have a different BSSID.
- The access points can still share the same SSID.

Seamless Roaming

- A user can move outside the range of one access point and into the range of another.

- The device can automatically connect to the other access point using the same wireless network.
- This allows continued use of:
    - Internet access
    - File transfers
    - Network services

Captive Portal

A **captive portal** is a screen that appears when connecting to certain wireless networks.

It may:

- Display information about the network.
- Require agreement to terms.
- Request a username.
- Request a password.
- Request another authentication factor.

Authentication Process

- A centralized access table identifies users or devices that have already authenticated.
- A new unauthenticated device is presented with the captive portal.
- After successful authentication, the information is added to the access table.
- Access can be provided for a predefined period.

Example:

- A captive portal may provide access for 24 hours.
- After 24 hours, the user must authenticate again.

Wireless Security

Wireless networks can carry:

- Private information

- Confidential information
- Sensitive information

Wireless security focuses on:

Authentication

- Controls who can access the network.
- Methods can include:
    - Username
    - Password
    - Multi-factor authentication

Encryption

- Provides confidentiality for information sent over the wireless network.

Message Integrity

- Allows the receiver to verify that the information received is the same information that was originally sent.

Wireless Security Standards

WEP

WEP stands for **Wired Equivalent Privacy**.

- It was one of the first encryption types used on wireless networks.
- It was discovered to have significant cryptographic vulnerabilities.

WPA

WPA stands for **Wi-Fi Protected Access**.

- It replaced WEP.
- It was introduced as a temporary security improvement.
- It was designed to work with the same hardware used by WEP.

WPA2

WPA2 stands for **Wi-Fi Protected Access version 2**.

- The text states that WPA2 has been available since 2004.
- WPA2 is commonly associated with **CCMP**.

- CCMP stands for:
    - Counter Mode with Cipher Block Chaining Message Authentication Code Protocol
- It combines encryption and integrity.
- AES is used for encryption.
- CBC-MAC provides the message integrity check.

WPA3

WPA3 stands for **Wi-Fi Protected Access version 3**.

- Introduced in 2018.
- Uses a different block cipher mode described in the text as **GCM**.
- Continues to use AES for data confidentiality.
- Uses **GMAC**, or Galois Message Authentication Code, for message integrity.

Security Recommendation from the Text

- Use the highest level of wireless security available on the access points and client devices.
- If possible, upgrade devices to use the most capable security method available.

Wireless Antennas

Wireless access points can use different antenna types.

Omnidirectional Antenna

- Common on consumer and home-based access points.
- Distributes the wireless signal around the antenna.
- Can provide coverage in multiple directions.
- An access point can be placed in a central location to provide coverage around the area.

Limitation

- If the access point is located in a corner, part of the signal may be sent toward areas where no users need connectivity.

Directional Antennas

A directional antenna focuses wireless transmission and reception in a particular direction.

Useful when:

- A device is located in a specific direction.
- Connecting different buildings.
- Focusing wireless communication over a particular area.

Antenna Gain

- Antennas can be described using decibels of gain.
- The text gives an example of a 3 dB increase.
- According to the text, every 3 dB effectively doubles the available power.

Yagi Antenna

- A directional antenna.
- Highly directional.
- Provides a high amount of gain.

Parabolic Antenna

- Another directional antenna type.
- Multiple signals can enter and reflect toward a single feed horn.
- Focuses signals into a single point.
- Especially useful for longer distances.

Autonomous and Lightweight Access Points

Autonomous Access Point

- Common in home environments.
- Operates as a standalone device.
- Does not require additional hardware or software to maintain wireless connectivity.

Lightweight Access Point

- Common in enterprise environments.

- Separates wireless functionality between the access point hardware and centralized management.
- Configuration and intelligence can be maintained through the network infrastructure.
- Allows access points to be managed centrally.

CAPWAP

CAPWAP is described in the text as a standard for managing wireless access points.

- It allows network administrators to manage and configure wireless access points from a central management station.
- It supports centralized wireless management.

Wireless LAN Controller

A **Wireless LAN Controller (WLC)** provides centralized management of wireless access points.

The controller can be used to:

- Deploy new access points.
- Monitor access point performance.
- Change wireless network configurations.
- Deploy configuration changes to some or all access points.
- Generate reports about wireless network usage and users.

The centralized management interface is described as a **single pane of glass**.

Wireless Mesh

A wireless mesh consists of multiple access points that can communicate with each other.

- Multiple access points are placed within a relatively close area.
- Other devices can connect to the access points.
- Additional access points can be added to the same mesh.

- Mesh devices can communicate with each other.
- If one device leaves the network, the mesh can identify that the device is no longer available.
- The network can self-heal to maintain connectivity for the remaining devices.

Point-to-Point Wireless

Point-to-point mode allows two locations to connect directly over wireless.

Example:

- One house has an access point connected to a switch.
- Another house has a similar configuration.
- The two networks can be connected through point-to-point wireless.

Requirements

Not every access point supports point-to-point mode.

The hardware and software must support:

- Point-to-point functionality
- Longer-distance transmission
- Appropriate antennas
- Appropriate power output
- Frequencies capable of communicating over the required distance

---

Infrastructure Mode

Infrastructure mode is one of the most common wireless network configurations.

- A centralized access point provides wireless connectivity.
- Wireless devices connect to the access point.
- Devices can communicate with:
    - The wired network
    - Other wireless devices through the access point

Some access points can:

- Allow wireless devices to communicate with each other.
- Isolate wireless devices so they can communicate only with the access point and networks beyond it.

The access point should be checked to ensure that the appropriate configuration and security are enabled.

Wireless Security Evolution

The wireless security progression described in the text is:

WEP → WPA → WPA2 → WPA3

WEP

- Had significant cryptographic vulnerabilities.

WPA

- Replaced WEP.
- Served as an initial improvement and temporary security solution.

WPA2

- Introduced stronger security.
- Uses AES and CCMP.
- Provides encryption and message integrity.

WPA3

- Introduced in 2018.
- Uses AES for confidentiality.
- Uses Galois-based message integrity mechanisms.
- Represents a more capable wireless security method described in the text.

Key Wireless Terms

- **IEEE 802.11:** Committee and standards for wireless networking.
- **SSID:** Name of the wireless network.
- **BSSID:** Hardware identifier of an access point.

- **ESSID:** Shared wireless network identification across multiple access points.
- **IBSS:** Independent Basic Service Set; ad hoc wireless connection.
- **Captive Portal:** Authentication or information screen presented before network access.
- **DFS:** Dynamic Frequency Selection.
- **TPC:** Transmit Power Control.
- **CAPWAP:** Standard for managing and configuring wireless access points.
- **WLC:** Wireless LAN Controller for centralized wireless management.
- **WEP:** Older wireless security method with significant vulnerabilities.
- **WPA:** Security method that replaced WEP.
- **WPA2:** Wi-Fi Protected Access version 2.
- **WPA3:** Wi-Fi Protected Access version 3.
- **PSK:** Pre-Shared Key.
- **CCMP:** Counter Mode with Cipher Block Chaining Message Authentication Code Protocol.
- **AES:** Encryption method used with WPA2 and WPA3 in the text.
- **GMAC:** Galois Message Authentication Code.
- **LACP:** Link Aggregation Control Protocol.
- **LAG:** Link Aggregation.
- **MTU:** Maximum Transmission Unit.
- **STP:** Spanning Tree Protocol.
- **RSTP:** Rapid Spanning Tree Protocol.
- **VLAN:** Virtual LAN configuration mentioned in interface configuration.
- **IoT:** Internet of Things.

Distribution Frames

A **distribution frame** is an area of the network where cables are passively terminated.

Common components

- **Punch-down blocks**

- **Patch panels**
- Other cable-termination methods

These are commonly found in:

- Data centers
- Large facilities
- Central network locations

They are commonly used for terminating:

- Local Area Network (LAN) connections
- Voice communication connections

Although the actual punch-down blocks and patch panels technically form the distribution frame, the **entire room** is often referred to as the distribution frame room.

Main Distribution Frame (MDF)

The **MDF (Main Distribution Frame)** is usually the primary or central distribution point of the network.

MDF characteristics

- Usually located in a single central room.
- Often located in or associated with the **data center**.
- Acts as a central point for network connectivity.
- May contain:
    - WAN connections
    - LAN connections
    - Punch-down blocks
    - Patch panels
    - Networking equipment
    - Computer equipment
    - Servers

WAN connections may be brought into the MDF and terminated there along with internal LAN connections.

MDF as a testing point

Because many connections meet at the MDF, it is a useful testing location.

It can be used to test:

- Internal LAN connectivity
- External WAN connectivity

Even when there are not large numbers of punch-down blocks, the central data-center room may still be referred to as the **MDF**.

Intermediate Distribution Frame (IDF)

An **IDF (Intermediate Distribution Frame)** is another type of distribution frame.

IDF characteristics

- Usually located on:
  - A different floor
  - A different building
- Often directly connected to the MDF.
- Modern IDFs commonly contain:
  - Switches
  - Routers
  - Other networking equipment

The IDF connects a particular area of the network back to the larger central MDF.

MDF and IDF relationship

This combination is common in medium- and large-scale environments where many remote areas need to connect back to a central data center.

Example:

Internet → Core Router → Core Switch → MDF → IDFs → Users

The MDF may contain:

- Internet connection
- Core router
- Core switch

- File servers
- Database servers

An IDF on another floor may contain a switch that connects users on that floor.

That IDF switch then connects back to the main core switch located in the MDF.

Network Racks

Racks are commonly found in both MDF and IDF environments.

Most racks follow standardized dimensions.

Rack Width

Most rack sizes are **19 inches wide**.

Equipment designed for rack mounting is generally designed to fit this standardized width.

Rack Unit (U)

Equipment can have different heights.

The standard measurement for rack height is called a **rack unit**, abbreviated as **U**.

- **1U = 1.75 inches**
- Equipment may require:
    - 1U
    - 2U
    - 3U
    - etc.

For example:

- A 2U device requires 2U of rack space.
- A 3U device requires 3U of rack space.

Most racks mentioned in the text are approximately **42U high**.

Rack Depth

Although rack width is standardized, rack **depth can vary**.

Some racks may be:

- Very shallow

- Deep enough to support longer equipment

When receiving rack-mountable equipment, you generally expect it to be about 19 inches wide, but you must ensure that the rack is deep enough for the equipment.

Efficient rack usage

Standardized rack dimensions allow equipment to be stacked vertically.

For example:

- One server may occupy approximately 3U.
- Another server can be installed directly above it.

This allows available rack space to be used efficiently.

Data Center Cooling and HVAC

Large amounts of equipment operate continuously in a data center.

This equipment generates significant heat.

A data center therefore requires an **HVAC system**.

HVAC

HVAC = Heating, Ventilating, and Air Conditioning

A data-center HVAC system is much more complex than simply using a window-mounted air conditioner.

The system must be engineered to handle:

- Heat generated by equipment
- Power requirements
- Environmental conditions
- Fire-system integration

The HVAC system may also need to shut down automatically when a fire is detected.

Hot Aisles and Cold Aisles

Data centers may divide equipment aisles into:

- **Cold aisles**

- **Hot aisles**

Equipment should be installed so that:

- Servers pull cool air from the cold aisle.
- Hot air is exhausted into the hot aisle.

This arrangement helps manage airflow and equipment temperature.

Raised-Floor Cooling

One example of data-center cooling uses a raised floor.

Cooling process

1. HVAC produces cold air.
2. Cold air travels underneath the raised floor.
3. Vents/openings in the floor allow cold air to enter the cold aisle.
4. The front of servers pulls in the cold air.
5. The air cools the internal components.
6. Heated air exits from the back of the servers.
7. The heated air enters the hot aisle.
8. Hot air rises toward the ceiling.
9. HVAC pulls the hot air back in.
10. The air is cooled again.
11. The cycle repeats.

Data Centers Without Raised Floors

Not every data center has a raised floor.

In some environments:

- There is a concrete floor.
- Cold air is blown directly into the cold aisle from above.
- Air moves through the equipment.
- Hot air enters the hot aisle.

- Hot air is pulled back toward the ceiling.
- The cooling process repeats.

Some cold aisles may also be enclosed with plastic coverings to keep the cold air contained.

Structured Network Cabling

One objective when installing network cabling is to avoid moving or touching permanent cabling after installation.

The goal is to create reliable permanent cable runs.

A common design is:

User Desk → Permanent Cable Run → IDF → Patch Panel → Switch

Each desk may have a dedicated cable run back to the IDF.

Patch Panels

A patch panel is commonly installed inside an IDF.

A patch panel may contain a **110 block** on the back where permanent cables are punched down.

The front provides **RJ45 modular connectors**.

Purpose

The permanent cable between the IDF and user's desk does not need to be repeatedly touched.

Instead:

Permanent Cable → Patch Panel → Patch Cable → Switch

The patch cable connects the patch panel to the appropriate switch port.

This makes network changes much easier.

Moves, Adds, and Changes

Patch panels make it easier to handle:

- **Moves**
- **Adds**

- **Changes**

Moving a user

If a user moves to another desk:

- The permanent cabling does not need to be changed.
- The patch cable can simply be moved to another appropriate switch connection.

Adding a new employee

If additional cables have already been installed:

- There is no need to run new permanent cabling.
- A new patch connection can be made between the patch panel and switch.

This provides flexibility while keeping permanent cabling untouched.

Numbered Patch Panel Connections

Patch panels commonly have numbered ports.

These numbers allow administrators to identify exactly which desk or network drop corresponds to a particular connection.

This makes it easier to:

- Connect users
- Disconnect users
- Troubleshoot connections
- Identify network drops

Fiber Optic Distribution Panels

Fiber-optic networks can use dedicated **fiber-optic distribution panels**.

These panels can receive fiber runs from:

- Another building
- Another floor
- Another part of the network

The fiber connections are terminated on the distribution panel.

Fiber Bend Radius

An important consideration when installing fiber optics is the **bend radius**.

Fiber should not be bent beyond its permitted bend radius.

Excessive bending can damage the fiber.

For this reason, fiber may be stored in a large loop inside the distribution panel rather than being bent sharply.

Fiber Service Loop

Extra fiber may be stored as a **service loop**.

A service loop provides additional fiber that can be used later.

It provides flexibility if:

- The distribution panel needs to be moved.
- Additional fiber needs to be installed.
- Changes need to be made.

The existing extra fiber can be extended rather than requiring the entire fiber connection to be rerun.

Locked Data Center Racks

Data-center racks may be completely enclosed and locked.

Purpose

Locked racks provide:

- Additional security
- Additional physical access control

They help prevent unauthorized people from accessing equipment inside the rack.

Rack ventilation

Locked racks still require ventilation.

Ventilation may exist:

- At the front
- At the top
- At the bottom

This allows equipment inside the rack to be properly cooled.

Electrical Safety

When working with electrical equipment, safety is critical.

You should never place yourself in a situation where you can come into contact with dangerous voltage.

Basic safety principle

Before working on equipment:

- Disconnect it from the power source.
- Avoid touching energized components.
- Do not intentionally connect yourself to electrical ground or other electrical components.

Stored electrical charge

Disconnecting a device does not necessarily mean that it is immediately safe.

Some devices can store electrical charge in **capacitors**.

Examples mentioned include:

- Older CRT monitors
- Laser printers
- Other electrical devices

These devices can potentially cause an electrical shock even after being disconnected from power.

The goal is to avoid allowing any part of your body to contact something that could be energized.

Ampere / Amp

An **ampere**, commonly called an **amp**, measures the rate at which electrons move past a particular point in one second.

The abbreviation is:

A = Ampere

Water-hose analogy

Electricity can be compared to water flowing through a hose.

The amount of water flowing through the hose is related to the hose's diameter.

The transcript uses this analogy to describe electrical current in terms of amps.

Voltage

Voltage describes the electrical pressure in a circuit.

The abbreviation is:

V = Volt

Water-hose analogy

Voltage can be compared to water pressure.

Increasing water pressure through a hose is similar to increasing electrical voltage.

Examples of electrical systems mentioned:

- 120 volts
- 240 volts

Watts

Watts describe how much power is being used.

The abbreviation is:

W = Watt

The relationship given in the transcript is:

Watts = Volts × Amps

Example

A device connected to:

- 120 volts
- 0.5 amps

uses:

$120 \times 0.5 = 60$ watts

Therefore, the device uses **60 watts**.

Alternating Current (AC)

The power normally received from wall outlets is **Alternating Current (AC)**.

AC is useful for distributing electricity over long distances.

The transcript states that AC is represented by a **wavy line**.

The wave represents the changing direction of electrical power as it moves through the wire.

Electrical standards mentioned

United States and Canada

- Approximately 110–120 volts AC
- 60 Hz
- 60 cycles per second

Europe

- Approximately 220–240 volts AC
- 50 Hz

Direct Current (DC)

Many electrical components use **Direct Current (DC)**.

With DC:

- Current moves in a single direction.
- Voltage remains constant.

The abbreviation is:

DC

The symbol is described as:

- A solid line on top
- Multiple lines underneath

AC-to-DC Conversion

Wall outlets provide AC power.

Many IT devices require DC power.

Therefore, devices such as:

- Routers
- Switches
- Other networking equipment

convert AC into DC using power supplies.

The conversion may occur:

- Inside the device's power supply
- Within the power cord/power adapter

Uninterruptible Power Supply (UPS)

Power is essential for IT systems.

A **UPS (Uninterruptible Power Supply)** helps protect systems from power interruptions.

A UPS generally contains batteries that provide power when the primary power source is lost.

A UPS can also help deal with changes in voltage.

Brownout

A **brownout** occurs when voltage drops.

Surge

A **surge** occurs when voltage spikes or becomes significantly higher.

A UPS can help compensate for these conditions and maintain power to connected equipment.

Types of UPS

Offline / Standby UPS

An **offline or standby UPS** waits until the main power fails before switching to battery power.

There can be a small gap before battery power becomes available.

For devices such as:

- Switches
- Routers

this gap can be enough to cause the device to reboot.

Line-Interactive UPS

A **line-interactive UPS** can compensate for voltage drops.

When a brownout occurs, it can increase the amount of power provided to compensate for the reduced voltage.

Online / Double-Conversion UPS

Many data centers prefer an **online or double-conversion UPS**.

In this design:

- Systems are continuously powered through the battery system.
- The batteries are constantly refreshed by the main power source.

If the main power source is lost:

- The connected systems are already running from the primary battery power.
- Therefore, the power loss has no effect on the connected systems.

Additional UPS Features

Different UPS devices may provide different features.

Examples include:

- Automatic system shutdown when batteries become very low
- Higher or lower battery capacity
- Different numbers of power outlets

- Additional interfaces for:
    - Phone lines
    - Cable modem connections

Power Distribution Unit (PDU)

A **PDU (Power Distribution Unit)** distributes power to equipment in a rack.

Although a PDU may look similar to a surge suppressor, the transcript describes it as an intelligent device.

PDU capabilities

A network-connected PDU can allow administrators to:

- Remotely connect to the device
- Manage individual interfaces
- Power devices on remotely
- Power devices off remotely

A rack-mountable PDU can be installed directly into a rack.

Network management

The PDU may have:

- Ethernet connectivity
- An assigned IP address
- An internal web server

This allows administrators to connect to and manage the PDU remotely.

Data Center Environmental Management

Data centers may operate:

24 hours a day, 7 days a week

The transcript states that approximately **2% of all U.S. power consumption** is estimated to be used by data centers.

A significant amount of this power is used for environmental management.

Environmental management includes:

- Humidity
- Temperature
- Other environmental controls

Data Center Humidity

Humidity must be carefully controlled.

High humidity

High humidity can cause:

- Condensation
- Water-related problems

Water in a data center can damage equipment.

Low humidity

Very low humidity can increase:

- Static discharge

Static discharge can potentially damage equipment.

Humidity range

The transcript states that many industry guidelines place ideal humidity around:

40%–60%

This range is described as:

- High enough to reduce static-discharge concerns
- Low enough to reduce condensation concerns

The exact level can vary depending on geography.

Data Center Temperature

Data centers need to maintain an appropriate temperature.

The transcript gives an approximate range of:

64°F–81°F

Cooling systems continuously work to maintain the desired temperature.

Temperature can be affected by:

- Outdoor temperature
- Increasing system load
- Additional heat generated by equipment

As equipment load increases, additional cooling may be required.

HVAC Environmental Monitoring

Data-center HVAC systems are designed to maintain:

- Temperature
- Humidity

Many HVAC systems use sensors throughout the data center.

These sensors help ensure that temperature and humidity remain consistent throughout the facility.

Data Center Fire Suppression

Fire is another major concern in a data center.

Data centers contain:

- Large amounts of electrical equipment
- Large physical areas

Water is normally used to suppress fires, but the transcript explains that other fire-suppression methods are desirable around large amounts of electrical equipment.

A data center may have a tank containing:

- Inert gas
- Chemical fire-suppression agents

These systems can suppress a fire or remove oxygen from the environment.

When the fire alarm is activated:

- The suppression chemicals may be distributed throughout the data center.
- This may damage equipment.

- However, it prevents the fire from spreading.

HVAC and Fire Suppression Integration

Fire-suppression systems may be integrated with HVAC systems.

When a fire is detected:

- HVAC can automatically shut down.
- This prevents the HVAC system from feeding additional oxygen to the fire.

Network Documentation

Network documentation is an important part of managing an infrastructure.

Different types of network documentation include:

- Physical network maps
- Logical network maps
- Rack diagrams
- Cable maps
- Layer 1/2/3 diagrams
- Asset documentation
- IP address management documentation
- Service Level Agreements
- Wireless site surveys

Physical Network Map

A **physical network map** shows the physical layout of:

- Equipment
- Cables
- Connections

Example from the transcript:

Internet → Internet Router → Internet Firewall → Core Router

The example includes:

- Internet router: **10.1.1.1**
- Core router: **10.1.10.1**

The physical map allows someone to physically trace a cable to the corresponding equipment.

Software for network maps

Examples mentioned:

- Visio
- OmniGraffle
- Gliffy.com

These tools can be used to create physical and logical network maps.

Logical Network Map

A **logical network map** provides a higher-level view of connectivity.

Unlike a physical map, it does not focus on:

- Individual interfaces
- Individual cables

Instead, it focuses on:

- How the overall network is connected
- Major network locations
- WAN connectivity

For example, multiple headquarters buildings may be shown as connecting to a central cloud representing the wider network.

Routers, switches, and servers may exist inside that cloud but are represented more broadly.

Uses

Logical maps can help when:

- Planning additional locations
- Understanding existing WAN connectivity

- Determining where another installation could be connected

Rack Diagram

A **rack diagram** provides a detailed physical representation of equipment inside a rack.

It is designed to represent the view you would have while standing in front of the rack.

Uses

Rack diagrams can identify:

- Exact rack
- Rack row
- Equipment location
- Rack unit position
- Power-strip location
- Server location

For example, documentation might specify:

- Row 3
- Rack W
- Unit 14
- Unit 15

This is useful when someone needs to work in a data center but does not have the same level of access or is not physically located in the same building.

Cable Map

A **cable map** documents where physical network cables are installed throughout an office.

Cable maps are commonly found in:

- IDFs
- MDFs

They may be displayed on a wall inside the wiring closet.

Cable map information

It can show:

- Where cables are physically routed
- Cables under the floor
- Cables above the ceiling
- Number assigned to each network drop

The numbered drops allow administrators to correlate:

- Network drops at desks
- Patch-panel connections inside the wiring closet

Troubleshooting

Cable maps are useful during troubleshooting because administrators can:

1. Identify the affected network drop.
2. Find the corresponding numbered connection.
3. Track the physical cable route.
4. Correlate the cable with the problem.

Layer 1, Layer 2, and Layer 3 Network Diagrams

Network diagrams can combine different networking layers.

Layer 1

Physical Layer

Shows:

- Physical interfaces
- Switches
- Routers
- Physical cables
- Physical connectivity

Layer 2

Data Link Layer

Can add:

- MAC addresses
- Layer 2 connectivity information

Layer 3

Network Layer

Can add:

- IP addresses
- Layer 3 addressing information

These layers can be overlaid so that administrators can identify the relationship between:

Physical interface → MAC address → IP address

Asset Tracking

Organizations need to track the assets they purchase and deploy.

Assets can include:

- Laptops
- Desktops
- Routers
- Firewalls
- Switches
- Other equipment

Each asset should have a **tag or label**.

Asset Tags

Asset tags provide a way to identify specific equipment.

For example:

Router → Asset Tag #42

The tag can be referenced in:

- Trouble tickets

- Support processes
- Financial records
- Warranty information

Financial uses

Asset tracking can help determine:

- When equipment was purchased
- Equipment depreciation
- Warranty status

Identification

Asset tags are usually visible so that support personnel can identify the device while communicating with a support team.

Tags may contain:

- Barcode
- RFID
- Additional identifying information

If a third party encounters the equipment, the asset information can also identify the organization that owns it.

Asset Database

Asset information can be stored in a centralized **asset database**.

The database can associate:

- Asset tag
- Device
- Components
- Assigned user
- Physical location
- Warranty information
- Purchase information

Location tracking

If the database shows which user has been assigned a device, and the user's location is known, the organization can determine the device's location.

Software tracking

Asset databases can also track software installed on devices.

This can help organizations determine:

- How many devices they own
- How many software licenses they need
- How many licenses are already in use

IP Address Management (IPAM)

Organizations need to track their IP addressing.

This can be done using an **IP Address Management (IPAM)** solution.

IPAM can help:

- Plan IP addressing
- Track IP addressing
- Configure DHCP and IP address schemes
- Manage IPv4 addressing
- Manage IPv6 addressing

IP Address Tracking

IPAM can document which users are using particular IP addresses.

Because dynamic IP addressing is commonly used, it can be useful to associate:

User + IP address + Date/Time

This allows administrators to determine which user was associated with an IP address at a particular time.

IPAM and Address Problems

IPAM can help identify:

- IP addressing problems

- IP address shortages
- DHCP configuration issues

Based on this information, administrators may need to:

- Modify IP address ranges
- Add additional IP addresses
- Modify DHCP infrastructure

IPAM provides centralized management of organizational IPv4 and IPv6 addressing.

IPAM Console

An IPAM console may display:

- Subnet details
- IP address statistics
- Change-log entries
- Warnings
- Error logs
- Other IP configuration information

This information helps administrators manage IP addressing.

Service Level Agreement (SLA)

Organizations often depend on service providers.

They need a way to verify that the provider delivers the expected level of service.

This is commonly handled through an:

SLA = Service Level Agreement

An SLA defines the minimum level of service required under a contract with a third-party organization.

SLA Examples

For a:

- WAN provider
- Internet provider

an organization may specify an availability expectation such as:

99.99% uptime

An SLA can also specify a maximum amount of unscheduled downtime.

Example from the transcript:

No more than four hours of unscheduled downtime

The service provider may therefore need processes for restoring service quickly.

Possible provider actions mentioned include:

- Quickly dispatching a technician
- Keeping replacement equipment at the customer's location
- Allowing the customer to install replacement equipment in the rack

---

Wireless Site Survey

As organizations deploy more:

- Wireless access points
- Wireless infrastructure

they need documentation to understand the wireless environment.

This is done using a **site survey**.

A site survey can identify:

- Access points
- Frequencies in use
- Wireless infrastructure
- Wireless signal behavior

The survey can include access points that the organization does not directly control.

Wireless Interference and Site Surveys

Once frequencies in use are understood, administrators can document them.

This helps configure wireless systems to minimize interference.

Site surveys can be particularly useful in:

- Buildings containing multiple companies
- Large campuses
- Areas containing many buildings
- Environments with many wireless networks

Frequent surveys may be useful where the wireless environment changes regularly.

Wireless Heat Maps

A common way to document wireless coverage is a **heat map**.

A heat map can be created by walking around a building with a mobile device and recording how the wireless signal propagates.

It can show how wireless signal strength changes depending on where the device is located inside the building.

End of Life (EOL)

Hardware and software eventually become outdated.

This is referred to as:

EOL = End of Life

When a manufacturer declares a product end-of-life:

- The manufacturer no longer intends to support the product in the same way.
- It may continue providing security patches and updates for some period.
- It will no longer provide new software versions or enhanced features.

Eventually, the product may also stop receiving security updates.

End of Support (EOS)

When a product reaches **End of Support (EOS)**, the manufacturer no longer provides:

- Security patches
- Updates
- New features
- Other associated support

This creates a significant security concern because the product will not receive future security updates.

Difference between EOL and EOS

EOL

- Product is no longer actively supported as a current product.
- Some updates or security patches may still continue.

EOS

- No further patches or updates are provided.

Planning for EOL and EOS

When a product reaches EOL:

- The organization can create a replacement plan.
- A budget can be established.
- Replacement can be performed over a period of time.

When the product reaches EOS:

- The security risk becomes much more significant.
- The organization knows that no further security updates will be received.

Software Updates and Patches

Updates are important for maintaining:

- Stability
- Availability
- Security

Updates may contain:

- Bug fixes

- Stability fixes
- Security fixes

Security updates can close security vulnerabilities.

Operating System Updates

Operating systems may receive regular updates.

The transcript mentions:

- Windows
- Linux
- iOS
- Android

Organizations may use recurring update schedules.

For example:

- Monthly updates
- Other regularly scheduled updates

A regular update schedule allows organizations to plan deployment.

Emergency / Out-of-Schedule Updates

Sometimes a manufacturer releases an update outside the normal update schedule.

This is often associated with:

- Security concerns
- Zero-day vulnerabilities
- Significant security events

These updates may need to be deployed outside the normal monthly update cycle.

Operating System Configuration Changes

Organizations do not only install manufacturer updates.

They may also change operating-system configurations.

Examples include:

- Modifying user accounts
- Increasing minimum password length
- Changing password complexity requirements
- Changing how users access the operating system
- Modifying built-in firewall configurations
- Restricting access based on IP address or URL
- Enhancing security features
- Updating embedded antimalware/antivirus

Firmware

Purpose-built hardware appliances often do not provide direct access to their underlying operating system.

These devices have their own embedded operating system and configuration.

The software running on this type of hardware is called **firmware**.

Examples mentioned include:

- Printers
- Cable modems
- Other devices with embedded operating systems

Firmware may occasionally need to be updated to close security vulnerabilities.

Firmware Updates

Firmware updates may be performed:

- Over the network
- By physically connecting to the device

Because hardware depends on its firmware to function correctly, organizations should have a plan for firmware upgrades.

That plan should include:

- Upgrade procedure
- Backup/fallback procedure
- Previous firmware version

Firmware Rollback

Before upgrading firmware, it is useful to keep previous firmware versions.

Firmware binaries should be stored in a safe location.

If a new firmware version causes a problem:

- The organization can access the previous version.
- The device can potentially be returned to the previous version.

This provides a way to **fall back** when an update creates problems.

Firmware Maintenance Challenges

Some companies are very good at manufacturing hardware but may not be as effective at maintaining the firmware that runs on those devices.

The transcript provides an example involving **Trane ComfortLink thermostats**.

The example states:

- Three vulnerabilities were presented to Trane in April 2014.
- Firmware updates were not received until April 2015.
- The final update was not available until January 2016.

The organization's security team would need to determine whether the device should:

- Remain on the network
- Be replaced
- Have another alternative used

Decommissioning

Another part of life-cycle management is **decommissioning**.

Decommissioning is the process of disposing of an asset when it is no longer needed.

This can apply to:

- Desktops
- Laptops
- Mobile devices
- Other outdated or unsupported devices

Data Sanitization During Decommissioning

Old equipment may contain sensitive data.

Therefore, organizations need a way to:

- Sanitize storage media
- Destroy the device when necessary

The objective is to prevent data from reaching unauthorized people.

Legal considerations

There may be legal requirements concerning certain types of data.

An organization may therefore need to:

- Store equipment securely
- Keep equipment off-site
- Wait until it can legally be destroyed or disposed of

Equipment should not simply be placed in normal trash.

Someone could recover the equipment and access the stored data.

Hardware Recycling

Organizations may want to recycle hardware.

However, recycling must be performed in a way that protects the **confidentiality of the data** stored on the devices.

Change Management

Organizations constantly make changes to their computing environments.

Examples include:

- Upgrading switch software
- Changing firewall configurations
- Modifying router tables
- Changing software
- Modifying device configurations
- Updating operating systems
- Patching devices
- Updating applications
- Installing network devices
- Installing new applications

These changes need to be controlled and tracked.

Change Management Process

A formal change-management process should define:

- When changes can occur
- Change windows
- How changes are installed
- How changes are documented
- How changes are monitored
- How changes can be rolled back

Organizations should have a centralized process.

Everyone making modifications to devices should follow that process.

Change Documentation

Change documentation becomes especially important if a system needs to be rebuilt.

For example, after a disaster, administrators may need to restore a system quickly.

Documentation showing:

- Initial installation
- Configuration changes
- Software changes
- Other modifications

can help rebuild the system.

Production Configuration

When a switch, router, or firewall is initially installed and configured correctly, it can have a defined **production configuration**.

This becomes the standard configuration used when that device is deployed into production.

Examples:

- Standard production switch configuration
- Standard production router configuration
- Standard production firewall configuration

The production configuration is not limited to individual settings.

It can include:

- Hardware versions
- Firmware versions
- Device drivers
- Software versions
- Other configuration components

Testing Before Production

A configuration should generally be tested before being deployed into production.

Organizations can test:

- Hardware
- Firmware
- Drivers
- Software
- Configurations

The objective is to identify problems before deployment.

However, not every possible scenario can be reproduced in a test lab.

Therefore, organizations still need a rollback plan.

Backup Before Changes

A common standard operating procedure is:

Back up the system before making changes.

This applies to:

- Firewalls
- Switches
- Routers
- Operating systems
- Other systems being modified

After the change:

- If everything works, the new configuration can remain.
- If problems occur, the previous backup can be restored.

Configuration File Backup

A backup does not always require backing up the entire system.

Sometimes the relevant configuration may be stored in:

- A configuration file
- Other files affected by the change

A copy of the previous file can be saved.

If the change causes problems:

- The previous file can be copied back.
- The system can return to the previous configuration.

Virtual Machine Snapshots

Virtual machines make rollback relatively easy.

A **VM snapshot** can preserve:

- Files
- Configuration
- Other VM state

at a particular point in time.

If a configuration change causes a problem:

- The VM can be reverted to the snapshot.
- The previous state can be restored.

Snapshots can also be useful when administrators later decide that they need to return to an earlier configuration.

Application Deployment

Deploying an application often involves more than installing a single executable on one server.

An application may require changes to multiple components.

These may include:

- User workstation configurations
- Firewall configurations
- Application server configuration
- Other application components

Therefore, the complete application installation needs to be documented.

Golden Configuration

A **golden configuration** documents the complete configuration required for an application to operate correctly.

It can serve as a standard configuration that confirms the application will work properly when the required configurations are present.

A golden configuration can include:

- Correct software
- Correct configurations
- Required system settings

Integrity Checks

Organizations can compare the current production configuration against the **baseline/golden configuration**.

This comparison can identify changes.

If differences are found, administrators can choose to:

Option 1

Change the production environment so that it matches the golden configuration.

Option 2

Update the baseline/golden configuration so that it matches the new production configuration.

Once the appropriate changes are made:

- The new baseline becomes the reference configuration.
- Future integrity checks can use the updated baseline.

Key Concepts

Distribution and Data Center Infrastructure

- Distribution Frame
- MDF
- IDF
- Patch Panel
- Fiber Distribution Panel

- Rack
- Rack Unit
- HVAC
- Hot Aisle
- Cold Aisle

Power

- Ampere
- Voltage
- Watt
- AC
- DC
- UPS
- Brownout
- Surge
- PDU

Documentation

- Physical Network Map
- Logical Network Map
- Rack Diagram
- Cable Map
- Layer 1/2/3 Diagram
- Asset Database
- IPAM
- SLA
- Wireless Site Survey
- Heat Map

Life Cycle Management

- EOL

- EOS
- Software Updates
- Firmware
- Decommissioning
- Data Sanitization
- Change Management
- Production Configuration
- Backup
- Rollback
- VM Snapshot
- Golden Configuration
- Integrity Check

Simple Network Management Protocol (SNMP)

Purpose of SNMP

- **SNMP (Simple Network Management Protocol)** is used to monitor and manage network devices such as:
    - Switches
    - Routers
    - Servers
    - Firewalls
    - Other network infrastructure equipment
- After installing network equipment, administrators need to ensure that devices are operating optimally and that errors or performance problems are identified.
- SNMP provides a standardized management interface for querying devices from different manufacturers.
- A network team commonly uses a **central Network Management Station (NMS)** to query SNMP-enabled devices and collect performance information.

Example

A network management station may query a switch to determine how many bytes have passed through a particular interface. The switch returns the requested value, and the management station stores the information for analysis.

Management Information Base (MIB)

- A **Management Information Base (MIB)** is a structured collection of management information associated with a network device.
- SNMP queries specific values stored in the MIB.
- Each individual value is identified using an **Object Identifier (OID)**.
- MIBs contain many different variables that administrators can query to monitor device performance and status.

Object Identifier (OID)

- An **Object Identifier (OID)** is a series of numbers used to identify a particular management variable.
- Each number represents a level in a hierarchical identification structure.
- The transcript gives an example of an OID beginning with 1.3.6.1.2.1 and ending in 28.0.
- The hierarchy identifies progressively more specific categories and variables, allowing administrators to retrieve individual values from a MIB.
- Hundreds or thousands of variables may exist within a MIB, and each can be queried using its OID.

Standard and manufacturer-specific OIDs

- Some OIDs are standardized across different devices.
- The **MIB-II standard** provides commonly used OIDs that can be shared across multiple devices.
- Manufacturers may also define unique OIDs for device-specific information.
- Administrators may need to consult manufacturer documentation to determine which OIDs correspond to particular values.
- Manufacturers may provide MIB files that can be imported into a network management station so that device-specific information can be identified and queried.

SNMP versions

SNMPv1

- The original version of SNMP.
- Allows management systems to query structured information stored in a device's MIB.
- Sends information without encryption.
- Data transmitted using SNMPv1 can be visible to someone monitoring the connection.

SNMPv2c

- An improved version of SNMP, commonly referred to as SNMPv2c.
- Supports more efficient querying of larger amounts of data.
- Communication remains unencrypted, meaning transmitted information can be observed by someone monitoring the connection.

SNMPv3

- Provides stronger security capabilities than SNMPv1 and SNMPv2c.
- Supports encryption, message integrity, and authentication.
- Uses user-based authentication rather than relying solely on community strings.
- Authentication information is protected using cryptographic mechanisms.

Key point: SNMPv3 provides security capabilities that are absent from the unencrypted SNMPv1 and SNMPv2c approaches described in the transcript.

SNMP ports

- **UDP port 161:** Used for SNMP queries and responses.
- **UDP port 162:** Used for SNMP traps sent to the network management station.

MIB Walker

- A **MIB Walker** is a tool that queries a device's MIB to discover and retrieve available management variables.
- It can walk through a range of OIDs and display the values returned by the device.
- A MIB browser is an example of software used for this purpose.

Example configuration shown in the transcript

The MIB browser is configured with:

- A target host at 10.1.1.1064 in the original transcript; this address appears malformed and should be checked against the original demonstration.
- UDP port 161.
- SNMP version 2c.
- Community string public.
- A root OID beginning with 1.3.6.

When the fetch operation begins, the tool queries available OIDs and displays the values returned by the device.

The collected information may include:

- System identification.
- Contact information.
- System name.
- SNMP-related variables.
- Counts of SNMP requests.

For example, the demonstration displays an SNMP Get Requests value of 4 and an SNMP Get Next value of 1,540.

These values can help administrators understand device activity and identify the device associated with the collected information.

SNMP polling and performance graphs

- SNMP polling involves querying devices at regular intervals.
- A management station might poll every minute, every five minutes, or at another configured interval.
- The collected information can be stored over time and converted into graphs or reports.
- These reports can display:
    - Response times.
    - Error counts.
    - Interface statistics.
    - Other performance information stored in the MIB.

A limitation of polling is that a problem may not be detected until the next scheduled query.

SNMP traps

- An **SNMP trap** is a notification sent proactively from a device to a network management station.
- Unlike regular polling, the management station does not need to query the device first.
- SNMP traps use **UDP port 162**.

Example

A switch can be configured to send a trap when its CRC error count increases beyond a specified threshold.

When the management station receives the trap, it may:

- Generate an alert.
- Notify network administrators.
- Trigger scripts that help investigate or resolve the problem.

SNMP community strings

- A **community string** is a shared string used to control access to SNMP information in SNMPv1 and SNMPv2c.
- It functions similarly to a simple password, but it does not provide the protection of modern user-based authentication.

Common examples mentioned in the transcript:

- public: Often used as a read-only community string.
- private: Often used as a read-write community string.
- A separate community string may be configured for other SNMP functions.

Because these versions do not encrypt the communication, community strings and other transmitted information can be exposed.

Security recommendation from the transcript: Use SNMPv3 when available to benefit from stronger authentication, integrity, and encryption capabilities.

NetFlow and Network Traffic Monitoring

Purpose of network monitoring

- Network devices operate continuously and must be monitored to ensure availability and performance.

- Administrators may need information about:
    - Device load.
    - CPU utilization.
    - Network utilization.
    - Authentication activity.
    - Log information.
    - Security-related events.
- A central network management console can collect and display information from multiple devices in one location.

NetFlow

- **NetFlow** summarizes statistics about traffic flows travelling across a network.
- It provides traffic metadata rather than requiring administrators to inspect every raw packet individually.
- Different manufacturers provide products that collect and analyze flow information.

NetFlow collection process

1. A probe receives traffic information from the network.
2. The probe may obtain traffic through a physical connection or a switch's port-mirroring feature.
3. The probe compiles statistics from the packets it observes.
4. The summarized information is sent to a central NetFlow collector.
5. The collector stores the information and may provide reports and visualizations.

Centralized collection means administrators can review information from multiple probes without accessing each probe individually.

NetFlow reports

A NetFlow console may display:

- **Top conversations:** The largest traffic exchanges between network endpoints, including ingress and egress traffic.
- **Top endpoints:** Devices identified by IP address or device name.

- **Traffic sources:** Devices or probes supplying NetFlow data.
- **Traffic events:** Events reported by NetFlow-enabled devices.
- **Application traffic:** Traffic summarized by port number or application category.

The transcript's example displays traffic categories such as SSL, SQL Server, web traffic, NetFlow UDP traffic, and port-zero traffic.

The collector can retain this information in a database and generate real-time or scheduled reports.

NetFlow versus protocol analysis

- NetFlow provides summarized metadata about traffic flows.
- A protocol analyzer provides detailed information about individual packets and frames.
- NetFlow helps administrators identify overall traffic patterns.
- Protocol analysis helps administrators inspect the exact packet-level communication.

Both methods support troubleshooting and network performance analysis.

Protocol Analyzers and Packet Capture

Protocol analyzer

- A **protocol analyzer** captures and displays network frames and packets.
- It can collect traffic from wired and wireless networks.
- Instead of requiring users to interpret raw hexadecimal data, it presents packet information in a readable format.
- Administrators can view a summary of conversations and inspect individual packets for further details.

Uses

Protocol analyzers can help administrators:

- Investigate unknown traffic.
- Troubleshoot slow applications.
- Examine conversations between devices.
- Identify communication problems.

- Review network activity in detail.

Long-term packet storage

- Some organizations capture and store network packets for extended periods.
- This may require large amounts of storage.
- Stored packets can be examined later during troubleshooting or forensic investigations.

Network Performance Baselines

Definition

- A **network performance baseline** is a collection of measurements describing normal network behavior over time.
- It provides a reference against which current performance can be compared.
- Baselines help determine whether current network utilization or performance is normal or unusual.

Creating a baseline

A baseline can be built by collecting information from:

- SNMP queries.
- NetFlow statistics.
- Protocol analyzers.
- Device logs.
- Network management consoles.
- Security information and event management systems.

The collected data is stored and analyzed over time.

Uses of a baseline

A baseline helps administrators:

- Determine normal network utilization.
- Compare current utilization with typical activity.
- Identify unusual changes in performance.

- Examine the performance of individual devices.
- Understand changes that occur during a normal workday.
- Support troubleshooting using historical measurements.

Syslog and Security Information and Event Management (SIEM)

Syslog

- **Syslog** is a standard used to transfer log messages from network devices and other systems to a centralized destination.
- Devices such as switches, routers, firewalls, and servers can send log information to a central system.
- Administrators configure the devices to forward their logs to the appropriate destination.

SIEM

- **SIEM** stands for **Security Information and Event Management**.
- A SIEM collects and analyzes security-related logs and events from multiple sources.
- It centralizes information so administrators can investigate activity across the network.

The transcript describes a SIEM as a central management system that collects log information and provides dashboards, alerts, and reports.

Information contained in log records

A SIEM may identify:

- The device that generated the log.
- The facility code, identifying the program or component that generated the message.
- The severity level associated with the log entry.
- The event details and timestamp.

SIEM dashboards and reporting

A SIEM can provide:

- Real-time dashboards.
- Alerts for failed authentication attempts.

- Notifications when devices go offline.
- Reports covering activity across multiple devices.
- Historical searches and comparisons.
- Information about application and network utilization.

Administrators can follow traffic and related events as communication moves through servers, switches, firewalls, and internet connections.

SIEM and forensic investigations

- Retained log information can support investigations into past events.
- Administrators may determine when a user authenticated and which services the user accessed.
- Historical logs can help reconstruct activity across multiple devices.

Log filtering and sensitivity settings

- A newly configured SIEM may receive a large volume of log messages.
- Not every message needs to be retained for a long period.
- Administrators can adjust filtering and sensitivity settings to determine which messages should be stored or highlighted.
- For example, they may prioritize warnings and urgent events rather than retaining every informational message in the same way.

Searching for failed authentication attempts

A SIEM can run queries against stored events.

The transcript demonstrates searching for:

- Records containing words such as fail, failure, or failed.
- Records containing password.

The results can identify unsuccessful authentication attempts, such as a Windows Server event reporting an unknown username or an incorrect password.

Administrators can review the results over time to determine whether failed authentication attempts are unusual.

A large number of failed attempts may indicate activity worth investigating, including:

- Brute-force attempts.
- Password-spraying attempts.

Additional security tools can then be used to investigate or help prevent further activity.

API Integration for Network Management

Why API integration is useful

- Traditionally, administrators connect to each switch, router, or firewall through SSH, run commands, make changes, and disconnect.
- Repeating these steps across hundreds or thousands of devices can be tedious and difficult to manage.
- API integration allows software to communicate directly with network devices.

Application Programming Interface (API)

- **API** stands for **Application Programming Interface**.
- An API allows an application or script to communicate with a device through an interface provided by that device.
- It can be used to:
    - Send commands.
    - Retrieve information.
    - Make configuration changes.
    - Automate repetitive administrative tasks.

Instead of manually interacting with an SSH prompt or web console, administrators can use applications and scripts to automate supported operations across multiple devices.

Port Mirroring and Network Taps

Port mirroring

- **Port mirroring** copies traffic from selected switch ports or VLANs and sends the copy to a monitoring device.
- It allows administrators to inspect network traffic without redirecting the original communication away from its intended destination.

Common names

Port mirroring may also be called:

- **SPAN (Switched Port Analyzer)**.
- A switch's port-mirroring feature.

A physical network tap is another method of obtaining a copy of traffic between two devices.

How port mirroring works

1. A device sends traffic through a switch.
2. The switch forwards the original traffic toward its intended destination.
3. The switch sends a copy of the selected traffic to a monitoring port.
4. A connected monitoring device analyzes the copied traffic.

Uses

Port mirroring can provide traffic to:

- Intrusion Detection Systems (IDS).
- Intrusion Prevention Systems (IPS).
- NetFlow probes or collectors.
- Protocol analyzers.
- Network performance monitoring systems.

Some switches can also mirror traffic to a different switch, allowing monitoring equipment in one location to inspect selected traffic from another location.

Administrators can configure mirroring for individual interfaces, multiple interfaces, or VLANs supported by the switch.

Network Discovery and Traffic Analysis

Network discovery

When troubleshooting a network problem, administrators may initially have limited information about the environment. Discovery techniques help identify devices and understand the network.

Common techniques described in the transcript include:

- **LLDP (Link Layer Discovery Protocol):** Used by network devices to discover neighboring devices.

- **CDP (Cisco Discovery Protocol):** Cisco's device-discovery protocol.
- **Ping scans:** Used to identify devices that respond to network reachability checks.
- **Port scans:** Used to identify accessible ports on devices.
- **Commercial network scanners:** Used to discover devices and gather network information.
- **SNMP:** Used to query management information from supported devices.

Some organizations run scans every night, while others run them when troubleshooting a particular problem.

Regular discovery and reporting can help administrators understand which devices exist and what traffic is present on the network.

Traffic analysis

- **Traffic analysis** examines the traffic flowing across a network, often at the frame or packet level.
- Detailed information may be summarized into easier-to-read reports so administrators do not have to inspect thousands of individual records.

Traffic logs may contain:

- Timestamp.
- Source IP address.
- Destination IP address.
- Protocol, such as TCP or UDP.
- Port number.
- Application information.
- Number of bytes transferred between devices.

DNS information may also help associate IP addresses with hostnames.

Historical traffic analysis

- Storing traffic information over time supports long-term reporting.
- Historical information can help administrators investigate earlier network activity.
- Firewall logs can reveal details about communication between devices and the amount of traffic transferred.

Performance Monitoring and Availability Monitoring

Performance monitoring

Performance monitoring examines how well a device or network is operating.

Information may include:

- Network utilization.
- Error counts.
- Device performance statistics.
- NetFlow statistics.
- SNMP data.
- Packet-level protocol analysis.
- Summaries supplied by software agents running on devices.

This information helps administrators identify heavily utilized devices and areas where errors may be occurring.

Availability monitoring

- **Availability monitoring** determines whether a device is up or down.
- A monitoring dashboard may use green to represent an available device and red to represent an unavailable device.
- Monitoring systems can generate alerts when a device becomes unavailable.

Possible automated responses include:

- Sending email notifications.
- Opening help-desk tickets.
- Sending alarms to the responsible team.

Historical availability reporting

- Availability information can be collected over time.
- Historical reports can show whether devices remained available throughout a particular period.
- Up/down monitoring is a basic metric; detailed performance issues may require additional SNMP or NetFlow information.

Network Configuration Files and Change Control

Configuration files

- Network devices commonly have configuration files that define how they operate.
- Depending on the device, configuration information may be stored in different locations or displayed through management interfaces.
- Many devices support backing up and restoring configurations.

Configuration backups and software versions

- Administrators should retain configuration backups for future restoration.
- A configuration file may depend on the software or firmware version running on the device.
- An older configuration file may not be compatible with a newer software version.
- In some cases, restoring an older configuration may require returning the device to a compatible software version.
- Maintaining relevant previous software or firmware versions can therefore be part of the recovery process described in the transcript.

Comparing configurations

- Configuration files from similar devices can be compared to identify differences.
- For example, multiple web servers intended to have similar configurations can be checked for consistency.
- This helps verify that devices follow the intended configuration.

Configuration monitoring and change control

- Organizations may continuously monitor configuration files.
- Alerts can be generated when a configuration changes without prior approval.
- Configuration monitoring may be integrated with an organization's change-control process.
- This process helps ensure that configuration changes are performed and documented in a standardized way.

Disaster Recovery Planning

Disaster Recovery Plan (DRP)

- A **Disaster Recovery Plan (DRP)** describes how an organization will respond to an outage or significant disruptive event.
- It documents the procedures and resources required to recover systems and resume operations.

A DRP may involve:

- Data backups.
- Off-site data replication.
- Cloud-based replacement servers.
- A separate remote recovery site.
- Third-party recovery facilities.
- External recovery services that help manage the recovery process.

Recovery Time Objective (RTO)

- **RTO (Recovery Time Objective)** describes the target time within which a service should be restored after an outage.
- It is measured in time.
- Organizations establish a normal service level and determine how quickly they need to return to that level following a disruption.
- The target is often kept as short as practical.

Example: If the target for restoring a failed web server is one hour, the RTO is one hour.

Recovery Point Objective (RPO)

- **RPO (Recovery Point Objective)** describes the acceptable amount of data loss measured in time.
- It indicates how far back in time recovered data may need to go relative to the disruption.
- The RPO depends on backup frequency, replication arrangements, available recovery resources, and business requirements.

Examples

- Banking transactions or patient information may require a short RPO because losing recent data could be unacceptable.
- Website updates or internal documents may have a longer RPO if backups are performed every one or two hours.

RTO versus RPO

- **RTO:** The target time for restoring service after an outage.
- **RPO:** The acceptable time gap between the disruption and the most recent recoverable data.

On a timeline:

1. Data is backed up or replicated.
2. An outage occurs later.
3. The gap between the outage and the most recent recoverable data represents the RPO.
4. Recovery work takes place.
5. The time from the outage until the service is restored represents the RTO.

Mean Time to Repair (MTTR)

- **MTTR (Mean Time to Repair)** is the average time required to repair a problem and restore operation.
- For example, it may represent the average time required to replace a failed router and bring the network back into service.

Mean Time Between Failures (MTBF)

- **MTBF (Mean Time Between Failures)** is a measure of the average operating time between failures of a repairable system or device.
- It is expressed as a time interval.
- The transcript gives the example of a firewall with an MTBF of 20 years.
- MTBF can be considered when planning equipment redundancy and recovery resources.

MTBF is a statistical measure, not a guarantee that a device will operate without failure for the entire stated period.

Site Resiliency and Disaster Recovery Sites

Site resiliency

- **Site resiliency** involves preparing to move operations from a primary location to an alternative location when necessary.
- The process may include preparing the recovery site, arranging power and hardware, transferring data, moving operations, and eventually returning to the primary site.
- Recovery may last an hour, a day, or months, depending on the disruption.

A recovery plan should address both:

- Moving operations to the alternative location.
- Returning operations to the original location after it becomes available.

Cold site

- A **cold site** is a recovery location with little or no equipment and no active copy of the organization's operational data.
- Equipment and backup data must be brought to the site.
- Staff may also need to travel to the location.
- Considerable work is required before operations can resume.
- A cold site is comparatively inexpensive to maintain.

Hot site

- A **hot site** is a prepared recovery location designed to closely replicate the primary data center.
- It has the necessary hardware and applications.
- Data is kept close to the primary site's current data through replication or ongoing backups.
- Because the infrastructure is already available, the organization can switch operations more quickly than it could with a cold site.

Warm site

- A **warm site** is between a cold site and a hot site in terms of preparation.
- It may have power, racks, infrastructure, and some hardware.
- The organization may still need to bring or restore data and complete some setup.

- The amount of equipment and data prepared in advance depends on the chosen service level.

Comparison

- **Cold site:** Minimal preparation and equipment; more work is required before resuming operations.
- **Warm site:** Some infrastructure is ready; additional equipment or data recovery may be required.
- **Hot site:** Most required infrastructure, applications, and replicated data are ready for faster recovery.

Disaster Recovery Testing

Importance of testing

- Organizations should practice and test disaster recovery procedures.
- Testing helps participants understand their responsibilities and identify weaknesses in the recovery process.
- A full disaster recovery test can be expensive and may take staff away from their regular work.

Tabletop exercise

- A **tabletop exercise** is a meeting in which key participants discuss how they would respond to a simulated disaster.
- Participants walk through the recovery plan without physically moving equipment or restoring the entire production environment.
- The exercise helps determine whether the required logistics, responsibilities, and procedures are in place.

Full disaster recovery validation

- Some organizations perform full disaster recovery tests annually or multiple times per year.
- These tests walk through the recovery process according to a particular scenario.
- The organization can test different situations, such as:
    - A fire destroying the primary data center.

    - An evacuation affecting the wider geographic area around the data center.

Different scenarios may require different recovery steps.

Documenting test results

After a test, organizations can document:

- What worked.
- Which steps failed or were inefficient.
- What needs to be corrected.
- What improvements should be made to the recovery procedures.

Ongoing testing and improvement help ensure that personnel know how to respond when an actual disaster occurs.

Network Redundancy: Active-Passive and Active-Active

Active-passive configuration

- **Active-passive redundancy** uses two devices, but only one actively handles traffic at a time.
- The primary device operates in active mode.
- The secondary device remains in standby or passive mode.
- The devices communicate and exchange status information.
- If the primary device fails, the secondary device takes over.

Configuration synchronization

For successful failover:

- Both devices need compatible or identical configurations.
- Changes made to the primary configuration must be copied to the secondary device.
- Relevant operational information, such as session tables and routing tables, may also need to be synchronized.

Firewall example

1. Two firewalls are installed.
2. One firewall actively processes traffic.

3. The second firewall remains on standby.
4. If the active firewall fails, the passive firewall detects the failure.
5. The passive firewall becomes active and handles subsequent traffic.

Active-active configuration

- **Active-active redundancy** allows both devices to process traffic simultaneously.
- It can make use of the processing capacity of both devices.
- However, it requires more design and configuration than simply powering on both devices.

Design considerations

- Traffic may travel through different devices in different directions.
- One part of a communication may pass through one firewall while return traffic passes through another.
- The system must account for traffic flows and relevant state information.
- Routing, switching, and traffic paths must be designed to support the configuration.

Failure behavior

- Both devices process traffic during normal operation.
- If one device fails, the remaining active device continues handling traffic.
- Because both devices were already active, the surviving device does not need to transition from passive to active mode.

Main difference

- **Active-passive:** One device handles traffic while the other waits to take over.
- **Active-active:** Both devices handle traffic simultaneously.

Bootstrap Protocol (BOOTP) and DHCP

Manual IP configuration

Historically, network administrators configured TCP/IP settings manually on each device, including:

- IP address.

- Subnet mask.
- DNS settings.
- Other required network configuration values.

Manual configuration becomes difficult to manage as the number of devices increases.

Bootstrap Protocol (BOOTP)

- **BOOTP (Bootstrap Protocol)** was introduced to help automate IP configuration.
- It reduced the need to configure every setting manually.
- However, the transcript describes limitations that led to the development of a newer protocol.

Dynamic Host Configuration Protocol (DHCP)

- **DHCP (Dynamic Host Configuration Protocol)** automatically provides IP configuration to network devices.
- It helps assign addresses to devices as they connect to the network.
- It also allows addresses to be made available for reuse when they are no longer assigned, according to the DHCP configuration and lease process.

DHCP is widely used to configure network settings automatically.

DHCP DORA Process

The four main stages of the DHCP address-assignment process are represented by **DORA**:

1. **D — Discover**
2. **O — Offer**
3. **R — Request**
4. **A — Acknowledgement**

Discover

- A client joins the network without an assigned IP address.
- It sends a DHCP Discover message to find available DHCP servers.
- The client's source IP address is 0.0.0.0.

- The destination IP address is the limited broadcast address 255.255.255.255.
- The message is sent from UDP port 68 to UDP port 67.
- The broadcast allows DHCP servers on the local subnet to receive the request.

Offer

- A DHCP server receives the Discover message and responds with a DHCP Offer.
- The offer contains the proposed IP configuration.
- The transcript's example DHCP server has IP address 10.10.10.99.
- The server sends from UDP port 67 to UDP port 68.
- In the initial exchange described, the Offer is broadcast to 255.255.255.255 because the client has not yet configured its IP address.
- If multiple DHCP servers are available, the client may receive multiple offers.

Request

- The client chooses one of the offers.
- It sends a DHCP Request indicating which offer it wants to use.
- The client's source IP address remains 0.0.0.0.
- The message is sent from UDP port 68 to UDP port 67.
- In the transcript's example, the request is broadcast to 255.255.255.255.

Acknowledgement

- The DHCP server responds with a DHCP Acknowledgement, commonly called a DHCP ACK.
- The server confirms the requested configuration.
- The message is sent from UDP port 67 to UDP port 68.
- In the transcript's example, it is broadcast to 255.255.255.255.

After receiving the acknowledgement, the client can configure its IP settings using the assigned information.

DHCP DORA summary

- **Discover:** Client searches for DHCP servers.

- **Offer:** Server offers an IP configuration.
- **Request:** Client requests the selected offer.
- **Acknowledgement:** Server confirms the assignment.

DHCP ports

- **UDP port 67:** DHCP server port.
- **UDP port 68:** DHCP client port.

DHCP Relay and DHCP Helper

Why DHCP relay is needed

- DHCP clients initially use broadcast messages to discover DHCP servers.
- Broadcasts remain within the local subnet and do not normally cross routers.
- As a result, a client on a subnet without a DHCP server cannot directly reach a DHCP server on another subnet through the initial broadcast alone.

This creates a challenge for organizations that want to centralize DHCP servers or distribute them across different network locations.

DHCP relay

- A **DHCP relay**, also known as a **DHCP helper**, allows DHCP messages to be forwarded between different subnets.
- A router configured for DHCP relay receives a client's local broadcast and forwards the request toward a configured DHCP server.
- The relay allows clients to use a DHCP server located on a different subnet.

DHCP relay example

The transcript describes the following devices:

- DHCP server: 10.10.10.99.
- Router interface on Jack's local subnet: 10.10.30.1.
- Jack's laptop: A client that needs an IP address.

Step 1: Client sends Discover

- Jack's laptop sends a DHCP Discover broadcast on its local subnet.
- The router receives the broadcast.

- The router is configured to relay DHCP requests to 10.10.10.99.

Step 2: Router forwards the request

- The router forwards the request toward the DHCP server.
- The relay uses the router's interface address, 10.10.30.1, to identify the client-side subnet in the relayed request.
- The destination becomes the DHCP server's IP address, 10.10.10.99.
- The request is forwarded as unicast rather than remaining a local-subnet broadcast.

Step 3: DHCP server sends Offer

- The DHCP server processes the request and sends its response toward the relay router.
- The response is directed to the relay's address on the client's subnet.

Step 4: Router relays the response

- The router forwards the response to the local subnet.
- In the transcript's example, the response is broadcast locally so the client can receive it.

Step 5: Request and Acknowledgement

- The client continues with the Request and Acknowledgement stages.
- The relay forwards the relevant messages between the client subnet and the DHCP server as needed.
- Jack's laptop can receive its IP configuration even though the DHCP server is located on a different subnet.

Benefits of DHCP relay

- Allows centralized DHCP services.
- Avoids the need for a separate DHCP server on every subnet.
- Enables clients on different subnets to obtain their IP configurations from a shared DHCP server.
- Helps support DHCP services across larger networks.

Quick Revision Points

SNMP and monitoring

- SNMP: Monitors and queries network devices.
- MIB: Stores structured management information.
- OID: Identifies an individual management variable.
- UDP 161: SNMP queries and responses.
- UDP 162: SNMP traps.
- SNMPv1: No encryption.
- SNMPv2c: More efficient queries, but no encryption.
- SNMPv3: Supports authentication, integrity, and encryption.
- MIB Walker: Queries available OIDs.
- NetFlow: Summarizes traffic-flow statistics.
- Protocol analyzer: Inspects individual packets and frames.
- Performance baseline: Describes normal network behavior.
- Syslog: Standard method for transferring log messages.
- SIEM: Collects and analyzes security information and events.
- API integration: Automates communication and administration of devices.
- Port mirroring/SPAN: Copies selected switch traffic for monitoring.

Troubleshooting and configuration

- LLDP: Link Layer Discovery Protocol.
- CDP: Cisco Discovery Protocol.
- Ping scans: Help identify responsive devices.
- Port scans: Identify accessible ports.
- Availability monitoring: Identifies whether devices are up or down.
- Configuration backups: Support restoration and comparison.
- Change control: Standardizes and monitors configuration changes.

Disaster recovery

- DRP: Disaster Recovery Plan.
- RTO: Target time to restore service.
- RPO: Acceptable time-based data loss.

- MTTR: Mean Time to Repair.
- MTBF: Mean Time Between Failures.
- Cold site: Minimal equipment and preparation.
- Warm site: Partially prepared recovery facility.
- Hot site: Highly prepared facility with equipment, applications, and replicated data.
- Tabletop exercise: Discusses a simulated disaster without performing the full recovery.
- Disaster recovery validation: Tests recovery procedures against defined scenarios.
- Site resiliency: Supports moving operations to and from an alternative location.

Redundancy

- Active-passive: One device handles traffic; another stands by.
- Active-active: Both devices handle traffic simultaneously.
- Failover: Transfer of service to another device when the active device fails.

DHCP

- BOOTP: Earlier protocol for assisting with IP configuration.
- DHCP: Automatically provides IP configuration.
- DORA: Discover, Offer, Request, Acknowledgement.
- UDP 67: DHCP server.
- UDP 68: DHCP client.
- 0.0.0.0: Source address used by a client that has not yet obtained an IP address in the described initial exchange.
- 255.255.255.255: Limited IPv4 broadcast address.
- DHCP relay/helper: Forwards DHCP messages between different subnets.

DHCP Server Configuration

DHCP Scope

- When a device requests an IP address, the DHCP server selects an address from a predefined pool within a **DHCP scope**.

- A DHCP scope commonly contains:
    - **IP address range:** The addresses that can be assigned to devices.
    - **Excluded addresses:** Addresses that must not be assigned dynamically.
    - **Subnet mask:** Identifies the subnet associated with the scope.
    - **Lease duration:** Determines how long a device can use an assigned IP address.
    - **DHCP options:** Additional network settings, such as DNS servers, default gateways, and Voice over IP (VoIP) server addresses.
- DHCP scopes are usually created separately for each subnet.
- The address pool is commonly a contiguous range, such as addresses 1–100 within a subnet.
- Exceptions to the pool can be configured by excluding specific addresses.

Managing DHCP Scopes

- On Windows Server, DHCP settings can be managed through Server Manager.
- A DHCP server may have one or more scopes, each responsible for providing addresses to a particular subnet.
- A scope can contain:
    - Address pools
    - Address leases
    - Reservations
    - Scope options
- Changes to the address pool, DNS configuration, and other scope settings are made centrally on the DHCP server.
- When a device requests an address, the server automatically selects an available address from the appropriate pool and assigns it to the device.

DHCP Leases

- DHCP-assigned IP addresses are generally **leased rather than permanently assigned**.
- A lease allows a device to use an address for a limited period.

- If the lease expires and is not renewed, the address can return to the available pool and be assigned to another device.
- DHCP servers track the relationship between devices' MAC addresses and their assigned IP addresses.
- When a device reconnects to the same network, the DHCP server may assign it the same previously used address if that address remains available.

DHCP Address Reservations

- Manually configuring static IP addresses on printers, servers, and other devices can become difficult to manage in large networks.
- Every configuration change would require administrators to update each device individually.
- **DHCP reservation** provides a centralized alternative.
- The administrator associates a device's MAC address with a specific IP address in the DHCP server.
- Whenever that device requests an address, the server reserves the configured address for that MAC address.
- Reservations may also be described as:
    - Static DHCP assignments
    - Static DHCP
    - IP reservations
- Example reservations from the transcript:
    - 192.168.1.6 — device named Prometheus
    - 192.168.1.9 — device named Odyssey
- Each reservation is associated with the corresponding device's MAC address.
- Reservations allow administrators to manage address assignments centrally while ensuring that particular devices receive consistent IP addresses through DHCP.

DHCP Lease Renewal Timers

Lease Duration

- The DHCP administrator determines the lease duration for the network.

- Lease times may be long or short, depending on the organization's requirements.
- A device's IP configuration can display its lease information.
- Restarting a device or disconnecting and reconnecting it may cause it to begin the DHCP process again.
- A device can also release its address when shutting down, allowing the address to become available for other devices.

T1 and T2 Timers

DHCP does not wait until the entire lease expires before attempting renewal. It uses two important timers.

T1 timer — Renewal timer

- By default, T1 occurs at **50% of the lease duration**.
- At T1, the client attempts to renew its lease with the original DHCP server.
- If the original server responds successfully, the lease is renewed and the timing process starts again.

T2 timer — Rebinding timer

- By default, T2 occurs at **87.5% of the lease duration**, or seven-eighths of the lease.
- If the original DHCP server has not renewed the lease, the client attempts to renew it with any available DHCP server.
- This allows the client to continue using its address if the original server is unavailable and another DHCP server can renew the lease.

Example: Eight-Day Lease

For a DHCP lease lasting eight days:

- **T1:** 50% of eight days = four days.
- **T2:** 87.5% of eight days = seven days.

The process is as follows:

1. The client uses its assigned address during the initial lease period.
2. At day four, the client attempts to renew the lease with the original DHCP server.
3. If the original server responds, the lease is renewed and the timing process restarts.

4. If the original server is unavailable, the client continues until the T2 rebinding stage.
5. At day seven, the client attempts to renew with any available DHCP server.
6. If another DHCP server successfully renews the lease, the lease timing process restarts.

DHCP Options

Purpose of DHCP Options

- DHCP can configure more than an IP address, subnet mask, and default gateway.
- Additional settings are called **DHCP options**.
- These options allow DHCP to distribute other TCP/IP-related configuration information to clients.
- DHCP options are defined in the relevant Request for Comments (RFCs).
- In the earlier BOOTP protocol, similar settings were called vendor extensions.
- DHCP options are identified by numbers; the transcript describes 254 usable options.
- Support for individual options depends on the DHCP server implementation.

Common DHCP Settings

Common configuration information includes:

- IP address
- Subnet mask
- Default gateway
- DNS server addresses

Additional options can configure services such as Voice over IP.

Examples stated in the transcript:

- **Option 129:** VoIP call server IP address.
- **Option 135:** HTTP proxy configuration.

The DHCP server must support the specific option for it to be configured and distributed.

Configuring DHCP Options

- On Windows Server, DHCP options can be configured through the DHCP Server settings.
- Options may be configured at the server level or within an individual scope.
- Once configured, supported options are provided to clients when they connect and receive their DHCP configuration.
- Clients can therefore receive their IP address and standard network settings together with any additional settings configured by the administrator.

IPv6 Address Configuration

DHCPv6

- DHCP also supports IPv6 through **DHCPv6**.
- DHCPv6 provides automated IPv6 address configuration through a process similar in purpose to DHCP for IPv4.
- DHCPv6 servers can be deployed redundantly for enterprise networks.
- Administrators manage DHCPv6 configuration and server settings.

However, IPv6 also supports a method that allows a device to configure its own address without a DHCP server.

Stateless Addressing

- Stateless addressing allows an IPv6 device to configure an address without a DHCP server managing the address assignment.
- The device does not need a DHCP server to allocate its address or maintain a DHCP lease.
- This process uses the **Neighbor Discovery Protocol (NDP)** and **Stateless Address Autoconfiguration (SLAAC)**.

Neighbor Discovery Protocol (NDP)

- IPv4 uses the Address Resolution Protocol (ARP) to help discover devices on a local network.
- ARP relies on broadcasts.
- IPv6 uses NDP, which uses multicast for relevant communications.

- NDP supports functions including:
    - Discovering neighboring devices.
    - Discovering routers on the local subnet.
    - Obtaining IPv6 network-prefix information.
    - Checking whether an IPv6 address is already in use.

Stateless Address Autoconfiguration (SLAAC)

SLAAC allows an IPv6 device to create its own address using information advertised by a router.

The process is:

1. **Router Solicitation (RS):** The device sends a multicast request asking routers on the local network to provide information.
2. **Router Advertisement (RA):** A router responds with information about the local network.
3. **Prefix information:** The router advertisement provides the IPv6 subnet prefix and its length, along with other configuration information that may be included.
4. **Interface ID creation:** The device creates the remaining part of its IPv6 address.
5. **Duplicate Address Detection:** The device checks whether another device is already using the proposed address.
6. **Address configuration:** If the address is not detected as a duplicate, the device can use it.

IPv6 Interface ID

- An IPv6 address contains 128 bits.
- In the example described, the router provides a 64-bit subnet prefix.
- The device creates the remaining 64 bits, known as the **interface ID**.
- Some devices derive the interface ID from a modified MAC address, placing FF:FE in the middle.
- Other devices generate a randomized interface ID.

Duplicate Address Detection (DAD)

- DAD stands for **Duplicate Address Detection**.

- It checks whether a proposed IPv6 address is already being used on the network.
- NDP is used to perform this check.
- This helps prevent two devices from configuring the same IPv6 address.

Router Solicitation and Router Advertisement

- A device can use router solicitation to ask whether a router is available on the local subnet.
- A router that receives the solicitation can respond with a router advertisement.
- Routers can also send advertisements without first receiving a solicitation.
- These are called unsolicited router advertisements.
- Router advertisements can provide:
    - IPv6 subnet prefix
    - Prefix length
    - DNS server configuration
    - Other IPv6 configuration parameters

When SLAAC is complete, the device has configured an IPv6 address using the information available on its local network.

Domain Name System (DNS)

Purpose of DNS

- DNS stands for **Domain Name System**.
- It translates human-readable domain names into IP addresses.
- For example, a user can enter www.professormesser.com into a browser instead of manually entering an IP address.
- DNS resolves the name to the corresponding IP address so that the device can connect to the server.

DNS Hierarchy

DNS uses a hierarchical structure.

The hierarchy begins with:

1. **Root (.):** The top of the DNS hierarchy.
2. **Top-Level Domain (TLD):** Examples include .com, .org, and .net.
3. **Domain name:** The name registered beneath a TLD, such as professormesser.com.
4. **Host or subdomain name:** A name such as www within the domain.

Country-code TLD examples include:

- .us — United States
- .ca — Canada
- .uk — United Kingdom

The transcript describes the root DNS system as consisting of 13 root-server identities or clusters, implemented across more than 1,000 actual servers.

Fully Qualified Domain Name (FQDN)

- An **FQDN** is a Fully Qualified Domain Name.
- It identifies a name within the DNS hierarchy.
- Example: www.professormesser.com.
- Organizations can structure their DNS names to represent different services, locations, and groups.
- For example, an organization might have separate east and west names, with additional server and service names beneath them.

DNS supports name resolution for many services, including:

- Web servers
- Active Directory
- Application access

Redundant DNS Servers

- Organizations often use two or more DNS servers to improve availability.
- A computer may be configured with a primary DNS server and a secondary DNS server.

Primary DNS server

- Holds the primary configuration for the DNS zone.
- Changes to the zone are made on the primary server.

Secondary DNS server

- Receives zone information from the primary server.
- Maintains a copy of the zone information in a read-only form.
- Is not normally where administrators make primary zone configuration changes.

From the user's perspective, both servers can provide the same DNS information.

Local Name Resolution and the Hosts File

- Sometimes a device needs to resolve a name locally rather than rely on a DNS server.
- This is called **local name resolution**.
- It can be useful when:
    - Testing a server before directing users to the production server.
    - Overriding incorrect DNS information on a particular machine.
- The hosts file is a text file containing IP addresses and associated host names.

Example from the transcript:

- IP address: 10.1.10.170
- Host name: www.professormesser.com

An entry in the hosts file can direct a device to this IP address when the specified name is used.

Some applications may not use the hosts file in the expected way, so the application's documentation may need to be checked.

Windows hosts file location:

C:\Windows\System32\drivers\etc\hosts

- The file may have read-only permissions.
- To modify it, the user may need to change the permissions or access it with the appropriate privileges.
- Entries in the hosts file can override the DNS result for a name on that particular machine.

Forward and Reverse DNS Lookups

Forward lookup

- A forward lookup provides a name and requests its associated IP address.
- Example: resolving www.professormesser.com to an IP address.

Reverse lookup

- A reverse lookup provides an IP address and requests the associated name.
- It requires the relevant reverse DNS information to be configured.
- The name returned by a reverse lookup may differ from the name used in a forward lookup.

Using dig for DNS Queries

- dig is a utility used to query DNS servers.
- A forward lookup can be performed by querying a domain name.
- The output can show:
    - The queried name
    - The answer section
    - One or more returned IP addresses
    - The DNS server used
    - The time taken to perform the query

The transcript's example returns three IP addresses for www.professormesser.com.

A reverse lookup can be performed with the -x option followed by an IP address.

Example from the transcript:

dig -x 104.22.72.108

The returned name is shown as cruz.ns.cloudflare.com, illustrating that a reverse lookup may reveal a name associated with the IP address used by a service's front end.

Authoritative and Nonauthoritative DNS Servers

Authoritative DNS server

- Holds authoritative DNS information for a particular zone.
- Provides the official DNS records for that zone.

Nonauthoritative DNS server

- Provides DNS information without being the authority for that zone.

- It may be a secondary server holding a copy of the zone or a server that has cached the result.

The transcript demonstrates a query using nslookup against www.professormesser.com. The result is identified as nonauthoritative because the responding server is returning information that it has cached rather than serving as the authoritative source for the zone.

DNS Caching and Time to Live (TTL)

- DNS servers cache query results to make later requests faster.
- Cached information can become outdated if records change on the authoritative server.
- **TTL** stands for **Time to Live**.
- TTL specifies how long a DNS record may remain cached.
- It is configured as a number of seconds.

Example:

- A TTL of 300 seconds equals five minutes.
- After the TTL expires, a cached record is no longer valid for continued use as an unexpired cache entry.
- A subsequent query may require the DNS information to be retrieved again.

The dig output in the transcript shows three IP addresses with a TTL of 300 seconds.

Recursive DNS Queries

- Recursive DNS resolution allows a local DNS server to obtain the information needed to answer a client's query.
- The client device is referred to in the transcript as the resolver.
- The local DNS server performs the necessary queries to locate the relevant DNS information.
- Once the result is obtained, the local DNS server returns the answer to the client and caches it.

Recursive resolution process

1. The client asks its local DNS server to resolve www.professormesser.com.
2. If the local server has no cached answer, it queries a root DNS server.

3. The root server directs it toward the appropriate TLD server, such as a .com server.
4. The local DNS server queries the .com server.
5. The .com server provides information directing the local server toward the domain's authoritative DNS server.
6. The local DNS server queries the authoritative server for the requested host name.
7. The authoritative server returns the relevant DNS record.
8. The local DNS server returns the answer to the client and caches the information.
9. Subsequent requests can be answered from the local cache while the cached information remains valid.

This process reduces repeated external queries and allows clients to receive cached answers more quickly.

DNS Security

DNS Security Challenges

- DNS was originally designed with limited security protections.
- Traditional DNS queries and responses are generally sent in cleartext.
- Someone monitoring the traffic may be able to see the names of services or devices being accessed.
- Traditional DNS responses are not inherently authenticated.
- A client may therefore have difficulty determining whether a response came from the legitimate DNS source or from a spoofed source.

DNSSEC

- DNSSEC stands for **Domain Name System Security Extensions**.
- It uses digital signatures to help validate DNS information.
- DNSSEC allows a resolver to verify that DNS data has not been modified and that the response can be traced to a trusted source through the DNSSEC validation process.
- DNSSEC requires additional DNS configuration.
- It improves the authenticity and integrity of DNS information.

Important distinction: DNSSEC does not encrypt traditional DNS query and response contents. The names being queried can still be visible to someone monitoring the traffic.

DNS over TLS (DoT)

- DoT stands for **DNS over TLS**.
- It encrypts DNS traffic using TLS.
- The default port described in the transcript is **TCP port 853**.
- This helps prevent observers from reading the contents of the DNS communication.

DNS over HTTPS (DoH)

- DoH stands for **DNS over HTTPS**.
- It sends DNS requests and responses through HTTPS.
- It uses encrypted communication similar to that used for secure web traffic.
- The transcript describes this traffic as using **TCP port 443**.
- In a packet capture, the traffic can resemble other encrypted HTTPS communication.
- Some modern browsers support or use DoH to encrypt DNS traffic.

Comparing DNS Security Methods

- **DNSSEC:** Helps verify DNS data authenticity and integrity; does not itself encrypt the DNS contents.
- **DoT:** Encrypts DNS communication using TLS, commonly over TCP port 853.
- **DoH:** Encrypts DNS communication within HTTPS, commonly over TCP port 443.

DNS Resource Records

What Are Resource Records?

- DNS configuration files contain records known as **resource records**.
- These records provide information used for name resolution and other DNS functions.

- Different record types store different kinds of information, including IP addresses, aliases, mail-server information, and name-server details.

SOA Record — Start of Authority

- SOA stands for **Start of Authority**.
- It provides important information about a DNS zone.
- Information may include:
    - Zone details
    - Serial number
    - Retry information
    - Expiration information
    - Timing information associated with zone data

A Record — IPv4 Address

- An **A record** associates a host name with an IPv4 address.
- It is used when a DNS client requests the IPv4 address of a name.

Example from the transcript:

- Name: www.professormesser.com
- IPv4 address: 162.159.246.164

When a client queries the DNS server for this name, the server can return the address stored in the A record.

AAAA Record — IPv6 Address

- An **AAAA record** associates a host name with an IPv6 address.
- It provides the IPv6 equivalent of an A record.
- The record type uses four A characters: AAAA.

CNAME Record — Canonical Name

- CNAME stands for **Canonical Name**.
- It creates an alias that points to another DNS name.
- Multiple aliases can point to the same canonical name.

Example from the transcript:

- chat.example.com → mail.example.com

- ftp.example.com → mail.example.com
- www.example.com → mail.example.com

When a client queries an alias, DNS can return the canonical name. If the client does not already know the canonical name's IP address, another DNS lookup may be needed to resolve it.

MX Record — Mail Exchanger

- MX stands for **Mail Exchanger**.
- It identifies the mail server responsible for receiving email for a domain.
- Example from the transcript: mail.example.com.
- The mail server's name must also be resolved to an IP address, normally using an appropriate address record.

TXT Record — Text

- A TXT record stores text information associated with a DNS name.
- It is used for a variety of purposes, including email security configuration.

SPF — Sender Policy Framework

- SPF stands for **Sender Policy Framework.**
- It identifies which mail servers are authorized to send email on behalf of a domain.
- Receiving mail servers can check the domain's SPF record to help determine whether a message came from an authorized source.

DKIM — DomainKeys Identified Mail

- DKIM stands for **DomainKeys Identified Mail**.
- It uses digital signatures associated with outgoing email.
- The receiving system can verify the signature.
- The public key needed for verification can be published in DNS as a TXT record.

The transcript shows SPF and DKIM information stored as text records, including a public key used to verify DKIM signatures.

NS Record — Name Server

- NS stands for **Name Server**.
- An NS record identifies a DNS server associated with a domain or zone.

- A domain may have multiple NS records.

Example from the transcript:

- ns1.example.com
- ns2.example.com

These records identify the name servers associated with the domain.

PTR Record — Pointer

- PTR stands for **Pointer**.
- A PTR record is used for reverse DNS lookups.
- It maps an IP address in the reverse DNS namespace to a name.

Examples from the transcript:

- 192.168.23.2 → joe.example.com
- 192.168.23.15 → www.example.com
- 192.168.23.17 → bill.example.com

The reverse DNS records use the IP address information in reverse order within the relevant reverse DNS namespace. A reverse lookup for 192.168.23.15 can therefore return www.example.com when the corresponding PTR record is configured.

DNS Record Quick Reference

- **SOA:** Start of Authority; zone information and timing details.
- **A:** Maps a name to an IPv4 address.
- **AAAA:** Maps a name to an IPv6 address.
- **CNAME:** Creates an alias to another DNS name.
- **MX:** Identifies a mail exchanger.
- **TXT:** Stores text information, including SPF and DKIM-related data.
- **NS:** Identifies name servers for a domain or zone.
- **PTR:** Maps an IP address to a name for reverse DNS resolution.

Network Time Protocol (NTP)

Purpose of NTP

- NTP stands for **Network Time Protocol**.

- It synchronizes clocks across network devices, including:
    - Laptops and desktop computers
    - Switches
    - Routers
    - Firewalls
    - Other networked systems
- Consistent timestamps are important when comparing logs from different devices.
- Time synchronization is generally automatic.
- Devices may be configured to update their clocks at different intervals, such as hourly or daily.
- The transcript notes that devices may differ by only milliseconds when synchronization is working properly.

NTP Servers and Clients

- An NTP server listens on **UDP port 123**.
- It responds to requests from NTP clients with time information.
- The NTP client is the software responsible for querying a server and obtaining the latest time information.
- An NTP server does not automatically correct its own time simply because it serves other clients; it needs its own time source or a separate NTP client process to obtain time from another server.
- A device can operate as both an NTP server and an NTP client.
    - Its server function supplies time information to other clients.
    - Its client function obtains time from another NTP server.

Internal and External NTP Servers

- An organization may obtain time information from an external NTP server.
- Alternatively, it may maintain internal NTP servers.
- Network devices and computers can then be configured to query the organization's internal time servers.

NTP and Security

- Traditional NTP commonly sends its time information without encryption.

- Although time is not generally private, incorrect time can create security and availability problems.
- The transcript gives Kerberos as an example: authentication may fail when client and server timestamps differ by more than five minutes.
- Incorrect time information could therefore cause authentication problems or contribute to a denial-of-service situation.

Network Time Security (NTS)

- NTS stands for **Network Time Security**.
- It adds authentication so that clients can establish trust in the time information they receive.
- NTS uses a Network Time Security Key Establishment (NTS-KE) server to help authenticate clients and provide cryptographic cookies.

NTS process

1. The client performs a TLS handshake with the NTS key establishment server.
2. The client requests authentication information.
3. The key establishment server provides a cookie.
4. The client sends an NTP request to an NTP server with the cookie.
5. The NTP server returns the time response.
6. The client can validate the response using the NTS security information.

This process helps the client determine whether the timestamp response can be trusted.

Precision Time Protocol (PTP)

- PTP stands for **Precision Time Protocol**.
- It is used when more precise time synchronization is required.
- The transcript describes PTP as providing time precision down to the nanosecond.
- PTP may use dedicated hardware separated from the main operating system and applications.
- This arrangement reduces the effect of delays caused by other processes.
- PTP is especially useful in industrial environments where devices require highly precise timestamps.

Virtual Private Networks (VPNs)

Purpose of a VPN

- VPN stands for **Virtual Private Network**.
- A VPN encrypts data that would otherwise be sent across a network without that VPN protection.
- The encrypted data can then travel across a public network, such as the internet.
- VPNs are used to provide protected communication between remote users and corporate networks or between different sites.

VPN Concentrator

- A VPN concentrator handles VPN encryption and decryption.
- This functionality is commonly integrated into modern firewalls.
- A concentrator may be:
    - Dedicated hardware designed for VPN processing.
    - Software running on an existing server or device.
- When encrypted traffic reaches the concentrator, it decrypts the traffic and forwards it toward its destination on the internal network.
- Traffic returning to the remote user is encrypted before being sent across the public network.

VPN Clients

- VPN clients are available for many operating systems.
- Some operating systems include VPN client functionality.
- A client may need to be installed separately, depending on the operating system and VPN configuration.

Client-to-Site VPNs

Purpose

- A **client-to-site VPN** connects an individual client device to a central network.
- VPN software runs on the client workstation.

- The client establishes an encrypted connection to a VPN concentrator at the organization.

Example:

- An employee uses a laptop at a coffee shop.
- The laptop establishes a VPN connection to the company's VPN concentrator.
- The client's traffic is encrypted while travelling across the public network.
- The concentrator decrypts the traffic and forwards it into the corporate network.

Manual and Always-On VPNs

A VPN can be configured in different ways.

Manually enabled VPN

- The user starts the VPN software and enables the connection when needed.
- The user may disable it when it is no longer required.

Always-on VPN

- The VPN client is configured to start automatically.
- When the user turns on the laptop and logs in, the client establishes a connection to the corporate network.
- Traffic sent through the VPN is encrypted and processed by the concentrator.

Return Traffic

- Traffic from the corporate network to the remote laptop follows the reverse process.
- The concentrator encrypts the traffic.
- The encrypted traffic crosses the public network.
- The VPN client decrypts it on the laptop.

Site-to-Site VPNs

Purpose

- A **site-to-site VPN** encrypts communication between two separate network locations.
- It is commonly configured as an always-on connection.
- Firewalls at both locations can provide VPN concentrator functionality.
- Traffic travelling between the sites is sent through the encrypted VPN tunnel.

How It Works

1. A device at one site sends traffic intended for the other site.
2. The local firewall encrypts the traffic.
3. The encrypted traffic travels across the public network through the VPN tunnel.
4. The firewall at the destination site decrypts the traffic.
5. The traffic is forwarded to the destination network.

- Users at either site may not need to interact with the VPN directly.
- The firewalls handle encryption and decryption between the locations.

Clientless VPNs

Purpose

- A **clientless VPN** provides VPN access without requiring a separately installed VPN client.
- The transcript describes this functionality as operating within a web browser using HTML5.
- HTML5 allows web applications to use browser APIs.
- The transcript specifically mentions the Web Cryptography API as part of the browser-based approach.

How It Works

1. The user opens the appropriate web page in a compatible browser.
2. The browser-based VPN functionality runs without a separate VPN application being installed.
3. The browser establishes the protected connection.

4. The VPN functionality provides encrypted access to the required resources.

- The administrator must configure the VPN system to support clientless access.
- The user needs an HTML5-compatible browser for the functionality described in the transcript.

Full-Tunnel and Split-Tunnel VPNs

Full-Tunnel VPN

- A **full-tunnel VPN** sends all of the client's traffic through the VPN connection.
- Traffic is encrypted before it crosses the public network.
- The VPN concentrator decrypts the traffic at the other end.
- The client does not selectively send some traffic through the VPN while bypassing it for other destinations.
- Internet-bound traffic may also pass through the corporate VPN concentrator before reaching its final destination.

Full-tunnel example

1. A remote user connects to the corporate VPN.
2. Traffic intended for the corporate network enters the VPN tunnel.
3. Traffic intended for an external website also enters the VPN tunnel.
4. The concentrator decrypts the traffic.
5. External traffic is then forwarded toward the internet.
6. Return traffic travels back through the concentrator and VPN tunnel to the client.

This arrangement means that even traffic intended for third-party websites may travel through the corporate network.

Split-Tunnel VPN

- A **split-tunnel VPN** divides traffic between two paths:
    - Traffic intended for the corporate network travels through the encrypted VPN tunnel.

    - Other traffic, such as traffic intended for an unrelated public website, travels directly through the normal internet connection.
- The VPN client determines which path to use according to the configured VPN routing rules.

Split-tunnel example

1. The remote user establishes a VPN connection to the corporate network.
2. Traffic intended for corporate resources travels through the VPN tunnel.
3. The corporate VPN concentrator decrypts that traffic and forwards it internally.
4. Traffic intended for an unrelated external website is sent directly from the remote workstation to that website.
5. The website's return traffic comes back directly to the remote workstation rather than travelling through the corporate VPN concentrator.

Full Tunnel vs. Split Tunnel

Full tunnel

- All client traffic travels through the VPN.
- External internet traffic is also routed through the concentrator.
- Communication follows the VPN path regardless of whether the destination is a corporate resource or an unrelated external website.

Split tunnel

- Corporate traffic uses the encrypted VPN connection.
- Selected non-corporate traffic uses a separate, direct internet path.
- Avoids routing unrelated internet traffic through the corporate VPN concentrator.

Remote Access and Device Management

Secure Shell (SSH)

- **SSH (Secure Shell)** is a protocol used to securely access the command-line interface of a remote device over a network.
- It is commonly used to manage switches, routers, firewalls, servers, and other network devices.

- SSH encrypts the traffic exchanged between the client and the remote device, protecting usernames, passwords, commands, and other transmitted information.
- SSH uses **TCP port 22**.
- It was designed to replace Telnet for secure remote command-line access.

Telnet

- Telnet provides a command-line interface for accessing and managing remote devices.
- It uses **TCP port 23**.
- Unlike SSH, Telnet does not encrypt its communication.
- Usernames, passwords, commands, and other transmitted data can be exposed if someone captures the traffic.
- **Best practice:** Use SSH instead of Telnet for remote administration.

Remote Desktop Protocol (RDP)

- **RDP (Remote Desktop Protocol)** is Microsoft's protocol for remotely accessing and controlling a Windows desktop.
- It allows a user to interact with a remote computer as though they were sitting in front of its monitor and keyboard.
- RDP can be used between Windows computers or through compatible clients on other operating systems.
- It is useful for help-desk staff and support teams that need to troubleshoot or manage remote computers through a graphical interface.

Virtual Network Computing (VNC)

- **VNC (Virtual Network Computing)** provides remote desktop access and control across different operating systems.
- It uses the **RFB (Remote Framebuffer)** protocol.
- VNC performs a function similar to RDP, allowing a remote user to view and interact with another computer's desktop.
- It is useful for help-desk and technical support teams that manage computers running different operating systems.

API-Based Device Management

- Managing hundreds or thousands of devices manually is time-consuming.

- Command-line scripts and batch files can automate repetitive tasks, but handling errors and complex workflows can be difficult.
- An **API (Application Programming Interface)** allows software to communicate with a device programmatically.
- APIs can be used to automate device configuration, management tasks, and error handling.
- Centralized automation can apply changes to many devices without requiring an administrator to connect to each device manually.

Console Connections

- A console connection provides direct access to a device through a dedicated console interface.
- Switches, routers, and other infrastructure devices may provide console connections through:
    - RJ45 serial connections.
    - DB9 serial connections.
    - USB connections.
- Console access is particularly useful when network connectivity is unavailable.
- If a device cannot be reached through ping or SSH, an administrator may still be able to connect directly through its console port.
- Console interfaces are commonly text-based, so the administrator needs to understand the device's command-line interface.
- A computer may require a physical serial port or a USB-to-serial adapter to connect to the device.

Jump Servers

- A **jump server** is an intermediary system used to access other devices within an organization's network.
- An administrator first connects to the jump server and then connects from it to internal systems.
- The initial connection may use a VPN, SSH, or another secure remote-access mechanism.
- This reduces the need to establish separate external connections to every internal device.

- Because a jump server may be accessible from an external network, it must be strongly secured.

Important security measures include:

- Enforcing strong authentication and multi-factor authentication (MFA).
- Keeping the system updated with security patches.
- Preventing unauthorized access through strong security controls.
- Using secure connections for remote administration.

In-Band and Out-of-Band Management

In-band management

- Uses the existing network to manage a device.
- The device is assigned a management IP address, subnet mask, and other required network settings.
- Administrators can connect using SSH or a web-based management interface.
- The management interface may be a dedicated network interface or an existing interface configured for management.

Out-of-band management

- Uses a separate management path that does not depend on the normal production network.
- Common methods include serial console connections and USB console connections.
- Some systems may use a modem and telephone line to provide remote access when the main network is unavailable.
- A communications server may allow administrators to connect to one system and then access other devices through that connection.

Key distinction: In-band management depends on the existing network, whereas out-of-band management provides a separate path for administration.

Data Security: Data in Transit and Data at Rest

Data in Transit

- **Data in transit**, also called data in motion, is information being transferred across a wired or wireless network.

- Switches and routers primarily forward traffic toward its destination.
- Additional security controls are needed to inspect traffic and determine whether it should be permitted or blocked.

Security mechanisms mentioned in the text include:

- **Firewalls:** Monitor traffic and determine whether it should be forwarded or blocked.
- **Intrusion Prevention Systems (IPS):** Inspect traffic for malicious activity and help prevent attacks.
- **Encryption:** Protects information while it travels across a network.

Two common encryption-related technologies are:

- **TLS (Transport Layer Security).**
- **IPsec (Internet Protocol Security).**

Data at Rest

- **Data at rest** is information stored on a hard drive, solid-state drive (SSD), or another storage device.
- Encryption can protect stored information if someone gains access to the storage medium.

Types of data-at-rest encryption include:

- **Full-disk encryption:** Encrypts the entire storage drive.
- **Database encryption:** Encrypts part or all of a database.
- **File and folder encryption:** Encrypts selected files or folders within a file system.

Access control lists (ACLs) can also restrict who may access stored data.

- ACLs define which users are permitted or denied access to particular information.
- Administrators or data owners can use these controls to manage access.
- These controls are commonly implemented through the operating system.

Public Key Infrastructure (PKI) and Digital Certificates

Public Key Infrastructure

- **PKI (Public Key Infrastructure)** is the collection of policies, processes, and controls used to manage digital certificates and cryptographic keys.
- Organizations use PKI to manage security assets associated with users, devices, and servers.
- PKI becomes important as an organization deploys encryption keys, creates certificates, and digitally signs certificates for users and devices.
- PKI policies may need to be updated as the organization's security requirements change.

Digital Certificates

- A **digital certificate** helps establish trust in a user, device, or other entity.
- Certificates are commonly digitally signed by a Certificate Authority.
- The signature allows a party examining the certificate to identify who issued it and determine whether that issuer is trusted.
- Certificates can be deployed to users, devices, and servers.

Certificate Authority (CA)

- A **Certificate Authority (CA)** is an entity responsible for issuing or digitally signing certificates.
- A CA establishes trust in certificates that it signs.
- Organizations may operate their own internal CA or use a trusted third-party CA.
- An internal CA can issue certificates for users and devices within the organization.
- For those certificates to be trusted, the relevant systems must trust the issuing CA.

A third-party CA can provide trust for certificates used beyond an organization's internal environment.

Web of Trust

- A **web of trust** is a distributed approach to establishing trust.
- Trust can be passed through relationships between entities.
- For example, if A trusts B and B trusts C, A may also choose to trust C through that relationship.

This differs from a centralized CA model, where trust is based on certificates signed by a recognized central authority.

Identity and Access Management (IAM)

What Is IAM?

- **IAM (Identity and Access Management)** manages identities, authentication, permissions, and access to organizational resources.
- Data may be stored on local computers, centralized servers, or in the cloud.
- Different users, customers, contractors, and employees may require different levels of access.
- IAM helps ensure that users can access the information necessary for their responsibilities while preventing unauthorized access.

IAM includes:

- Managing user identities and access permissions.
- Authenticating users.
- Authorizing access to resources.
- Tracking and monitoring access.
- Auditing who accessed information and when.

Principle of Least Privilege

- The **principle of least privilege** means giving users and applications only the permissions required to perform their assigned tasks.
- Additional access should not be granted without a legitimate requirement.
- Giving every user administrator access would allow excessive control over systems and data, creating a significant security risk.
- Least privilege limits the potential impact of unauthorized or inappropriate access.

Role-Based Access Control (RBAC)

- **RBAC (Role-Based Access Control)** assigns permissions according to a user's organizational role.
- Administrators define roles and the permissions associated with each role.
- Users are then assigned to the appropriate roles.

Example roles from the text:

- Shipping and receiving employee.
- Shipping and receiving manager.
- Vice president responsible for multiple departments, such as shipping and receiving and accounting.

Each role receives the permissions needed for its responsibilities.

In Windows, RBAC can be implemented using groups:

1. Create groups representing the different roles.
2. Assign appropriate permissions to each group.
3. Add users to the relevant groups.

Geographic Restrictions and Geofencing

- Geographic restrictions can be used to apply different access permissions based on a user's location.
- An organization may use a user's IP address, GPS information, or the location of a wireless access point to help determine where the user is connecting from.
- A VPN connection may provide a user with an internal IP address, allowing access similar to that available on the organization's internal network.

Geofencing uses physical location to allow or deny access to information.

Example:

- Highly sensitive organizational information may only be accessible from the corporate headquarters.
- Access may be denied when the user is outside the permitted physical area.

IP addresses may not always accurately identify a user's physical location, so other location information may be used.

Physical Security

Closed-Circuit Television (CCTV)

- **CCTV (Closed-Circuit Television)** is used to monitor physical areas within a building or campus.

- Cameras may support motion detection and other identification features.
- Some modern cameras may be capable of reading license plates or performing facial recognition.
- Multiple cameras can be connected to a central system that stores video recordings.
- Stored recordings allow administrators to review footage from a particular location and time.

Door Locks and Access Control

Physical access controls include:

- **Traditional locks:** Require a physical key and may include a deadbolt.
- **Electronic readers:** Allow entry using a personal identification number (PIN).
- **RFID badges:** Use a token or badge presented to an electronic reader.
- **Biometric systems:** May use fingerprints, handprints, or retina scans.

Combining Authentication Factors

- Organizations can combine multiple access methods to strengthen authentication.
- For example, a user may need both an RFID badge and a PIN to enter a restricted area.
- A badge alone could be used by someone who finds or steals it, but combining it with a PIN adds another authentication requirement.

Authentication, Authorization, and Accounting (AAA)

Identification

- **Identification** is the information a user provides to identify themselves.
- Common examples include a username or email address.
- Identification information is often publicly known or discoverable.
- Knowing someone's username or email address does not, by itself, prove their identity or grant access.

The Three As of AAA

1. Authentication

- Verifies that a user is who they claim to be.
- Commonly uses a password or another authentication factor.

2. Authorization

- Determines what an authenticated user is permitted to access.
- Permissions may apply to files, directories, applications, or areas of a network.

3. Accounting

- Records activity associated with access to a system.
- May document successful logins, logouts, failed authentication attempts, and other relevant events.
- Helps organizations track and audit user activity.

How AAA Works

A common example is a user connecting to a VPN concentrator from a laptop over the internet.

1. The user provides a username, password, or other authentication factor.
2. The VPN concentrator sends the authentication information to an AAA server.
3. The AAA server checks the submitted information against its stored authentication records.
4. If the information is valid, the server returns an approval response.
5. The user can then access resources permitted by their authorization.

The AAA server stores authentication information in a protected form and uses it to help verify the user's identity.

Single Sign-On (SSO)

- **SSO (Single Sign-On)** allows a user to authenticate once and then access multiple permitted resources without repeatedly entering their credentials.
- For example, an employee may sign in at the beginning of the workday and access the resources available to them.
- SSO sessions are not necessarily permanent; they may expire after a configured period.

- The user may need to authenticate again when the session expires or when starting a new session.
- Not every authentication method supports SSO, so support depends on the method being used.

Authentication Protocols and Directory Services

RADIUS

- **RADIUS (Remote Authentication Dial-In User Service)** is an established authentication protocol supported by many operating systems and network devices.
- Despite the reference to dial-in access in its name, it is also used on modern networks.
- A VPN concentrator or firewall may communicate with a RADIUS server to authenticate users.
- RADIUS can also be used for network access authentication, including wireless access involving **IEEE 802.1X**.
- Its broad support makes it a common option for network authentication.

Lightweight Directory Access Protocol (LDAP)

- **LDAP (Lightweight Directory Access Protocol)** is used to access and manage information in a centralized directory.
- A directory can store information about users, devices, departments, and other organizational objects.
- LDAP is similar in purpose to a directory or phone listing because it allows information to be searched and retrieved.
- It is associated with the X.500 directory standards.
- LDAP is used with directory services, including Windows Active Directory and other directory systems.

LDAP directories can provide additional context about a user or device beyond a simple username and password.

X.500 Attributes and Directory Hierarchies

- X.500 provides a framework for representing directory information and attributes.

- Attributes can describe a user or device, including its name, organizational unit, organization, and location.
- For example, a server named widget-web could have additional information indicating that it belongs to the marketing department, is part of the widget organization, and is associated with London and the widget.com domain.

Directory information can be organized hierarchically:

- **Containers:** Higher-level objects that organize directory information.
- **Leaf objects:** Individual entries, such as users or devices.

An organization might organize its directory by country, department, and then individual users or devices.

This hierarchy provides context about the objects involved in authentication and access management.

Security Assertion Markup Language (SAML)

- **SAML (Security Assertion Markup Language)** is an open standard used to exchange authentication and authorization information.
- It allows authentication information to be used across different applications.
- A typical flow involves three components:
    - **Client:** Usually the user interacting through a browser.
    - **Resource server:** The server hosting the requested resource.
    - **Authorization or identity service:** The service involved in authenticating the user and providing an assertion or token.

A simplified flow described in the text is:

1. The user requests a protected resource.
2. The resource server redirects the client to the appropriate authentication service.
3. The user provides authentication information.
4. The information is validated, potentially using a directory service or RADIUS.
5. A signed assertion or token is provided to the client.
6. The client presents the assertion or token to the resource server.

7. The resource server verifies it and grants access if the requirements are satisfied.

TACACS+

- **TACACS+ (Terminal Access Controller Access-Control System Plus)** is an authentication protocol commonly associated with network-device administration.
- The original TACACS family was used to control access to systems such as dial-up connections.
- TACACS+ has historically been associated with Cisco devices, although aspects of the protocol have been made public.
- It can be integrated into authentication systems to support access control.

Multi-Factor Authentication (MFA)

What Is MFA?

- **MFA (Multi-Factor Authentication)** requires users to provide multiple authentication factors.
- A username and password alone may not be sufficient because credentials can be stolen or exposed.
- Additional factors provide more evidence that the person attempting to sign in is the legitimate user.

Types of Authentication Factors

Something you know

- A password or PIN.

Something you have

- A mobile phone or an application that generates authentication codes.

Something you are

- A biometric characteristic, such as a fingerprint.

Somewhere you are

- Location information, such as a GPS location, where the authentication system supports location-based checks.

One or more of these factors may be used to strengthen authentication.

Time-Based One-Time Password (TOTP)

- **TOTP (Time-Based One-Time Password)** is a method that generates temporary authentication codes using a shared secret and the current time.
- It is commonly integrated into an authenticator application on a mobile device.
- The code changes periodically, often every 30 seconds.
- The code is not truly random: it is generated using a process that allows the user's application and the authentication server to calculate the expected value.
- Both sides must have the appropriate shared secret and time synchronization.

A typical login may require:

1. Username.
2. Password.
3. A code displayed by the authenticator application.

The text identifies this method as a common way to add an additional authentication factor.

Honeypots and Honeynets

Honeypot

- A **honeypot** is a deliberately deployed system intended to attract attackers and allow defenders to observe their activity.
- Many attacks are performed by automated scripts or programs rather than by a person manually interacting with a keyboard.
- A honeypot may imitate an attractive target so that attackers interact with it.
- Security teams can observe the techniques and actions used during those interactions.
- Honeypots need to appear realistic because attackers may attempt to identify fake or deliberately exposed systems.
- Open-source honeypot implementations are available for organizations that want to build their own.

Honeynet

- A **honeynet** is a larger deception environment made up of multiple systems.
- It may include:
  - Workstations.
  - Servers.
  - Routers.
  - Firewalls.
  - Switches.
  - Proxy servers.
- Multiple interconnected systems can make the environment appear more like a real network.
- Defenders can observe whether an attacker moves from one system to another within the honeynet.
- Honeynets provide a broader environment for observing attacker behavior than a single honeypot.

Risk, Vulnerabilities, Threats, and Exploits

Risk

- **Risk** describes exposure to something potentially harmful or dangerous.
- Organizations face risk when expanding, deploying applications, changing configurations, or making other business decisions.
- Identifying risk helps decision-makers determine whether to proceed with an activity or implement additional security controls.
- As organizations grow and their systems become more complex, they may face additional risks.

Vulnerability

- A **vulnerability** is a weakness in a system, application, configuration, authentication process, or other component that could be exploited.
- Vulnerabilities may expose systems or data to unauthorized access.
- Examples described in the text include:

    - Weaknesses in how data is handled or injected into a data stream.
    - Flaws in an operating system's authentication process.
    - Exposed data.
    - Security misconfigurations.
- A vulnerability can exist even before researchers discover it.
- Software updates and vulnerability reports may identify weaknesses that need to be addressed.

Exploit

- An **exploit** is a method or action that takes advantage of a vulnerability.
- Exploitation may involve a simple technique or a sequence of more complex actions.
- An attacker may use an exploit to gain access to a system or obtain data.

Threat and Threat Agent

- A **threat** is a potential source of harm to a system, organization, or its information.
- Threats are not limited to malicious attackers; events such as fires and floods can also threaten an organization.
- A **threat agent** is an entity that may carry out an attack or take advantage of a weakness.
- A **threat action** is an action that exploits a vulnerability or otherwise causes harm.

A simplified attack sequence is:

1. A system contains a vulnerability.
2. A threat agent identifies or takes advantage of the weakness.
3. A threat action exploits the vulnerability.
4. The system or its data is affected.

Possible consequences include:

- System unavailability.
- Unauthorized access to private information.
- Disclosure of information to other parties.

The CIA Triad

The **CIA Triad** describes three fundamental principles of information security. It is also called the AIC Triad in some contexts to distinguish it from the U.S. Central Intelligence Agency.

Confidentiality

- Confidentiality means protecting information from unauthorized access or disclosure.
- Access restrictions and encryption are examples of controls that help maintain confidentiality.

Integrity

- Integrity means protecting information against unauthorized or improper modification.
- Systems need mechanisms to detect or prevent changes made by unauthorized parties.
- Digital signatures are one mechanism associated with integrity.

Availability

- Availability means ensuring that systems and information remain accessible to authorized users when needed.
- Security measures should not unnecessarily prevent legitimate users from accessing systems.
- Maintaining system availability is an important responsibility of IT and security teams.

Regulatory Compliance

Compliance Requirements

- IT professionals must follow applicable laws, policies, procedures, standards, and other rules.
- Requirements may come from national, state, local, or international authorities, as well as from industry standards.
- Failure to comply may result in fines, legal consequences, loss of employment, or other penalties.

- Organizations must understand which requirements apply to their activities and how those requirements affect data storage, processing, and movement.

Data Localization

- **Data localization** refers to requirements that certain data be stored or maintained within a specified geographic area.
- Organizations must understand where data is required to reside and whether it may be transferred elsewhere.
- These requirements can be associated with national or regional regulations.

General Data Protection Regulation (GDPR)

- **GDPR (General Data Protection Regulation)** is a European Union regulation concerning the protection and privacy of personal data.
- The text gives examples of personal data, including:
    - Names and addresses.
    - Photographs.
    - Email addresses.
    - Banking information.
    - Information about websites a person visits.
    - Other data associated with an individual.
- GDPR establishes requirements for how covered personal data is handled and protected.
- It gives individuals rights concerning their personal data, including rights relating to access and deletion in applicable circumstances.
- The text describes these rights using the phrase **"right to be forgotten."**

Payment Card Industry Data Security Standard (PCI DSS)

- **PCI DSS (Payment Card Industry Data Security Standard)** is a standard intended to protect payment-card information.
- It is particularly relevant to organizations that store, process, or transmit credit-card information.

The six areas of focus described in the text are:

1. **Build and maintain secure networks and systems.**

2. **Protect cardholder data.**
3. **Maintain a vulnerability management program.**
4. **Implement strong access control measures.**
5. **Regularly monitor and test networks.**
6. **Maintain an information security policy.**

- Organizations may be audited to determine whether they follow PCI DSS requirements.
- Failure to comply can affect an organization's ability to process payment cards.

Network Segmentation

What Is Network Segmentation?

- **Network segmentation** separates devices or systems into distinct physical or logical network sections.
- Physical segmentation uses separate physical infrastructure.
- Logical segmentation can use technologies such as VLANs.
- Segmentation may improve performance by separating applications that transfer large amounts of data.
- From a security perspective, it limits which devices and systems can communicate with one another.
- Some compliance requirements, including those associated with payment-card data, may require segmentation.

Internet of Things (IoT)

- **IoT (Internet of Things)** refers to connected devices such as:
    - Heating and cooling sensors.
    - Lighting systems.
    - Home automation devices.
    - Video doorbells and cameras.
    - Smartwatches and health monitors.
- Some IoT devices may not have security controls that are as strong as those found in other systems.

- Placing IoT devices on a separate network can help isolate them from sensitive data and internal systems.

Industrial Internet of Things (IIoT)

- **IIoT (Industrial Internet of Things)** involves connected industrial devices and machines that communicate with one another.
- These systems may support manufacturing, energy, and other industrial operations.
- Communication failures or disruptions can have serious consequences because multiple machines may depend on one another.
- Segmentation can help prevent unrelated or external devices from disrupting critical industrial communications.
- The text highlights examples involving oil and gas systems and medical devices that must communicate reliably.

SCADA and Industrial Control Systems

- **SCADA** stands for **Supervisory Control and Data Acquisition**.
- **ICS** stands for **Industrial Control Systems**.
- These systems are used in industrial environments, including manufacturing and power generation.
- They require monitoring and control over network connections.
- Access to these systems may be restricted to authorized people in approved locations.
- Segmentation can isolate SCADA and ICS environments from other networks.

Operational Technology (OT)

- **OT (Operational Technology)** refers to systems used to monitor or control physical processes and equipment.
- Examples include:
    - Electric-grid operations.
    - Traffic-control systems.
    - Manufacturing plants.
- Failures can have significant effects, such as traffic lights not working, manufacturing shutdowns, or problems distributing electrical power.

- Network segmentation helps isolate these systems and protect their availability and communications.

Guest Networks

- A guest network allows visitors to access the internet without giving them access to internal organizational services.
- Wireless access points may provide a separate guest SSID.
- Access may be controlled with a passphrase or a login process.
- Proper segmentation allows guests to use the internet while keeping internal services isolated.

Bring Your Own Device (BYOD)

- **BYOD (Bring Your Own Device)** means employees use personal devices for work purposes.
- It may also be described as bringing your own technology.
- Personal devices must meet the organization's security requirements.
- Segmentation can separate company information from personal information on the same device.
- Administrators can manage and protect the organization's data.
- If an employee leaves the organization, company data can be removed while personal data remains on the device.

Spoofing

Definition:

Spoofing is a technique in which one person or device pretends to be another person or device.

Examples include:

- A fake web server pretending to be a legitimate web server.
- A fake DNS server pretending to be a legitimate DNS server.
- An email that appears to come from another person but actually comes from someone else.
- A phone call where the displayed caller ID is not the actual caller's number.

Spoofing can allow an attacker to gain access to a system or person that they would not normally be able to access.

Spoofing is commonly associated with **on-path attacks**, where an attacker positions themselves between two communicating devices and can monitor or potentially modify the communication.

ARP Poisoning

Address Resolution Protocol (ARP)

ARP (Address Resolution Protocol) is used when a device knows the IP address of another device but needs to determine its MAC address.

For example:

- Device: 192.168.1.9
- Router: 192.168.1.1

If 192.168.1.9 needs to communicate with 192.168.1.1, it first needs the router's MAC address.

Normal ARP Process

The process works as follows:

1. The device sends an ARP request as a **broadcast**.
2. The request asks which device has the IP address 192.168.1.1.
3. The device using 192.168.1.1 responds with its MAC address.
4. The requesting device receives the response.
5. The IP-to-MAC information is stored in the local **ARP cache**.

Lack of ARP Authentication

The ARP process does not have authentication or special security associated with it.

This lack of security can be exploited by an attacker.

ARP Poisoning Example

Consider:

- Victim: 192.168.1.9
- Legitimate router: 192.168.1.1
- Attacker: 192.168.1.14

The attacker sends an ARP response to the victim even though the victim did not originally request it.

The attacker claims:

"I am 192.168.1.1, and this is the MAC address associated with that IP."

However, the MAC address in the response is actually the **attacker's MAC address**, not the router's legitimate MAC address.

Effect on the ARP Cache

When the victim receives the forged ARP response:

1. The ARP cache is updated.
2. 192.168.1.1 is now associated with the attacker's MAC address.
3. The victim believes that the router is located at the attacker's MAC address.

From that point forward:

Victim → Attacker → Legitimate Router

Traffic intended for the router is first sent to the attacker.

The attacker can then forward the traffic to the legitimate router.

This allows the attacker to sit in the middle of the communication without the victim or router necessarily realizing that an attacker is intercepting and forwarding the traffic.

DNS Spoofing / DNS Poisoning

Definition

DNS spoofing, also called **DNS poisoning**, involves modifying DNS information or modifying DNS responses so that a victim receives an incorrect IP address.

The modification can occur by:

- Changing information on the DNS server itself.
- Modifying DNS responses as they are sent between the DNS server and the victim.
- Modifying the client's **hosts file**.

Hosts File

The hosts file on a client can contain hostname-to-IP-address information.

According to the transcript, information in the hosts file has a higher priority than DNS responses, so entries in the hosts file take precedence over information received from a DNS server.

DNS Poisoning Through Fake Responses

DNS poisoning can work by sending **fake responses to legitimate DNS requests**.

If the DNS server itself is not modified, the attacker must change the information while it is being transmitted in real time between the DNS server and the victim.

This makes an **on-path attack** useful for DNS poisoning.

DNS Poisoning Example

Consider a network containing:

- User 1
- User 2
- DNS server
- Attacker

The legitimate IP address of professormesser.com is stored on the DNS server.

User 1

User 1 requests the IP address of professormesser.com.

The DNS server responds with the legitimate IP address.

User 1 stores this information in its cache.

At this point:

DNS server → Legitimate IP address

and

User 1 cache → Legitimate IP address

Attacker Modifies the DNS Information

The attacker can:

1. Use ARP poisoning to position themselves between the DNS server and another device.
2. Or compromise the DNS server and modify its configuration.

The attacker changes the IP address associated with professormesser.com to the attacker's own IP address.

User 2

User 2 requests professormesser.com.

The DNS server now returns the poisoned IP address.

User 2 stores this address in its cache.

When User 2 attempts to access professormesser.com:

User 2 → Attacker's IP address

instead of:

User 2 → Legitimate web server

The user is therefore redirected to a web server running on the attacker's IP address.

DHCP and Rogue DHCP Servers

Dynamic Host Configuration Protocol (DHCP)

DHCP (Dynamic Host Configuration Protocol) provides devices with network configuration information such as:

- IP address
- Subnet mask
- Default gateway
- Other IP configuration details

Lack of DHCP Security

The DHCP protocol does not have security built into it.

A DHCP request sent by a client can potentially be answered by any device on the network.

That device could be:

- A legitimate DHCP server
- An illegitimate or **rogue DHCP server**

Rogue DHCP Server

An attacker could create their own DHCP server and begin handing out IP addresses.

This could result in:

- Duplicate IP addresses
- Invalid IP addresses
- Problems communicating with the Internet
- Network duplication problems
- Network-wide communication failures

A rogue DHCP server could therefore effectively shut down the network through incorrect configuration.

DHCP Snooping

Enterprise switches can provide a feature called **DHCP snooping**.

DHCP snooping:

- Examines DHCP requests.
- Allows DHCP responses only from legitimate DHCP servers.

This helps prevent an unauthorized device from operating as a DHCP server on the network.

The transcript also describes a feature in **Microsoft Active Directory** that identifies authorized DHCP servers and allows only those servers to distribute IP addresses.

Responding to a Rogue DHCP Server

If a rogue DHCP server is discovered:

1. Remove it from the network.
2. Prevent it from responding to DHCP requests.
3. Renew IP addresses across the network.
4. Ensure devices receive legitimate IP configuration.

Rogue Access Points

Definition:

A **rogue access point** is an unauthorized wireless access point connected to the network.

Access points are relatively inexpensive and can be connected to an available Ethernet connection.

How Rogue Access Points Can Appear

A rogue access point does not necessarily have to be installed by a malicious attacker.

For example, an employee might install one to:

- Improve wireless coverage.
- Expand an existing wireless network.

Even if the employee has no malicious intent, the unauthorized access point can create security problems.

If the access point does not have appropriate security controls, unauthorized users could potentially gain access to the network.

Installing a Rogue Access Point

A rogue access point can be created simply by:

1. Connecting an access point to an available Ethernet connection.
2. Configuring it as a wireless network.
3. Allowing devices to connect through it.

Some operating systems also provide **wireless sharing** capabilities.

This can allow an entire computer to function as an access point.

Detecting Rogue Access Points

Organizations can periodically scan the network to identify unauthorized access points.

Another approach is to physically walk around the facility using a **wireless analyzer** to identify wireless access points that should not be present.

Preventing Rogue Access Points

Organizations can use **802.1X** on network switches.

802.1X is also described in the transcript as **network access control**.

It requires users or devices to authenticate before receiving access to the network.

Therefore:

1. An unauthorized person connects an access point.
2. The access point still needs network access.
3. Network authentication is required.
4. Without the proper authentication, the rogue access point cannot gain access to the network.

Wireless Evil Twin

Definition:

A **wireless evil twin** is a malicious wireless access point designed to look like an existing legitimate wireless network.

It combines techniques associated with **phishing** with wireless networking technologies.

Characteristics

An evil twin may imitate:

- The same or a very similar **SSID**.
- Similar wireless security settings.
- An identical or similar captive portal configuration.

Increasing Signal Strength

An attacker may increase the output power of the wireless radio.

This can allow the evil twin to overpower nearby legitimate access points.

As a result, the evil twin may become the primary access point for users attempting to connect to the network.

Protecting Against Wireless Evil Twins

One way to reduce the impact of an evil twin is to always use **encrypted traffic**.

The transcript specifically mentions:

- Keeping a **VPN** enabled.
- Using **HTTPS** when communicating with a web server.

Even if a user connects to an evil twin, encryption helps protect the information being transmitted.

On-Path Attack

Definition:

An **on-path attack** occurs when an attacker positions themselves between two communicating devices.

It is also referred to as a **man-in-the-middle attack**.

The attacker can:

1. Receive information from one device.
2. Examine the information.
3. Potentially modify the information.
4. Forward the information to the destination.

The source and destination devices may not realize that another device is positioned between them.

Examples Mentioned

The transcript identifies several types of on-path attacks:

- Wireless evil twin
- ARP poisoning
- Session hijacking
- HTTPS spoofing
- Wi-Fi eavesdropping

Encryption as Protection

Encryption can prevent much of the damage associated with an on-path attack.

Even if an attacker is positioned in the middle of the communication, encrypted information prevents the attacker from understanding the information being transferred between the two devices.

Phishing

Definition:

Phishing is described as a form of **social engineering combined with spoofing**.

The attacker pretends to be someone else in an attempt to convince the victim to provide personal information.

Phishing commonly uses messages that appear to come from a trusted person or organization.

Identifying Phishing Emails

Several indicators can reveal that an email may be a phishing attempt.

Check the URL

Examine the URL associated with a link.

The domain should correspond to the organization that supposedly sent the message.

Check Graphics

Phishing pages may contain graphics that look slightly different from the legitimate website.

Check Spelling

Spelling mistakes can be an indication of phishing.

Check Grammar

Unusual or incorrect grammar can also indicate a phishing attempt.

Check the Sender's Email Address

The displayed sender name may appear legitimate while the actual email address belongs to another domain.

Phishing Website Example

The transcript describes a phishing page designed to look like the **Rackspace Technology webmail login page**.

The phishing page appeared very similar to the legitimate page, but some differences could be noticed:

- Graphics were slightly different.

- The sender's email address came from icloud.com instead of the expected Rackspace domain.
- The email used different fonts.
- The message contained unusual wording and punctuation.

The message began with:

"Dear user, we notice your email has not been confirmed for the new upgraded service."

The lack of expected punctuation and other inconsistencies should encourage the recipient to investigate the message more carefully.

The transcript emphasizes that users should **never click links inside suspicious email messages**.

Phishing Credential Theft

A phishing website is designed to look sufficiently similar to a legitimate login page that the victim may enter their credentials.

The attacker wants the victim to:

1. Open the phishing page.
2. Enter their email address.
3. Enter their password.
4. Click the login button.

The credentials are then sent directly to the attacker.

The transcript demonstrates this using an isolated virtual machine so that the phishing link could be examined without affecting other systems.

Shoulder Surfing

Definition:

Shoulder surfing is a social engineering technique where an attacker observes sensitive information displayed on someone's screen.

People commonly use laptops and mobile devices in public places such as:

- Airports
- Restaurants

- Coffee shops

Someone sitting or standing nearby may be able to see information displayed on the screen.

Sensitive Information

The information viewed could include sensitive data that others would find valuable.

Examples mentioned in the transcript include information related to:

- Payroll
- Social Security numbers

Long-Distance Shoulder Surfing

Shoulder surfing does not always require the attacker to stand directly behind the victim.

The transcript describes situations where people have viewed computer screens from another building using:

- Binoculars
- A telescope

Camera-Based Observation

A more advanced version involves malware placed on a computer to enable its camera.

This could allow an attacker to observe what the person is doing while sitting at the computer.

Preventing Shoulder Surfing

Several measures can reduce the risk.

Be Aware of Your Environment

Consider where you are and who may be able to see your screen.

For example:

- In a coffee shop, position your back toward a wall.

- Avoid viewing highly sensitive information in public areas.

Use a Privacy Filter

A **privacy filter** can restrict the viewing angle of an LCD screen.

When viewed directly from the front, the user can see the display.

People viewing from the side may see only a black screen.

Position the Computer Carefully

If a computer is near a window:

- Turn the monitor away from the window.
- Prevent people outside from viewing the screen.
- This can also help reduce glare from the window.

The general principle is to consider **what information is displayed and where the user is located** so that sensitive information is not exposed to third parties.

Physical Security and Building Access

A significant amount of physical security is concentrated at the **front door** of a building.

Once an attacker gets through the initial door, they may have access to many other areas where doors may already be open.

Attackers therefore look for ways to enter buildings without authorization.

Tailgating

Definition

Tailgating occurs when an unauthorized person follows an authorized person into a building without the authorized person's knowledge or assistance.

Example

1. An authorized employee uses their badge.
2. The door unlocks.
3. The employee enters.
4. The employee walks away.
5. The unauthorized person prevents the door from locking.

6. The unauthorized person enters the building.

The authorized person may not even realize that someone followed them inside.

Piggybacking

Definition:

Piggybacking is similar to tailgating, but the authorized person knowingly allows the other person to enter.

Example

An unauthorized person could:

1. Carry lunch or boxes of donuts.
2. Ask an authorized employee to hold the door.
3. The employee knowingly holds the door.
4. The unauthorized person enters the building.

Because the authorized person is aware of and assists with the entry, this is described as **piggybacking**.

Preventing Tailgating and Piggybacking

Organizations should encourage employees to look for people who may not belong in the building.

Visitor Badges

Visitors can be given clearly identifiable visitor badges.

Employees should notice people who:

- Are walking around without a badge.
- Appear unfamiliar.
- Cannot explain why they are in the building.

If someone cannot provide a good explanation, security should be contacted.

One-Person-at-a-Time Entry

Some organizations require:

1. One person badges in.

2. That person walks through the door.
3. The person physically closes the door.
4. The next person badges in separately.

This helps prevent unauthorized people from entering behind authorized users.

Access Control Vestibule / Airlock

An **access control vestibule**, or **airlock**, can mechanically enforce the one-person-at-a-time rule.

It allows only one person to enter the building at a time.

This changes the security policy from simply requiring people to enter individually into a physical mechanism that prevents multiple people from entering together.

Security Awareness

Employees should be trained to identify unusual situations.

For example, employees may ask an unfamiliar person:

- Whether they have a visitor badge.
- Which organization or group they are with.

The transcript emphasizes that organizations can train employees to recognize things that are out of the ordinary and address them before they become larger security problems.

Dumpster Diving

Definition:

Dumpster diving is searching through discarded garbage to find useful or sensitive information.

The transcript notes that garbage bins may also be called:

- Dumpsters
- Rubbish skips

The term "dumpster" is also described as a brand name for a particular type of unit.

Information Found in Garbage

Sensitive information may be discarded without proper protection.

Attackers may find:

- Names
- Contact information
- Information useful for impersonation
- Information useful for phishing or other attacks

Attackers may choose specific times to search through garbage, such as:

- End of a quarter
- End of a major project

These periods may result in larger amounts of potentially useful information being discarded.

Legal Considerations of Dumpster Diving

The transcript discusses the legality of searching through another person's garbage in the United States.

It states that garbage may effectively be available for others to search once it has been discarded by the original owner, but this is **not always the case**.

Local or state regulations may restrict access.

If the garbage is located on private property and there are signs such as:

- No trespassing
- No visitors allowed

then access to the garbage may not be permitted.

The transcript recommends contacting a legal professional to understand the applicable rules in a particular geographic area.

Preventing Dumpster Diving

Several physical security measures can help protect discarded information.

Lock the Garbage

Restrict access to garbage bins.

Use Fencing

A fence can prevent unauthorized people from physically reaching the garbage.

Use Monitoring Cameras

Cameras can monitor the area around the garbage.

Shred Sensitive Information

Sensitive information should be shredded before disposal.

Some organizations use a third-party shredding service that comes to the organization's location and shreds sensitive information.

Destroy Sensitive Information

The transcript also describes some governments destroying sensitive information by burning it.

The principle is to make sensitive information inaccessible before it is discarded.

Key Concepts

Concept	Meaning
Spoofing	Pretending to be another person or device
ARP	Address Resolution Protocol used to determine a MAC address from an IP address
ARP poisoning	Using false ARP information to associate an IP address with the attacker's MAC address
DNS spoofing	Modifying DNS information or responses to provide an incorrect destination
DNS poisoning	Another term used for DNS spoofing
DHCP	Dynamic Host Configuration Protocol that provides IP configuration information
Rogue DHCP server	An unauthorized DHCP server providing potentially incorrect network configuration
DHCP snooping	Switch feature that restricts DHCP responses to legitimate DHCP servers

Concept	Meaning
Rogue access point	Unauthorized wireless access point connected to a network
802.1X	Network access control requiring authentication before network access
Wireless evil twin	Malicious wireless access point designed to imitate a legitimate one
On-path attack	Attack where an attacker positions themselves between communicating devices
Man-in-the-middle	Another term used for an on-path attack
Phishing	Social engineering using spoofing to trick users into providing information
Shoulder surfing	Viewing sensitive information on another person's screen
Privacy filter	Screen filter that limits visibility from side viewing angles
Tailgating	Entering a secured building by following an authorized person without their assistance
Piggybacking	Entering a secured building with the knowledge and assistance of an authorized person
Access control vestibule / airlock	Physical access system that allows one person to enter at a time
Dumpster diving	Searching discarded garbage for useful or sensitive information

Important Relationships

ARP Poisoning → On-Path Attack

ARP poisoning can associate the attacker's MAC address with another device's IP address, causing traffic to pass through the attacker and enabling an **on-path attack**.

DNS Poisoning → Incorrect Destination

DNS poisoning changes the DNS information provided to a user, causing the user to connect to an incorrect or attacker-controlled IP address.

Rogue DHCP → Network Configuration Problems

A **rogue DHCP server** can provide incorrect IP configuration and potentially disrupt network communication.

Rogue AP → Unauthorized Wireless Access

A **rogue access point** can introduce an unauthorized wireless entry point into the network.

Evil Twin → On-Path Attack

A **wireless evil twin** can imitate a legitimate wireless network and position the attacker between users and the network, enabling an **on-path attack**.

Phishing → Credential Theft

A phishing message can direct a victim to a fake login page designed to capture the victim's credentials.

Tailgating / Piggybacking → Physical Unauthorized Access

Both techniques exploit authorized physical access to allow an unauthorized person into a secured building.

Dumpster Diving → Information Gathering

Discarded information can provide names, contact details, and other information that may support impersonation, phishing, or additional attacks.

Malware

Definition:

Malware is a broad term for malicious software that can:

- Cause problems on a computer.
- Steal data.
- Disrupt an entire computer.
- Capture information being typed.
- Capture images from a webcam.
- Infect multiple computers and make them work together as a **botnet**.
- Display advertisements to generate money for the attacker or malware author.

- Replicate and spread to other devices.

Different types of malware can work together to compromise a system.

Virus

A **virus** is a broad type of malware that uses **human intervention to replicate and spread**.

Key characteristic

- Requires some action from a person to move from one machine to another.
- For example, a user may need to click something that causes the malware to execute.

Important distinction

A virus depends on human intervention, while a worm can replicate without human intervention.

Worm

A **worm** is malware that can replicate itself from one computer to another **without human intervention**.

Virus vs. Worm

Characteristic	Virus	Worm
Replication	Requires human intervention	Does not require human intervention
User action	Often required	Not required
Example behavior	User clicks something	Malware automatically moves between systems
Relative danger in the transcript	Less dangerous	Considered more dangerous

A user can potentially avoid a virus by choosing not to click a particular link. A worm can replicate without the user clicking anything.

The transcript notes that worms are relatively rare and it is unusual to find one capable of moving automatically from system to system.

Ransomware

Ransomware is malware that encrypts personal data on a system.

How ransomware works

1. Malware gets onto the system.
2. It begins encrypting personal data.
3. The attacker offers a **decryption key**.
4. The victim is asked to pay a ransom.
5. Cryptocurrency or another type of currency may be requested.
6. The attacker provides the key to unlock the encrypted data.

Why the data is valuable

Personal data may be valuable to the victim, making the victim more likely to pay to regain access.

Ransomware in organizations

The stakes can be higher when ransomware affects a company because the encrypted information may include:

- Company planning documents.
- Employee details.
- Personally Identifiable Information (**PII**).
- Private information.
- Financial details.

The value of this information can be associated with a monetary value, and organizations or individuals may work with malware authors to provide the requested ransom in exchange for the data.

Trojan Horse

A **Trojan horse** is malware that gets onto a system by **pretending to be something else**.

Example

A user may see an executable that appears to be a game.

When the executable is opened:

- It appears to install the game.
- Behind the scenes, malware is also installed.
- The system becomes infected.

The key characteristic is that the malware disguises itself as something legitimate.

Rootkit

A **rootkit** is malware that can hide within the operating system itself.

Characteristics

- Difficult to detect.
- Integrated into the operating system itself.
- Difficult to eradicate.

Keylogger

A **keylogger** records every key pressed on an infected system.

This is especially important during login because the keylogger can record:

- Username.
- Password.

The recorded information can then be sent to the attacker, potentially giving the attacker access to the same systems as the victim.

Adware and Spyware

The transcript describes **adware and spyware** as being especially useful for presenting advertisements on the screen.

The displayed advertisements can generate money for the malware author.

Bloatware

Bloatware is not directly malicious, but it can cause problems on a system.

Characteristics

- Usually preloaded on a system.
- Often software that the user will never use.
- Takes up storage space.
- May have processes actively running in system memory.

Removing bloatware

Removing unnecessary bloatware can:

- Free storage space.
- Free system memory/resources.
- Make resources available for legitimate applications.

Logic Bomb

A **logic bomb** waits for a particular event to occur and then executes.

The triggering event may be:

- A particular time of day.
- A particular day of the week.
- A particular date or time.

Once the specified condition is reached, the malware begins executing on the system.

Malware Working Together

Multiple types of malware can work together during an attack.

Example scenario

1. A **worm** identifies a vulnerability on a system.
2. The vulnerability exists because the system was not updated with the latest security patches.
3. The worm gets onto the system.
4. It embeds a **remote-access backdoor**.
5. The backdoor allows access to the system from the malware author.
6. The attacker may install a **bot**.

7. The bot allows the attacker to view and control the computer remotely.

This demonstrates how several malware capabilities can be combined.

How Malware Gets Executed

For malware to perform its actions, an executable generally needs to run on the system.

Possible methods mentioned in the transcript

- A user clicks something.
- A user clicks a link embedded in an email.
- A user clicks a website pop-up.
- A **drive-by download** downloads an executable without the user clicking anything on a particular web page.
- A worm automatically executes and spreads without human interaction.

Best practice

The transcript emphasizes that you should **never click a link embedded within an email** on your system.

Keeping Systems Updated

Keeping systems updated with the latest security patches is an important defense against malware.

Why?

If vulnerabilities are patched:

- The security holes can be closed.
- Malware may be prevented from embedding itself into the system.

Updates should apply to:

- The operating system.
- Applications running on the system.

How Malware Authors Make Money

Malware authors ultimately want to make money from compromised systems.

One method is taking valuable data.

Ransomware example

1. Personal or organizational data has value.
2. Malware encrypts the information.
3. The victim wants the information back.
4. The attacker requests a ransom.
5. The victim may provide the ransom in exchange for the data.

Device Security

Open Ports and Network Services

Port numbers are extensively used to transfer information across networks.

They can also provide an **entry point into a system**.

Each network-based service running on a system can have an open port number that can be seen by others on the network.

Open ports

- If a service is actively being used, its port may need to remain open.
- If the service is no longer needed, the service should be closed.
- The associated port should no longer be accessible from outside.

Port range

The transcript identifies the available port-number range as:

0–65,535

Any service running on a system can use one of these port numbers to provide network access.

Firewalls

A **firewall** can control access to open ports.

A firewall can determine:

- Which devices can connect to a computer.
- Whether users inside the network can access a port.

- Whether access should be available to a wider audience.

Unknown open ports

Sometimes a port is open even though the administrator does not know why.

This may happen because:

- A service is running on the system.
- The administrator is unaware that the service is running.
- The service has opened a port externally.

Nmap

Nmap can be used to scan a system and determine which port numbers are advertised to the world.

The results can help determine:

- Which ports are open.
- Which services may need to remain available.
- Which services can be shut down.

If an unnecessary service is shut down, its associated port can be removed from external access.

Default Credentials

Networking devices such as:

- Switches.
- Routers.
- Firewalls.
- Other networking devices.

often have standard credentials for logging in.

Some devices require the administrator to change these credentials during the first login. Other devices may continue to use default credentials.

Security risk

If default credentials remain unchanged:

- Anyone who knows the default username and password may potentially access the system.
- Default credentials may provide administrative access.
- This could give an attacker complete control over the system.

The transcript specifically mentions **routerpasswords.com** as a database containing default usernames and passwords for devices.

Switch Port Security

Administrators cannot physically monitor every interface on every switch all the time.

Port security provides additional protection.

Purpose

Port security helps prevent someone from:

1. Unplugging an authorized system.
2. Plugging their own system into the same switch interface.
3. Communicating through that interface.

The switch identifies devices using their **MAC address (Media Access Control address)**.

MAC Address and Port Security

A MAC address describes the physical interface on a network interface card.

Port security can be configured to control which MAC addresses are allowed on a particular interface.

Port-security configuration can specify:

- How many MAC addresses can be associated with an interface.
- Whether only one MAC address is allowed.
- Whether multiple MAC addresses are allowed.
- The exact MAC address allowed on an interface.

The switch monitors devices connected to the interface and records their MAC addresses.

If an unexpected device connects, port security can activate.

Typical response mentioned

On many switches, the default behavior is to:

- Immediately disable the interface.
- Send a message to the network administrator.

Disable Unused Switch Interfaces

Unused switch interfaces should be disabled.

Example

Conference rooms and break rooms may have Ethernet interfaces on walls.

If those interfaces are not currently needed, they should be disabled.

Advantage

An attacker cannot simply connect a device to an unused but active network port.

Administrative cost

This requires:

- Tracking which interfaces are enabled.
- Tracking which interfaces are disabled.
- Enabling an interface when someone needs it.

Although this requires additional administration, it increases network security.

Network Access Control (NAC)

Some organizations require authentication before a device can communicate on the network.

This is called **Network Access Control (NAC)**.

The transcript identifies **802.1X** as a common implementation.

Process

1. A device connects to the network or wireless network.
2. The device is prompted for authentication.
3. The user provides a username and password.

4. Only after authentication is the device allowed to communicate.

MAC Address Filtering

MAC address filtering can be used to prevent unauthorized devices from accessing the network.

Because a MAC address identifies the hardware address of a device, access can be limited based on the device's MAC address.

Challenge

When a new device appears or a visitor needs access:

- Their MAC address may need to be added to the filter.

MAC Filtering and Security Through Obscurity

A MAC address can be administratively changed through the software configuration of a network adapter.

An attacker could potentially:

1. Perform a packet capture.
2. Identify MAC addresses being used on the network.
3. Wait for an authorized device to leave.
4. Change their own MAC address to match the authorized address.
5. Attempt to gain access through the MAC address filter.

For this reason, the transcript categorizes MAC filtering as **security through obscurity**.

Security through obscurity

The security depends heavily on the method not being known.

Once someone knows the security method, it may become easy to circumvent.

Key Management

Managing Authentication Keys and Certificates

IT environments often have many security-related items to manage, including:

- Authentication details.

- Certificates.
- Encryption keys.
- Other security keys.

Third-party software can be used to manage these keys from a central console.

Key Management System

A key management system can provide a centralized location where administrators can:

- Create keys.
- Associate keys with a specific service.
- Associate keys with a cloud provider.
- Associate keys with a user.
- Associate keys with a service.
- Monitor expiration dates.
- Renew keys.
- Revoke keys.
- Monitor when keys are being used.
- Determine when someone may be using a key to access another system.

Key Management Dashboard

The transcript describes a key-management dashboard containing items such as:

- Certificates.
- Keys.
- Certificate authorities.
- Vulnerabilities.
- SSH keys.
- License details.

SSL Key Management

Organizations with many web servers may need to manage many SSL keys.

A key-management system can provide a central location to:

- List keys.
- Determine when keys expire.
- Renew keys.

SSH Key Management

SSH keys can be used to log into:

- Linux servers.
- Other SSH-compatible services.

A centralized key-management system can show:

- SSH keys.
- Users accessing the keys.
- Devices being authenticated to.

Because the keys are centrally managed, reports can be created about their use.

Examples mentioned:

- SSH reports.
- Landing server reports.
- Private key reports.

Access Control Lists

ACL — Access Control List

An **Access Control List (ACL)** is a generic term for a list describing:

- Traffic that is allowed.
- Traffic that is disallowed.

ACLs can combine multiple criteria to create complex access rules.

Possible criteria

- Source IP address.

- Destination IP address.
- Port number.
- Time of day.
- Application.

These criteria can be combined to determine whether traffic is allowed or denied.

ACL Groups

ACLs can also refer to groups of criteria.

For example:

- One group of IP addresses may be allowed.
- Another group of IP addresses may be denied.

ACLs can be found on:

- Routers.
- Firewalls.
- Operating systems.
- Other devices that need to make access decisions.

Firewall security policies are described as a complex form of ACL.

Firewall Security Policies

A firewall security policy may contain:

- Rule name.
- Source zone.
- Destination zone.
- Source address.
- Destination address.
- Destination port.
- Username.
- Other criteria.

This allows very specific and fine-grained security controls.

Firewall Rule Processing

Most firewall rules are evaluated from **top to bottom**.

Process

1. Start with rule number one.
2. Compare the traffic against the rule.
3. Continue down the list until a matching rule is found.
4. Apply that rule's action.

Rule ordering

More specific rules are generally placed near the top so that they are matched before more general rules.

Implicit Deny

If traffic reaches the bottom of the firewall rule set without matching any rule, the traffic is automatically denied.

This is called **implicit deny**.

There does not need to be an explicit rule denying the traffic.

Conceptually, the bottom of the rule set can be viewed as:

Deny all traffic that did not match rules 1–7.

Example Firewall Rules

The transcript provides a firewall rule set.

Rule	Traffic	Action	Purpose
1	Any remote IP/port → TCP 22	Allow	SSH
2	Any remote IP/port → TCP 80	Allow	Web traffic
3	Any remote IP/port → TCP 443	Allow	Web traffic
4	Any remote IP/port → TCP 3389	Allow	RDP

5	Any remote IP → UDP 53	Allow	DNS
6	Any remote IP → UDP 123	Allow	NTP
7	Any remote IP → ICMP	Deny	Prevent ICMP/ping
Bottom	Any unmatched traffic	Deny	Implicit deny

Ports mentioned

- **22/TCP** — SSH
- **80/TCP** — Web traffic
- **443/TCP** — Web traffic
- **3389/TCP** — Microsoft Remote Desktop Protocol (RDP)
- **53/UDP** — DNS
- **123/UDP** — Network Time Protocol (NTP)
- **ICMP** — Denied in the example

URL Filtering

URL filtering controls which URLs users can access.

A system may allow or block:

- A specific URL.
- A category of URLs.

The transcript also mentions the term **URI (Uniform Resource Identifier)**.

URL Categories

Instead of maintaining hundreds or thousands of individual URLs, URLs can be grouped into categories.

Examples mentioned:

- Auction sites.
- Hacking sites.
- Travel sites.
- Recreation sites.

Users may attempt to find ways around URL filtering.

For this reason, URL filtering is often combined with firewall rules to help prevent circumvention.

URL Filtering in Next-Generation Firewalls

The transcript states that many **next-generation firewalls** have URL filtering built into their software.

Administrators can:

- Enable or disable categories.
- Include URL filtering within existing firewall rules.

URL filtering is a form of **content filtering** because it controls what content can pass through the network.

Content Filtering

Content Filtering

Content filtering can examine network traffic for particular types of information.

Examples mentioned:

- Internal company documents.
- Financial information.
- Unsafe-for-work content.
- Malicious software.

Possible uses

Organizations can use content filtering to prevent sensitive information from leaving the network.

It can also be used:

- To prevent unsafe-for-work content from being displayed.
- At home as parental controls.
- In antivirus and antimalware software to identify malicious software being transferred across the network.

Screened Subnet

A **screened subnet** creates a separate area of the network for services that need to be available to visitors or the public.

Purpose

It allows public access to services without providing direct access to the internal network.

Examples of public services:

- Public web servers.
- Public email servers.

Basic structure

Internet → Screened Subnet → Public Services

while the internal network remains separate.

The internal network can communicate with the Internet, but users accessing public services are directed toward the screened subnet instead of internal services.

Security Zones

Zone-Based Firewalling

Security zones allow broad references to be used within firewall rules.

Instead of specifying individual:

- IP addresses.
- Port numbers.
- Applications.

a firewall rule can refer to a **security zone**.

Examples of Security Zones

A network might contain:

- Trusted zone.
- Untrusted zone.
- Internal zone.

- External zone.

For greater granularity, it might contain:

- Inside zone.
- Internet zone.
- Server zone.
- Database zone.
- Screened subnet zone.

Advantages of Security Zones

Zones simplify firewall policies.

For example:

Trusted Zone → Untrusted Zone → Allow

The administrator does not need to specify every IP address or port.

Other examples:

- Allow untrusted traffic to the screened subnet.
- Prevent untrusted traffic from reaching the trusted zone.

Example Zone-Based Network

A network can contain:

- Internet connection.
- Firewall.
- Router.
- Another firewall.
- Honeypots.
- Internal network.

Zones can be overlaid onto this architecture.

For example:

- **Untrusted zone** — Internet.

- **Trusted zone** — Internal network.
- **Screened subnet** — Public-facing services.
- **Internet zone** — External traffic.

Additional zones provide greater control and granularity in firewall rules.

Network Troubleshooting

Network Troubleshooting Process

The troubleshooting process described in the transcript consists of:

1. **Identify the problem**
2. **Establish a theory**
3. **Test the theory**
4. **Establish a plan of action**
5. **Implement the plan**
6. **Verify full system functionality**
7. **Document findings**

Step 1 — Identify the Problem

The first step is determining what the actual problem is.

Some problems are obvious.

Example:

- A network cable is physically broken into two pieces.

Other problems are more difficult to identify.

Useful activities

- Attempt to duplicate the issue.
- Talk to users.
- Ask what they experienced.
- Review statistics and metrics from routers and switches.
- Identify multiple symptoms.

- Determine what has changed.

Questions to ask

- What changed from today compared with yesterday?
- Did someone move cables in the wiring closet?
- Was a system powered off?
- Was a configuration changed?

Duplicate the Problem

Attempting to reproduce the problem can help confirm that the actual problem has been identified.

Network professionals may build a **lab environment** where they can attempt to duplicate the problem.

If the problem can be reproduced:

- Finding the root cause becomes easier.
- Testing possible solutions becomes easier.

If the problem is too large:

- Break it into smaller pieces.
- Examine one component at a time.

Step 2 — Establish a Theory

Once information has been gathered, establish theories about what could be causing the problem.

Start with:

- Obvious problems.
- Problems that can be fixed quickly.

Example:

- If a cable might be causing the issue, swap the cable and see if the problem changes.

For more complex problems, examine the environment in greater detail.

OSI Troubleshooting Approaches

Two approaches mentioned are:

Top-down

Start with:

- Application.
- Then work down through the network.

Bottom-up

Start with:

- Lower-level network functionality.
- Then work toward the application.

A bottom-up approach may be useful for a new network implementation.

Eliminating different parts of the environment can simplify troubleshooting.

Example

If a problem occurs:

- In one operating system.
- And also in another operating system.

then the problem is obviously not directly associated with the operating system.

Step 3 — Test the Theory

After establishing a probable cause, create steps to test it.

Example

Theory:

A configuration is causing the problem.

Test:

1. Change the configuration in the lab.
2. Evaluate the result.
3. Determine whether the problem improves.
4. Determine whether the problem remains.

If the configuration change does not solve the problem:

- The issue is not fixed.
- Establish a new theory.
- Repeat the troubleshooting process.

Step 4 — Establish a Plan of Action

Once the correct fix has been identified in the lab, create a plan for implementing it in production.

Some changes can be performed during the day.

However, many production networks cannot be changed during the day.

In those environments:

- Schedule a **change-control window**.
- Prepare the implementation plan.
- Prepare for unexpected problems.

Plan B, Plan C, and Rollback

Production changes should include contingency planning.

The plan may include:

- **Plan A** — Primary implementation.
- **Plan B** — Alternative if Plan A fails.
- **Plan C** — Further alternative if Plan B fails.
- **Rollback process** — Return to the previous configuration.

The rollback process is especially important if the change causes additional problems.

Step 5 — Implement the Fix

Once:

- The solution has been identified.
- Change control has been completed.

- A maintenance window is available.

the change can be implemented in production.

In some organizations:

- One team determines what needs to change.
- Another operations team implements the change.

Step 6 — Verify Full System Functionality

A change is not considered confirmed until users verify that the original problem has been resolved.

Verification may involve:

- End users.
- The people who originally identified the problem.

This is also an opportunity to discuss how the problem could be prevented in the future.

Step 7 — Document Everything

The process is not complete until the work is documented.

Documentation should identify:

- The troubleshooting process.
- The change that was made.
- The solution that resolved the issue.

This is useful if the same problem occurs again later.

The information can be stored in:

- Help desk databases.
- Knowledge bases.

Complete Troubleshooting Flow

1. Identify the Problem

- Gather information, symptoms, user experiences, statistics, metrics, and recent changes.

2. Establish a Theory

- Develop a possible explanation for the problem.

3. Test the Theory

- Test whether the theory explains the problem.
- If unsuccessful, establish a new theory and test it again.
- If successful, proceed to the production plan.

4. Create a Production Plan

- Prepare a plan to implement the solution.
- Follow the change-control process.

5. Implement the Fix

- Apply the planned solution.

6. Verify Full System Functionality

- Confirm that the system is working correctly.
- Obtain user validation.

7. Document the Solution

- Record the problem, solution, and results for future reference.

Fiber Optics

Multimode Fiber

Multimode fiber allows light to use multiple modes or paths through the fiber.

The light can travel along different paths from one end to the other.

The transcript mentions two common core sizes:

- **50 microns**
- **62.5 microns**

The cladding brings the total fiber size to **125 microns**.

Single-Mode Fiber

Single-mode fiber uses a single path for the light to travel through the fiber.

The transcript identifies the fiber core as approximately:

- **9 microns**

The cladding brings the total size to:

- **125 microns**

Although the core sizes differ considerably, the external size can appear the same when holding the fibers.

Multimode vs. Single-Mode Fiber

Feature	Multimode	Single-mode
Light path	Multiple paths	Single path
Core size mentioned	50 or 62.5 microns	Approximately 9 microns
Total size with cladding	125 microns	125 microns

Using the wrong type of fiber can result in:

- Signal errors.
- Difficulty communicating across the link.

Fiber colors can help identify fiber types, but the transcript warns that colors cannot always be relied upon.

Documentation and writing on the outside of the fiber should be checked to determine the fiber type and size.

Copper Cabling

Cable Categories

Copper cables can look very similar externally while having different constructions.

The **Telecommunications Industry Association (TIA)** establishes standards for cable construction and testing.

Once cables are manufactured and tested, they qualify as particular cable categories.

Examples:

- Category 5.
- Category 6.
- Category 7.

Different categories support different types of signals.

Ethernet Standards and Cable Requirements

IEEE Ethernet standards specify minimum cable types for particular standards.

Examples from the transcript:

Ethernet Standard	Minimum Cable Mentioned	Distance Mentioned
1000BASE-T	Category 5	100 meters
10GBASE-T	Category 6 / Category 6A depending on distance	55 meters for Category 6 UTP

The standards specify:

- Signal speed.
- Required cable type.
- Maximum distances.

Bandwidth

Bandwidth is the theoretical maximum data rate available across a link.

It is often measured in:

- Bits per second.

Throughput

Throughput is the total amount of data actually being transmitted across a link during a particular period.

It may be measured in:

- Bits per second.

- Bytes per second.

Bandwidth vs. Throughput

Term	Meaning
Bandwidth	Theoretical maximum data rate
Throughput	Actual amount of data being transferred during a time period

---

Cable Testing

Newly installed cables should be tested.

Testing verifies that:

- The cable was installed correctly.
- The cable can support the intended signals.
- The cable matches the expected category.

A cable tester can perform tests similar to those used to qualify cable categories.

If the wrong category of cable is used:

- Errors can occur.
- Link performance can slow down.

Signal Attenuation

Physical cable errors can be caused by **signal attenuation**.

Possible results include:

- Complete loss of signal.
- CRC errors.
- Slower communication.

UTP — Unshielded Twisted Pair

UTP stands for **Unshielded Twisted Pair**.

Characteristics:

- Four pairs of wires.

- No shield around the wires.
- Each pair is twisted.

The twisting gives the cable its name:
Unshielded Twisted Pair.

STP — Shielded Twisted Pair

STP stands for **Shielded Twisted Pair**.

It is more resistant to electrical interference.

The shield may be:

- Around the entire cable.
- Around individual pairs.

STP also has a grounding wire so the shield can be grounded.

The transcript gives an example where:

- A braid surrounds the entire group of cables.
- Each individual pair is also shielded.

These shields may need to be pulled back when:

- Installing an RJ45 connector.
- Punching down the cable into a punch-down block.

Crosstalk

The wires inside copper cables are close to each other.

An electrical signal traveling through one pair can affect another pair.

This interference is called **crosstalk (XT)**.

Crosstalk describes how much signal leaks from one wire or pair into another.

NEXT — Near-End Crosstalk

NEXT stands for **Near-End Crosstalk**.

It measures signal leakage into other wires at the end closest to the testing device.

The signal is usually strongest at this point, so NEXT is an important crosstalk measurement.

Far-End Crosstalk

Far-end crosstalk measures crosstalk at the far end of the cable.

NEXT and far-end crosstalk provide information about how much signal is being shared between wires as the signal travels through the cable.

Alien Crosstalk

Crosstalk can also originate from another cable.

This is called **alien crosstalk**.

It describes interference coming from other cables.

ACR — Attenuation-to-Crosstalk Ratio

ACR stands for **Attenuation-to-Crosstalk Ratio**.

It compares:

- Attenuation / signal loss.
- Near-end crosstalk.

This provides a way to compare:

- How much signal is being lost.
- How much signal is leaking between wires.

The transcript relates this to the concept of a **signal-to-noise ratio**.

Signal-to-Noise Ratio

A signal-to-noise ratio compares useful signal against interference.

Example: 10:1

- Signal is 10 times stronger than the noise.

Example: 1:1

- Signal and noise have the same strength.

A larger ratio is desirable.

Reducing Crosstalk

High crosstalk may be related to:

- Punch-down connections.
- Cable connectors.
- Improper cable termination.

Possible actions:

- Reconnect the cable.
- Maintain as much twisting as possible.
- Avoid unnecessarily untwisting the cable.

Some untwisting is necessary at the end of a cable for:

- Punch-down blocks.
- RJ45 connectors.

However, as much twisting as possible should be maintained.

Category 6A and Crosstalk

Category 6A cables have increased cable size.

Many Category 6A cables contain a **spacer** between wires.

The spacer physically separates the wires and helps minimize crosstalk.

Electromagnetic Interference

EMI

EMI stands for **Electromagnetic Interference**.

It can affect:

- Unshielded cables.
- Shielded cables.

Shielded cables are designed to resist this type of interference.

Maintaining Cable Shields

The shield on a shielded cable should be maintained throughout the cable run.

If there is a break in the shield:

- Electromagnetic interference may enter the wires.

Cable Installation Practices

When installing cables:

Avoid excessive bending

Check the cable documentation for the minimum bend radius.

Avoid staples

Do not use staples on these cables.

Limit plastic cable ties

Plastic cable ties can crimp the cable.

Use removable cable ties

The transcript recommends a removable cable tie such as **Velcro**.

Sources of Interference

Network cables should be kept away from external sources of interference such as:

- Power cords.
- Fluorescent lights.
- Electrical systems.
- Generators.
- Fire prevention components.

These can produce electromagnetic interference that may enter network cables.

Cable testing can be used to determine whether interference is affecting the signal-to-noise ratio.

Attenuation

Attenuation describes the loss of signal strength over distance.

It can occur in:

- Copper cables.
- Fiber optic connections.
- Wireless networks.

As the signal travels farther from its source:

- Signal strength decreases.
- Communication can eventually be lost.

This contributes to the distance limitations specified in networking standards.

Cable Termination

Cable termination can strongly affect network performance.

Problems can occur because:

- Connectors can look similar but have different specifications.
- Different cable installers may produce different-quality punch-downs.
- Poor termination can affect network communication.

It is useful to have someone familiar with cable-installation standards perform the work.

Pin Matching

In a normal cable connection, pins should match between both ends.

For example:

- Pin 1 → Pin 1
- Pin 2 → Pin 2
- Pin 3 → Pin 3
- And so on.

Incorrect pin connections can cause:

- Lower connection speeds.

- Complete loss of signal.

Example

A connection expected to provide **1 Gbps** may only support **100 Mbps** if the cable is not properly terminated.

Pin Reversal

A simple termination mistake could produce:

- Pin 1 → Pin 2
- Pin 2 → Pin 1

This can result in:

- No signal.

Crossed Pairs

A cable may have **crossed pairs**.

For example:

- Pin 1 → Pin 3
- Pin 3 → Pin 1

The transcript describes a half-crossed view where:

- Four wires are crossed.
- Four wires remain straight through.

This can occur:

- At RJ45 connectors.
- At punch-down blocks.

A cable tester can identify these problems.

Auto-MDIX

Some Ethernet chipsets can detect a cable cross and electronically reverse the connection.

This capability is called **Auto-MDIX**.

It can effectively turn a crossed connection back into a straight-through connection.

However:

- It is not always enabled.
- It does not work across every Ethernet chipset.

Therefore, the best practice is to install a properly terminated cable instead of relying on Auto-MDIX.

Interface Monitoring

Monitoring Network Interfaces

Network administrators spend significant time monitoring interfaces on important devices.

Monitoring can identify:

- Bad cables.
- Bad interfaces.
- Developing problems.
- Congestion.
- Network overutilization.

Early detection may allow administrators to resolve issues before they become outages.

Interface statistics can also help determine whether network design changes are necessary to deal with oversaturation.

Operating-System Interface Statistics

Operating systems can provide information about network-interface performance.

The exact location depends on the operating system.

Network administrators can use these statistics to understand how an interface is performing.

SNMP

Administrators cannot manually log into every system to monitor every interface.

The process can be automated using:

SNMP — Simple Network Management Protocol

Many devices support standard SNMP statistics.

MIB-2

Many devices support a standard set of SNMP statistics known as **MIB-2**.

MIB stands for **Management Information Base**.

MIB-2 provides standard statistics that are common across many devices.

Proprietary MIBs

A firewall or switch may have statistics unique to that particular device.

A **proprietary MIB** can be integrated into the monitoring process to provide additional visibility.

Link Status

One useful piece of information is whether a network link is:

- **Up**
- **Down**

This is called **link status**.

A down link may indicate:

- Cable problem.
- Interface problem.
- Device rebooting.

Utilization

Utilization is an important metric when monitoring total network throughput.

Administrators need to ensure there is enough bandwidth for the services using a network connection.

If the expected throughput is unclear, bandwidth tests can be performed to determine the throughput available over the link.

Interface Errors

Interface error counters can provide warnings that problems are developing.

The transcript specifically mentions:

- CRC errors.
- Runts.
- Giants.
- Drops.
- Other interface errors.

These errors may indicate:

- Cable problems.
- Interface problems.

Additional troubleshooting may be required to determine the exact cause.

Ethernet Frame Structure

Ethernet Frame

The transcript describes the major parts of an Ethernet frame:

1. Preamble
2. Start Frame Delimiter (SFD)
3. Destination MAC address
4. Source MAC address
5. EtherType
6. Payload
7. Frame Check Sequence (FCS)

Preamble and SFD

The **Preamble** and **Start Frame Delimiter (SFD)** identify the beginning of an Ethernet frame.

They indicate that the information following the SFD is a normal Ethernet frame.

When capturing frames with a packet analyzer, the Preamble and SFD may not be visible.

Destination and Source MAC Addresses

The frame contains:

Destination MAC

Identifies where the frame should go.

Switches use the destination MAC address to determine where to forward the frame.

Source MAC

Identifies the device that sent the frame.

EtherType

The **EtherType** field identifies the type of data expected in the remainder of the Ethernet frame.

Payload

The **payload** contains the actual data carried by the Ethernet frame.

Frame Check Sequence (FCS)

The **Frame Check Sequence (FCS)** is a checksum used to determine whether the received frame has been corrupted.

The Ethernet adapter:

1. Calculates the CRC from the received frame.
2. Compares the calculated value with the checksum contained in the FCS.
3. If they match, the frame was received without the type of corruption being checked for.

4. If they do not match, an error has occurred.

A mismatch causes the CRC error counter to increase.

CRC Errors

A **CRC error** is an early warning that there may be a problem with the signal on the connection.

Possible areas to investigate include:

- Cable.
- Network interface.

Runts

A **runt** is a frame smaller than **64 bytes**.

The transcript identifies 64 bytes as the minimum Ethernet frame size.

Runts are less commonly seen on full-duplex switch networks.

They may occur on a half-duplex network when a collision happens.

Giants

The default maximum Ethernet frame size described in the transcript is:

1,518 bytes

A frame larger than this is described as a **giant**.

Jumbo Frames

Jumbo frames can be larger than 1,518 bytes.

However, their maximum size is determined by the switches and devices being used.

If a jumbo frame exceeds the configured maximum frame size, it is also treated as a giant.

Dropped Frames

Frames can be lost because of:

- Contention.
- Buffering problems.

If there is not enough buffer space to hold frames, the **drop counter** may increase.

An increasing drop counter can indicate a communication problem on the network.

Interface Error Counters

Network devices and operating systems often provide interface error counters.

A Cisco device may display statistics such as:

- Runts.
- Giants.
- CRCs.
- Other errors.

Increasing CRC errors can indicate a problem with:

- Cable.
- Interface.

Downtime may be required to replace the affected component.

Error-Disabled Interfaces

Administrative Interface Disable

If an interface is causing problems, an administrator can manually disable it.

This:

- Does not necessarily fix the underlying problem.
- Addresses the immediate symptom.
- Prevents the interface from causing additional network problems.

Error-Disabled State

It would be preferable for a device to automatically recognize certain problems and disable the interface.

This state is called:

Error-disabled

The switch automatically disables the interface without human intervention.

Interface Flapping

An interface is **flapping** when it repeatedly changes state:

Up → Down → Up → Down

A switch may error-disable a flapping interface to prevent it from causing additional problems with:

- Switching functions.
- Spanning Tree functions.

Port Security and Error-Disabled State

Error-disabled state can also occur when port security detects an unauthorized device.

Example:

1. An authorized device is connected.
2. Someone unplugs it.
3. Someone connects another laptop.
4. Port security detects the unexpected device.
5. The switch moves the interface into an error-disabled state.

Other Causes of Error-Disabled State

The transcript also identifies:

- Configuration problems.
- Increased numbers of errors.

as possible causes of an interface entering an error-disabled state.

Re-enabling an Error-Disabled Interface

When a switch automatically disables an interface because of errors, it does **not** automatically turn the interface back on.

An administrator must:

1. Log into the switch.
2. Administratively re-enable the interface.
3. Allow it to operate again.

If the original problem occurs again, the interface may return to the error-disabled state.

Administratively Down

An interface marked **administratively down** means an administrator intentionally disabled it.

This differs from error-disabled:

State	Meaning
Error-disabled	Device automatically disabled the interface because of a detected problem
Administratively down	Administrator intentionally disabled the interface

To use an administratively down interface again, an administrator must manually enable it.

Suspended Port

Some switches have a **suspended** state that is similar to error-disabled.

A suspended port indicates that an error occurs immediately when the interface is enabled because its configuration is incompatible with the connected configuration.

Example: LACP

The transcript gives link aggregation as an example:

- LACP is configured on one side.

- LACP is not configured on the other side.
- The interface may immediately enter a **suspended** state.

LACP = Link Aggregation Control Protocol.

Final Review — Key Concepts

Topic	Key Point
Malware	Broad category of malicious software
Virus	Malware requiring human intervention to replicate
Worm	Malware capable of replicating without human intervention
Ransomware	Encrypts data and demands ransom for decryption
Trojan horse	Pretends to be legitimate software
Rootkit	Hides within the operating system
Keylogger	Records keystrokes
Adware/Spyware	Can display advertisements that generate money
Bloatware	Unnecessary preloaded software consuming resources
Logic bomb	Executes when a particular event/time occurs
Open port	Entry point to a network service
Firewall	Controls network access
Nmap	Scans systems for advertised/open ports
Port security	Controls devices/MAC addresses allowed on switch interfaces
NAC	Requires authentication before network communication
MAC filtering	Filters access based on MAC addresses
Key management	Centralizes management of certificates and keys
ACL	Defines allowed and denied traffic

Implicit deny	Unmatched firewall traffic is denied
URL filtering	Controls access to URLs/categories
Content filtering	Controls specific types of content
Screened subnet	Separates public services from the internal network
Security zones	Simplify and organize firewall rules
Troubleshooting	Identify → theory → test → plan → implement → verify → document
Multimode fiber	Multiple light paths
Single-mode fiber	Single light path
Bandwidth	Theoretical maximum data rate
Throughput	Actual data transferred over time
UTP	Unshielded twisted pair
STP	Shielded twisted pair
Crosstalk	Signal interference between wires
NEXT	Near-end crosstalk
Alien crosstalk	Interference from other cables
ACR	Attenuation-to-crosstalk ratio
EMI	Electromagnetic interference
Attenuation	Signal loss over distance
Auto-MDIX	Can electronically correct certain cable crosses
SNMP	Automates network-device monitoring
MIB-2	Standard set of SNMP statistics
CRC error	Indicates frame-check mismatch
Runt	Frame smaller than 64 bytes
Giant	Frame larger than the configured maximum

Drop	Frame lost because of issues such as buffering
Error-disabled	Device automatically disables problematic interface
Administratively down	Administrator intentionally disables interface
Suspended	Interface configuration is incompatible

Power over Ethernet (PoE)

What is PoE?

Power over Ethernet (PoE) allows a network cable to carry both:

- Network data
- Electrical power

This simplifies installation because a separate power cable is not required for devices such as:

- Telephones
- Wireless access points
- Cameras

Sources of PoE Power

Power can come from two different sources:

Method	Description
Endspan	Power is built directly into the network switch. A PoE switch provides both data and power.
Midspan	A separate device called a **PoE injector** sits between the switch and the PoE device and injects power onto the Ethernet connection.

---

2. PoE Standards

The transcript describes three major PoE power levels.

Standard	DC Power	Maximum Current	Typical Use

PoE	15.4 W	350 mA	Small devices such as simple telephones or small access points
PoE+	25.5 W	600 mA	Larger telephones, cameras, and other devices
PoE++ Type 3	Up to 51 W	600 mA	Higher-power devices
PoE++ Type 4	Up to 71.3 W	960 mA	Much larger devices

PoE++ can provide enough power for devices such as:

- Some laptops
- Large cameras
- Pan-tilt-zoom cameras

The newer PoE standard also provides support for additional Ethernet speeds, including:

- 2.5 Gbit
- 5 Gbit
- 10 Gbit

PoE Compatibility

When installing a PoE switch, the **switch and powered device must be compatible**.

For example:

- A PoE+ switch cannot power a PoE++ device.
- Therefore, always verify the PoE capabilities of both the switch and the device.

A switch may have:

- PoE on every interface
- Some interfaces without PoE
- Some interfaces supporting PoE+
- Other interfaces supporting PoE++

The switch itself usually identifies which interfaces support PoE.

PoE Power Budget

The switch also has a maximum total PoE power capacity.

Examples from the transcript:

- 200 W
- 720 W

To determine whether a switch can support all connected devices:

1. Determine the maximum power requirement of every PoE device.
2. Add those power requirements together.
3. Compare the total against the switch's maximum PoE capacity.
4. Make sure the total required power is below the switch's available capacity.

Ethernet Transceivers

Transceivers provide a **modular connection** for Ethernet devices.

However, the correct transceiver must be used for the specific connection.

Fiber Transceiver Compatibility

When using a fiber transceiver, its specifications must match the fiber being connected.

Transceivers normally have a wavelength marking, such as:

- **850 nm**
- **1310 nm**

The wavelength specifications need to match throughout the link.

Problems Caused by Incorrect Transceivers

Using an incorrect transceiver can result in:

- Signal loss
- Increasing error counters
- Loss of signal
- Reduced network efficiency
- Slowdowns

Two transceivers may look almost identical while having different wavelength specifications.

The markings can be difficult to read, especially after installing the transceiver in a switch.

You may need to:

- Check the switch specifications
- Check the transceiver specifications
- Physically remove the transceiver to read its label

Fiber Power Budget

When connecting fiber devices, enough signal must reach the receiving device.

This becomes especially important when:

- The fiber run is very long.
- There are many connections.
- There are connectors or splices along the path.

Sensitivity Level

Each device has a **sensitivity level** that specifies how much signal it can receive while still properly interpreting the signal.

This value is normally included in the device specifications.

Calculating the Power Budget

The calculation involves several factors:

1. Determine the transmitted power.
2. Determine the length of the fiber connection.
3. Determine how much signal is lost over that distance.
4. Account for connectors.
5. Account for splices.
6. Calculate the total signal loss.
7. Subtract the total loss from the transmitted power.
8. Determine the expected received power.
9. Compare the received power with the receiver's sensitivity value.

The final calculated value represents the expected power received at the other end.

Example

The transcript gives an SFP transceiver operating at **850 nm** with a receiver sensitivity of:

-17 dBm

If the calculated received power is:

-17 dBm or higher

the signal is considered good according to the example.

If the calculated received power is:

-20 dBm

there is not enough power for the transceiver to properly receive the signal.

Switching Loops and Spanning Tree Protocol

At the MAC address level, there is no mechanism that counts how many times a frame has been seen.

Therefore, a switching loop can cause frames to continue circulating around the network.

This can continue until one of the connections is removed.

Why Loops Are Dangerous

Consider two switches connected by two different paths.

If traffic enters the loop:

1. The frame travels through the network.
2. It encounters the second connection.
3. It loops back.
4. The traffic continues circulating.
5. Additional traffic creates even more looping.
6. The switches eventually become overwhelmed.
7. The network can come to a complete halt.

This is why **Spanning Tree Protocol (STP)** is important.

IP TTL vs. Switching Loops

IP packets have a **Time to Live (TTL)** field.

TTL helps prevent an IP packet from circulating indefinitely.

However, there is nothing equivalent inside a switch frame that automatically prevents this type of Layer 2 loop.

Therefore, switching networks need a mechanism such as STP to prevent loops.

Broadcasts and Multicasts

Switches normally forward traffic based on the destination MAC address.

Unicast

A unicast frame has a single destination.

Broadcast and Multicast

Broadcasts and multicasts can be sent to multiple devices.

A single frame can enter a switch and be transmitted out multiple interfaces.

Because of this behavior, loops involving broadcast or multicast traffic can quickly create large amounts of traffic.

Bridge Protocol Data Units (BPDUs)

One mechanism used by STP is the:

Bridge Protocol Data Unit (BPDU)

Switches send BPDUs using MAC-layer multicast.

Other switches on the local broadcast domain can receive these frames.

BPDUs contain configuration information used by STP.

BPDU Timing

According to the transcript:

- BPDUs are normally sent every **2 seconds**.
- Switches receive updates from other switches every 2 seconds.
- If updates stop arriving, the switch waits for additional intervals.

- After **three missed updates**, the link is considered down.

STP then recognizes a topology change and redesigns the network to prevent loops.

STP Root Bridge

STP uses the concept of a **root bridge**.

The root bridge is elected when the network starts.

All bridges participate in the root bridge election.

Bridge ID

The Bridge ID determines which switch becomes the root bridge.

The transcript states that Bridge IDs can range from:

0 to 61,240

The switch with the **lowest Bridge ID** becomes the root bridge.

If multiple switches have the same Bridge ID, the switch with the **lowest MAC address** becomes the root bridge.

Root Port

After the root bridge is elected:

- Other switches determine how to reach the root bridge.
- Each non-root switch has a **root port**.
- The root port provides the path toward the root bridge.

Designated Ports

Switches may also have **designated ports** that are active and passing traffic.

Blocked Ports

STP can block certain ports to prevent loops.

A blocked port does not forward traffic through that path.

STP Topology Changes

STP can dynamically change the network configuration when a connection fails.

Example:

1. Network A normally communicates through Bridge 6.
2. Another path exists but is blocked by STP.
3. Connectivity between Network A and Bridge 6 fails.
4. STP notices that the expected hello updates are missing.
5. After the required missed updates, STP recognizes the topology change.
6. Previously blocked ports can be reconfigured.
7. Traffic can use the alternate path.
8. Communication is restored without creating a loop.

This allows STP to change the network configuration in real time.

STP Port States

The transcript describes several STP interface states.

Blocking / Discarding

The interface does not pass traffic.

This prevents a loop from forming.

Listening

The interface listens for other switches on the local broadcast domain.

It uses information from those switches to determine its configuration.

Learning

The switch begins learning information and adding information to its MAC address table.

Forwarding

The interface is enabled and can forward traffic.

Disabled

An administrator can administratively turn off an interface.

If an interface should never be used for forwarding, it can be placed into a disabled state.

VLAN Configuration Troubleshooting

A device may successfully receive an IP address but still be unable to communicate with other devices on the same network.

One possible reason is an incorrect **VLAN configuration**.

Every switch interface is associated with a VLAN.

Access Ports

An **access port** is associated with a particular VLAN.

Each access port is assigned a VLAN ID.

Example VLANs from the transcript:

- VLAN 254
- VLAN 100

Troubleshooting an Incorrect VLAN

Suppose:

- Switch interface 4 is expected to use VLAN 254.
- Interface 4 is actually configured for another VLAN.

Possible solutions:

1. Change the VLAN configuration of interface 4.
2. Move the device to another interface that is already configured for VLAN 254.

This can be a common issue on networks containing many VLANs.

Access Control Lists (ACLs)

Even if:

- VLAN configuration is correct
- IP addressing is correct
- Routing tables appear correct

traffic may still be blocked by an **Access Control List (ACL)**.

ACLs can be configured on switches and routers.

ACL Troubleshooting

If packets are being dropped:

1. Check the switches and routers along the path.
2. Check whether ACLs are configured.
3. Check the interfaces where those ACLs are applied.
4. Determine whether the ACL is blocking the traffic.

This should be part of normal troubleshooting, especially when traffic cannot communicate between networks.

ACL Processing

ACLs work similarly to firewall rule bases.

The ACL stops evaluating rules after it finds a match.

Therefore:

- More granular rules should generally be placed toward the top.
- More common matches can be placed higher for efficiency.
- Broader controls can be placed lower.

ACL Safety

When changing an ACL, consider disabling the ACL functionality first.

A poorly designed ACL could remove your own access to the switch.

Also remember that many devices use an implicit/default deny behavior when traffic is not specifically permitted by the ACL.

Therefore:

An empty ACL could effectively block all communication through the device.

Routing Table Troubleshooting

Routers use **routing tables** to determine the best next hop when forwarding traffic.

The routing table acts as a map showing where traffic should be forwarded.

It may contain:

- Directly connected routes
- Static routes
- Other learned routes

- A default route

Troubleshooting Routing

When troubleshooting a routing problem:

1. Check the routing table of the first router.
2. Check the routing table of every router along the path.
3. Verify that a route exists toward the destination.
4. Verify that a return route exists toward the original workstation.

Both directions are important.

If the router does not have a route for a destination, it may simply drop the traffic.

It may also send an:

ICMP Host Unreachable

message to the device that originally sent the traffic.

Example of Routing Table Troubleshooting

The transcript describes:

- A laptop on one side of the network.
- Another laptop on the other side.
- Three routers between them.
- Switches connecting the routers.

If the source laptop attempts to ping:

10.3.1.2

the router must have a route toward the appropriate destination subnet.

The administrator should:

1. Open Router 1's routing table.
2. Look for a route toward the destination network.
3. Verify that the route exists.
4. Continue checking the routers along the path.
5. Verify return routing as well.

If the destination route is missing, the ping may return a message indicating that the destination host is unreachable.

Gateway of Last Resort

Large networks can have very large routing tables.

A router can use a **default route**, also called the:

Gateway of Last Resort

The Gateway of Last Resort is used when no other route in the routing table matches the destination.

According to the transcript, it is commonly represented as:

0.0.0.0/0

This encompasses every host and network.

Example

A router may have routes for:

- 10.10.10.0/24
- 10.10.40.0/24
- 10.10.50.0/24
- A static route to 10.10.20.0/24
- A route learned through RIP to 10.10.30.0/24

If none of these routes match the destination, the router can use the Gateway of Last Resort.

The transcript gives an example where the default route sends traffic through:

10.10.50.2

to the next router.

APIPA and DHCP Address Exhaustion

If a computer starts without receiving a DHCP address and instead receives an **APIPA** address, one possible cause is that the DHCP server has run out of available addresses.

DHCP Address Pool Exhaustion

If the DHCP address pool is exhausted:

- The system may receive an APIPA address.
- The APIPA address allows communication with devices on the local subnet.
- It is non-routable.
- It cannot communicate outside the local subnet.

Troubleshooting

Check the DHCP server for:

- Available addresses
- Address pool size
- Address availability

If necessary, additional addresses can be added to the DHCP server.

IP Address Management (IPAM)

Organizations may use an **IP Address Management (IPAM)** system to monitor DHCP servers.

IPAM can provide visibility into:

- Available IP addresses
- Address pools
- Address availability
- Multiple DHCP servers

DHCP Lease Time

In networks where users are present for short periods of time, administrators may decrease the DHCP lease time.

This allows addresses to become available again more quickly.

It can reduce the chance of address pool exhaustion.

Incorrect DHCP Configuration

DHCP normally provides network configuration information such as:

- IP address

- Subnet mask
- Default gateway

Sometimes a device can receive incorrect information.

For example:

- Incorrect IP address
- Incorrect subnet mask
- Incorrect default gateway

Troubleshooting

You can:

1. Confirm the received information with the network administrator.
2. Perform a packet capture.
3. Examine other devices on the same subnet.
4. Check another router or workstation already connected to the network.
5. Compare its IP configuration with the configuration received by the affected device.

Ping-Based Troubleshooting

A basic troubleshooting process is:

Ping local address → Ping default gateway → Ping an IP address beyond the default gateway

Using:

- ping
- traceroute

can help build a picture of where the device is located in the network and determine whether the assigned IP configuration is correct.

Duplicate IP Addresses

Even properly configured DHCP environments can occasionally experience duplicate IP addresses.

Possible causes mentioned in the transcript include:

- A manually configured device using an address inside an existing DHCP pool.
- Multiple DHCP servers with overlapping address pools.
- A device with DHCP enabled that begins handing out unexpected addresses.

Modern operating systems can detect duplicate IP addresses when connecting to the network and prevent the duplication.

Troubleshooting a Duplicate IP

If the address is manually configured:

1. Check the configuration.
2. Verify the IP address.
3. Verify the subnet mask.
4. Verify the default gateway.

Before statically configuring an address, another workstation can be used to ping that address.

If a response is received before the address is assigned, another device is already using it.

Finding the Device

If an unexpected device responds to the ping:

1. Ping the IP address.
2. Check the ARP table.
3. Obtain the device's MAC address.
4. Check the switch's MAC address table.
5. Determine which switch interface the device is connected to.

If duplicate addresses are being distributed by DHCP servers, a packet capture can help determine exactly what the DHCP servers are offering.

Network Congestion and Bottlenecks

Networks operate at predefined speeds.

For example:

1 Gbit/s Ethernet → 1 Gbit/s maximum transmission rate

If two 1-Gbit/s connections both attempt to send 1 Gbit/s toward the same 1-Gbit/s destination link:

2 Gbit/s of traffic → 1 Gbit/s link

The destination link cannot carry all of the traffic simultaneously.

This creates **congestion**.

What Happens During Congestion?

Switches and routers have buffers.

These buffers temporarily hold packets.

However, buffers are relatively small.

When the buffers become full:

- Packets must be discarded.
- Information is lost.
- Applications may experience delays.

Resolving Congestion

Two general approaches are:

1. Increase the speed or capacity of the network.
2. Reduce the amount of traffic crossing the network.

Network Bottlenecks

When someone says:

"The network is slow."

the actual problem may be a **bottleneck** somewhere in the communication path.

The bottleneck could involve:

- System bus
- CPU speed inside a switch
- CPU speed inside a router
- Hard drive
- SSD

- Different network links
- Other resources between devices

Therefore, troubleshooting may require examining the different technologies and resources involved in the communication path.

Example: Database Bottleneck

The transcript gives a web transaction example:

- Response times were approximately **1500–1750 milliseconds**.
- This represented almost 2 seconds of delay.
- A significant amount of time was being spent by the database.
- Configuration changes were made to the database server.
- Response time dropped to approximately **500 milliseconds**.

This demonstrates that what appears to be a network slowdown may actually be caused by another component.

Bandwidth

Bandwidth percentage measures how much of a network's available capacity is being used over a particular period.

It is usually represented as a percentage.

This helps determine how heavily a network is being utilized.

Bandwidth statistics can be collected from:

- Switches
- SNMP
- NetFlow

Throughput

Throughput measures how much data can actually be moved through the network during a particular period.

It provides information about:

- How much data is being transferred
- How long the transfer takes

When multiple links exist between two devices, the **slowest link** can limit the throughput of the overall path.

Latency

Latency is the delay between a request and its response.

There will always be some amount of latency because information takes time to travel between devices.

Measuring Latency

Ideally, latency should be measured at every point along the path.

This can reveal:

- How quickly data moves from one segment to another.
- How long a packet stays in a device.
- How long it takes to traverse a network.
- How long it takes to reach the next segment.

Packet capture provides very detailed measurements because individual packets can be examined.

Packet Loss and Discards

Network administrators want to avoid packet loss.

Ideally, every packet should reach its destination.

A **packet loss/discard** means that the packet was not delivered because of some reason other than a packet error.

Possible reasons mentioned include:

- Network outage
- Contention
- Insufficient bandwidth

Corrupted Packets

A packet may also become corrupted during transmission.

Possible causes mentioned include:

- Poor wireless connection

- Bad cable

When corruption is detected at the receiving side:

1. The corrupted information is discarded.
2. The traffic must be resent.
3. Additional time and resources are required.
4. Applications may experience delays.

Voice and Video Sensitivity

Voice-over-IP calls and live video streams are especially sensitive to:

- Delay
- Packet loss
- Irregular packet arrival

Ideally, packets should arrive at regular and predictable intervals.

If a packet is lost during a live conversation or stream:

- The conversation cannot be rewound.
- The live stream cannot be rewound.
- The missing packet is simply discarded.
- The communication continues.

This can cause:

- Clicking or interruptions during phone calls
- Small stutters in live video

Jitter

Jitter is the variation in the time between arriving frames.

A network ideally has a consistent amount of delay between packets.

Low Jitter

With low jitter:

- Packets arrive at relatively regular intervals.

- There is only a small amount of variation.
- Voice and video operate more smoothly.

High Jitter

With high jitter:

1. Several packets may arrive quickly.
2. A long delay occurs.
3. Several more packets arrive quickly.
4. Another delay occurs.

High jitter can cause:

- Problems hearing a phone call
- Stuttering during live video

Wireless Frequency and Interference

Wireless networks have a limited number of frequencies available.

Only one device can communicate at a time on the same wireless network.

Additionally, there may be many wireless networks in the same physical area.

Therefore, access points must be configured carefully to avoid interference.

Modern wireless standards provide mechanisms to automatically identify suitable frequencies and avoid frequencies already being used.

Access points can also be configured manually.

Many access points can automatically:

- Identify interference.
- Select a better frequency.
- Change their configuration.

2.4 GHz, 5 GHz, and 6 GHz

2.4 GHz

The transcript states that in North America there are **three non-overlapping channels** available.

In a busy area with many access points, these channels may not be sufficient to avoid interference.

5 GHz

The 5-GHz band provides many more available frequencies.

This gives a better opportunity to find a frequency with less interference.

6 GHz

The 6-GHz band provides even more available frequencies, helping reduce frequency conflicts and interference.

Wireless Performance Best Practices

Disable Legacy/Low-Speed Support

Access points may have options allowing older wireless technologies to operate.

According to the transcript, enabling legacy support can:

- Reduce wireless efficiency.
- Slow down the overall network.

If all devices support the newest standard, legacy support can be disabled.

Review Manually Configured Channels

If a channel was manually configured:

- Periodically verify that it is not conflicting with nearby networks.
- Another access point may appear later and use the same or overlapping frequency.

Automatic channel selection can allow an access point to move to a more appropriate channel.

Wireless Output Power

Interference can come from other access points.

An access point may cover only a small area while transmitting at maximum power.

If appropriate, reducing output power can:

- Continue providing wireless access to nearby devices.

- Reduce interference with other access points.

Wireless coverage can also be divided among multiple access points using different frequencies.

This can distribute the load while limiting interference.

Wireless Channel Overlap

The transcript provides an example using the 2.4-GHz band:

- One network uses Channel 6.
- Another network uses Channel 11.
- Channel 1 is also available without conflicting with those networks.

If another access point is configured on **Channel 8**, its frequency overlaps with:

- Channel 6
- Channel 11

This creates interference and reduces throughput.

A better configuration would be to use Channel 1.

Wireless Attenuation

As you move farther away from an access point, the wireless signal becomes weaker.

This is called:

Attenuation

Administrators can use a Wi-Fi analyzer to measure signal strength throughout a facility.

They can identify:

- Strong signal areas
- Weak signal areas

and then make changes to improve coverage.

Improving Wireless Coverage

Possible methods mentioned include:

Increase Access Point Power

If the access point supports adjustable output power, the administrator can increase it to cover the required area.

Use an External Antenna

An external antenna can provide additional gain.

This can increase the ability of the access point to transmit over a larger distance without necessarily increasing the amount of power transmitted by the access point.

Minimize Coaxial Cable

When an antenna is connected using coaxial cable:

- Some signal is lost in the cable.
- Loss can be especially important at higher frequencies.

Therefore, the cable between the access point and antenna should be kept as short as practical.

The cable should also be checked for damage because damaged cable can cause signal leakage.

Wireless Site Surveys

Determining the correct wireless coverage can be challenging.

Administrators should identify:

- Where users access the wireless network.
- How much signal is available at those locations.
- Where nearby access points are located.
- Which frequencies nearby access points are using.

This helps prevent interference with nearby organizations and networks.

Repeated Site Surveys

Wireless environments can change over time.

Therefore, it can be useful to perform multiple site surveys throughout the year.

This helps identify changes in wireless infrastructure.

Wireless Heat Maps

A **wireless heat map** provides a visual representation of wireless signal strength.

The administrator walks around with a Wi-Fi analyzer.

The analyzer measures signal strength at different locations.

The resulting heat map shows:

- Stronger signal → brighter colors
- Weaker signal → cooler colors

This makes it easier to identify locations where access points may need to be added or repositioned.

Wireless Client Disassociation Attacks

Older wireless networks may be vulnerable to a denial-of-service attack involving **client disassociation**.

The attack takes advantage of older 802.11 management frames that do not have protection associated with them.

An attacker can use these frames to remove devices from the network.

Symptoms

A device may:

- Constantly disconnect and reconnect.
- Never successfully connect to the wireless network.

Detection

An administrator can use an **802.11 packet capture device** to capture wireless frames.

The administrator can look for:

Client disassociation frames

This can help identify the attacking device.

Prevention

The transcript recommends upgrading to one of the latest 802.11 standards that prevents this type of attack.

SSID and Multiple Access Points

Multiple access points can be configured with the same:

SSID — Service Set Identifier

Using the same SSID allows administrators to create one large wireless network using multiple access points.

As a user moves through:

- A building
- A campus

the device can move from one access point to another.

Configuration Consistency

For seamless operation, the access points need to have the same configuration.

This includes:

- Security features
- Other configuration details

If the new access point has a different configuration, the user may be dropped from the network and required to reconnect.

Protocol Analyzers

When someone reports:

"The network is slow."

the network may actually be operating normally, with a problem located somewhere within the communication path.

A **protocol analyzer** can help identify where the problem exists.

What a Protocol Analyzer Does

It captures frames from:

- Wired networks
- Wireless networks

and presents the captured information in a human-readable format.

Capture functionality may be built into:

- Switches
- Routers
- Firewalls

The captured file can then be viewed using a protocol analyzer such as:

Wireshark

Wireshark can provide:

- Protocol decodes
- Additional protocol details
- Information about network conversations

Packet Capture for Later Analysis

Some network administrators capture all packets and store them for later analysis.

This is described as a **big data analysis** capability.

Large amounts of information can be stored and later searched to find specific information.

Although packet capture is widely used for IT security, network administrators can also use it for network troubleshooting and analysis.

Nmap — Network Mapper

The transcript identifies **Nmap** as:

Network Mapper

Nmap can gather information about:

- Open ports
- Operating systems
- Operating system versions
- Application versions
- Services running on ports

- Active devices in an IP address range

Open Port Scanning

Nmap can scan a device and identify open ports.

Example ports mentioned:

Port	Service
22	SSH
80	HTTP
443	HTTPS
548	Apple Filing Protocol
2049	NFS

If Nmap identifies an unexpected open port, additional research may be needed to determine what service is running there.

Operating System Identification

Nmap can identify:

- The operating system on a device.
- The operating system version.

This can be done without logging into the device.

Service Identification

Nmap can identify services running on open ports.

Therefore, you can determine not only that a port is open, but also what service is associated with it.

Nmap Scripting Engine

Nmap includes a scripting engine.

This allows administrators to:

- Write scripts.
- Extend Nmap's capabilities.

Active Scanning

Nmap is an **active scanning tool**.

It:

1. Sends queries to a device.
2. Examines the responses.
3. Uses those results to determine information about the device.

This can include:

- Open ports
- Operating system information
- Services

Network Range Scanning

Nmap can scan an entire IP address range.

This allows an administrator to identify:

- Active devices
- IP addresses
- Operating systems
- Services
- Additional device information

It can also help identify devices that were not expected to exist on the network.

The transcript also notes that Nmap can perform scans on the same subnet using Layer 2, making it difficult for a rogue device to hide from the scan.

Cisco Discovery Protocol (CDP)

CDP — Cisco Discovery Protocol

is a proprietary protocol from Cisco.

It can provide information about Cisco devices without requiring the administrator to log directly into the device.

CDP can provide information about:

- IP addresses
- Interfaces

- Software
- Native VLAN
- Hardware-level interface configuration

Example

A CDP protocol decode can show:

- An address associated with an interface.
- The port associated with that address.
- Software information.
- Native VLAN information.
- Interface configuration details.

Link Layer Discovery Protocol (LLDP)

LLDP — Link Layer Discovery Protocol

is a more vendor-neutral discovery protocol.

The transcript states that almost all switches support LLDP.

LLDP can provide information about how a device is configured.

Information can include:

- MAC address
- Interface
- Device name
- VLAN ID

CDP and LLDP Together

Using CDP and LLDP can provide significant information about switches without logging into them.

Third-party applications and devices can use these protocols to collect and present the information in a consolidated form.

Internet Bandwidth Testing

When connecting to the Internet, it can be useful to determine how much bandwidth is available.

One way to test bandwidth is to:

1. Send a large amount of information.
2. Measure how long it takes to download.
3. Measure how long it takes to upload.

This can be done using Internet speed-test sites.

Before and After Network Changes

Speed tests can be performed:

Before configuration change → Make change → After configuration change

This allows administrators to compare the results and determine whether the network changed in performance.

Testing at Different Times

Bandwidth availability can vary throughout the day.

For example:

- Off-hours may have more available bandwidth.
- The middle of a workday may have less available bandwidth.

Therefore, testing at different times can provide a better understanding of network performance.

Speed-Test Server Location

Different speed-test sites can produce different results because:

- Some servers are farther away.
- Some servers may have bandwidth limitations.
- Different network paths may be used.

The transcript states that local ISP-provided speed-test sites may provide some of the most accurate representations of the available local network speed.

Examples given:

- Xfinity speed test for Xfinity users
- AT&T speed test for AT&T users

Other third-party speed-test sites mentioned are:

- Fast.com
- SpeedOf.Me
- Speedtest.net
- TestMy.net

Quick Troubleshooting Reference

Problem	Possible Cause / Check
PoE device has no power	Check PoE compatibility and switch power budget
Fiber connection has signal loss	Check transceiver wavelength and fiber specifications
Network loop	Check STP and blocked/forwarding ports
Device gets IP but cannot communicate	Check VLAN configuration
Traffic between networks is blocked	Check ACLs
Router cannot reach destination	Check routing tables
No route matches destination	Check Gateway of Last Resort
Computer receives APIPA	Check DHCP address pool
Duplicate IP address	Check static configuration, DHCP pools, ARP, and MAC tables
Network is slow	Investigate congestion and bottlenecks
Packets are discarded	Check congestion, outages, wireless issues, and corrupted transmission
Voice/video stutters	Check jitter, packet loss, and latency
Wireless interference	Check channels, nearby APs, output power, and site survey

Weak Wi-Fi signal	Check attenuation, AP power, antenna, and cable
Wireless devices repeatedly disconnect	Check for client disassociation attacks
Unknown network device	Use Nmap, CDP, LLDP, or packet capture
Unexpected open port	Investigate the service associated with the port
INTERNET SEEMS SLOW	PERFORM BANDWIDTH/SPEED TESTING AT DIFFERENT TIMES

KEY CONCEPTS TO REMEMBER

- **POE** CARRIES POWER AND DATA OVER THE SAME ETHERNET CABLE.
- **ENDSPAN** PROVIDES POE DIRECTLY FROM THE SWITCH.
- **MIDSPAN** USES A SEPARATE POE INJECTOR.
- POE POWER CAPACITY MUST BE COMPATIBLE WITH THE CONNECTED DEVICES.
- FIBER TRANSCEIVERS MUST MATCH THE FIBER AND WAVELENGTH REQUIREMENTS.
- **POWER BUDGET** DETERMINES WHETHER ENOUGH OPTICAL SIGNAL REACHES THE RECEIVER.
- **STP** PREVENTS SWITCHING LOOPS.
- **BPDUS** ALLOW SWITCHES TO EXCHANGE STP INFORMATION.
- STP ELECTS A **ROOT BRIDGE** AND CAN BLOCK REDUNDANT PATHS.
- AN INCORRECT **VLAN ASSIGNMENT** CAN PREVENT COMMUNICATION EVEN WHEN A DEVICE RECEIVES AN IP ADDRESS.
- **ACLS** CAN BLOCK TRAFFIC EVEN WHEN SWITCHING, ROUTING, AND ADDRESSING ARE CORRECTLY CONFIGURED.
- **ROUTING TABLES** DETERMINE WHERE ROUTERS FORWARD TRAFFIC.
- A **GATEWAY OF LAST RESORT** PROVIDES A DEFAULT PATH WHEN NO MORE SPECIFIC ROUTE EXISTS.
- **APIPA** CAN APPEAR WHEN DHCP ADDRESSING IS UNAVAILABLE, INCLUDING WHEN THE DHCP POOL IS EXHAUSTED.
- DUPLICATE IP ADDRESSES CAN RESULT FROM STATIC CONFIGURATION OR OVERLAPPING DHCP SERVICES.

- **CONGESTION** OCCURS WHEN MORE TRAFFIC ATTEMPTS TO USE A LINK THAN THE LINK CAN CARRY.
- **BANDWIDTH** DESCRIBES NETWORK UTILIZATION/CAPACITY USAGE, WHILE **THROUGHPUT** DESCRIBES THE AMOUNT OF DATA ACTUALLY MOVED OVER TIME.
- **LATENCY** IS DELAY BETWEEN REQUEST AND RESPONSE.
- **PACKET LOSS** MEANS TRAFFIC DOES NOT SUCCESSFULLY REACH ITS DESTINATION.
- **JITTER** IS VARIATION IN PACKET/FRAME ARRIVAL TIMING.
- WIRELESS PERFORMANCE DEPENDS HEAVILY ON FREQUENCY SELECTION AND INTERFERENCE.
- **ATTENUATION** CAUSES WIRELESS SIGNAL STRENGTH TO DECREASE WITH DISTANCE.
- **HEAT MAPS** VISUALLY SHOW WIRELESS SIGNAL STRENGTH.
- OLDER WIRELESS NETWORKS CAN BE AFFECTED BY CLIENT DISASSOCIATION ATTACKS.
- MULTIPLE APS CAN USE THE SAME **SSID** TO PROVIDE A LARGER WIRELESS NETWORK.
- **WIRESHARK** CAN ANALYZE CAPTURED NETWORK TRAFFIC.
- **NMAP** ACTIVELY SCANS DEVICES, PORTS, OPERATING SYSTEMS, AND SERVICES.
- **CDP** PROVIDES CISCO DEVICE DISCOVERY INFORMATION.
- **LLDP** PROVIDES VENDOR-NEUTRAL DEVICE DISCOVERY INFORMATION.
- INTERNET SPEED TESTS CAN BE USED BEFORE AND AFTER NETWORK CHANGES TO COMPARE PERFORMANCE.

PING COMMAND

WHAT IS PING?

- **PING** IS ONE OF THE MOST COMMONLY USED COMMAND-LINE UTILITIES FOR NETWORK TROUBLESHOOTING.
- IT TELLS YOU WHETHER A DEVICE ON THE NETWORK IS **REACHABLE**.

- IF THE DEVICE IS NOT REACHABLE, THE RESPONSE CAN PROVIDE INFORMATION THAT MAY HELP DETERMINE WHY.
- PING USES **ICMP (INTERNET CONTROL MESSAGE PROTOCOL)** TO QUERY A DEVICE AND RECEIVE A RESPONSE.
- PING IS USUALLY ONE OF THE **FIRST TROUBLESHOOTING STEPS** WHEN CHECKING WHETHER A DEVICE IS REACHABLE.

BASIC PING PROCESS

THE BASIC TROUBLESHOOTING QUESTION IS:

CAN YOU PING THAT HOST?

EXAMPLE:

PING 1.1.1.1

- 1.1.1.1 IS IDENTIFIED IN THE TRANSCRIPT AS A DNS SERVER AT **CLOUDFLARE**.
- PING WORKS SIMILARLY IN:
    - MACOS
    - LINUX
    - WINDOWS

SUCCESSFUL PING RESPONSE

A SUCCESSFUL RESPONSE INCLUDES INFORMATION SUCH AS:

- NUMBER OF BYTES SENT AND RECEIVED
- ICMP SEQUENCE NUMBER
- TIME TO LIVE (TTL)
- ROUND-TRIP TIME

EXAMPLE INFORMATION DESCRIBED IN THE TRANSCRIPT:

64 BYTES RECEIVED

SEQUENCE NUMBER

TTL

ROUND-TRIP TIME

PING TIMEOUT

IF THE DEVICE CANNOT BE REACHED:

- PING MAY SHOW THAT THE REQUEST HAS **TIMED OUT**.
- THIS INDICATES THAT YOU ARE NO LONGER ABLE TO ACCESS THAT PARTICULAR DEVICE THROUGH THE PING TEST.

IF THE NETWORK IS RESTORED:

- PING RESPONSES WILL BEGIN APPEARING AGAIN.
- RESPONSE TIMES WILL AGAIN BE DISPLAYED FOR INDIVIDUAL PINGS.

STOPPING CONTINUOUS PING

IF PING IS CONFIGURED TO RUN CONTINUOUSLY:

CTRL + C

- STOPS THE PING PROCESS.
- DISPLAYS STATISTICS FOR THE PING SESSION.

TRACEROUTE

WHAT IS TRACEROUTE?

TRACEROUTE IS ANOTHER IMPORTANT NETWORK TROUBLESHOOTING UTILITY.

ITS PURPOSE IS TO:

- TRACE THE ROUTE BETWEEN YOUR DEVICE AND A DESTINATION.
- IDENTIFY THE ROUTERS BETWEEN THE SOURCE AND DESTINATION.
- MAP THE PATH TAKEN THROUGH THE NETWORK.

COMMAND NAMES

OPERATING SYSTEM	COMMAND
LINUX	TRACEROUTE
MACOS	TRACEROUTE
WINDOWS	TRACERT

HOW TRACEROUTE USES ICMP

TRACEROUTE ALSO USES **ICMP**, BUT IT USES A DIFFERENT PART OF ICMP THAN THE PING COMMAND.

IT RELIES ON THE:

ICMP TIME EXCEEDED MESSAGE.

THIS MESSAGE IS SENT BACK TO THE ORIGINATING DEVICE WHEN THE **TTL HAS BEEN EXCEEDED**.

TTL IN TRACEROUTE

IN IP, TTL REFERS TO THE NUMBER OF **HOPS/ROUTERS** THROUGH WHICH THE PACKET CAN TRAVEL.

THE TRANSCRIPT SPECIFICALLY EMPHASIZES THAT TTL DOES **NOT** REPRESENT:

- SECONDS
- MINUTES

TRACEROUTE MANIPULATES THE TTL VALUE TO DISCOVER EACH ROUTER ALONG THE PATH.

HOW TRACEROUTE DISCOVERS ROUTERS

TRACEROUTE STARTS WITH A SMALL TTL AND PROGRESSIVELY INCREASES IT.

TTL = 1

1. TRACEROUTE SENDS A PACKET WITH TTL SET TO 1.
2. THE FIRST ROUTER RECEIVES THE PACKET.
3. THE ROUTER DECREASES THE TTL BY 1.
4. TTL BECOMES 0.
5. TTL IS EXCEEDED.
6. THE ROUTER SENDS AN **ICMP TIME EXCEEDED** MESSAGE BACK.
7. TRACEROUTE RECORDS THE ROUTER'S INFORMATION.

THIS IDENTIFIES THE **FIRST HOP**.

TTL = 2

1. TRACEROUTE SENDS ANOTHER PACKET WITH TTL 2.
2. ROUTER 1 DECREASES TTL FROM 2 TO 1.
3. THE PACKET REACHES ROUTER 2.
4. ROUTER 2 DECREASES TTL FROM 1 TO 0.

5. TTL IS EXCEEDED.
6. ROUTER 2 SENDS THE ICMP TIME EXCEEDED MESSAGE BACK.
7. TRACEROUTE RECORDS ROUTER 2.

TTL = 3

THE SAME PROCESS CONTINUES:

1. SOURCE SENDS A PACKET TOWARD THE DESTINATION.
2. ROUTER 1 FORWARDS THE PACKET AND DECREASES ITS TTL (TIME TO LIVE) BY 1.
3. ROUTER 2 FORWARDS THE PACKET AND DECREASES ITS TTL BY 1.
4. ROUTER 3 DECREASES THE TTL TO 0.
5. ROUTER 3 DISCARDS THE PACKET BECAUSE THE TTL HAS EXPIRED.
6. ROUTER 3 SENDS AN ICMP TIME EXCEEDED MESSAGE BACK TO THE SOURCE.
7. SOURCE RECEIVES THE ICMP MESSAGE, IDENTIFYING THE ROUTER WHERE THE PACKET'S TTL EXPIRED.

TRACEROUTE THEN RECORDS ROUTER 3.

INCREASING TTL

TRACEROUTE CONTINUES INCREASING THE TTL:

TTL 1 → DISCOVERS HOP 1

TTL 2 → DISCOVERS HOP 2

TTL 3 → DISCOVERS HOP 3

TTL 4 → DISCOVERS HOP 4

...

THIS ALLOWS TRACEROUTE TO PROGRESSIVELY IDENTIFY THE ROUTERS ALONG THE PATH.

TRACEROUTE AND ICMP FILTERING

ONE CHALLENGE WITH ICMP IS THAT:

- FIREWALLS AND OTHER DEVICES MAY DISABLE OR FILTER ICMP MESSAGES.

- A TTL MAY BE EXCEEDED, BUT THE ICMP RESPONSE MAY NOT REACH THE SOURCE BECAUSE IT IS FILTERED.

HOW FILTERING APPEARS

IN TRACEROUTE OUTPUT, ICMP FILTERING CAN APPEAR AS:

*

INSTEAD OF NORMAL TRACEROUTE STATISTICS.

THIS INDICATES THAT THE EXPECTED INFORMATION DID NOT MAKE IT BACK TO THE CONSOLE.

OPERATING-SYSTEM DIFFERENCES IN TRACEROUTE

TRACEROUTE CAN OPERATE DIFFERENTLY DEPENDING ON THE OPERATING SYSTEM.

WINDOWS

WINDOWS USES:

- ICMP ECHO REQUESTS
- ICMP TIME EXCEEDED MESSAGES
- ICMP ECHO REPLY WHEN THE DESTINATION IS FINALLY REACHED

LINUX AND MACOS

DIFFERENT VERSIONS OF TRACEROUTE MAY PROVIDE OPTIONS FOR CHANGING THE TYPE OF PAYLOAD BEING SENT.

THEREFORE, LINUX AND MACOS MAY PROVIDE COMMAND-LINE OPTIONS FOR CUSTOMIZING TRACEROUTE MESSAGES.

TRACEROUTE EXAMPLE: SAM TO JACK

CONSIDER A NETWORK WHERE:

1. SAM (SOURCE) SENDS A PACKET TOWARD JACK (DESTINATION).
2. THE PACKET REACHES ROUTER 1, WHERE THE TTL DECREASES BY 1.
3. THE PACKET REACHES ROUTER 2, WHERE THE TTL DECREASES BY 1.
4. THE PACKET REACHES ROUTER 3, WHERE THE TTL DECREASES BY 1.

5. THE PACKET CONTINUES THROUGH ADDITIONAL ROUTERS UNTIL IT REACHES JACK.
6. JACK (DESTINATION) RECEIVES THE PACKET IF THE ROUTING PROCESS SUCCEEDS AND THE TTL DOES NOT EXPIRE.

FIRST TEST — TTL = 1

SAM SENDS THE FIRST TRACEROUTE PACKET:

TTL = 1

ROUTER 1 RECEIVES IT AND DECREASES THE TTL:

1 → 0

THE TTL IS EXCEEDED.

ROUTER 1 SENDS AN ICMP MESSAGE BACK TO SAM.

THE ROUTER IN THE EXAMPLE IS:

10.10.10.1

THE RESPONSE TIME IS SHOWN AS APPROXIMATELY:

2 MS

SECOND TEST — TTL = 2

SAM SENDS ANOTHER PACKET:

TTL = 2

PROCESS:

1. SAM (SOURCE) SENDS A PACKET WITH A TTL OF 2.
2. ROUTER 1 RECEIVES THE PACKET AND DECREASES THE TTL FROM 2 TO 1.
3. ROUTER 2 RECEIVES THE PACKET AND DECREASES THE TTL FROM 1 TO 0.
4. ROUTER 2 DISCARDS THE PACKET BECAUSE THE TTL HAS EXPIRED.
5. ROUTER 2 SENDS AN ICMP TIME EXCEEDED MESSAGE BACK TO SAM.
6. SAM RECEIVES THE ICMP MESSAGE, INDICATING THAT THE PACKET'S TTL EXPIRED AT ROUTER 2.

ROUTER 2 IS THEREFORE IDENTIFIED AS THE NEXT HOP.

THIRD TEST — TTL = 3

THE PROCESS CONTINUES:

ROUTER 1 → TTL 3 → 2

ROUTER 2 → TTL 2 → 1

ROUTER 3 → TTL 1 → 0

ROUTER 3 SENDS THE TTL EXCEEDED MESSAGE BACK TO SAM.

TRACEROUTE RECORDS THAT INFORMATION.

TTL = 4 AND BEYOND

THE SAME PROCESS CONTINUES WITH:

TTL = 4

TTL = 5

TTL = 6

...

THE TTL IS REDUCED BY EACH ROUTER UNTIL THE DESTINATION IS EVENTUALLY REACHED.

---

7. READING TRACEROUTE OUTPUT

THE TRANSCRIPT GIVES AN EXAMPLE OF TRACING TO:

8.8.8.8

THIS IS IDENTIFIED AS A **GOOGLE DNS SERVER IP ADDRESS**.

THE OUTPUT CONTAINS MULTIPLE HOPS.

EXAMPLE STRUCTURE:

HOP	ROUTER/IP	RESPONSE TIMES
1	10.1.10.1	THREE MEASUREMENTS

2	96.120.58.137	THREE MEASUREMENTS
3	NEXT ROUTER	THREE MEASUREMENTS
4	NEXT ROUTER	THREE MEASUREMENTS
...	...	...
10	8.8.8.8	RESPONSE MEASUREMENTS

THREE MEASUREMENTS PER HOP

BY DEFAULT, TRACEROUTE PERFORMS EACH TEST **THREE TIMES** IN THE EXAMPLE DESCRIBED.

THEREFORE, EACH HOP DISPLAYS THREE RESPONSE-TIME MEASUREMENTS.

WHY TRACEROUTE OUTPUT IS USEFUL

TRACEROUTE SHOWS:

- EACH ROUTER/HOP ALONG THE PATH.
- RESPONSE TIMES.
- THE PATH BETWEEN THE LOCAL DEVICE AND DESTINATION.

IF COMMUNICATION LATER FAILS:

1. RUN TRACEROUTE AGAIN.
2. COMPARE THE NEW RESULTS WITH THE PREVIOUS RESULTS.
3. IDENTIFY WHERE A ROUTE MAY BE MISSING.
4. IDENTIFY WHICH ROUTER MAY BE PREVENTING THE ROUTE FROM CONTINUING.

DNS TROUBLESHOOTING UTILITIES

WHEN TROUBLESHOOTING A NETWORK, YOU MAY NEED TO PERFORM **NAME RESOLUTION** BETWEEN:

- AN IP ADDRESS
- A FULLY QUALIFIED DOMAIN NAME (FQDN)

DNS NORMALLY PERFORMS THIS PROCESS BEHIND THE SCENES WHEN:

- USING A BROWSER
- REFERENCING A FULLY QUALIFIED DOMAIN NAME

YOU CAN ALSO QUERY THE DNS SERVER DIRECTLY.

TWO USEFUL UTILITIES ARE:

- **NSLOOKUP**
- **DIG**

NSLOOKUP

PURPOSE

NSLOOKUP ALLOWS YOU TO QUERY A DNS SERVER AND RECEIVE DNS INFORMATION.

IT CAN BE USED TO EXAMINE INFORMATION SUCH AS:

- CANONICAL NAMES
- IP ADDRESSES
- CACHE TIMERS
- TEXT RECORDS
- OTHER INFORMATION STORED ON THE DNS SERVER

AVAILABILITY

THE TRANSCRIPT STATES THAT NSLOOKUP IS INCLUDED WITH:

- WINDOWS
- LINUX
- MACOS

DEPRECATION

THE TRANSCRIPT DESCRIBES NSLOOKUP AS **DEPRECATED**, MEANING THE GOAL IS NOT TO USE IT IN THE FUTURE.

INSTEAD, THE TRANSCRIPT RECOMMENDS USING:

DIG

DIG

DIG IS ANOTHER DNS QUERY UTILITY.

THE TRANSCRIPT STATES THAT IT IS INCLUDED WITH:

- LINUX
- MACOS

SOME VERSIONS CAN ALSO BE DOWNLOADED AND USED IN WINDOWS.

AN OLDER VERSION OF THE **BIND** PACKAGE INCLUDES AN EXECUTABLE FOR DIG.

THE TRANSCRIPT MENTIONS THAT THIS WAS AVAILABLE FROM:

ISC.ORG

NEWER VERSIONS OF THE BIND UTILITY NO LONGER INCLUDE THIS EXECUTABLE IN THE PACKAGE ACCORDING TO THE TRANSCRIPT.

NSLOOKUP EXAMPLE

EXAMPLE:

NSLOOKUP WWW.PROFESSORMESSER.COM

THE DNS SERVER USED IN THE EXAMPLE IS:

1.1.1.1

THE QUERY RETURNS THREE IP ADDRESSES:

104.22.73.108

104.22.72.108

172.67.41.114

WHY MULTIPLE IP ADDRESSES?

THE TRANSCRIPT EXPLAINS THAT REDUNDANCY IS BUILT INTO THESE SERVERS.

IF ONE IP ADDRESS IS UNAVAILABLE, OTHER IP ADDRESSES CAN BE USED TO ACCESS THE WEB SERVER.

DIG EXAMPLE

THE SAME DNS QUERY CAN BE PERFORMED WITH:

DIG WWW.PROFESSORMESSER.COM

THE OUTPUT SHOWS:

- THE QUERY BEING PERFORMED.

- THE QUESTION FOR WWW.PROFESSORMESSER.COM.
- THE REQUESTED INTERNET ADDRESS, OR **A RECORD**.
- THREE RESPONSES.

THE THREE RESPONSES ARE:

172.67.41.114

104.22.73.108

104.22.72.108

THESE MATCH THE RESULTS FROM NSLOOKUP.

NSLOOKUP VS DIG

UTILITY	PURPOSE
NSLOOKUP	QUERY DNS INFORMATION
DIG	QUERY DNS INFORMATION

THE INFORMATION OBTAINED IS ESSENTIALLY THE SAME IN THE EXAMPLE, BUT THE **OUTPUT FORMAT IS DIFFERENT**.

TCPDUMP

WHAT IS TCPDUMP?

TCPDUMP IS A COMMAND-LINE UTILITY FOR CAPTURING PACKETS FROM THE NETWORK.

IT ALLOWS YOU TO:

- CAPTURE NETWORK PACKETS.
- VIEW PACKETS DIRECTLY ON THE SCREEN.
- SAVE CAPTURED PACKETS TO A FILE.
- APPLY FILTERS.
- EXAMINE NETWORK TRAFFIC IN REAL TIME.

AVAILABILITY

THE TRANSCRIPT STATES THAT TCPDUMP IS USUALLY INCLUDED WITH:

- LINUX
- MACOS

FOR WINDOWS, A SIMILAR UTILITY IS:

WINDUMP

TCPDUMP AND PACKET CAPTURE

YOU CAN:

- CAPTURE EVERY PACKET TRAVERSING THE NETWORK.
- APPLY FILTERS.
- VIEW PACKET INFORMATION ON THE SCREEN.
- SAVE PACKET CAPTURES TO A FILE.

A LARGE AMOUNT OF PACKET INFORMATION MAY APPEAR ON THE SCREEN.

THEREFORE, YOU MAY CHOOSE TO SAVE THE CAPTURE TO A FILE AND LATER LOAD IT INTO **WIRESHARK**.

PCAP FORMAT

TCPDUMP SAVES CAPTURES IN:

PCAP — PACKET CAPTURE FORMAT

PCAP FILES CAN BE READ BY:

- WIRESHARK
- OTHER THIRD-PARTY UTILITIES

PACKET CAPTURE ANALYSIS

A PACKET CAPTURE CAN CONTAIN:

- HUNDREDS OF PACKETS
- THOUSANDS OF PACKETS

THIS CAN INITIALLY BE OVERWHELMING.

LEARNING HOW TO PARSE AND READ PACKET CAPTURES ALLOWS YOU TO GATHER INFORMATION ABOUT:

- NETWORK OPERATION
- APPLICATION OPERATION

EXAMPLE TCPDUMP OUTPUT

THE TRANSCRIPT DESCRIBES A LINUX COMMAND-LINE PACKET CAPTURE.

EACH LINE REPRESENTS A SEPARATE PACKET.

THE CAPTURE CONTAINS:

- IPV6 PACKETS
- IPV4 PACKETS

THE CAPTURE ALSO SHOWS COMMUNICATION WITH A **JABBER CLIENT**, WHICH MAY BE AN INSTANT MESSAGING APPLICATION.

IT ALSO SHOWS DNS INFORMATION WHERE A DEVICE IS PERFORMING A DNS QUERY TO THE GOOGLE DNS SERVER.

NETSTAT

WHAT IS NETSTAT?

NETSTAT STANDS FOR **NETWORK STATISTICS**.

IT ALLOWS YOU TO SEE:

- WHO YOUR SYSTEM IS COMMUNICATING WITH.
- WHO MAY BE COMMUNICATING WITH YOUR SYSTEM.
- ACTIVE NETWORK CONNECTIONS.

NETSTAT -A

THE TRANSCRIPT DESCRIBES:

NETSTAT -A

AS SHOWING:

- ALL ACTIVE CONNECTIONS COMING TO YOUR MACHINE.
- CONNECTIONS THAT YOUR MACHINE IS MAKING TO OTHER DEVICES.

NETSTAT -B

IN WINDOWS:

NETSTAT -B

SHOWS THE WINDOWS EXECUTABLE BEING USED TO MAKE THE NETWORK CONNECTION.

EXAMPLES FROM THE TRANSCRIPT INCLUDE:

- SERVICEHOST.EXE
- MICROSOFT EDGE WEBVIEW2.EXE
- HXOUTLOOK.EXE
- APPLE MOBILE DEVICE SERVICE.EXE

NETSTAT -N

NETSTAT -N

SHOWS THE SAME TYPE OF CONNECTION INFORMATION BUT DISPLAYS **IP ADDRESSES INSTEAD OF RESOLVING NAMES THROUGH DNS**.

READING NETSTAT OUTPUT

THE OUTPUT CAN SHOW:

- PROTOCOL
- LOCAL IP ADDRESS
- LOCAL TCP PORT
- REMOTE IP ADDRESSES
- CONNECTION STATUS

EXAMPLE LOCAL IP ADDRESS:

10.1.1.102

POSSIBLE CONNECTION STATES DESCRIBED IN THE TRANSCRIPT INCLUDE:

- ESTABLISHED
- CLOSING
- WAITING

USING NETSTAT -B

USING:

NETSTAT -B

ALLOWS YOU TO IDENTIFY THE EXECUTABLE ASSOCIATED WITH CONNECTIONS.

WHEN GOOGLE CHROME IS STARTED, THE OUTPUT CAN SHOW:

CHROME.EXE

ALONG WITH:

- IP ADDRESSES BEING CONTACTED.
- STATUS OF INDIVIDUAL SESSIONS.

LOCAL IP CONFIGURATION

WHEN TROUBLESHOOTING A COMPUTER, ONE OF THE FIRST USEFUL PIECES OF INFORMATION IS ITS **LOCAL IP CONFIGURATION**.

THE EXACT COMMAND DEPENDS ON THE OPERATING SYSTEM.

OPERATING SYSTEM	COMMAND
WINDOWS	IPCONFIG
LINUX	IFCONFIG
MACOS	IFCONFIG
LINUX/MACOS NEWER COMMAND	IP ADDRESS

WINDOWS IPCONFIG

RUN:

IPCONFIG

IT CAN SHOW:

- IPV6 INFORMATION
- IPV4 ADDRESS
- SUBNET MASK
- DEFAULT GATEWAY

THIS INFORMATION CAN THEN BE USED FOR NETWORK TROUBLESHOOTING.

WINDOWS IPCONFIG /ALL

FOR ADDITIONAL INFORMATION:

IPCONFIG /ALL

THE TRANSCRIPT STATES THAT THIS CAN DISPLAY:

- DEVICE NAME
- NODE TYPE
- MAC ADDRESS
- ADAPTER INFORMATION
- IP ADDRESS INFORMATION
- DHCP DETAILS
- DNS SERVERS

THIS PROVIDES A MORE DETAILED VIEW OF THE SYSTEM'S NETWORK CONFIGURATION.

LINUX/MACOS IFCONFIG

ON LINUX OR MACOS:

IFCONFIG

YOU CAN SPECIFY AN ADAPTER.

EXAMPLE:

IFCONFIG EN0

THE TRANSCRIPT IDENTIFIES EN0 AS AN ETHERNET ADAPTER.

THE OUTPUT CAN INCLUDE:

- MAC ADDRESS
- IPV6 ADDRESS
- IPV4 ADDRESS
- OTHER ADAPTER CONFIGURATION DETAILS

EXAMPLE IPV4 ADDRESS:

10.1.10.249

ARP CACHE

KNOWING THE IP ADDRESS OF A DEVICE IS USEFUL, BUT SOMETIMES YOU ALSO NEED TO KNOW ITS **MAC ADDRESS**.

THIS CAN BE PARTICULARLY USEFUL WHEN LOOKING FOR A DEVICE IN THE MAC ADDRESS TABLE OF A SWITCH.

ONE PLACE TO FIND THE IP-TO-MAC RELATIONSHIP IS THE LOCAL **ARP CACHE**.

ARP

ARP = ADDRESS RESOLUTION PROTOCOL

THE ARP CACHE IS MAINTAINED ON THE LOCAL MACHINE.

ARP -A

TO VIEW THE LOCAL ARP CACHE:

ARP -A

THIS DISPLAYS:

- IP ADDRESSES
- CORRESPONDING MAC ADDRESSES

FOR DEVICES ON THE LOCAL SUBNET THAT HAVE PREVIOUSLY BEEN RESOLVED.

THE COMMAND WORKS IN:

- LINUX
- MACOS
- WINDOWS

ARP CACHE EXAMPLE

SUPPOSE:

10.1.10.234

IS NOT CURRENTLY PRESENT IN THE ARP CACHE.

THE DEVICE CAN BE CONTACTED:

PING 10.1.10.234

AFTER RECEIVING A RESPONSE, RUNNING:

ARP -A

AGAIN SHOWS:

10.1.10.234

ALONG WITH ITS ASSOCIATED MAC ADDRESS.

WHY DID IT APPEAR?

THE DEVICE WAS NOT PREVIOUSLY IN THE ARP CACHE BECAUSE THERE HAD BEEN NO RECENT COMMUNICATION WITH IT.

AFTER COMMUNICATING WITH THE DEVICE:

1. THE DEVICE RESPONDS.
2. ITS IP-TO-MAC RELATIONSHIP IS RESOLVED.
3. THE INFORMATION IS STORED LOCALLY.
4. ARP -A CAN DISPLAY THE NEWLY STORED ENTRY.

TONE GENERATOR AND INDUCTIVE PROBE

NETWORK ADMINISTRATORS MAY WORK WITH HUNDREDS OF CABLES, ESPECIALLY IN:

- DATA CENTERS
- IDFS
- MDFS
- PATCH PANELS
- PUNCH-DOWN LOCATIONS

FINDING ONE SPECIFIC CABLE INSIDE A LARGE BUNDLE CAN BE DIFFICULT.

A **TONE GENERATOR** AND **INDUCTIVE PROBE** CAN MAKE THIS PROCESS EASIER.

TONE GENERATOR

A TONE GENERATOR:

- PLACES AN ANALOG TONE ONTO THE WIRE.
- ALLOWS YOU TO IDENTIFY THE CABLE CARRYING THAT TONE.

THE TONE GENERATOR CAN BE CONNECTED THROUGH:

- A MODULAR JACK
- COAX CONNECTIONS
- A PUNCH-DOWN BLOCK
- ALLIGATOR CLIPS

INDUCTIVE PROBE

AN **INDUCTIVE PROBE** LISTENS FOR THE TONE.

IT DOES NOT REQUIRE DIRECT CONNECTION TO THE CABLE.

YOU CAN MOVE THE PROBE NEAR CABLES IN A LARGE GROUP AND LISTEN FOR THE TONE.

BASIC PROCESS

1. TONE GENERATOR: SENDS A TONE SIGNAL THROUGH THE TARGET CABLE.
2. TARGET CABLE: CARRIES THE TONE SIGNAL ALONG THE CABLE.
3. LARGE CABLE BUNDLE: THE TARGET CABLE IS LOCATED AMONG MANY BUNDLED CABLES.
4. INDUCTIVE PROBE: DETECTS THE TONE SIGNAL TO HELP IDENTIFY AND TRACE THE CORRECT CABLE.

STEPS:

1. CONNECT THE TONE GENERATOR TO ONE END OF THE TARGET CABLE.
2. MOVE THE INDUCTIVE PROBE AROUND THE CABLE BUNDLE.
3. LISTEN FOR THE TONE.
4. THE STRONGEST/IDENTIFIED TONE INDICATES THE CABLE CONNECTED TO THE TONE GENERATOR.
5. FOLLOW THAT CABLE TO LOCATE ITS END.

CABLE TESTER

A **CABLE TESTER** PERFORMS A SIMPLE CONTINUITY TEST.

IT VERIFIES WHETHER:

PIN 1 → PIN 1

PIN 2 → PIN 2

...

ARE PROPERLY CONNECTED FROM ONE END OF THE CABLE TO THE OTHER.

CABLE TESTER CAN IDENTIFY

- UNCONNECTED PINS
- SHORTS
- CROSSED WIRES

EXAMPLE:

PIN 1 → PIN 3

PIN 2 → PIN 6

WOULD INDICATE CROSSED WIRING.

LIMITATIONS

THE SIMPLE CONTINUITY TEST DOES **NOT** PROVIDE INFORMATION ABOUT:

- HOW MUCH SIGNAL THE CABLE CAN SUPPORT.
- WHAT CATEGORY OF CABLE IS BEING USED.

HOWEVER, IT CAN VERIFY:

- CORRECT PUNCH-DOWN
- CORRECT CONNECTORS
- CONTINUITY BETWEEN BOTH ENDS

TONE GENERATOR AS A CABLE TESTER

THE TONE GENERATOR AND INDUCTIVE PROBE DESCRIBED EARLIER CAN ALSO FUNCTION AS A CABLE TESTER.

THE INDUCTIVE PROBE HAS AN **RJ45 CONNECTOR**.

WHEN THE CABLE IS CONNECTED AND THE TESTER IS PLACED INTO CABLE-TESTING MODE:

PINS 1 → 8

SHOULD LIGHT UP IN ORDER.

IF PINS 1 THROUGH 8 LIGHT UP IN THE CORRECT ORDER, THIS INDICATES:

- CORRECT WIRING
- CONTINUITY ACROSS ALL EIGHT CONNECTIONS

NETWORK TAP

A **NETWORK TAP** CAN BE USED WHEN YOU NEED TO CAPTURE DATA FROM THE NETWORK.

ITS PURPOSE IS TO:

- INTERCEPT NETWORK TRAFFIC.
- SEND A COPY OF THE TRAFFIC TO A PROTOCOL ANALYZER OR PACKET-CAPTURE DEVICE.

PHYSICAL TAP

A PHYSICAL TAP IS PLACED DIRECTLY INTO THE NETWORK CONNECTION.

BASIC STRUCTURE:

DEVICE A (SENDER) SENDS NETWORK TRAFFIC.

TAP (NETWORK TEST ACCESS POINT) COPIES THE TRAFFIC FOR MONITORING.

DEVICE B (RECEIVER) RECEIVES THE ORIGINAL NETWORK TRAFFIC.

MONITORING DEVICE CONNECTS TO THE TAP'S MONITORING PORT TO CAPTURE AND ANALYZE A COPY OF THE TRAFFIC.

INSTALLING A PHYSICAL TAP MAY REQUIRE SOME INITIAL DOWNTIME BECAUSE THE CONNECTION MUST BE PHYSICALLY BROKEN TO INSERT THE TAP.

PASSIVE TAP

A PASSIVE TAP:

- DOES NOT REQUIRE POWER.

- IS COMMONLY USED WITH OPTICAL FIBER ACCORDING TO THE TRANSCRIPT.

ACTIVE TAP

AN ACTIVE TAP:

- REQUIRES ADDITIONAL POWER.
- REGENERATES THE SIGNAL THROUGH THE TAP.

PORT MIRRORING / SPAN

IF A PHYSICAL TAP IS UNAVAILABLE, SIMILAR FUNCTIONALITY MAY BE AVAILABLE DIRECTLY INSIDE A SWITCH.

MANY SWITCHES SUPPORT:

- **PORT MIRRORING**
- **SPAN**

SPAN MEANS:

SWITCHED PORT ANALYZER

THE SWITCH CAN BE CONFIGURED TO:

1. TAKE TRAFFIC FROM ONE OR MORE INTERFACES.
2. MAKE A COPY OF THAT TRAFFIC.
3. SEND THE COPIED TRAFFIC TO ANOTHER SWITCH INTERFACE.
4. CONNECT A PROTOCOL ANALYZER TO THAT DESTINATION INTERFACE.

BASIC STRUCTURE

NETWORK TRAFFIC: DATA FLOWS BETWEEN DEVICES ACROSS THE NETWORK.

SWITCH: FORWARDS NETWORK TRAFFIC TO THE APPROPRIATE PORTS.

PORT MIRROR / SPAN: COPIES TRAFFIC FROM SELECTED SWITCH PORTS OR VLANS.

ANALYZER PORT: RECEIVES THE MIRRORED TRAFFIC.

PROTOCOL ANALYZER: ANALYZES CAPTURED PACKETS USING A TOOL SUCH AS WIRESHARK.

LIMITATION OF PORT MIRRORING

ONE CHALLENGE WITH PORT MIRRORING IS **BANDWIDTH LIMITATION**.

EXAMPLE FROM THE TRANSCRIPT:

- SOURCE CONNECTION CAN CARRY **2 GBIT/S TOTAL**.
- THIS COULD MEAN:
    - 1 GBIT/S IN ONE DIRECTION
    - 1 GBIT/S IN THE OTHER DIRECTION
- THE ANALYZER CONNECTION HAS ONLY **1 GBIT/S** AVAILABLE.

THEREFORE, A PORT MIRROR MAY NOT BE SUITABLE FOR VERY LARGE-BANDWIDTH TRAFFIC.

FOR VERY LARGE BANDWIDTH USE, THE TRANSCRIPT SUGGESTS USING A **PHYSICAL TAP** INSTEAD.

PHYSICAL TAP EXAMPLE

A TAP CAN BE USED WITH:

- COAX
- FIBER
- TWISTED-PAIR COPPER

THE NETWORK DEVICES CONNECT ON BOTH SIDES OF THE TAP.

THE TAP HAS:

- TRANSMIT CONNECTIONS
- RECEIVE CONNECTIONS
- NETWORK CONNECTIONS
- MONITOR PORTS

THE TAP COMPLETES THE NETWORK CIRCUIT WHILE PROVIDING A COPY OF THE TRAFFIC THROUGH THE MONITOR PORTS.

THE MONITORING PORTS CAN BE CONNECTED TO:

- PROTOCOL ANALYZERS

- ANALYSIS TOOLS
- PACKET-CAPTURE DEVICES

WIRELESS SURVEY TOOLS

WIRELESS NETWORKS REQUIRE DIFFERENT TROUBLESHOOTING TOOLS BECAUSE COMMUNICATION OCCURS OVER THE AIR.

A **WIRELESS SURVEY TOOL** CAN PROVIDE INFORMATION ABOUT:

- WIRELESS COVERAGE
- RECEIVED SIGNAL
- POTENTIAL INTERFERENCE
- NEARBY ACCESS POINTS

THE TOOL MAY BE:

- BUILT INTO AN OPERATING SYSTEM
- BUILT INTO A WIRELESS ACCESS POINT
- A THIRD-PARTY APPLICATION

WI-FI ANALYZER

FOR MORE DETAILED WIRELESS INFORMATION, SPECIALIZED HARDWARE MAY BE REQUIRED.

A **WI-FI ANALYZER** CAN PROVIDE INFORMATION ABOUT:

- 802.11 TRAFFIC
- CHANNEL STRENGTH
- CHANNELS CURRENTLY IN USE
- ACCESS POINTS IN THE AREA
- POTENTIAL INTERFERENCE SOURCES

SPECTRUM ANALYZER

MORE ADVANCED WI-FI ANALYZERS MAY INCLUDE A **SPECTRUM ANALYZER**.

A SPECTRUM ANALYZER ALLOWS YOU TO VIEW:

- AN ENTIRE RANGE OF FREQUENCIES
- SIGNALS BEING BROADCAST ACROSS THOSE FREQUENCIES

THIS INCLUDES SIGNALS ASSOCIATED WITH:

- 802.11 WIRELESS NETWORKS
- OTHER SOURCES

INTERFERENCE INVESTIGATION

IF INTERFERENCE MAY BE COMING FROM:

- THIRD-PARTY EQUIPMENT
- A MICROWAVE OVEN

A SPECTRUM ANALYZER CAN HELP VERIFY THE SOURCE OF THE INTERFERENCE.

SIGNAL-TO-NOISE RATIO

A WI-FI ANALYZER CAN DISPLAY THE RELATIONSHIP BETWEEN:

- WIRELESS SIGNAL
- NOISE

THE TRANSCRIPT GIVES TWO EXAMPLES.

EXAMPLE 1

THERE IS:

- SOME NOISE AT THE BOTTOM.
- SOME WIRELESS SIGNAL ABOVE IT.

THE SIGNAL AND NOISE ARE RELATIVELY CLOSE TOGETHER.

THE TRANSCRIPT DESCRIBES THE **SIGNAL-TO-NOISE RATIO** AS NARROW.

EXAMPLE 2

ANOTHER VIEW SHOWS:

- A MUCH STRONGER SIGNAL.
- MUCH MORE SIGNAL THAN NOISE.

THIS REPRESENTS A STRONGER WIRELESS ENVIRONMENT COMPARED WITH THE FIRST EXAMPLE.

WIRELESS ANALYSIS INFORMATION

A WI-FI ANALYZER CAN PROVIDE SIGNIFICANT INFORMATION ABOUT HOW WELL A WIRELESS NETWORK IS OPERATING.

IT CAN HELP IDENTIFY:

- CHANNEL USAGE
- SIGNAL STRENGTHS
- ACCESS POINTS
- POTENTIAL INTERFERENCE
- OTHER SIGNALS SHARING FREQUENCIES

VISUAL FAULT LOCATOR

FIBER-OPTIC NETWORKS MAY DEVELOP PROBLEMS BECAUSE OF:

- BREAKS IN THE FIBER.
- EXCESSIVE BENDS.
- FIBER BEING IN POOR CONDITION.

A **VISUAL FAULT LOCATOR (VFL)** CAN BE USED TO IDENTIFY THESE PROBLEMS.

WHAT IS A VISUAL FAULT LOCATOR?

THE TRANSCRIPT DESCRIBES IT AS SIMILAR TO A FLASHLIGHT SPECIFICALLY DESIGNED FOR OPTICAL FIBER.

IT SHINES LIGHT THROUGH THE FIBER.

IF THERE IS A BREAK:

- LIGHT LEAKS OUT AROUND THE DAMAGED AREA.
- THE LEAK CAN REVEAL THE LOCATION OF THE FAULT.

TURNING OFF THE ROOM LIGHTS CAN MAKE THE LEAKING LIGHT EASIER TO SEE.

ADVANTAGES

THE VFL IS RELATIVELY LOW-TECH BUT EFFECTIVE FOR FINDING FIBER PROBLEMS.

IT CAN HELP IDENTIFY FAULTS BEFORE INSTALLING A PARTICULAR FIBER PATCH INTO THE NETWORK.

NETWORK EQUIPMENT COMMAND-LINE INTERFACES

NETWORK ADMINISTRATORS OFTEN WORK WITH EQUIPMENT FROM DIFFERENT MANUFACTURERS.

FORTUNATELY:

- COMMAND-LINE CONCEPTS ARE OFTEN SIMILAR.
- COMMANDS MAY BE SIMILAR BETWEEN MANUFACTURERS.
- OUTPUT OFTEN CONTAINS SIMILAR INFORMATION.

HOWEVER:

THE SYNTAX IS NOT NECESSARILY EXACTLY THE SAME BETWEEN MANUFACTURERS.

LEARNING THE TECHNOLOGY BEHIND THE COMMANDS MAKES IT EASIER TO MOVE BETWEEN DIFFERENT MANUFACTURERS' EQUIPMENT.

MAC ADDRESS TABLE

A SWITCH MAINTAINS A **MAC ADDRESS TABLE**.

THE TABLE CONTAINS:

- MAC ADDRESSES LEARNED BY THE SWITCH.
- THE PORTS ON WHICH THOSE MAC ADDRESSES WERE LEARNED.

WHEN THE SWITCH RECEIVES TRAFFIC:

1. IT EXAMINES THE DESTINATION MAC ADDRESS.
2. IT LOOKS UP THAT MAC ADDRESS IN THE TABLE.
3. IT IDENTIFIES THE ASSOCIATED PORT.
4. IT SENDS THE TRAFFIC DIRECTLY THROUGH THAT PORT.

SHOW MAC ADDRESS TABLE

ON CISCO EQUIPMENT, THE COMMAND DESCRIBED IS:

SHOW MAC ADDRESS TABLE

IT CAN BE USEFUL WHEN TROUBLESHOOTING:

- TRAFFIC BEING SENT OUT EVERY INTERFACE.
- MAXIMUM MAC ADDRESS TABLE ENTRIES.
- MAC ADDRESSES LEARNED BY THE SWITCH.

SHOW ROUTE

ROUTING INFORMATION CAN BE VIEWED USING:

SHOW ROUTE

THIS DISPLAYS ROUTES INSIDE THE ROUTING TABLE.

IT CAN BE USED TO DETERMINE:

- WHERE ROUTES EXIST.
- WHICH ROUTE APPLIES TO SPECIFIC TRAFFIC.
- WHICH INTERFACE SHOULD BE USED.
- THE PATH TRAFFIC SHOULD TAKE THROUGH THE NETWORK.

ROUTE TROUBLESHOOTING PROCESS

1. VIEW THE ROUTING TABLE.
2. FIND THE ROUTE APPLICABLE TO THE TRAFFIC.
3. IDENTIFY THE NEXT HOP.
4. IDENTIFY THE OUTGOING INTERFACE.
5. MOVE TO THE NEXT ROUTER.
6. REPEAT THE PROCESS THROUGH THE NETWORK.

THIS ALLOWS YOU TO TRACK TRAFFIC FROM ONE END OF THE NETWORK TO ANOTHER.

ROUTING TABLE EXAMPLE

THE EXAMPLE USES A CISCO ROUTER.

THE ROUTING TABLE BEGINS WITH ABBREVIATIONS THAT DESCRIBE HOW ROUTES WERE LEARNED.

TWO ABBREVIATIONS SHOWN IN THE EXAMPLE ARE:

CODE	MEANING
R	RIP
C	CONNECTED

R — RIP

R INDICATES A ROUTE LEARNED THROUGH THE **RIP PROTOCOL**.

C — CONNECTED

C INDICATES THAT THE NETWORK IS DIRECTLY CONNECTED TO AN INTERFACE OF THE ROUTER.

EXAMPLE ROUTE

THE EXAMPLE CONTAINS A ROUTE FOR:

1.0.0.0/8

TRAFFIC FOR THIS NETWORK IS SENT THROUGH:

20.20.0.2

THE NEXT HOP IS REACHED THROUGH:

SERIAL 3/0

MORE SPECIFIC ROUTES

WHEN TROUBLESHOOTING ROUTING:

- THERE MAY BE MULTIPLE ROUTES.
- A MORE SPECIFIC ROUTE MAY APPLY TO THE TRAFFIC.
- EACH ROUTE SHOULD BE EVALUATED TO DETERMINE WHICH ONE APPLIES.

SHOW INTERFACE

IF THE PROBLEM IS NOT WITH ROUTING OR SWITCHING, THE ISSUE MAY INVOLVE AN INDIVIDUAL INTERFACE.

THE COMMAND:

SHOW INTERFACE

CAN PROVIDE INFORMATION ABOUT THE INTERFACE.

IT CAN SHOW WHETHER THE INTERFACE IS:

- UP
- DOWN
- CONNECTED
- ADMINISTRATIVELY DISABLED

IT CAN ALSO PROVIDE INFORMATION ABOUT:

- SPEED
- MTU
- ENCAPSULATION
- CRC ERRORS
- DROPPED FRAMES
- INPUT ERRORS
- OUTPUT ERRORS
- NUMBER OF FRAMES
- NUMBER OF BROADCASTS
- INPUT/OUTPUT RATES

SHOW INTERFACE EXAMPLE

THE EXAMPLE USES:

FASTETHERNET 0/0

THIS INDICATES:

- A 100-MEGABIT INTERFACE.
- SLOT 0
- PORT 0

THE INTERFACE IS SHOWN AS:

INTERFACE: UP

LINE PROTOCOL: UP

THIS INDICATES THAT THE INTERFACE IS RECEIVING SIGNAL.

THE EXAMPLE ALSO SHOWS:

- FULL DUPLEX
- 100 MBIT/S
- RJ45 MEDIA TYPE
- INPUT RATES
- OUTPUT RATES
- CRC ERRORS
- OTHER INTERFACE PERFORMANCE INFORMATION

CONFIGURATION VIEWING

NETWORK DEVICES MAY PROVIDE CONFIGURATION INFORMATION THROUGH:

- A WEB INTERFACE
- A TEXT-BASED COMMAND-LINE INTERFACE

SOME DEVICES ONLY SUPPORT A TEXT-BASED CONFIGURATION VIEW.

THE TRANSCRIPT DESCRIBES COMMANDS SUCH AS:

SHOW CONFIG

AND, ON THE CISCO DEVICE EXAMPLE:

SHOW RUNNING-CONFIG

SHOW RUNNING-CONFIG

THE COMMAND:

SHOW RUNNING-CONFIG

SHOWS THE CONFIGURATION CURRENTLY ACTIVE INSIDE THE ROUTER.

THE CONFIGURATION CAN CONTAIN:

- DEVICE INFORMATION
- VERSION INFORMATION
- TIMESTAMPS
- ROUTER INFORMATION

- IP ADDRESSES
- SUBNET MASKS
- PORT/INTERFACE INFORMATION
- OTHER CONFIGURATION DETAILS

THE EXAMPLE CONFIGURATION IS **830 BYTES**.

MANUFACTURERS USE DIFFERENT:

- FORMATS
- SYNTAX
- CONFIGURATION LAYOUTS

THEREFORE, YOU NEED TO BECOME FAMILIAR WITH THE PARTICULAR MANUFACTURER'S CONFIGURATION FORMAT.

ARP CACHE ON SWITCHES AND ROUTERS

ARP INFORMATION CAN ALSO BE VIEWED ON SWITCHES AND ROUTERS.

THE COMMAND DESCRIBED IS:

SHOW ARP

THIS DISPLAYS THE **ADDRESS RESOLUTION PROTOCOL CACHE** INSIDE THE DEVICE.

IT IS SIMILAR TO THE ARP CACHE MAINTAINED BY AN OPERATING SYSTEM.

THE OUTPUT CAN SHOW:

- PROTOCOL
- IP ADDRESS
- HARDWARE/MAC ADDRESS
- INTERFACE USED TO IDENTIFY THE PAIR

THIS CAN HELP DETERMINE WHETHER A SWITCH OR ROUTER HAS IDENTIFIED A PARTICULAR MAC ADDRESS ASSOCIATED WITH AN IP ADDRESS.

VLAN INFORMATION

SWITCHES COMMONLY USE VLAN CONFIGURATIONS.

TO VIEW VLAN INFORMATION, THE TRANSCRIPT DESCRIBES:

SHOW VLAN

THIS CAN LIST:

- VLANS ON THE DEVICE.
- VLAN IDS.
- INTERFACES ASSIGNED TO EACH VLAN.
- DEFAULT VLAN.
- INTERFACES BELONGING TO THE DEFAULT VLAN.
- INTERFACES ASSIGNED TO OTHER VLANS.

VLAN TROUBLESHOOTING USE

IF YOU ARE TROUBLESHOOTING A VLAN-RELATED PROBLEM, THIS COMMAND ALLOWS YOU TO VERIFY WHICH INTERFACES ARE ASSOCIATED WITH WHICH VLANS.

POE MONITORING

MANY SWITCHES SUPPORT **POWER OVER ETHERNET (POE)**.

POE ALLOWS ETHERNET CABLES TO PROVIDE POWER TO DEVICES CONNECTED TO THE SWITCH.

BECAUSE THE SWITCH HAS A LIMITED POWER CAPACITY, IT IS IMPORTANT TO MONITOR:

- POE STATUS
- POWER CONSUMPTION
- REMAINING POWER CAPACITY

THE COMMAND DESCRIBED IN THE TRANSCRIPT IS:

SHOW POWER

SHOW POWER

THE COMMAND CAN DISPLAY:

- SWITCH INTERFACES
- WHETHER POE IS ENABLED OR DISABLED

- POWER BEING USED BY EACH CONNECTED DEVICE
- REMAINING POWER CAPACITY

EXAMPLE

THE EXAMPLE SWITCH SUPPORTS:

370 W

CURRENT USAGE:

40 W

REMAINING CAPACITY:

330 W

THEREFORE, ACCORDING TO THE EXAMPLE, THERE IS ENOUGH REMAINING POWER TO ADD ADDITIONAL POE DEVICES.

COMMAND-LINE TROUBLESHOOTING REFERENCE

PURPOSE	COMMAND / UTILITY
TEST DEVICE REACHABILITY	PING
TRACE NETWORK PATH	TRACEROUTE
WINDOWS TRACEROUTE	TRACERT
QUERY DNS	NSLOOKUP
QUERY DNS	DIG
CAPTURE PACKETS	TCPDUMP
WINDOWS PACKET CAPTURE UTILITY MENTIONED	WINDUMP
VIEW NETWORK STATISTICS/CONNECTIONS	NETSTAT
SHOW ACTIVE CONNECTIONS	NETSTAT -A
SHOW EXECUTABLE ASSOCIATED WITH CONNECTION IN WINDOWS	NETSTAT -B
SHOW IP ADDRESSES WITHOUT NAME RESOLUTION	NETSTAT -N
WINDOWS IP CONFIGURATION	IPCONFIG

DETAILED WINDOWS IP CONFIGURATION	IPCONFIG /ALL
LINUX/MACOS INTERFACE CONFIGURATION	IFCONFIG
NEWER LINUX/MACOS IP COMMAND MENTIONED	IP ADDRESS
VIEW LOCAL ARP CACHE	ARP -A
VIEW SWITCH MAC ADDRESS TABLE	SHOW MAC ADDRESS TABLE
VIEW ROUTING TABLE	SHOW ROUTE
VIEW INTERFACE INFORMATION	SHOW INTERFACE
VIEW CONFIGURATION	SHOW CONFIG
VIEW ACTIVE CISCO CONFIGURATION	SHOW RUNNING-CONFIG
VIEW ARP CACHE ON NETWORK DEVICE	SHOW ARP
VIEW VLAN INFORMATION	SHOW VLAN
VIEW POE INFORMATION	SHOW POWER

OVERALL TROUBLESHOOTING WORKFLOW FROM THE TRANSCRIPT

WHEN TROUBLESHOOTING A NETWORK DEVICE OR CONNECTION, THE UTILITIES AND COMMANDS DISCUSSED CAN BE USED TO PROGRESSIVELY GATHER INFORMATION.

STEP 1 — CHECK REACHABILITY

USE:

PING

DETERMINE WHETHER THE DESTINATION RESPONDS.

STEP 2 — IDENTIFY THE NETWORK PATH

USE:

TRACEROUTE

OR ON WINDOWS:

TRACERT

DETERMINE THE ROUTERS/HOPS BETWEEN THE SOURCE AND DESTINATION.

STEP 3 — CHECK DNS

USE:

NSLOOKUP

OR:

DIG

CHECK HOW A HOSTNAME RESOLVES AND INSPECT DNS INFORMATION.

STEP 4 — CAPTURE NETWORK TRAFFIC

USE:

TCPDUMP

OR A SIMILAR PACKET-CAPTURE UTILITY.

SAVE CAPTURES AS **PCAP** WHEN NEEDED FOR LATER ANALYSIS WITH WIRESHARK.

STEP 5 — CHECK LOCAL CONNECTIONS

USE:

NETSTAT

DETERMINE:

- LOCAL CONNECTIONS
- REMOTE CONNECTIONS
- PORTS
- CONNECTION STATES
- EXECUTABLES ASSOCIATED WITH CONNECTIONS WHEN USING -B IN WINDOWS

STEP 6 — CHECK LOCAL IP CONFIGURATION

USE:

IPCONFIG

OR:

IFCONFIG

CHECK:

- IP ADDRESS
- SUBNET MASK
- DEFAULT GATEWAY
- DNS
- DHCP INFORMATION WHEN AVAILABLE

STEP 7 — CHECK ARP

USE:

ARP -A

DETERMINE THE RELATIONSHIP BETWEEN:

IP ADDRESS ↔ MAC ADDRESS

STEP 8 — CHECK PHYSICAL CABLING

USE:

- TONE GENERATOR
- INDUCTIVE PROBE
- CABLE TESTER

THESE HELP LOCATE CABLES AND VERIFY CONTINUITY/WIRING.

STEP 9 — CAPTURE TRAFFIC FROM THE NETWORK

USE:

- PHYSICAL TAP
- PORT MIRRORING/SPAN

SEND TRAFFIC TO A PROTOCOL ANALYZER OR PACKET-CAPTURE DEVICE.

STEP 10 — TROUBLESHOOT WIRELESS

USE:

- WIRELESS SURVEY TOOLS
- WI-FI ANALYZER
- SPECTRUM ANALYZER

CHECK:

- COVERAGE

- SIGNAL
- CHANNELS
- ACCESS POINTS
- INTERFERENCE
- SIGNAL-TO-NOISE RELATIONSHIP

STEP 11 — TROUBLESHOOT FIBER

USE A:

VISUAL FAULT LOCATOR

TO IDENTIFY BREAKS OR FAULTS WHERE LIGHT IS LEAKING FROM THE FIBER.

STEP 12 — TROUBLESHOOT NETWORK DEVICES

USE COMMANDS SUCH AS:

SHOW MAC ADDRESS TABLE

SHOW ROUTE

SHOW INTERFACE

SHOW RUNNING-CONFIG

SHOW ARP

SHOW VLAN

SHOW POWER

THESE PROVIDE INFORMATION ABOUT:

- MAC ADDRESS LEARNING
- ROUTING
- INTERFACES
- CONFIGURATION
- ARP
- VLANS
- POE POWER USAGE

KEY CONCEPTS TO REMEMBER

PING

TESTS WHETHER A DEVICE IS REACHABLE USING ICMP.

TRACEROUTE

MAPS THE PATH BETWEEN A SOURCE AND DESTINATION BY MANIPULATING TTL VALUES AND OBSERVING ICMP RESPONSES.

NSLOOKUP / DIG

QUERY DNS INFORMATION.

TCPDUMP

CAPTURES NETWORK PACKETS FROM THE COMMAND LINE AND CAN SAVE THEM AS PCAP FILES.

netstat

Shows network connections and statistics.

ipconfig / ifconfig

Shows local IP configuration.

ARP

Provides the relationship between IP addresses and MAC addresses.

Tone Generator + Inductive Probe

Helps locate a specific cable within a large group of cables.

Cable Tester

Checks cable continuity and wiring.

Network TAP

Copies network traffic for analysis.

SPAN / Port Mirroring

Copies switch traffic to another interface for analysis.

Wireless Survey Tool

Provides information about wireless coverage and potential interference.

Wi-Fi Analyzer

Provides detailed wireless and channel information.

Spectrum Analyzer

Shows signals across a range of frequencies and can help identify interference sources.

Visual Fault Locator

Uses light to identify breaks in fiber.

show mac address table

Displays MAC addresses learned by a switch and their associated ports.

show route

Displays routing information.

show interface

Displays interface status, speed, duplex, errors, and performance information.

show running-config

Displays the currently active device configuration.

show arp

Displays the ARP cache on a network device.

show vlan

Displays VLANs and their assigned interfaces.

show power

Displays PoE status and power consumption.